

Positive Grassmannians and amplituhedra: canonical forms and refinement

Lattice duality and the Yang–Mills mass gap
in Triadic Modular Holography

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Abstract

Triadic Modular Holography constructs the joint discrete structure of the continuum through APP, TRIT and the TPK's transport with memory. Compatible cylinder reduction produces the arithmetic realization, while the conjugate projection preserves exceptional incidence. Within this genealogy, a collection of recoverable positive realizations is established. The uniform charge and two nonadic memories determine the twelve-coordinate record; its ordered separations produce a planar boundary measurement and a Grassmannian representation with a marked inverse. Readers that factor through the record are transported to this representation and preserved under coordinate changes of the same point. Nonadic refinement constructs positive flags and amplituhedral points in every admissible finite rank and external dimension. In rank one, complete simplicial coverage and cancellation of interior residues are obtained in every finite dimension. Forms valued in the action line satisfy an exact gluing condition on common boundaries. Cell averages preserve a chronological measure and decompose moment increments into orthogonal details; their limit determines a Jacobi operator whose finite compressions have simple spectrum, positive residues and continued-fraction resolvents. Schur reduction preserves the effective energy of the partition, with reconstruction of the detail and a limiting Sobolev realization. The record flow also induces a twenty-four-dimensional lattice deformation that preserves volume and ternary symmetry, and converts parameter inversion into lattice duality and theta reciprocity. The elliptic, hyperbolic and parabolic Hamiltonians, their symplectic charts, connection terms and Legendre transforms are brought together with the corresponding material readers. The Yang–Mills development incorporates the common measure, the areal–volumetric scale, uniform coercivity, the persistent spectral witness and Hamiltonian reconstruction. The articulation of these results distinguishes and relates Plücker positivity, moment positivity, constitutive positivity and spectral positivity through explicit operations on the same enriched state. The incidence representation preserves all twelve coordinates through a weighted isometry with an explicit inverse. The hexad–octad transition distinguishes total area from sector composition and reconstructs occupations; its purification determines an entanglement entropy with a finite modular law. The longitudinal composition of marked return determines the areal scale and preserves its naturality under inherited refinements; canonical forms with an area-operator coefficient transmit additivity and boundary residues.

Keywords: positive Grassmannians; planar networks; Plücker coordinates; incidence geometry; Triadic Modular Holography.

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1 Introduction: genealogy of form and conservation under refinement

The guiding question of this work is the conservation of a generated structure when its mathematical realization changes. Triadic Modular Holography produces, through lifted arithmetic operations, states that simultaneously retain orientation, incidence, boundary and memory. Its discrete structure of the continuum assembles the compatible prolongations of these states. The Archimedean realization yields coordinates through cylinder refinement; the conjugate realization preserves exceptional incidence relations. Genealogy thus precedes both the values and the forms in which they are represented. This is the initial thesis that the following formal development sets out with its operations and dependencies.

The article's specific geometric problem begins when a positive output of that construction becomes a separation between nodes, a network weight, a Plücker coordinate, a moment of a measure, an energy coefficient or a direction of lattice deformation. Justifying this passage requires determining what information each representation recovers, how it compares with the others and what its refinement preserves. The significance of a positive realization increases decisively when it admits a marked inverse: geometry then ceases to provide only an evaluation of the state and also preserves the information used by its readers. The marking comprises the order of positions, the total scale and the enriched data required by the particular reader.

The central contribution is organized around three operations. The first is reconstruction: charge and memory recover the coordinates of the record, and ratios of minors recover its separations. The second is compatible refinement: a partition is subdivided while preserving its endpoints, its markings and an exact record of the detail. The third is transport of structures: a form, an operator or a reader is transferred to another representation by an explicit map. These operations establish conservation of residues, effective energy, moments and spectra in the specific domains where each identity has meaning. The adjective positive thus acquires several precise mathematical realizations, articulated through reconstruction of the same datum.

Positive geometry has its own antecedents, which allow the level of this contribution to be identified precisely. Boundary measurements of directed networks parametrize cells of the totally nonnegative Grassmannian in Postnikov's work [10]; Scott describes the cluster-algebra structure of the Grassmannian coordinate ring [11]. Arkani-Hamed and Trnka introduce the amplituhedron in the geometric description of amplitudes in supersymmetric $\mathcal{N} = 4$ Yang–Mills theory in the planar limit [12]. The formulation of positive geometries subsequently characterizes their canonical forms by logarithmic poles and recursive boundary residues [13]. The present development composes these languages of realization with data previously generated by HMT. Its distinctive result is a marked, recoverable and refinable collection, together with the maps linking its positive form and its material readings.

Refinement also extends the dimensional scope. The number of nodes grows through nonadic subdivisions and permits construction of positive matrices of every finite rank and external dimension compatible with that number. The proof rests on Vandermonde determinants and full-rank moment forms. In rank one, the complete convex geometry is obtained, with simplicial coverage and residues in every finite dimension. For higher ranks, an explicit collection of amplituhedral images is constructed and its marked flags are preserved. Global coverage of a higher-rank amplituhedron is a statement of a different scope from the existence and compatibility theorem for this collection; the subsequent proofs preserve that distinction in their formulation.

The physical dimension is introduced through the corresponding operators. The tritic calendar distinguishes elliptic rotation, hyperbolic opening and parabolic memory transport. Their Hamiltonian realizations preserve the symplectic form and allow calculation of the Legendre transform and the connection terms arising from a variable chart. In the gauge sector, the areal–volumetric scale, common measure and reflection forms precede reconstruction of the Yang–Mills Hamiltonian. The spectral argument is presented together with uniform coercivity, the witness attaining the first positive level and the intertwiners that preserve it. The relation with canonical forms is established through their readers

and residues, while the gap belongs to the Hamiltonian and its physical space. This hierarchy places each result where its proof actually produces the conclusion.

The exposition first incorporates the formal construction required to obtain the state and its record. It then develops positive realizations, dimensional extension, energy and lattice duality; finally, it brings together the Hamiltonian realizations and the gauge development. Verification programs accompany specified identities and explicit negative controls. Symbolic proof retains the general scope; exact execution checks the declared specializations and their correspondence with the printed formulas.

1.1 Arithmetic genealogy and the system of geometric realizations

The construction begins with the two arithmetic sheets of APP. For $p, q \in \{1, \dots, 9\}$, each operation preserves a complete division,

$$p + q = R_{\Sigma} + 9q_{\Sigma}, \quad pq = R_{\Pi} + 9q_{\Pi}.$$

TRIT adds the local regime and orientation. The tritic selector of operative phase chooses the operation compatible with that state, while both sheets remain in the support. On the enriched state space, the Topological Phase Kernel composes selection, transport and update:

$$U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

The record preserves the residues, quotients, carries, sheets, routes, boundaries, memory and survival used by each prolongation. The nonadic holonomy $\Gamma_9 = U_{t+8} \circ \dots \circ U_t$ belongs to this enriched dynamics. Its reduced memory readout is obtained afterwards. Phase return preserves the advancement of the record and thus distinguishes states at different depths.

The prolongation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

organizes compatible refinements of the same state. The five constructions of the continuum emerge jointly, with their spectral, solenoidal, gauge, cohomological and determinantal aspects. Readers of this state yield the correlated outputs π, φ, e, α . Their common origin admits an internal evaluation order: the already generated regional coordinates act together with the dodecaphasic record in the closure readers for α . In this article, Grassmannians, polygons and differential forms are subsequent realizations of data generated by this chain. Their parameters are fixed before any geometric or metrological comparison.

The specific problem is to describe what a positive realization preserves and what may vary when its form changes. The same positive list admits several mathematical uses: edge weights, interval lengths, coefficients of a quadratic form or coordinates of a material reader. Each use has its own domain, operation and invariants. The term *multimorphology* denotes here the system of these realizations with their explicit compatibility maps. An identity between their numbers requires less verification than an identity between their operators; the following development states and proves each level separately.

The source basis is the article on coinductive generation of the constants and the algebraic structure of the electron, together with the development of the holographic extreme and mean ratio. Their prior constructions fix the record used below. The geometric results in this text begin with that finite output and prove its consequences using explicit data. The material-transfer propositions additionally declare the readers they use. This organization preserves the specific scope of each proof.

Part I

Arithmetic construction, enriched state and joint continuum

2 Formal kernel of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information needed to prolong them. Numerical realizations are formed only afterwards. A constant recognized at the end of this process does not replace the rule that produced its region or the history determining its evaluation.

The name *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not assume that its arithmetic law is the Feynman path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory updating, and prolongation. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks on a line graduated from zero to ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must be distinguished from the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions leads to the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself it does not preserve the integer before reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulation of i repeated j times produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ exchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, rather than a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation preserving both sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^-, q_{ij}^-)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

The term *sheet* here denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the remainders are reduced.

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

One support: 81 positions.

Two evaluations:
 $S(i, j) = i + j$, $P(i, j) = ij$.

Each row is generated by accumulation:
 $P(i, j + 1) - P(i, j) = i$.

The chart represents $0 \pmod{9}$ by 9.

The highlighted block is
 $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$,
 with eigenvalues 9 and 5.

Figure 1: APP multiplication table in positive representatives. Non-unit rows and columns are distinguished by gray shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representation edges do not constitute an intrinsic topological boundary of the torus.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The remainders alone do not determine these integer evaluations or their quotients.*

Proof. The first assertion is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same remainder and different quotients. On the two-dimensional support, positions (5, 5) and (8, 2) give the same sum and the same product remainder, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residue projection identifies data that the lifted evaluation distinguishes. $\square$

2.1.1 Cayley graph, toroidal topology, and winding number

In row and column coordinates, the cardinal translations are given by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southwards and the second eastwards. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic grid: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not make every row of APP into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When a path closes, its integer displacement before reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Path reversal changes the sign of this vector. Returning to the same residue position therefore does not require zero winding. This is the first example of a distinction that will reappear in phase return: one coordinate can close while another records nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called a *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of addition of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the schedule into phase and window will use representatives from zero to eight.

The totals of the two evaluations distinguish the visible contribution from the contribution retained in the quotient:

$$\begin{aligned} \sum S = 810, \quad \sum P = 2025, \quad \sum \Sigma = 405, \quad \sum \Pi = 459, \\ \sum q^+ = 45, \quad \sum q^\times = 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Removing it would change the arithmetic being transported, even if the residue table remained identical.

2.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row has this property exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Multiplicative folding therefore already distinguishes units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readouts: its direct reduction modulo three is 0, 0, 0, whereas extraction of the factor three followed by reduction of its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as direct reduction of the original number.

2.1.4 Latin realization of the additive evaluation

The difference between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in every row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a + b$ permutes the nine residues. This gives a realization of a Latin square by rank-one projectors, in the terminology of quantum Latin squares [19, 21]. This construction starts from the additive sheet already defined and makes the map to its operator representation explicit.

The multiplicative evaluation preserves different information: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

Multiplicity is recorded in the operator and characterizes the multiplicative radical. The toroidal support, the Latin property, and the operator resolution of the identity are therefore properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries[20]. This reference provides a precise language for comparing realizations of APP.

2.2 TRIT: orientation and quadratic regimes

Definition 2.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1), (0, 0), (0, +1)$ may have the same zero remainder and different carries.

The signature does not replace the trajectory. A zero remainder does not mean absence of orientation, phase, or quotient information. Nor does it introduce a physical quantity by itself. Its function is to distinguish algebraically the transport classes that will be realized below.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 2.5 (Exponentiation of the three types). *In each algebra (2.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	-I	$\cos t I + \sin t J_\tau$
0	0	$I + t J_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, all terms of degree at least two vanish. The other two cases recover the trigonometric or hyperbolic series. $\square$

The exponential is common to the three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau) \exp(-tJ_\tau) = I$. The same algebraic principle preserves the unit through realizations differing in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction appear jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an order of parametrization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of lifts

Ternary reduction expresses an integer APP evaluation through a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the notation for an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes the compatibility of this rule with sign inversion explicit.

Definition 2.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r+3c = s+3d$, then $r-s$ is a multiple of 3 belonging to the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the datum needed to reconstruct the preceding one. The quotient is not a perturbation of the remainder: it is its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values +1, -1, 0. The TPK selector defined below uses these classes to determine the operating mode. Reading an evaluation and selecting a mode thus have the same ternary reduction but different domains and functions.

Proposition 2.7 (Arithmetic of balanced pairs). *Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define*

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to pairs through

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product reconstruct as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also lies in the prolongation coordinate, not only in the remainder.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity ensures that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component suppresses this accounting: $1 + 1 = -1 + 3$, whereas the residue readout preserves only -1 . The removed information is determined in this case by $\chi(1, 1) = 1$.

The relation to modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images first fixes r by reduction modulo 3, and then $\bar{c}$. Addition transported through this bijection contains the term $\chi(r, s)$ from (2.7), now reduced modulo 3. The two ternary levels therefore do not compose by independent coordinatewise addition: normalization of the first level modifies the second.

In particular, the classes 3, 6, and 0 of $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving that coordinate prevents the zero-residue sector from being artificially reduced to a single position. Modulus 1000, used later to organize decimal blocks, has a different positional-representation function; its quotient is preserved by the corresponding division. A change of base preserves the integer when it retains both remainder and quotient, whereas agreement of remainders in different bases does not by itself identify their transport states.

2.2.3 Inversion, residue class, and nonvisible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum amplitude 1, these are the three possibilities in the visible fibre over zero; in a general integer lift, this fibre contains every pair $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, remainder, and quotient while differing in orientation, mode sequence, or winding. The decomposition $\mathcal{B}$ is therefore integrated into the transport state rather than replacing it. Arithmetic memory of a division and memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented inversion is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reversed order. In the adopted parametrization, it exchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while preserving the threshold 0. Transport of the remaining trajectory data is specified through the corresponding TPK involution.

This inversion of signatures must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed below, acts within the type already fixed. This distinction prevents algebraic equivalence between different regimes from being attributed to sign inversion.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (2.9) does not replace (2.8); it expresses a different operation in the specified quadratic codomain.

Definition 2.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm is nonzero, the stated formula provides the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm both vanish. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. The algebraic norm here is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

The three types admit faithful realizations through

$$aI + bJ_\tau \longmapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. The complex structure, the dual-number structure, and the split real structure are thus represented by a common matrix expression. Each preserves its law and its zero divisors. The form of signature $(1, 1)$ in the last case has a Lorentzian interpretation in dimension $1 + 1$; identifying a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of elements of norm one:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau) \exp(sJ_\tau) = \exp((t + s)J_\tau).$$

Comparing the two components yields the addition laws

$$C_\tau(t + s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.10)$$

$$S_\tau(t + s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.11)$$

Conjugation replaces t with $-t$ and agrees with inversion within this subgroup. It does not change τ : it reverses an evolution in the chosen regime.

In the elliptic type, the conserved form is $a^2 + b^2$, and the exponential trajectory follows the norm-one circle. In the hyperbolic type, the eigen-coordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not state that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

Evolution does not stop because $J_0^2 = 0$. Nilpotency eliminates terms of order at least two but preserves the linear term transporting the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is needed to describe the threshold without identifying it with absence of state or dynamics.

Proposition 2.10 (Composition of affine coordinates). *In the chart where the scalar component is nonzero, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Division by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio between components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a well-defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the complete algebra; a coordinate singularity does not automatically amount to a singularity of the object. This formula will reappear in the elliptic regional readout. Its use requires preserving the denominator and, where appropriate, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. The change of signature organizing a route therefore cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, the domains, and the effective operator acting at each transition.

Within the electron chapter, the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution of square I and two nilpotent directions. The joint realization will be proved on this specific pair of projections; the preceding classification explains why the three quadratic relations can occur in a single matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. This relation equips the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its connection to a real operator square root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of this same relation. Each preserves its characteristic, dimension, and incidence information. The development of the respective operators establishes the connection, rather than equality of their cardinalities.

Norm-one conservation therefore describes the quadratic realizations of evolutions. Conservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties participate in the construction with different functions: one determines the identities of the representations; the other permits composition, reconstruction, and prolongation of the operations from which they originate.

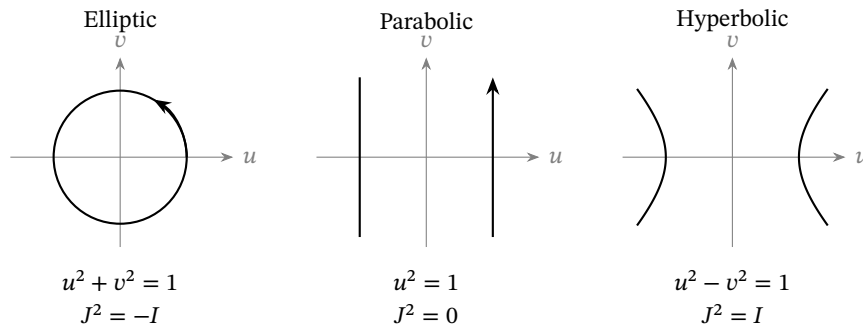


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. Parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: topological phase kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation operates. Cardinal transport moves the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore different functions: a ternary sign is not a displacement on the torus.

We denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of the two cursors, the active mode, the TRIT signature, the phase, the remainders and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object with a reduced version preserving only the visible symbol.

Definition 2.11 (TPK update). Let Sel_t be mode selection through M_{ph} , Tra_t lifted cardinal transport, and Mem_t record updating. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.12)$$

Each map preserves the fields it does not modify and updates the remaining fields through Euclidean division and trajectory concatenation.

The notation (2.12) expresses the order of composition; it does not replace the effective rules. The next section fully specifies its finite two-cursor realization. Subsequent prolongation will be obtained by transporting compatible prefixes and boundaries, rather than by erasing memory and repeating the first emission. In this way, APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamical state, observables, and dependence of operations

The topological phase kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations making prolongations compatible. Equation (2.12) acquires operational content when the coordinates received and modified by each of its factors are specified. Position belongs to the toroidal support; phase belongs to the schedule; orientation determines transport; quotients record the arithmetic evaluation before reduction. These coordinates are related through the transition and preserve different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.13)$$

The subscripts distinguish the additive and multiplicative cursors. The fibres of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their remainders and quotients, and the orientation and boundary records. Precision prolongation adds the prefix and compatible cylinder. Two states with the same projection q can therefore retain different records.

The dependence of operations has three levels. At the local level, the selector activates a sheet, transport modifies its position, and updating calculates the new arithmetic coordinates. At the trajectory level, these transitions compose and preserve the records of preceding steps. At the refinement level, extension relations determine which continuation preserves the prefix, signature, and boundary of its antecedent. The finite schedule describes the phase of these operations; holonomy describes their composition on the history.

2.3.2 Tritic selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the representative of the schedule. The operating phase and dodecaphasic sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.14)$$

The selector already defined can be written as $\varepsilon_9 = M_{\text{ph}}$. Its value vector, in the ordered chart of D_9 , is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_\times(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_\times(\theta)v_{d_\times} \quad \text{in } X_9. \quad (2.15)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event forms part of the chronology and its record. The cardinal directions determine the vectors v_d , whereas the phase signature determines which of these vectors is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses the positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records preceding crossings, the update is $z' = z + b_d(x, a)$. Thus,

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport datum remains reconstructible after position is reduced on the torus.

Proposition 2.12 (Conservation of lifted displacement). *For a cardinal trajectory of n steps with activation indicators a_t ,*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved when two compatible trajectories are concatenated.

Proof. The one-step identity is (2.16). Summing it over the n steps cancels the intermediate coordinates. Splitting the summation interval into two parts produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the winding number defined previously. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its record simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial crossing memory z_t and the arithmetic quotient q_t have different definitions; their joint preservation retains the geometric and arithmetic provenance of each transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j exchanges N with S and E with O . After (2.15) is applied, both directions are reversed at the end of a block of 27 steps. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)} d_+, x'_\times, j^{h(\theta)} d_\times, \theta + 1). \quad (2.17)$$

The readout forming the emission uses positions before transport, according to the explicit recurrence of Section 4. Observable updating and emission therefore compose in a fixed order.

Proposition 2.13 (Reversibility and period of the observable schedule). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. A block of 54 steps restores positions and directions; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, first recover the preceding phase by subtracting one in Θ_{108} . That phase determines whether the directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector recovers the initial state uniquely.

Each interval of 27 steps contains nine activations of each cursor. The following block preserves the activations and reverses the directions, so the integer displacements cancel over 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any origin of the schedule. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.13). Projection of the history onto the schedule permits this finite check. The trajectory record, its evaluations, and new prefixes are preserved in the dynamical state on which $\mathcal{U}_t$ acts; observable equality does not reinitialize them.

2.3.4 Affine memory transport and composition of states

The observable projection admits fibres of arithmetic and geometric data. Over a configuration q , write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , remainders and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q and S_q denote, respectively, the remainder–quotient data sets, the abelian group of transport memory, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified componentwise in the regional and terminal constructions. This dependence prevents the signature from being treated as a free parameter once the data producing it are fixed.

The elementary remainder–quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives of D_9 congruent modulo 9 are equal. This chart provides local integer reconstruction. The precision towers additionally incorporate truncation maps and the cylinders developed in 4 and 6.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fibre transport is expressed through maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

These maps respect identities and composition, and $C(\gamma)$ is a homomorphism of abelian groups. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.18}$$

The symbol k_{tr} here denotes the transport cocycle; the rational coordinate κ of record K retains its notation in the chapter on dodecaphasic closure.

The transportable coordinates update by

$$\begin{aligned} h' &= H(\gamma)h, \quad c' = C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, \quad \sigma' = \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.19}$$

The second equation is an affine action: the linear term transports the preceding memory and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$ and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.16). This concrete instance links the abstract law to the cardinal operations already defined.

Proposition 2.14 (Compatibility of updating with concatenation). *Updating according to (2.19) by γ_1 , followed by updating by γ_2 , agrees with updating by $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The assertion for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. The states, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves arithmetic transport and boundary prolongability simultaneously. This is the structure in which returns with memory and successive refinements take place.

2.3.5 Dodecaphasic record, phase, and accumulated memory

The 108 transitions are distributed among the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The record preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = \left((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m) \right)_{m=1}^{12},$$

where τ_m is the window subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence record. We adopt the chart developed in 9.2; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal

records. Composition of segments and their increments is governed by (2.19). The group C_{12} , by contrast, describes translation of the window index; Θ_{108} preserves the step schedule. This distinction specifies what transforms and what is observed upon completing a cycle.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, an advance of 9 steps preserves r and increases b by one. Phase addition introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite schedule preserves $[b]_{12}$; a revolution record preserves $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4 proves the nonsplit exact sequence (4.5). Its function is to describe the relation between these coordinates before they are used to construct the signed state.

The signed record receives phase and memory data through the composition developed in Section 9:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between this signed state and K , together with the rational readout of K , is proved in the corresponding chapter. Recording operations precede these transformations. Reconstruction of a record from its cyclic differences verifies its recoverability; its dynamical provenance must be preserved through the events that produced it.

For the coded trace (d_t) preserved by the record, the event reader counts the codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are distinct coordinates of the record. Chapter 9 develops this reader, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data this chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal construction of the channels

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases and let $O_{\text{ph}} = D_9 \setminus \{9\}$ be the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and the schedule origin already fixed; subsequent exceptional realizations will transport these incidences into their own domains.

Proposition 2.15 (Cardinalities and oriented phase incidence). *The sets in (2.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneous translation of the phase origin and the marked return preserves all these cardinalities and their inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Products by 6, 8, 9 give 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A schedule translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve incidence. $\square$

The cardinality 271 here has a realization through phase incidence. The cubic completion of Section 4 has its own construction of the shell $1000 - 729$. The correspondence between these realizations must preserve the maps and marks identifying them, as well as cardinality. Similarly, the selector ε_9 , the incidence $(90, 120)$, and the extractor $\mathcal{E}_{108}^{90,120}$ preserve the domains and operations just distinguished.

2.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; Jd_n, \dots, Jd_1).$$

The origin of the reversed trajectory is the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and the integer displacement changes sign. These identities follow by reversing step order and replacing each cardinal vector with its opposite. For a loop, this gives $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On records, reversal also requires transport of component order and event orientation. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete record. In particular, the regional involution exchanging the closure and propagation components preserves the self-scaling axis; its formula is presented in Section 6.13. This regional operation acts on a different domain from reversal of a cardinal trajectory.

Extensions over a route $r : q \rightarrow q'$ are expressed through relations between fibres of compatible states. If $\mathcal{F}_q$ denotes the fibre of complete states over q , require

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

A phase revolution composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \dots \circ \mathcal{U}_k. \quad (2.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the complete domain, we retain $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$, with the admissible prolongations developed below. The cyclic base has nine phases, whereas the fibre preserves trajectory, quotients, and boundary. Monodromy is the action of one revolution of this base on the fibres; holonomy $\Gamma_{9,k}$ expresses its realization through the effective TPK transitions.

If g observes phase and M the number of preserved revolutions,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure and the second identifies the new temporal depth. Coinductive prolongation additionally preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section brings together its antecedents without replacing them with the latest observation. This arithmetic of histories and its unit are developed in 4.8–4.10; the regions and their readers are constructed in Section 6.

Phase transfer on already constructed emissions has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the alphabets of additive and multiplicative signatures, of cardinalities 18 and 26. Their product identifies the catalogue $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 4. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for the remaining phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. History transport remains given by (2.22), with its memory and boundary.

The resulting hierarchy establishes the function of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphasic record determines the data entering K and the closure of α . Numerical realization and incidence realization receive this preceding structure. This order permits each consequence to be presented in its chapter while preserving, from the formal kernel, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

Calendar return does not remove increments produced during the journey. To express this distinction in operator form, an accumulated trajectory-event record is constructed first, followed by its generating series. Reduction of memory variables will be performed on this transport, also preserving the source and output reader.

2.4.1 Oriented events and record update

Let (d_t) be the sequence of integer direction codes retained by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification of their alphabets is assumed. We number the windows starting from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphase record. Its orientation is

$$\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1).$$

For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign retained in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.23)$$

Each summand is incorporated when its transition occurs. This definition counts trace events; it does not select directions through a value that is to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$ and $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated record in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at a different origin, z_0 retains the preceding record. The frame accompanying the calendar is $y_m = S^{-m}z_m$.

Proposition 2.16 (Update and return with memory). *The record in the moving frame satisfies*

$$y_{m+1} = Ry_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.26)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1}y_m + a_m S^{-1}\mathbf{e}_0$, since $S^{-m}\mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (2.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j}\eta_j = a_j R^{12-j}\mathbf{e}_0 = a_j \mathbf{e}_j$, which proves (2.26). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^\top D_3 + D_4^\top D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^\top B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the record; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.17 (Event-induced balance). *Update (2.24) satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (By_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \quad (2.27)$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The coordinate sum is invariant under permutation R , giving the second equality. $\square$

2.4.2 Generating series and uniform memory control

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)) \quad (2.28)$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq Lr^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right). \quad (2.29)$$

The derivative series converges uniformly on every disk $|u| \leq r < 1$.

Proof. Multiplying (2.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} mr^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves its uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with another reader's variable requires preserving that clock and the map relating the two readings.

2.4.3 Memory elimination: operator, source, and reader

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.19 (Schur reduction with source and reader). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.30)$$

Its complete reading satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C) \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting it into the first gives (2.30); applying ℓ_0 and ℓ_1 to the two components gives (2.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2B_1D_uC = \sum_{k \geq 0} u^{k+2}B_1D^kC$ describes departure into memory, k steps within it, and a return. By contrast, $u\ell_1D_uC$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: the source and reader must also be transformed.

The distinction is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.32)$$

Indeed, $R^n\mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example concerns the twelve-window record, not the entire TPK state.

2.4.4 Segment composition and coherence of reduction

Let three composable affine transports be $F_i(y) = R_iy + b_i$, with compatible domains and codomains. The memory produced by the complete journey is

$$F_3F_2F_1(y) = R_3R_2R_1y + b_3 + R_3b_2 + R_3R_2b_1. \quad (2.33)$$

The proof is successive substitution of the three affine formulas. Grouping the first two segments first, or the last two, produces the same expression; the factors transport each increment from its point of origin. In (2.24), $R_i = R$ and b_i is the event of its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2Q_2C_2, & B'_1 &= B_1 + uB_2Q_2D_{21}, \\ C'_1 &= C_1 + uD_{12}Q_2C_2, & D'_{11} &= D_{11} + uD_{12}Q_2D_{21}. \end{aligned} \quad (2.34)$$

Proposition 2.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - uA' - u^2B'_1(I - uD'_{11})^{-1}C'_1 &= I - uA \\ &\quad - u^2(B_1 \ B_2)(I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (2.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting it into the first two produces (2.34). Eliminating the second component then gives the left-hand side of (2.35). Solving the two memory equations together gives the right-hand side, by Proposition 2.19. The formal solution is unique because $I - uT$ has constant term I . Both procedures also transform the source and reader according to (2.30)–(2.31). $\square$

Balance (2.27) describes record increments. The preceding identities allow them to be transported when memory is reused in dodecaphase closure; they do not by themselves determine additional analytic coefficients.

3 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

3.1 Phase selection, transport and enriched state

3.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (3.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t-1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (3.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b-1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

3.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i+1, j+1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (3.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

3.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (3.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

3.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (3.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (3.6)$$

Proposition 3.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = \lfloor [\nu/4] \rfloor_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (3.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

3.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = \lfloor (n - 1)/9 \rfloor, \quad n = \rho_9(n) + 9q_9(n). \quad (3.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

3.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (3.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

3.3.2 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 3.2 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_9^2 is crossed.

3.4 Six windows and decimal block emission

3.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 3.3 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (3.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

3.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \tag{3.9}$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \tag{3.10}$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 3.4 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (3.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

3.4.3 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (3.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 3.5 (Positional recovery of the emissions). *The encoding (3.11) determines the complete word by*

$$U_m = \left\lfloor \left\lfloor \frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right\rfloor \right\rfloor_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

3.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_{\times} = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 1: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

Table 2: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

The resulting word and its signatures are

$$\mathbf{U} = (501, 614, 498, 169, 272, 272),$$

$$\mathbf{s} = (5, 6, 4, 1, 2, 2),$$

$$\mathbf{e} = (5, 5, 5, 5, 5, 5).$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

3.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (3.12)$$

where $G^{(6)}$ applies (3.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

3.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 3: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 4: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 3.6 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 3.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 3.4 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (3.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 3.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

3.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (3.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$ \Psi^{-1}(w) $	1	2	3	4	5	6
number of words w	132	66	18	3	6	18.

(3.14)

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 3.7 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (3.13).

3.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (3.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (3.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 3.8 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (3.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 3.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 3.9 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 5; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 5: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

3.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $v = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (3.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (3.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (3.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 5.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (3.12); its scope is the same finite domain.

3.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (3.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

4 Development of the formal kernel: initial conditions, geometry, and refinement

The eighty-one-position support does not coincide with the domain of initial conditions for the dynamics. An initial condition must also specify an orientation, and the additive and multiplicative sheets have independent cursors. This distinction allows exhaustive reconstruction of the emission without selecting a numerical region in advance.

4.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (4.1)$$

The census of residual-emitter signatures in Section 39 enumerates these conditions by identifiers and presents their fibers; the law transforming them is defined here. The rows of an emission table are fibers of this initial-condition domain.

The finite realization uses six windows of nine transitions. At each instant t , the phase signature is $M_{\text{ph}}(1 + ((t-1) \bmod 9))$. If its value is $+1$, the positive residue $\Sigma(i+1, j+1) = \rho_9(i+j+2)$ of the first cursor is read and that cursor is then moved; if its value is -1 , the positive residue $\Pi(i+1, j+1) = \rho_9((i+1)(j+1))$ of the second cursor is read and the second cursor is then moved. Zero increments the neutral counter. At the end of transitions twenty-seven and fifty-four, cardinal orientations are reversed. Integer evaluations and their quotients are retained in the record, but the emitter uses the residue readings specified here.

In the unit sector, the multiplicative reader uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For finite emission it is extended by zero to the non-unit sector. This extension is an observable, not an identification of the elements 3, 6, 9 with the multiplicative identity. The sheet and its membership in the radical sector remain in the trajectory record.

Let S_m be the sum of the three positive additive residues Σ in window m , E_m the sum of the three discrete exponents of the multiplicative residues Π , and $Z_m = 3$ the number of neutral events. Thus S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the record. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (4.2)$$

The integer coefficients and calendar of this recurrence are explicit parts of the declared emission law. Values of π, φ, e or α are not received. Distinguishing an emission rule from a constant recognized from its output prevents the genealogy from being hidden in a decimal table.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule is written

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (4.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalog adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (4.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transport of subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with TRIT is performed after (4.4).

Proposition 4.1 (Injective factorization of emission). *The word $U = (U_1, \dots, U_6)$ determines both signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient set of 729 words.*

Proof. The hundreds digit of U_m recovers s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence of (4.3). Hence coupling is injective on the signature product.

Enumerating the 324 conditions for one cursor through the preceding recurrence produces eighteen additive signatures, each with eighteen preimages, and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. Section 39 collects the signatures, their multiplicities, and the exact rule recovering their initial conditions. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\begin{array}{ccc} 432 & 18 & 18 \\ \text{fibers of } 144, & \text{fibers of } 432, & \text{fibers of } 1944. \\ 18 \cdot 24 & & \end{array}$$

The sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (4.4) to the 468 emissions produces exactly 243 words. This last number is the cardinality of the image, not of the space $\mathbb{F}_3^6$. $\square$

This factorization explains the difference between finite exhaustiveness and unlimited depth. Domain (4.1) is complete for the declared support and emitter; it does not contain every prefix of every prolongation in advance. Depth belongs to the system of compatible extensions constructed over those conditions.

4.2 Calendar closure and memory advance

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (4.5)$$

If $t = 9b + r$ with $0 \leq r < 9$, adding two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 4.2 (Nonsplitting of the calendar). *Sequence (4.5) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{lcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring a prolongation prefix. Nonadic holonomy Γ_9 denotes transport through one turn on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (4.6)$$

Phase equality and memory increment are compatible and constitute two observations of the same transport.

4.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by TPK operations on $\hat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \hat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t . The compatibility relation precedes the choice of parameter: the order allows its transitions to be expressed as an evolution. When the term section is used, the projection $p : E \rightarrow \mathcal{T}$ and condition $p \circ \gamma = \text{id}_{\mathcal{T}}$ are also declared. The state space, temporal order, and transported information are thereby specified.

The signature τ selects a quadratic regime and admits a temporal reading relative to this parametrization. Type $+1$, with $J_{+1}^2 = -I$, represents return and memory; type 0 , with $J_0^2 = 0$, represents the transition threshold; type -1 , with $J_{-1}^2 = I$, represents opening in two eigendirections. In the model's temporal terminology, these functions correspond, respectively, to past, present, and future. The correspondence concerns the operational position of a transition within a history: retained antecedent, update, and admissible prolongation.

This interpretation acquires content through three distinct operations. Restriction of a trajectory recovers the prefix already constructed. The update $\mathcal{U}_t$ composes selection, transport, and recording in the observed transition. Prolongation considers continuations that preserve the boundary conditions of that prefix. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x , deleting the last step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unlimited continuation is a collection $(\gamma_n)_{n \geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the restrictions each depth imposes on preceding depths. Existence of a compatible collection is established in the corresponding prolongation domain; merely specifying the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the signature sequence, the involution C_{rev} reverses order and changes the sign of each component. The trajectory is also subject to the transformations of position, sheet, and boundary specified by transport. The reversed signature is an observation of the conjugate trajectory, not a replacement for its other coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical record is nonzero.

The relation with exponential geometries is equally precise. In each fixed regime, $\exp(tJ_t)$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to that local representation. Phase return of Γ_0 , by contrast, acts on the trajectory with memory, according to (4.6). A trajectory can recover its phase while retaining a new antecedent: historical data have changed although the phase observation is the same. The parabolic threshold retains a linear displacement; hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, whereas the TPK determines their selection and effective concatenation.

The aperiodic suspension in Section 6 will make this distinction explicit through a finite phase factor, an irrational rotation, and a memory degree. The link between TRIT and temporality is thus expressed by operations on trajectories and their quadratic representations. Its content exceeds a classification of signs: it preserves the regime, the direction of transport, and the relation between antecedent, transition, and continuation.

The monodromic representation of this return is developed in Section 6.10. There, the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension in Section 6.12 defines the aperiodic time-crystal model, and Section 6.14 composes this temporality with the three Euler realizations.

4.3 Observable compression and orthogonal conservation

An isometric realization provides a quantitative expression of what a reduced observation loses. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of transport through one turn, $U^*U = I$, and $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (4.7)$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 4.3 (Finite-horizon conservation). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r} D^* D T^r + T^{*n} T^n. \quad (4.8)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r , then summing, produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n} T^n$. $\square$

The final summand preserves the terminal state. Removing it before justifying a limit would alter the equality. This distinction is the operational content of conservation of unity under compression: the norm is distributed among the visible component, memory, and terminal, rather than being replaced by a single residue.

4.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonals modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$, one obtains $B_c = 9P_+ + 5P_-$. The unity of stochastic normalization and conservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \quad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2} \log \frac{9}{5}.$$

For $Q = \text{diag}(1, -1)$, the second matrix satisfies $B_c^T Q B_c = 45Q$. This yields a local realization in $SO^+(1, 1)$; no prior introduction of four-dimensional spacetime is required to prove it.

Uniform averaging over the classes $C_1 = \{1, 4, 7\}$, $C_2 = \{2, 5, 8\}$, and $C_0 = \{3, 6, 9\}$ gives

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_{Π} is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

so its quadratic form has signature $(2, 1)$. The local spectral gap $9 - 5 = 4$, the coefficient 2, the aggregate difference $51 - 45 = 6$, and the integrated defect $9 \cdot 6 = 54$ describe different levels of the same arithmetic comparison. They are not interchangeable indices of a single operator.

Discrete curvature can be described directly before this spectral realization. In residues, let $S(x, y) = x + y$, $P(x, y) = xy$, and let Δ_x, Δ_y be forward differences. Link variables are differences of potentials, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and we set

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}. \quad (4.9)$$

Sum and product, taken separately, induce flat connections; their ordered mixture changes the sign of plaquette curvature. Summing over a rectangle of sides a, b cancels interior edges, and the oriented boundary sum equals $\pm ab$ modulo nine. An exact relation thus appears between composition order and boundary orientation.

4.5 Cubic completion and two notions of boundary

Expanding the cubic support of side nine to that of side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (4.10)$$

The first term corresponds to three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The expansion corona contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (4.10) by 1000 gives a normalized partition of unity. Cube incidence distinguishes six faces, twelve edges, and eight vertices. The subsequent connection to Steiner hexads and octads requires the corresponding combinatorial maps; cube cardinalities do not replace them. Here this completion supplies the relation between bases 729 and 1000, while exceptional prolongation will be developed after numerical generation, the dodecaphasic record, and the constructions of α .

4.5.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description making its operation explicit. Let E_9 be the set of nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, extending E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without changing the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 4.4 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 4.5 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (4.11)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (4.12)$$

The punctured expansion contains 270 elements in addition to the initial cube; the complete expansion contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning one of the nine classes of E_9 to each remaining coordinate. Classification by the number of additional coordinates is exhaustive and its classes are disjoint. The sole element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH in the corpus, here denotes this articulation of the initial body, expansion strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure through restoration of the apex. The added unit therefore has an explicit incidence location.

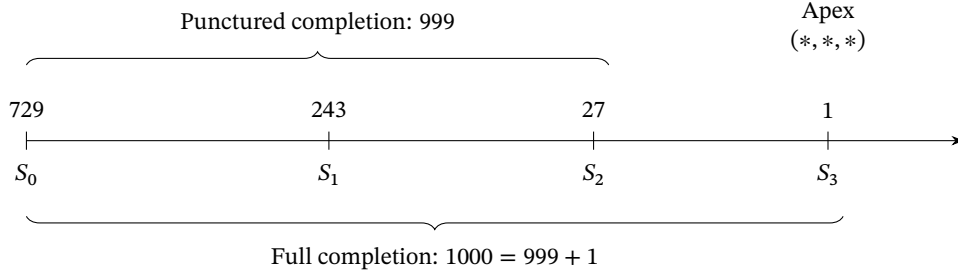


Figure 3: Strata of the cubic completion. The arrangement represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

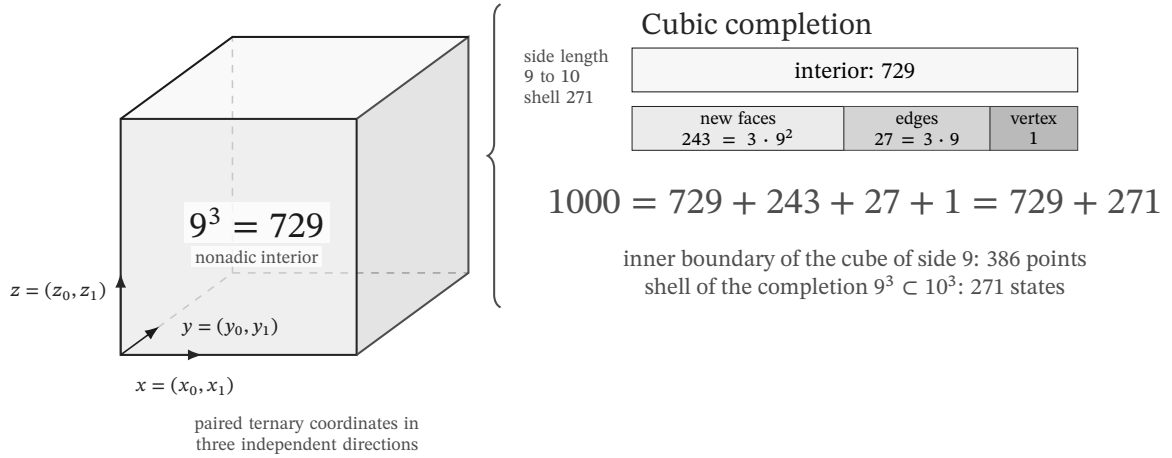


Figure 4: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 3; graphical spacing is schematic.

4.5.2 Conservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\hat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \hat{E}^{d+1} \rightarrow \hat{E}^d$ deletes the last coordinate.

Proposition 4.6 (Projective compatibility of normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\widehat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r}/10^d$, and the masses of all strata sum to one.

Proof. Each point of $\widehat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (4.11); the binomial $(9 + 1)^d$ gives total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define a different measure: the distinction is operationally relevant. In particular, conservation of probability mass, conservation of transport records, and thermodynamic dissipation concern different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamic and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\widehat{C} \setminus C$ belongs to its expansion. Cubic completion, base-thousand positional expansion, and exceptional-incidence reading share support data; their respective maps determine which information from these data each representation uses.

4.6 Cubic incidence and Lorentzian realization

Cubic completion supplies a relation between interior and boundary. Its geometry can be developed through incidence of the six faces with the eight vertices. Label faces by (i, ε) , $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and vertices by $\nu \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),\nu} = \mathbf{1}_{\{\nu_i=\varepsilon\}}.$$

Let $u = (1, \dots, 1)^T \in \mathbb{R}^6$, let R_F denote face opposition, and let $P_u = uu^T/6$, $P_- = (I - R_F)/2$.

Proposition 4.7 (Four-dimensional incidence quotient). *The incidence Gram matrix satisfies*

$$MM^T = 12P_u + 4P_-.$$

Its eigenvalues 12, 4, 0 have multiplicities 1, 3, 2. Consequently,

$$Q_{\square} = \mathbb{R}^6 / \ker(MM^T) \simeq \mathbb{R}u \oplus E_-, \quad E_- = \text{im } P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct non-opposite faces share two vertices. The matrix $12P_u + 4P_-$ has exactly these entries. The uniform subspace has dimension one; differences between opposite faces generate a three-dimensional space orthogonal to it. Its complement has dimension two and belongs to the kernel. $\square$

The decomposition $1 + 3$ comes from incidence. The temporal choice is declared through the bilinear form

$$B_{\square}([x], [y]) = \langle x, (P_- - P_u)y \rangle.$$

It is well defined on the quotient because both projectors annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \quad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\text{diag}(-1, 1, 1, 1)$. The domain, decomposition, and sign convention producing the Minkowski realization are thus explicit. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_{\square} = \sqrt{6}P_u + \sqrt{2}P_v, \quad W_{\square}e_{i,\varepsilon} = t + \varepsilon d_i$$

sends the six face labels to null directions, since $B_{\square}(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Direction reversal and face opposition thus acquire a concrete geometric realization.

4.6.1 Soldering, reversal, and subdivision

The cellular realization uses an invertible transport $U_m(a)$, preserving the Lorentzian form, between the endpoint fibers of an oriented dual edge a , a face label $\lambda_m(a)$, and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the soldering map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a)\beta_m(a).$$

Identify the source fibers at two consecutive levels through the declared refinement isometry. If the nine subsegments $a_1, \dots, a_9$, at level $m+1$, preserve label and orientation when transported to that common fiber, then

$$\beta_m(a) = \sum_{j=1}^9 U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \quad (4.13)$$

Each transported term equals $\ell_m/9$ times the same null direction, proving the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation brings together direction, scale, and transport, rather than limiting geometric correspondence to a dimension count.

With orientation and the Lorentzian form also fixed, Hodge duality on $\Lambda^2 Q_{\square}^*$ satisfies $\star^2 = -I$. In dimension four, the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, the eigenspaces with eigenvalues $+i$ and $-i$. The electromagnetic realization will use this domain after constructing the electronic section and declaring its coupling.

4.7 Compatibility of modular and positional reductions

The block-decimal base enters through Euclidean division preserving residue and quotient simultaneously:

$$n = 1000q + r, \quad 0 \leq r < 1000.$$

Since $1000 \equiv 1 \pmod{9}$ and $1000 \equiv 1 \pmod{3}$,

$$n \equiv q + r \pmod{9}, \quad n \equiv q + r \pmod{3}.$$

Reduction of a block therefore requires the quotient's contribution to recover the integer's class. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and 3. This is an operational reason for preserving memory through a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated through a single integer remainder. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D 3^L.$$

On adding six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \leq \nu < 729$. If the decimal prefix remains fixed, one obtains

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \quad (4.14)$$

Both identities follow by substitution of the definitions. Comparing endpoints is thus reduced to integer operations retaining each prolongation's contribution. This compatibility makes it possible to refine a region before evaluating its real coordinate.

4.8 Number as a compatible section and value as a reading

Distinguishing state from observation specifies what object is prolonged as resolution increases. At depth n , let X_n be the set of admissible states produced by APP–TRIT–TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. These are not independent coordinates: earlier arithmetic evaluations and transports impose their relations. A prolongation preserves these relations in each antecedent, even while adding new steps and memory data. This is the meaning of the arithmetic of histories developed in (Sections 4 and 6.10).

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined in it.

Definition 4.8 (Compatible numerical section). An HMT number is a compatible history of the state system:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (4.15)$$

A reader is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the reader. The pair consisting of a section and its reader constitutes a read presentation, not a redefinition of the section's identity.

On the Archimedean domain X_∞^{Arch} , the reader associates a closed cylinder with each prefix, preserves nesting, and makes its diameter tend to zero. It thus determines

$$\text{ev}_\mathbb{R} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_\mathbb{R}(x)\} = \bigcap_n I_n(x). \quad (4.16)$$

The concrete construction of these cylinders and the determination of their boundaries are developed in Section 6. Definition (4.15) does not by itself assert that the image of (4.16) is the entire line: a coverage claim requires its own theorem. Nor does equality of values establish equality of histories without an injectivity result. This separation allows precision to be discussed as depth while preserving the distinction between the arithmetic antecedent and its scalar realization.

4.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are its admissible histories. Composition is written $vu = v \circ u$: first traverse u , then v . Elementary steps come from TPK selection, transport, and updating. A local tritic fiber admits the notation $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the transported state; a transition between regimes does not amount to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of this composition. For each object x , a unital associative real algebra A_x , possibly noncommutative, is given. For each $u : x \rightarrow y$, a unital homomorphism $\rho_u : A_x \rightarrow A_y$ is given. Assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (4.17)$$

For each composable pair $u : x \rightarrow y, v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. Require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for every composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (4.18)$$

These are explicit hypotheses on transports and their coefficients, not a consequence of calling the dynamics holonomic. Applying this presentation to a TPK sector requires identifying its ρ_u and $\Omega(v, u)$. Transitions expressed through bimodules require preserving that structure and are not replaced by (4.17).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{t(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (4.19)$$

when the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. Addition gathers histories with coefficients; multiplication concatenates histories while preserving the record multiplier.

Proposition 4.9 (Sectorial associativity with a central multiplier). *Under (4.17) and (4.18), product (4.19) is associative. If the object set is finite, its identity is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to move to the right of that factor; (4.18) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero because of their initial and terminal states. Bilinearity completes the proof. Local identities and normalization of Ω verify the stated identity element. $\square$

Calendar carry provides an effective example. In the category with one object and arrows C_9 , take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

The two carry sums of a triple are $\lfloor (a+b+c)/9 \rfloor$; hence (4.18) holds. The formal variable z records the turn without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphase record of calendar (4.5); retaining z without that relation distinguishes successive turns. The example realizes this calendar's memory without identifying it with all TPK memory.

4.10 Double variance of refinement and conservation of unity

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (4.20)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ gathers finite-depth observations. This algebra is not identified here with the whole route algebra: prolonging transport operators also requires their compatibility with refinement.

Lemma 4.10 (Identity and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (4.21)$$

and, for every $f \in D_n, y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m . Their indicators are orthogonal and sum to 1_m , proving (4.21). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. $\square$

The global identity is the constant function one on all possibilities; an individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement may increase the number of extensions and the depth of each history while preserving the identity and all evaluations of its antecedents.

4.11 Phase return, Weyl representation, and two readings

Law (4.6) acquires an operator realization in the bilateral suspension constructed in Section 6. There, T is one-step advance, V_9 observes the nonadic phase, W_δ the rotation phase, and D_M the memory degree. Writing $\hat{\Gamma}_9 = T^9$ for the turn in this representation, the Weyl relations (6.17) imply, on finitely supported vectors,

$$\begin{aligned} V_9 \hat{\Gamma}_9 &= \hat{\Gamma}_9 V_9, & W_\delta \hat{\Gamma}_9 &= e^{2\pi i 9\delta} \hat{\Gamma}_9 W_\delta, \\ [D_M, \hat{\Gamma}_9] &= \hat{\Gamma}_9. \end{aligned} \tag{4.22}$$

The first equality follows from $\zeta_9^9 = 1$; the second from nine applications of the covariance of W_δ ; the third expresses the unit memory increment. The finite phase returns while irrational rotation and memory distinguish the next state. Characters and parameter δ are realized subsequently from the numerical construction specified there: they do not select the seeds retrospectively. This representation describes suspension observables and translations; it does not replace all TPK coordinates or operators.

The double reading extends the same distinction between history and observable. On its Archimedean domain, (4.16) applies; on the calibrated incidence domain, the maps Q_n of Section 11.1 preserve words and boundary frames. The equality proved there, $r_{n,m} Q_n = Q_m p_{n,m}$, determines their prolongation Q_∞ . On the common domain of both readings, $\text{ev}_{\mathbb{R}}(x)$ and $Q_\infty(x)$ can be considered jointly: the former retains the value; the latter, the transported incidence preserved by its codomain. The section precedes both. This articulation makes it possible to pass from the arithmetic of histories to constants and exceptional reading without turning their realizations into new initial data.

5 Correlative construction of the continuum: histories, transport, incidence and terminal

The arithmetic expressed by a normal form preserves a precise readout of the state producing it. To situate that readout within HMT, the histories, sheets and operations acting on them must also be maintained. This chapter brings together that common substrate. Its different constructions are obtained from the same APP-TRIT-TPK execution tree; expository divisions allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented earlier. An admissible emission contains the APP operation, TRIT orientation, sheet change, residue and quotient, together with the effective TPK memory update. When only a residue is expressed, the execution history still belongs to the starting object. Two executions with identical final readouts are not identified on that ground alone.

5.1 Admissible histories and memory by composition

Let X_k be the set of admissible histories of length k . Its elements are constructed by successive emissions from the kernel's initial domain. Write $h\varepsilon$ for prolongation of h by an admissible emission ε , and $\rho_k(h\varepsilon) = h$ for deletion of the last emission. The emission label is preserved, even when different labels produce the same expressed state.

On the two arithmetic sheets, local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \quad 0 \leq r_\diamond < 9, \quad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate likewise admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preservation of the quotient allows these representations to be composed without replacing an emission by its isolated residue.

In a memory chart, an emission's update is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a trajectory $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of these updates. Its linear part and translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \quad \kappa_\alpha = \sum_{j=1}^k C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 5.1 (Affine compatibility of concatenation). *If α precedes β , then*

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β produces $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the trajectory. $\square$

This cocycle identity is the precise content of accumulated memory. The quantity transported by a second trajectory depends on the chart update it performs; it is not, in general, the untransported sum of two registers.

Nonadic return preserves a second distinction. In phase and winding coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its representation is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

During nine steps, exactly one crossing from 8 to 0 occurs. Phase returns and the winding coordinate increases. Enriched holonomy also retains the memory updates accompanying this route.

5.2 Tree refinement and measure of histories

Fix the finite nonempty initial set of admissible seeds, with strictly positive initial masses summing to 1. In this subsection each history has a finite positive number of prolongations. Deletion of emissions defines the refinement maps. Kernel admissibility supplies the prolongations; when a residence uses a neutral emission, that emission explicitly gives a section of ρ_k . Existence of that section concerns the tree including this emission, not every arbitrary restriction of routes.

Let $d(h) \geq 1$ be the number of admissible emissions at h . Equiprobable choice among its successors defines

$$\mu(h\varepsilon) = \frac{\mu(h)}{d(h)}.$$

More generally, a positive local distribution $p(\varepsilon \mid h)$, summing to 1, defines $\mu(h\varepsilon) = \mu(h)p(\varepsilon \mid h)$. This second formulation preserves the readout's weights when numbers of prolongations are not uniform.

Proposition 5.2 (Cylindrical compatibility and conditional expectation). *At each level one has*

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

On the spaces $L^2(X_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \quad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^*$, $E_k J_k = I$ and $\|J_k f\| = \|f\|$.

Proof. The first identity is normalization of local probabilities. Applying it to $|f(x)|^2$ gives the isometry. Grouping by predecessors the sum defining $\langle J_k f, g \rangle$ yields the formula for E_k ; on a function constant on each fibre, that formula returns its value. $\square$

Compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(h)$. The measure extends from the cylinder algebra, and uniqueness follows because the cylinders generate the sigma-algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under emission deletion is nonempty. In the cylindrical topology, every nonempty cylinder has positive measure.

This measure distinguishes histories converging to the same representation. The uniform measure on terminal values would be a different readout, since it would remove the multiplicities just preserved by the construction.

5.3 Sheetwise transport and exact accounting of outputs

First consider the space whose orthonormal basis consists of histories at level k . Transport preserving the label of each prolongation is

$$\mathcal{U}_k e_h = \sum_{\varepsilon \text{ admissible at } h} \sqrt{p(\varepsilon | h)} e_{h\varepsilon}.$$

The successor sets of distinct predecessors are disjoint. Normalization of p therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation using history masses is related to this one by the diagonal basis change incorporating $\sqrt{\mu(h)}$.

The sheet label belongs to a history's complete register. Write

$$\hat{h} = ((h_0, s_0), \dots, (h_k, s_k)), \quad s_j \in \{\Sigma, \Pi\},$$

where s_j is the active sheet at step j . Prolongation adds the emission and sheet s_{k+1} determined by the TPK operational selector, preserving the entire prefix, including the initial sheet. In the basis of these lifted histories,

$$\mathcal{U}_k e_{\hat{h}} = \sum_{\varepsilon \text{ admissible at } \hat{h}} \sqrt{p(\varepsilon | \hat{h})} e_{\hat{h}\varepsilon}, \quad R_k e_{\hat{h}} = \mathbf{1}_{s_k=\Sigma} e_{\hat{h}}, \quad Q_k = I - R_k.$$

Thus, R_k reads the current sheet, while the distinction between histories is maintained by their complete prefixes. Two initial histories reaching the same active sheet still have distinct successors. The isometry proof using disjoint collections of prolongations applies verbatim to this basis. Tric orientation is another register coordinate: the value 0 preserves the threshold readout on each sheet and does not add a third sheet to this decomposition. We now separate sheet preservation from sheet interchange:

$$A_k = R_{k+1} \mathcal{U}_k R_k + Q_{k+1} \mathcal{U}_k Q_k, \quad \eta_k = Q_{k+1} \mathcal{U}_k R_k + R_{k+1} \mathcal{U}_k Q_k.$$

Proposition 5.3 (Sheetwise Gram identity). *For a linear transport U , whether or not it is isometric, the preceding decomposition satisfies*

$$A^*A + \eta^*\eta = PU^*UP + QU^*UQ.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^*A_k + \eta_k^*\eta_k = I.$$

Proof. In A^*A and $\eta^*\eta$, mixed terms containing $P_{k+1}Q_{k+1}$ vanish. Summing the remaining terms yields $P_k U^*(P_{k+1} + Q_{k+1}) U P_k$ and its analogue with Q_k . The first identity follows from complementarity of the projectors; the second additionally uses $U^*U = I$. $\square$

The first identity is the general formula. Replacing its right-hand side by U^*U requires the cross-blocks of that Gram operator to vanish. Isometric preservation by history transport provides this condition here; it must not be attributed automatically to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$ and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 5.4 (Finite decomposition with terminal residual). *For each horizon N ,*

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing and has a positive strong limit T_∞ . The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 5.3 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. Telescoping summation proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a bounded positive form and, by the representation theorem for bounded forms, an operator T_∞ . Convergence is strong because, for a positive operator $S \leq I$, $\|Sv\|^2 \leq \langle v, Sv \rangle$; apply this to $S = T_N - T_\infty$. $\square$

The terminal residual is preserved until its extinction is proved in the domain considered. This distinction prevents phase return alone from becoming a hypothesis that all histories are exhausted.

5.4 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and the counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)(N_\Sigma(h) - N_\Pi(h)), \quad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a readout expresses the average over prolongations, define $b_c = Eb_f$. Orientation remains in the argument of b_f ; it is not removed before averaging.

Proposition 5.5 (Exact cancellation defect). *The quantity*

$$\kappa_{\text{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\text{can}}, \quad E(b_f^-) = (b_c)^- + \kappa_{\text{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \leq E|b_f|$. Writing the positive and negative parts as $(|b| \pm b)/2$ yields both equalities. $\square$

The information omitted when only the balance is expressed is the common cancellation of opposite contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fibre contains nine equally weighted successors; conditional expectation is the rule that remains valid for general branching and weights.

5.5 Ternary sheet incidence and rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They yield six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus the counts 4, 6, 8 belong to distinct operations on a single sheet module.

On the six pair positions, define

$$(Ba)_{\{L,L'\}} = a_L + a_{L'}, \quad s(q) = \sum_{\{L,L'\}} q_{\{L,L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The readout $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{2}{9}, \quad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fibre contains 3^5 elements. The readout equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These averages describe the uniform ambient space; the distribution induced by a subset of histories must be computed with its own multiplicities, already preserved in X_k .

To fix the ports without losing multiplicities, at depth k retain the labelled readouts $\lambda_\Sigma(h) = (h, \Sigma)$ and $\lambda_\Pi(h) = (h, \Pi)$. Their sets are

$$I_k = \{\lambda_\Sigma(h) : h \in X_k\}, \quad J_k = \{\lambda_\Pi(h) : h \in X_k\}.$$

They are finite and nonempty because X_k is. Values, quotients and residues are expressed on these ports; equality of values does not identify their labels. The product $I_k \times J_k$ is the rectangular-readout domain, within which the support of admissible incidences is recorded.

With this level fixed, write $I = I_k, J = J_k$ and $A_N = \mathbb{Z}/3^N\mathbb{Z}$. Choose reference ports i_0, j_0 . This choice merely provides coordinates for reconstruction formulas. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \quad (\delta w)_{ij} = w_{ij} - w_{i j_0} - w_{i_0 j} + w_{i_0 j_0}$$

define the following sequence, valid over the finite ring A_N :

$$0 \longrightarrow A_N \longrightarrow A_N^I \oplus A_N^J \xrightarrow{\kappa} A_N^{I \times J} \xrightarrow{\delta} A_N^{(I \setminus \{i_0\}) \times (J \setminus \{j_0\})} \longrightarrow 0.$$

Proposition 5.6 (Exactness of rectangular incidence). *The preceding sequence is exact, with first arrow given by the constant diagonal.*

Proof. The equality $u_i - v_j = 0$ for every pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{i j_0}$ and $v_j = w_{i_0 j_0} - w_{i_0 j}$. Then $u_i - v_j = w_{ij}$. Finally, any matrix in the last term extends with zero reference row and column; its image under δ is the initial matrix. The proof uses only addition and subtraction, so it requires no division by a ring unit. $\square$

Reduction from A_{N+1} to A_N , with k, I_k, J_k, i_0, j_0 fixed, commutes with both maps. Reconstruction formulas are the same at each modular depth. This is the arithmetic compatibility proved in this subsection. Prolongation of histories $k \rightarrow k + 1$ preserves prefix labels; its maps between port sets are distinct data from the ring change $N \rightarrow N + 1$.

These maps are obtained from emission deletion:

$$\rho_k^\Sigma(h\varepsilon, \Sigma) = (h, \Sigma), \quad \rho_k^\Pi(h\varepsilon, \Pi) = (h, \Pi).$$

Fix a reference history with compatible prolongations, whose existence was proved for the tree without terminal leaves. Its two labels define compatible reference ports at every level. Coordinate transport is precomposition:

$$(L_k u)_{i'} = u_{\rho_k^\Sigma(i')}, \quad (L_k v)_{j'} = v_{\rho_k^\Pi(j')}, \quad (L_k w)_{i'j'} = w_{\rho_k^\Sigma(i'), \rho_k^\Pi(j')}.$$

For the sequence's last module, first extend the matrix by zero in the reference row and column, apply this same formula and restrict to the new complement of the reference ports. This rule defines L_k even when a new port has the marked port as its predecessor.

Proposition 5.7 (Naturality of the rectangular complex under prolongation). *The preceding maps satisfy $L_k \kappa_k = \kappa_{k+1} L_k$ and $L_k \delta_k = \delta_{k+1} L_k$. They also commute with coefficient reduction $A_{N+1} \rightarrow A_N$ and compose according to prefix deletion.*

Proof. The first identity follows from $u_{\rho_k^\Sigma(i')} - v_{\rho_k^\Pi(j')}$. In the second, each of the four terms defining δ is transported by the same precomposition. Compatibility of the marked ports identifies their rows and columns; when a predecessor is the marked port, the four terms cancel and agree with extension by zero. All formulas consist of additions and subtractions with integer coefficients, so they commute with modular reduction. Composition of precompositions is precomposition with the composite prefix, proving the last statement. $\square$

5.6 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, at a residence y , defines a free integer complex. Its reduction to A_N uses the same integer register before reduction; successive depths are therefore compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \quad \partial e_y = g_y, \quad \partial b_y = -3c_y g_y, \quad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y through $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a trajectory $\alpha : y \rightarrow y'$ updates memory, its transport in the complex sends the identically named generators to their new representatives, except that

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 5.8 (Naturality of the memory complex). *One has $\partial \Psi_\alpha = \Psi_\alpha \partial$ and $\Psi_{\beta\alpha} = \Psi_\beta \Psi_\alpha$. The maps extend linearly over the chosen coefficient ring.*

Proof. On f_y , the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'} e_{y'} + b_{y'}$. On b_y , the boundary of its image is $-3c_{y'} g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. In a concatenation, memory differences sum telescopically, proving the second identity. $\square$

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v , reduced in the same ring when appropriate, let

$$z^- = r_y, \quad z^+ = r_y + 3c_y v e_y + v b_y, \quad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport also updating v , the correction $K_\alpha = (v_{y'} - v_y)f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α of this local homotopy do not denote the dodecaphasic seal K : they

belong to another domain and their role is specified by the preceding identity. That role can also be checked in the equality

$$z_{y'}^+ - \Psi_\alpha z_y^+ = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'}e_{y'} + b_{y'})$.

Proposition 5.9 (Homology and memory of the local complex). *Over $\mathbb{Z}$, or after reduction to A_N , one has*

$$H_0(\Gamma) = H_2(\Gamma) = 0, \quad H_1(\Gamma) \cong A_N[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integer realization.

Proof. Define the degree-1 homotopy by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be projection onto the summand generated by r_y , and ι its inclusion. Checking on each generator gives

$$\partial h + h \partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex homotopy retracts onto the module generated by r_y , concentrated in degree 1, proving the assertion. $\square$

The transport Ψ_α is a determinant-1 shear. Thus this complex explicitly preserves how c_y changes at chain level, but its local homology does not distinguish values of that coordinate. Complete memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinantal readout in addition to this local complex.

5.7 Assembly with terminal paths

For a finite horizon, residences and their prolongations form a path category. To each residence ν , assign the free module $W(\nu) = A_N[\text{Path}(\nu, \text{Term})]$, preserving each terminal continuation as a generator. An initial trajectory acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding subsection acts covariantly through Ψ .

Assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \rightarrow \dots \rightarrow \nu_q} W(\nu_q) \otimes_{A_N} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary deletes an intermediate residence by composing adjacent arrows. At the first endpoint it transports Γ ; at the last it precomposes W . Alternating signs give the path boundary d_{bar} . The total differential is

$$D = d_{\text{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 in pairs. Naturality from Proposition 5.8 implies that d_{bar} commutes with ∂_Γ ; the factor $(-1)^q$ cancels the cross-terms of the square. Thus $D^2 = 0$. Compatibility is proved on path generators and is not introduced through a label asserting membership in a complex.

Terminal evaluation does not require choosing a particular history either. For each continuation h , the effective composition of its updates is applied. The result preserves the pair consisting of h and its readout, with its multiplicity. If $k < \ell < N$, continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N . Proposition 5.1 proves that direct evaluation and evaluation of these two parts produce the same readout. This gives the terminal naturality square.

Path modules, incidences, memory transport and their differentials arise from the same prolongations. Their assembly does not consist of adding five coordinates to a state that would then be preserved by definition: their relations are constructed through sums, boundaries and effective compositions, and verified on their generators.

5.8 Determinants, limit and arithmetic readout

At each residence, the free local complex Γ admits its graded determinant line

$$\text{Det } \Gamma = \bigotimes_r \left(\bigwedge^{\text{rk } \Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The total assembly of the preceding subsection, in turn, has modules $\text{Tot}_d \mathcal{B} = \bigoplus_{q+r=d} \mathcal{B}_{q,r}$. At finite horizon and over the same ring, these are free of finite rank and only finitely many are nonzero. Its line is

$$\text{Det}(\text{Tot } \mathcal{B}) = \bigotimes_{q,r} \left(\bigwedge^{\text{rk } \mathcal{B}_{q,r}} \mathcal{B}_{q,r} \right)^{(-1)^{q+r}}.$$

This formula follows from direct-sum decomposition in each degree. It distinguishes the local complex's line from that of assembly with all paths; their bases are ordered through the preserved labels. Modular reduction commutes with these exterior powers because the modules and their bases arise from the same free integer modules at fixed horizon. The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_N ; a negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, its bases and the relevant homology identifications. To compute Smith factors, the integer matrix produced before reduction is used, not an arbitrary lift of a modular matrix. A square integer block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_N only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences allows stability of the relevant invariant factors to be formulated and checked; these conditions do not follow from existence of the determinant line.

Passage to the limit preserves compatible inverse systems of residues, histories and their multiplicities. The squares proved in the preceding subsections remain commutative because their maps are given by the same formulas at every level. For an Archimedean coordinate, the additional step is to express nonempty cylindrical intervals, compatible by inclusion and with diameter tending to zero. Their nested compact closures have a one-point intersection: existence follows from compactness and uniqueness from the diameter bound. This is the value of the limit readout. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional expression. The value does not replace the history producing it. The other representation preserves the incidences of the same state. The two destinations share their origin without thereby receiving the same information.

In the following chapters, the coinductive regions, dodecaphasic register and exceptional incidence are expressed from that substrate through their respective maps. The Hilbert-space realization subsequently preserves phase labels and specifies its own unital inclusion; duality and the screens act on the codomains constructed there. The joint operations brought together here transmit histories, sheets, orientation and memory to these representations. Identifying a concrete functional with an external analytic form in turn requires writing its action on generators and verifying its inner products: the conservation identities of this chapter do not replace that composition.

5.9 Provenance, verification and scope

The five operations are recognized by their action on the same histories and by the relations proved in the preceding subsections. The affine memory transport U_α and linear history transport $\mathcal{U}_k$ have the respective domains of Sections 5.1 and 5.3. The sheet grading is $\sigma_k = P_k - Q_k$. For a history $h_j = (\varepsilon_1, \dots, \varepsilon_j)$, accumulation of the balance is made explicit by

$$M_0^\pm = 0, \quad M_k^\pm(h_k) = \sum_{j=1}^k b^\pm(h_j), \quad M_{k+1}^\pm(h_k \varepsilon) = M_k^\pm(h_k) + b^\pm(h_k \varepsilon).$$

Each summand is the signed readout of the corresponding prefix, defined in Section 5.4. Separating the last summand proves the update; the identities $M_k^+ - M_k^- = \sum_{j=1}^k b(h_j)$ and $M_k^+ + M_k^- = \sum_{j=1}^k |b(h_j)|$ follow from the definitions of the positive and negative parts. Accumulation and conditional cancellation are thus distinct operations on the same contributions.

Joint operation	Objects and effective action	Development and proof
Sheet transport and grading	$\mathcal{U}_k : \ell^2(\widehat{X}_k) \rightarrow \ell^2(\widehat{X}_{k+1})$, with $\widehat{X}_k$ consisting of lifted histories; $\sigma_k = P_k - Q_k$, A_k and η_k preserve or interchange the current sheet.	Section 5.3; Proposition 5.3 and Theorem 5.4.
Balance, cancellation and memory	$b : X_k \rightarrow \mathbb{Z}$, $b^\pm : X_k \rightarrow \mathbb{N}_0$; accumulation $M_k^\pm$ over prefixes and cancellation $\kappa_{\text{can}} = (E b_f - Eb_f)/2$ when publishing the conditional average.	Section 5.4; Proposition 5.5.
Ternary sheet incidence	$B : \mathbb{F}_3^4 \rightarrow \mathbb{F}_3^6$, $(Ba)_{\{L,L'\}} = a_L + a_{L'}$; the full ambient space of q is preserved, with $sB = 0$ and $W(q + Ba) = W(q)$.	First part of Section 5.5.
Ports and rectangular complex	$\kappa : A_N^{I_k} \oplus A_N^{J_k} \rightarrow A_N^{I_k \times J_k}$, $\kappa(u, v)_{ij} = u_i - v_j$; the difference δ and precomposition L_k preserve labels and their multiplicities.	Propositions 5.6 and 5.7.
Cellular memory, filling and determinant line	$\Psi_\alpha : \Gamma_y \rightarrow \Gamma_{y'}, K_\alpha \in \Gamma_{y',2}$, $\partial h_* = z^- - z^+$; the lines $\text{Det } \Gamma$ and $\text{Det}(\text{Tot } \mathcal{B})$ are formed on the corresponding based complexes.	Sections 5.6–5.8; Propositions 5.8 and 5.9.

The table identifies five actions of one construction: each prefix preserves its sheets, orientation, residues, quotients, memory and terminal prolongations. Port precomposition takes coordinates from one level to its refinement; coefficient reduction takes A_{N+1} to A_N . Their directions and domains are distinct, and their commutative squares are made explicit in Proposition 5.7. The assembly in Section 5.7 simultaneously uses terminal continuations and transport of the memory complex; its composition formulas bring together the table's actions on the same history system. Reading a determinant line as a scalar preserves the conditions on bases, homology and nonsingularity in Section 5.8.

This development brings together the constructions of state, tree and common measure; the correlative intrinsic operations; and terminal assembly with naturality and limit proved in this section. Their balance, incidence and memory identities are incorporated, with proofs within this chapter. The five continuum constructions are maintained as operations on the same tree. The exposition extends their local proofs and makes explicit the conditions of each specialization: isometric transport, uniform weights, terminal extinction and nonsingularity of a determinantal block.

The accompanying checker compares exact instances of the cocycles, weighted cancellation, rectangular sequence and memory complex. These checks accompany the text's general proofs and verify the operators defined here in their respective domains. The subsequent passage towards exceptional representations and the Hilbert-space realization preserves their actions on generators: Section 12 specifies the phase fibre, Weyl operators and inclusions determining their closures. Documentary preservation and finite checks retain their function of traceability and comparison for these proofs.

6 Coinductive generation of π, φ, e at arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its function is to produce regions with multiplicity, orientation, and transport rules. Prolongation must preserve these data and determine a compatible collection of prefixes of arbitrary length. The word *generation* denotes this operation; *genealogy* denotes reconstruction of its dependencies. Both notions are involved, but they have different functions.

6.1 Orbits, regions, and characters of closure, propagation, and self-scaling

On a word $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; these permutations therefore realize the dihedral group D_3 . The action preserves the catalogue and multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fibre predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. The catalogue contains exactly one orbit with these properties. To specify the other two selections, write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature preserves this class and the orientation *ES* or *NO*. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the orbits of size six, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, equal to that of closure, selects propagation uniquely; the census $\nu = (2, 2, 2)$ selects self-scaling uniquely. The diagonal support cannot be omitted: without it there are four balanced singleton-fibre orbits of multiplicity 144, rather than one. Finally, the existence of an emission with $d = 6$ and orientation *ES* selects one representative of each orbit and yields

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (6.1)$$

These symbols first denote regions of the catalogue. Scalar identification will be proved through their readers; they are not chosen because they agree with digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transverse	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and constant signature 555555 characterize the transverse region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position of the support would erase the distinction between a numerical region and the electron structure.

The microfibre also has a precise bipartite morphology. Each emission is a product of an additive-cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transverse region has a product 18×24 ; the three positive regions together have another 18×24 ; and the return has a product 18×8 . Decomposition by emissions has five parts, whereas decomposition into connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing form to cardinality.

6.2 Finite transport and prolongation boundary

The endomorphisms L_0, L_1 of the ternary word space act before the Archimedean realization. Their determination begins with transitions of the arithmetic state, rather than with two prescribed matrices. If U_t is the composite step of selection, transport, and updating, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl}, \text{pr}, \text{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The source and target matrix pairs obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each six-digit string represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ determine $L_a = X_a^{-1} Y_a$ uniquely. In this row-vector chart, they give

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The schedule (L_0, L_0, L_1, L_1) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace production of those transitions by U_t . The provenance of their record and examination of the sixteen binary schedules are documented in the technical supplement. The stated schedule is the only one of those sixteen that jointly reproduces the three sequences of five blocks.

6.3 Five closure regions and angular compensation

The closure region contains one transverse component and four accompanying components. The symmetric reader acts on the space of its five positions through

$$P_5 = \frac{1}{5} \mathbf{1} \mathbf{1}^\top, \quad \mathbf{1} = (1, 1, 1, 1, 1)^\top.$$

Invariance $P_5 \lambda = \lambda$ and conservation $\mathbf{1}^\top \lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes the unit among reader positions; it does not replace the seed multiplicities 432, 144, 144, 144, 144 with uniform frequencies. The reader's elliptic chart realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. This makes explicit the rule passing from a normalized coordinate to a slope, rather than identifying the two notions without a map. The reader composes the four accompanying positions by the law

$$a \oplus b = \frac{a+b}{1-ab}. \quad (6.2)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. Slope composition is therefore a realization of the TRIT quadratic structure, rather than ordinary addition of four fractions.

Proposition 6.1 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to slope one determines a unique positive compensator of the form $1/q$, with $q > 1$, and gives $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second gives $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, so $q = 239$. $\square$

The resulting closure character has realization

$$\Pi_{\text{cl}} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (6.3)$$

Machin's angular identity recognizes $\Pi_{\text{cl}} = \pi$. The direction of the construction is precise: the slope and number of compositions are specified by the penta fibre reader; the compensation equation forces 239. Once this character has been produced, its evaluation by alternating series does not reselect the seed orbit.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes in (6.3), they provide rational endpoints with explicit error. The numerical value is thus calculated as a consequence of the exact character, without introducing a target prefix.

6.4 Propagation reader and normalization of the unit

The propagation region is realized through a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not only its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

normalization determines $a_0 = a_1 = 1$. Comparison of the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (6.4)$$

The character at one unit of parameter is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

Normalization of the generator matters. Without this datum, the semigroup law would admit $P_t = e^{ct}$ for different c ; the condition fixing the unit of the parameter in the reader is therefore retained. This condition precedes decimal comparison and belongs to the exposition of the operational rule, rather than to the choice of an experimental value.

Proposition 6.2 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Every subsequent ratio is at most $1/(n+2)$. The geometric majorant of the tail sums to

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

Strict inequality follows from the decrease of successive ratios. $\square$

6.5 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. On the modes Σ, Π , the rule is $+1 : m \mapsto \Sigma, -1 : m \mapsto \Pi, 0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi, \Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Its incidence, in that order of modes, is therefore

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (6.5)$$

The four entries are obtained from the schedule, rather than from an equation that has previously received the value of φ . Repetition of the schedule gives counts $3F_{\text{av}}, 9F_{\text{av}}, 36F_{\text{av}}$ at lengths nine, twenty-seven, and one hundred and eight. These counts are not powers of the matrix.

Theorem 6.3 (Positive self-scaling character). *The incidence (6.5) has a unique strictly positive eigenray. When normalized as $(1, x)^T$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and is $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ produces the polynomial. Only one of its two roots is positive. Since F_{av}^2 has all entries positive, the positive ray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in production of F_{av} . $\square$

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The consecutive ratios F_{n+1}/F_n alternate around φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily narrow.

The measure on all compatible histories does not coincide with the mode frequency of the three-event schedule. The chronological frequency is $(1/3, 2/3)$. The measure of maximal entropy of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2} . The latter enters the areal–volumetric readout; it is not obtained by simply replacing one counting frequency with another.

6.5.1 Trajectory space and invariant measure between sheets

The toroidal topology of APP fixes the support of cardinal translations and their winding classes. The correspondence between sheets arises from another operation on that same support: temporal selection of the additive or multiplicative evaluation. Its geometric typing is

$$\eta_{\text{av}}(\Sigma) = \mathcal{H}_{\text{ar}}, \quad \eta_{\text{av}}(\Pi) = \mathcal{H}_{\text{vol}}.$$

The first sheet corresponds to the additive reading of the boundary; the second, to the multiplicative reading of the interior. This map transports mode labels and preserves the incidence (6.5). The square of the self-scaling ratio is determined through the trajectories of that incidence.

Definition 6.4 (Memory-one symbolic envelope). Let

$$X_{\text{av}} = \{(s_k)_{k \in \mathbb{Z}} \in \{\mathcal{H}_{\text{ar}}, \mathcal{H}_{\text{vol}}\}^{\mathbb{Z}} : (F_{\text{av}})_{s_k, s_{k+1}} = 1 \text{ for every } k\}.$$

Index shift acts on this space. It is the smallest memory-one symbolic system containing the modal sequences and preserving their consecutive pairs: it admits the two cross-transitions and volumetric self-retention. Phase, cardinal orientation, and the trajectory records remain in the TPK state from which this envelope is obtained.

Proposition 6.5 (Weighting trajectories between sheets). *In areal–volumetric order, the invariant measure of maximal entropy of X_{av} has transition matrix and stationary distribution*

$$P_{\text{av}} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \quad (\mu_{\text{ar}}, \mu_{\text{vol}}) = \frac{(1, \varphi^2)}{1 + \varphi^2}. \quad (6.6)$$

In particular, $\mu_{\text{ar}}/\mu_{\text{vol}} = \varphi^{-2}$.

Proof. The already constructed incidence has positive eigenvector $r = (1, \varphi)^T$ and eigenvalue φ . The normalization

$$(P_{\text{av}})_{ij} = \frac{(F_{\text{av}})_{ij} r_j}{\varphi r_i}$$

gives the stated matrix. Its rows sum to one because $\varphi^{-2} + \varphi^{-1} = 1$. The symmetry of F_{av} implies that weights proportional to r_i^2 satisfy detailed balance; therefore, $\mu P_{\text{av}} = \mu$. Irreducibility and volumetric self-retention determine a unique stationary distribution.

To verify maximality, every allowed transition satisfies $\log(P_{\text{av}})_{ij} = \log r_j - \log r_i - \log \varphi$. Under any invariant measure, the endpoint terms cancel on average. If ν_i are its vertex weights and $q_i(j) = \nu(s_1 = j \mid s_0 = i)$, for $\nu_i > 0$, then

$$h_\nu \leq H_\nu(s_1 \mid s_0) = \log \varphi - \sum_{i: \nu_i > 0} \nu_i D(q_i \parallel R_i) \leq \log \varphi,$$

where $D(q \parallel p) = \sum_j q_j \log(q_j/p_j)$ is the relative entropy on allowed pairs and $0 \log 0 = 0$ is adopted. The Markov measure in (6.6) attains the bound. Equality in the first inequality requires the Markov property; equality in the second requires $q_i = R_i$. Stationarity and irreducibility then fix μ , establishing uniqueness. This is the Parry measure of the defined envelope. $\square$

The frequency $(1/3, 2/3)$ counts the instants of the primitive calendar; μ weights the trajectories of the envelope. Its quadratic ratio expresses the product of the incoming and outgoing incidence amplitudes. TPK memory and phase restrictions retain their own domain; the entropy of the envelope corresponds to the symbolic system specified in this definition.

6.5.2 Closure operator and limit of marked returns

The closure coordinate π , generated by the regional reader, enters after the construction of modal incidence. Let $e_{\text{ar}} = (1, 0)^\top$, $e_{\text{vol}} = (0, 1)^\top$, and $P_i = e_i e_i^\top$ be the sheet projectors. The reader's conditions are preservation of both sheets, unit volumetric normalization, and an areal mark equal to the closure coordinate. These conditions determine the positive operator

$$D_\pi = \pi P_{\text{ar}} + P_{\text{vol}} = \begin{pmatrix} \pi & 0 \\ 0 & 1 \end{pmatrix}. \quad (6.7)$$

Diagonality expresses preservation of sheets, and the two entries express their normalizations; this makes explicit the uniqueness of the operator for the fixed reader.

Proposition 6.6 (Ratio of boundary returns). *For $n \geq 1$, the marked evaluation*

$$\Theta_n = \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle} = \pi \frac{F_{n-1}}{F_{n+1}} \quad (6.8)$$

satisfies

$$\lim_{n \rightarrow \infty} \Theta_n = \pi \frac{\mu_{\text{ar}}}{\mu_{\text{vol}}} = \frac{\pi}{\varphi^2}. \quad (6.9)$$

Proof. The diagonal entries of F_{av}^n count the paths of length n returning to their initial sheet. The already proved power formula provides F_{n-1} and F_{n+1} . The operator D_π multiplies the first count by π . Finally, F_{n-1}/F_{n+1} is the product of two consecutive ratios converging to φ^{-1} ; its limit is φ^{-2} . The identity with the weight ratio follows from (6.6). $\square$

The closure operator appears only once, in the terminal evaluation of the return. The transfer F_{av} remains determined by the calendar. Information from two readers of the same state is thus composed: modal incidence determines the quadratic weighting, and the closure region contributes its coordinate to the boundary.

The composition also preserves refinement. For positive cylinders $I = [a, b]$ and $J = [c, d]$, the image under the reader $\mathcal{R}(x, y) = x/y^2$ is the interval

$$\mathcal{Q}(I, J) = \left[\frac{a}{d^2}, \frac{b}{c^2} \right].$$

Monotonicity in each argument proves that the nested closure and self-scaling cylinders produce nested intervals for the ratio; positivity of their limits and contraction of their diameters determine π/φ^2 . The scalar expression thus retains a chain of operations compatible with depth, which is applied next in its two orientations.

After the characters have been generated, the two orientations of the reader

$$R(x, y) = \frac{x}{y^2}$$

provide

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (6.10)$$

The second expression enters the exponent of the electron construction developed in [2], in the section on the electronic centre and registers. The first is also obtained as the limit of the marked readout $\pi F_{n-1}/F_{n+1}$. Exchanging the arguments relates the two, but does not turn a reparametrization of the electron equation into a second independent derivation of all its coefficients.

6.6 Exponential realizations of the generated coordinates

The regional coordinates already determined allow recognition of concrete realizations of the TRIT quadratic collection. The order is substantive: construction of the regions precedes evaluation of an exponential at parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not participate in selecting the prefixes that produce them.

6.6.1 Ternary cycle, nilpotent transport, and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, consisting of integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. On residue evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. Quotients of the integer lift are retained in the corresponding arithmetic record.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (6.5) gives

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the nilpotent-transport plane, and the modal-incidence plane. Sharing a matrix dimension does not identify these spaces or their coordinates.

Proposition 6.7 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 6.8 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\boxed{\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.} \quad (6.11)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotency removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

Thus the exponential equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the complete turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use different parameters of one common exponential collection.

In this collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI)\exp(tI) = \exp((s+t)I)$. Its role is not restricted to the parabolic type. The coordinate e , previously obtained from its region, is recognized here as the unit evaluation of the exponential; this identity does not replace its coinductive genealogy.

6.6.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

Using Q keeps this operator square distinct from the dodecaphase record K .

Proposition 6.9 (Polynomial regime projectors). *The maps*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I , and recover the three summands of the representation.

Proof. The three polynomials take values $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

One thus obtains

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos tI + \sin t\mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh tI + \sinh t\mathbb{J}). \end{aligned} \tag{6.12}$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The total representation gathers the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation, and memory. The exponential articulation allows recognition of their types without replacing those dynamics by an identification of fibers.

6.7 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A sequence of blocks defines integer prefixes by $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting at $N_0 = m$, where m is the integer part determined by the reader. The preceding brackets place the closure, propagation, and

self-scaling characters in $(3, 4)$, $(2, 3)$, and $(1, 2)$, respectively; their initial values are therefore $m = 3, 2, 1$. The subsequent blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \quad (6.13)$$

The half-open convention selects a representative when a value lies on a boundary. Passage to the limit uses the closures of these intervals and identifies the two equivalent boundary expansions. This precision prevents the assumption that every intersection of nested half-open intervals is automatically nonempty.

Theorem 6.10 (Determination of the scalar realization). *A compatible block history determines a unique limit of the left endpoints of (6.13). If an exact character admits rational brackets of arbitrarily small width and is separated from the boundary of every decimal cylinder examined, each decimal prefix of finite length is determined after a finite calculation. The prefixes obtained in this way are mutually compatible.*

Proof. Extension by a block preserves the preceding prefix through Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints form a Cauchy sequence and have a unique limit. For a fixed decimal length, a value separated from the partition boundaries has positive distance from them. A sufficiently narrow bracket lies in a single cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; this decision does not follow from numerical width alone. $\square$

Corollary 6.11 (Finite determination and compatibility of regional prefixes). *For each realization π, φ, e and every finite decimal length, the constructed brackets determine the corresponding prefix by a finite calculation. Prefixes obtained at different depths are compatible under truncation.*

Proof. The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, irrationality of π, e, φ ensures their separation from the rational boundaries of every finite decimal partition. Apply the preceding theorem to each character. $\square$

Arbitrary depth expresses the quantifier in the corollary: for every finite horizon, there is a calculation determining its prefix. The mathematical object is the complete compatible collection; each execution obtains a finite projection of it.

6.8 Mirror involution and synchronization of regions

The nonadic connection transports the three characters jointly. In the ninth-phase chart, the self-scaling axis and the ternary sum of closure and propagation remain fixed under mirror exchange. Helical inversion changes the direction of transport; the mirror of the two channels changes their relative orientation. These are different involutions, although both act on the same system of trajectories.

The agreements of counts illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality of π , e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. In phases eight and nine the relative component vanishes, but the diagonal component changes from (59, 59, 59) to (43, 43, 43). Relative synchronization coexists with a displacement of -16 per channel. The closure leading to α must preserve both kinds of information: relative agreement and evolution of memory.

6.9 Sturmian helical suspension

The relation between bases nine and ten in the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \quad (6.14)$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic readout of refinement has an exact consequence.

Lemma 6.12 (Irrationality of the slope). *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. The factorization of the left-hand side contains the prime three, whereas that of the right-hand side contains only the primes two and five. Equality is impossible. $\square$

The lower mechanical word, with intercept θ , is

$$z_n(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod{1}$ through the interval $[1 - \delta, 1)$. The event sequence has density δ , and the sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, event separations are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word, of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first, rather than being a second free parameter.

6.9.1 Capacity index, vacancies, and induction of the tritic regime

The vacancy sequence is linked to refinement through an increment identity. To exhibit it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \quad \mathcal{C}_t = \lfloor (t+1)\rho \rfloor \quad (t \geq 0).$$

The index $\mathcal{C}_t$ expresses the decimal capacity of refinement in the normalization comparing a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphase record constructed later.

Proposition 6.13 (Complementarity of capacity and vacancy). *For $t \geq 1$,*

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t(0). \quad (6.15)$$

Consequently, if $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t(0) = b - a.$$

Proof. Irrationality of δ gives

$$\mathcal{C}_t = t + 1 - \lfloor (t+1)\delta \rfloor = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting consecutive terms produces (6.15). Summation over the interval telescopes. $\square$

Each transition increases the block counter by one. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity remains unchanged while a transition is incorporated into the history. Here a vacancy is a capacity-retention event, with a determined position and antecedent. The preceding balance preserves length exactly between acquired capacity and recorded vacancies. The observed phase $g_t^{\text{cap}} = [\mathcal{C}_t]_9$ satisfies

$$g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

This capacity phase and chronological phase $[t]_9$ are two different observations. Local capacity retention and a complete holonomy turn have distinct mechanisms, although both allow the phase observation to be preserved while the state changes.

The relation with TRIT can be followed in the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = \mathcal{C}_{p_j}, \quad h_j = p_{j+1} - p_j.$$

The inequality $p_j \delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j . Since $0 < \delta < 1$, it also gives $\lfloor (p_j + 1)\delta \rfloor = j$, and hence

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$ and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (6.16)$$

An interval of length 22 preserves the class; one of length 21 shifts it by one in the negative direction. The canonical selector is applied to the positive phase $\tau_j = 1 + [v_j]_9$:

$$\tau_j = M_{\text{ph}}(\tau_j).$$

A tritic regime is thus determined by the capacity index retained at each event, in addition to the vacancy's binary encoding.

The positions of the short intervals form the induced mechanical coding of slope $22 - 1/\delta$. Their separations 6 and 7 determine the lengths of plateaus of the class $[v_j]_3$, once the origin and endpoint convention are fixed. The first classes are 2 for 6 events, 1 for 7, and 0 for 7; the first segment of 6 is the initial restriction of a plateau. Their signatures traverse 0, -1 , $+1$. Equations (6.15) and (6.16) make explicit the link between refinement, vacancies, and regime selection. The quadratic reading of TRIT subsequently realizes these types through their respective generators; the temporal event and its geometric representation thus retain an identifiable dependence.

Inverting the monotone index supplies another useful observation:

$$\mathcal{T}(k) = \min\{t : \mathcal{C}_t = k\} = \left\lfloor \frac{k}{\rho} \right\rfloor - 1 \quad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks needed to gain m units of capacity is

$$\mathcal{T}(k + m) - \mathcal{T}(k) = m + \lfloor (k + m)\sigma \rfloor - \lfloor k\sigma \rfloor \in \{m + \lfloor m\sigma \rfloor, m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, whereas return time measured by the other index admits two values. This distinction belongs to refinement dynamics and is not an a posteriori correction of its numerical values.

Theorem 6.14 (Complexity of phase readouts). *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. Preserving the phase $n \bmod 9$ gives complexity $p_9(m) = 9(m + 1)$; preserving only its ternary class gives $p_3(m) = 3(m + 1)$. All three topological entropies vanish.*

Proof. For factors of length m , the predecessors of the two endpoints of the coding interval form $m + 1$ distinct points on the circle: agreement between shifted endpoints removes duplications, and irrationality prevents further agreements. These points determine $m + 1$ distinct intervals of constant itinerary. Every irrational orbit visits each interval, giving $p(m) = m + 1$.

For a fixed phase modulo nine, the positions of that phase follow a rotation of step 9δ , which is also irrational. Density makes every factor occur in each phase, and the first coordinate distinguishes its nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m+1)) \rightarrow 0$ for $c = 1, 3, 9$. $\square$

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has characters with frequencies

$$\frac{a}{9} + k\delta \pmod{1}, \quad a \in \mathbb{Z}/9\mathbb{Z}, k \in \mathbb{Z}.$$

The spectral module is not $\log 10$: it is the additive module generated by $1/9$ and δ , modulo the integers. The factor $\log 10$ belongs to the normalization in (6.14).

An operator presentation makes memory explicit. On $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n+1\rangle$, $V_\delta|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n \delta}|n\rangle$, and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finitely supported vectors,

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = e^{2\pi i \delta} T W_\delta, \quad [D_M, T^9] = T^9. \quad (6.17)$$

The first two relations are Weyl covariances; the third expresses that one nonadic revolution increases memory degree. The base rotation is abelian, whereas the algebra of observables and translations is not. The inversion $|n\rangle \mapsto |-n\rangle$ conjugates forward and backward motion. This realization provides an aperiodic helical suspension with finite phase return, rather than a periodic word of length nine.

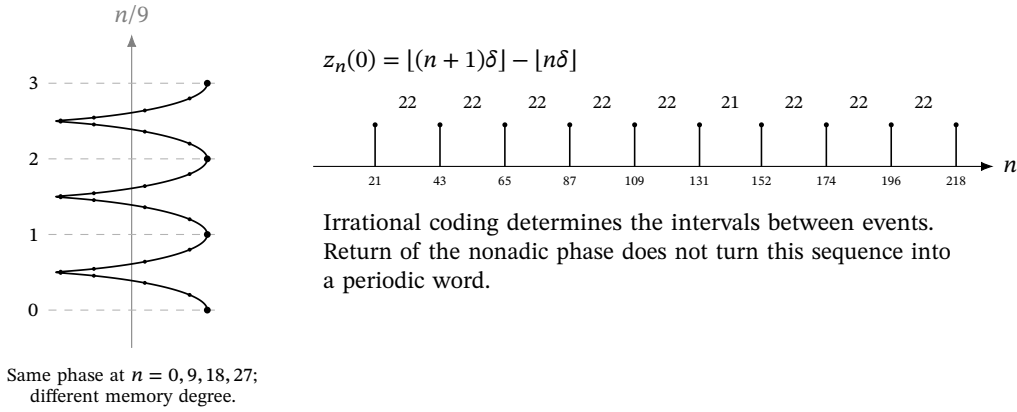


Figure 5: Complementary representations of transport. Left: projection of the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$: advancing nine steps recovers the phase and increases the height. Right: the first ten events of the lower word for $\delta = \log_{10}(10/9)$, with exact separations 21 or 22. The helical representation introduces no additional physical metric.

For this count, fix the intercept $\theta = 0$ and the cumulative endpoints

$$(N_0, \dots, N_4) = (0, 729, 972, 999, 1000),$$

corresponding to the cubic order $(729, 243, 27, 1)$. The lower counts are $\lfloor N_j \delta \rfloor - \lfloor N_{j-1} \delta \rfloor$, giving $(33, 11, 1, 0)$. The upper counts are $\lceil N_j \delta \rceil - \lceil N_{j-1} \delta \rceil$, with the same initial boundary zero, giving $(34, 11, 1, 0)$. These values depend on the intercept and the ordering of the strata; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in a common count preceding any unit chart; identifying them with physical exponents additionally requires the corresponding dimensional derivation.

The term *aperiodic crystal* is used here for this symbolic order, with specified complexity, spectrum, and return. Zero topological entropy does not by itself amount to zero physical dissipation: a thermodynamic realization needs defined energy dynamics and an erasure operation. Likewise, this Sturmian readout does not replace all TPK operators or by itself prove periodicity or aperiodicity of the complete output of α . Its function is to make explicit an order structure that simple schedule return does not describe.

6.10 Nonadic connection, holonomy, and monodromic representation

The already constructed regional connection is now studied through three objects: transport of the complete state, the representation of its turns, and the suspension of its itineraries. The APP-TRIT-TPK operations and the initial catalog retain their preceding definitions. The necessary distinction here concerns what each representation retains of the same history (Sections 4 and 6).

At prolongation index k , a state chart preserves

$$\hat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k brings together the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry record, Σ_k the boundary signature, and W_k the incidence information. The phase is $g_k = 1 + [(k-1)]_9$; m_k counts complete turns. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphase position, whereas the integer m_k distinguishes its successive turns.

Definition 6.15 (Transport through one nonadic turn). Let $\mathcal{R}_k \subseteq \hat{X}_k \times \hat{X}_{k+1}$ be the admissible-extension relations of the TPK. The realization of holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation to s turns is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves admissible continuations. On an individualized history, each arrow leads to the state actually prolonged; the isolated visible block may still admit more than one continuation. This difference is precisely the function of memory and boundary conditions in determining the trajectory.

Proposition 6.16 (Iteration of return and accumulated memory). *In every admissible composition of s turns,*

$$g_{k+9s} = g_k, \quad m_{k+9s} = m_k + s, \quad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-turn rule preserves phase and increases the counter by one unit. Applying it to the already transported state repeats exactly those two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. $\square$

The counter thus provides an integer witness to the difference between turns. After 12 turns the pair of finite phases is restored, but the counter increases by 12. The nonsplit extension (4.5) expresses the composition of the calendar through its cocycle; the trajectory additionally retains the integer lift of that datum. Conservation here means transport of previous contributions, not immobility of every coordinate.

6.10.1 Bilateral representation of the return index

The shift T in (6.17) admits an expression in phase and turn coordinates. We number phases from 0 to 8, by shifting the origin relative to g_k , and write $k = 9m + r$, with $0 \leq r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\text{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ expresses that same unitary shift

$$T|r, m\rangle = \begin{cases} |r+1, m\rangle, & r < 8, \\ |0, m+1\rangle, & r = 8. \end{cases}$$

The inverse reduces r by one unit when $r > 0$, and sends $|0, m\rangle$ to $|8, m-1\rangle$. Passage through the phase origin therefore incorporates the exact quotient unit.

Proposition 6.17 (Representation of the monodromy of the cycle). *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m+1\rangle, \quad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\text{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the turn group of the cycle.

Proof. Nine shifts preserve the remainder of division by 9 and increase the quotient. The powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m+s\rangle$ distinguishes every integer s . A permutation of an orthonormal basis is unitary. $\square$

This representation corresponds to the covering of the cycle, whose turn group is $\mathbb{Z}$, and represents phase together with its counter. The TPK state additionally contains the trajectory coordinates already described. The bilateral extension admits negative indices as coordinates of the covering; a trajectory starting from an initial condition uses its admissible half-axis. This provides an operator inverse of the index without turning every extension relation into a bijection of the complete state.

With $D_M|k\rangle = \lfloor k/9 \rfloor |k\rangle$, the identity $[D_M, S] = S$ expresses the memory increment on the dense domain of finitely supported vectors. If $\mathcal{I}|k\rangle = |-k\rangle$ and P_0 projects onto indices divisible by 9, one additionally obtains

$$\mathcal{I}T\mathcal{I} = T^{-1}, \quad \mathcal{I}D_M\mathcal{I} = -D_M - (I - P_0). \quad (6.18)$$

Indeed, for $k = 9m + r$, $\lfloor -k/9 \rfloor = -m$ if $r = 0$ and equals $-m - 1$ if $r > 0$. Inversion thus preserves the boundary term of Euclidean division.

6.11 Block clock and capacity clock

The index of an application of refinement and the number of available decimal positions have related but different laws. To express them, we reserve n for the ternary-block index and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho, \quad \mathcal{C}_n = \lfloor (n+1)\rho \rfloor, \quad n \geq 0.$$

The letter $\mathcal{C}_n$ here designates capacity, not the dodecaphase record K . At the canonical phase origin, the preceding mechanical word provides

$$Z_n = \sum_{j=0}^n z_j(0) = \lfloor (n+1)\delta \rfloor.$$

Proposition 6.18 (Exact balance between capacity and vacancies). *For $n \geq 0$ and, in the second identity, $n \geq 1$,*

$$\mathcal{C}_n = n - Z_n, \quad \mathcal{C}_n - \mathcal{C}_{n-1} = 1 - z_n(0).$$

Proof. The irrationality of δ implies $\lfloor (n+1)\delta \rfloor = \lfloor (n+1)\delta \rfloor + 1$. Then

$$\lfloor (n+1)(1-\delta) \rfloor = n+1 - \lfloor (n+1)\delta \rfloor = n - Z_n.$$

Subtracting two consecutive identities completes the proof. $\square$

An event $z_n = 1$ adds a block to the history while maintaining capacity $\mathcal{C}_n$; an event $z_n = 0$ increases both counters. In any interval, capacity increments and vacancies sum to exactly its length. Reducing modulo q , for $q = 9$ or $q = 108$, gives

$$[\mathcal{C}_n]_q = [n]_q - [Z_n]_q.$$

The uniform block phase and the capacity phase are related through the integrated memory of vacancies. In particular, a retention of capacity and a complete turn of the connection have different mechanisms, although both may preserve a phase observation.

Monotone inversion of the clock gives a first-arrival time:

$$T_{\text{cap}}(a) = \min\{n : \mathcal{C}_n = a\} = \left\lceil \frac{a}{\rho} \right\rceil - 1, \quad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time taken to gain b units of capacity is

$$T_{\text{cap}}(a + b) - T_{\text{cap}}(a) = b + [(a + b)\sigma] - [a\sigma].$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. Thus 9 or 10 blocks are obtained to gain 9 units of capacity, and 113 or 114 to gain 108. These are times of the inverse clock, measured from a first arrival; the turn counter is always interpreted in the index to which it belongs.

6.12 Symbolic suspension and aperiodic time crystal

Rotation coding combines phase periodicity, local recurrence, and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0, 1\}^{\mathbb{Z}}$ be the closure, in the product topology, of the translates of the bilateral word $z_n(\theta)$. Denote by σ_{sh} the shift of its sequences. Retaining a nonadic phase, the step is

$$F(r, z) = ([r + 1]_9, \sigma_{\text{sh}} z), \quad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 6.19 (Symbolic aperiodic time crystal). The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = ((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}) / \sim, \quad (y, t + 1) \sim (Fy, t),$$

with flow $\Phi_s[y, t] = [y, t + s]$. This model of temporal order, together with its vacancy coding and representation of the turn counter, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and its return section; aperiodicity characterizes its itineraries. The frequency of ones in every sequence is δ : the uniform discrepancy bound of less than one, obtained by telescoping in any window, persists on taking the closure. A periodic sequence would have rational frequency. Therefore, $\mathcal{X}_\delta$ contains no periodic orbits. The suspension also has no closed orbits: a return of the flow would require an integer time and periodicity of the itinerary.

Finite words do recur. Itineraries of length m arise from a partition of the circle by $m + 1$ points; each interval is visited recurrently by the irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why every word appears in all 9 phases, and explains the already established complexities

$$p(m) = m + 1, \quad p_9(m) = 9(m + 1), \quad p_3(m) = 3(m + 1).$$

Recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

Realization through the observables V_δ , W_δ and shift T expresses the same order through the covariances (6.17). Frequencies of the uniform clock belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[\mathcal{C}_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by the uniform phase when interpreting returns. The term *time crystal* is thus defined through a mathematical dynamical system. Its physical realization requires specification of an energy dynamics, its temporal observables, and its stability, data distinct from the present symbolic construction.

6.13 Specular symmetry of the regions and double realization

Specular exchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$cB = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixation refers to the involution within a fiber; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$, and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the distribution that the aggregate preserves without distinguishing it. The inversion $\mathcal{I}$ of the temporal index and the channel involution $\mathfrak{c}$ consequently act on different data.

Phase return has an explicit witness:

$$B_1 = (012222 \mid 121221 \mid 112202),$$

$$B_{10} = (101222 \mid 112221 \mid 112002).$$

Both positions have phase 1; the block, axis, and memory have changed. Likewise, synchronization of the censuses at phases 8 and 9 cancels their relative differences, while the diagonal component passes from $(59, 59, 59)$ to $(43, 43, 43)$. Equality of a relative observable does not exhaust the state determining it.

Double realization preserves that architecture. In Section 11.1, the same prefix determines nested cylinders on the one hand and a sequence of coded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞ . Figure 6 anticipates these two maps. The subsequent incidence development constructs codes, lattices, and the realization of Moonshine in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

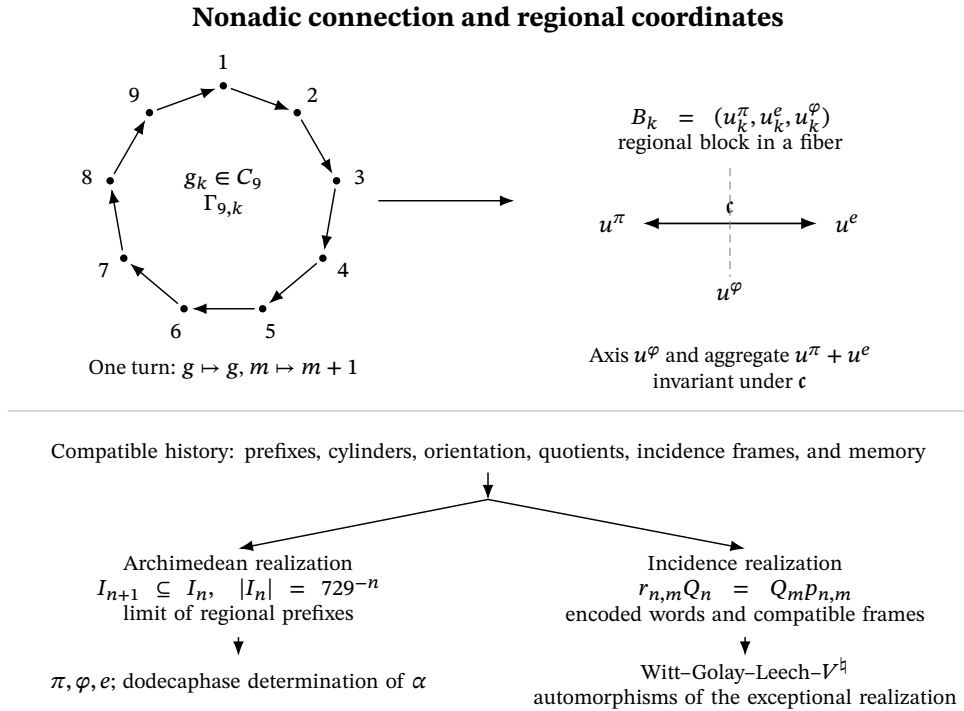


Figure 6: Nine phases of a connection and two realizations of a compatible history. The cycle represents phase; each turn increments the counter. The regional diagram represents the involution $\mathfrak{c}$, which interchanges closure and propagation and fixes the self-scaling axis in a fiber. This invariance does not require the axis to remain constant during transport. The two lower maps use the same prefix: one determines its numerical limit, while the other preserves words and incidence frames. The maps of the second realization are specified in Section 11.1; the exceptional construction is developed afterwards.

6.13.1 Rectilinear representation of phases and return

The representation on the segment $[0, 10]$ preserves the positional origin of APP and orders its nine interior marks. In this diagram, the integers $1, \dots, 9$ are phase representatives. Horizontal position indicates the order of the charts; upper labels specify regional operations. Figure 7 thus brings together the initial line, fixation of the self-scaling axis, the specular distribution of the two channels, and return with memory (Section 6.10).

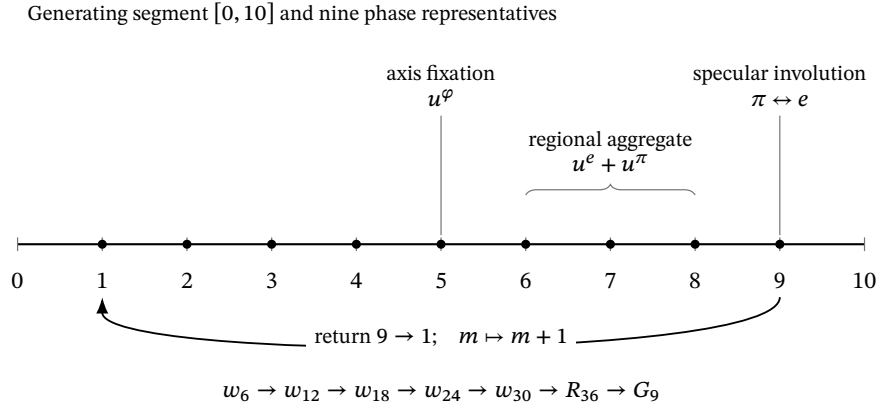


Figure 7: Rectilinear representation of the nonadic connection. The mark 5 locates the first strong saturation and the regional self-scaling axis; the brace 6–8 identifies the ternary aggregate fixed within each fiber; phase 9 selects the closure orientation. Return preserves phase and increases memory. The abscissae number operational positions, while the labels u^π, u^e, u^φ designate blocks of $\mathbb{F}_3^6$.

Fixation of the axis is formulated in the fiber of a given boundary. If $q = u^\pi + u^e + u^\varphi$ and $a = u^\pi + u^e - u^\varphi$, then

$$u^\varphi = 2(q - a), \quad s = u^\pi + u^e = q - u^\varphi \quad \text{in } \mathbb{F}_3^6.$$

The data (q, a) determine the axis and aggregate; the ordered distribution of s retains the specular freedom resolved by prolongation. The fifth phase has a local solution and an extension, with boundary datum $c = 111111$ and residual signature $\rho_W = 000$. This is the meaning of its first strong saturation. The symbol φ placed above the mark 5 identifies the regional axis entering that operation.

The intermediate phases preserve this decomposition within each fiber, while their boundary data evolve. The complete census of local multiplicities and extensions of the distinguished branch is

phase	1	2	3	4	5	6	7	8	9
N_{loc}	3	2	5	4	1	1	3	3	2
N_{ext}	2	5	4	1	1	3	3	2	1

Phase 6 combines local uniqueness and three prolongations; phase 7 resolves a triple specular fiber; phase 8 synchronizes the three regional censuses. Phase 9 preserves two local distributions with the same axis and aggregate, one of which prolongs. These counts express different functions within a single connection and justify the marks on the line.

In particular, for phases 6, 7, 8, the regional sums are respectively 012100, 101001, and 112000. The conservation represented by the brace is an invariance under the specular distribution within each fiber. Subsequent real evaluation applies the readers of compatible cylinders to complete histories; the internal sum represented in the figure belongs to the ternary domain.

Finally, the arc $9 \rightarrow 1$ represents passage from one turn to the next. The formulas of Section 6.10 preserve the phase remainder and increase the turn counter. The rectilinear diagram and the cyclic diagram in Figure 6 therefore express two complementary aspects of the same holonomy: the order of positions and composition of the return on the state with memory.

6.14 Temporal transport and tritic exponential composition

The Euler relations of Section 6.6 describe local realizations of the three regimes. For a fixed signature τ , the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation sends J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau) \exp(tJ_\tau) = \exp((s+t)J_\tau), \quad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is unity. In the elliptic type this is realized through $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, through $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I + tN)(I - tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

The elliptic realization closes a turn; the nilpotent one preserves a linear displacement; the hyperbolic one separates expansion and contraction with unit determinant. At the already generated regional parameters,

$$\exp(\pi J_e) = -I, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.$$

The evaluation $\exp(I) = eI$ recognizes unity in the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transport of a local realization by an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U \exp(tJ) = \exp(tJ')U$. The identity transports the regime and its norm. A transition between different quadratic types belongs, by contrast, to the TPK selection of sheets and regimes; it is not a conjugation changing the value of J^2 . The temporal product therefore preserves the order of transitions and the information at their endpoints.

This articulation distinguishes closure of a local representation from holonomy of a history. The identity $\exp(2\pi J_e) = I$ may hold at the same time as $S|r, m\rangle = |r, m+1\rangle$. The first equality restores the orientation of the regime; the second records another turn. The discrete structure preserves both: local return and the accumulated provenance of the state realizing it.

6.15 Multiplicative irreducibility and return periods

The distinction between structure, evaluation, and phase also appears in the multiplicative sector. APP's local product contains both residue and quotient; iterating it constructs normal forms and factorizations before their decimal periods are examined. This development places the expression *prime mode* in a precise domain and relates irreducibility to the operations already defined, rather than to an isolated coincidence of periods (Section 6.15).

Let

$$\mathcal{W}_9 = \{\emptyset\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \dots, 9\}^\ell.$$

Sequences are ordered from the least significant digit. The evaluation $\text{ev}_9(u) = \sum_j u_j 9^j$, with $\text{ev}_9(\emptyset) = 0$, is bijective onto the nonnegative integers: for a positive integer, division of $n - 1$ by 9 uniquely determines the first digit and continuation. This is bijective nonadic numeration; it retains the boundary representative 9.

Addition $\boxplus_\#$ is obtained by applying APP's additive sheet to the digits at each position and propagating their quotients. If there is only one digit, it is the provisional residue; a nonzero incoming quotient is normalized through a second additive operation. The sum of quotients is transported to the next position. To multiply a sequence by a digit d , the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated by

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j > 0. \end{cases}$$

Start with $c_0 = 0$ and incorporate the final quotient when nonzero. During addition, $c_j \leq 2$; during multiplication by a digit, $c_j \leq 9$. Local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_\# u$, with $d \odot_\# \emptyset = \emptyset$.

For $v = (v_0, \dots, v_{m-1})$, the complete product is constructed through the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot_{\#} a_{j+1}) \boxplus_{\#} (v_j \odot_{\#} u), \quad u \boxtimes_{\#} v = a_0.$$

The rule composes both local sheets. Its evaluation satisfies

$$\text{ev}_9(u \boxtimes_{\#} v) = \text{ev}_9(u) \text{ev}_9(v). \quad (6.19)$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the power of 9. The recurrence then gives $\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u)$, whose iteration proves (6.19). Uniqueness of normal form transports associativity and commutativity to the constructed product; its identity is (1).

Definition 6.20 (Reduced multiplicative coproduct). On the finitely supported vector space generated by nonempty normal forms, define

$$\tilde{\Delta}_{\#} e_w = \sum_{\substack{u \boxtimes_{\#} v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A non-unit normal form is irreducible when its image under $\tilde{\Delta}_{\#}$ is zero.

Factorization fibers are finite by (6.19). In the completion $\mathcal{H}_{\#} = \ell^2(\mathcal{W}_9 \setminus \{\emptyset\})$, images of distinct forms have disjoint supports. If $d_{\#}(w)$ counts ordered non-unit factorizations, the domain of the closure is

$$\text{Dom}(\overline{\tilde{\Delta}_{\#}}) = \left\{ x \in \mathcal{H}_{\#} : \sum_w d_{\#}(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is this weighted sum. Hence the operator

$$A_{\#} = (\overline{\tilde{\Delta}_{\#}})^* \overline{\tilde{\Delta}_{\#}}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered non-unit factorizations of w . The projection

$$P_{\text{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|) \mathbf{1}_{\{0\}}(A_{\#})$$

selects exactly the irreducible forms. Through multiplicative evaluation, these correspond to ordinary primes. Selection depends on factorization structure and keeps the unit state separate.

For a value $n \geq 2$ coprime to 10, decimal division induces the residue permutation $r \mapsto 10r \pmod{n}$. Return of residue 1 has period

$$\tau_n = \text{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \text{lcm}_{p^{a_p} \parallel n} \text{ord}_{p^{a_p}}(10).$$

Composite return synchronizes the returns of prime powers. For a prime $p \notin \{2, 3, 5\}$,

$$\tau_p \mid p-1, \quad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \quad M_p \equiv 0 \pmod{9}.$$

The last congruence expresses nonadic closure of decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for every n coprime to 90. In particular, 7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This differentiation specifies the relation with aperiodic refinement. The vacancy sequence determines increments of the capacity index; factorization determines multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, while quotient and factorization retain information that this closure does not distinguish. The connection to the treatise's spectral constructions resides in these operators and their intertwining. Proof of their subsequent analytic results retains its own development; it is not replaced by periodicity of a residue observation.

6.15.1 Vacancy discrepancy and Dirichlet readings

The connection also admits an exact analytic expression. Retain the same coding $z_n = z_n(0)$, for $n \geq 1$, and consider two subsequent observables: the series over all indices and the product over indices selected by irreducibility. For $\operatorname{Re} s > 1$,

$$Z_{\text{vac}}^+(s) = \sum_{n \geq 1} \frac{z_n}{n^s}, \quad Z_{\text{vac}}^\times(s) = \prod_p (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in this half-plane. Their common factor is density δ , but their discrepancies differ.

Proposition 6.21 (Separation of vacancy density). *We have*

$$Z_{\text{vac}}^+(s) = \delta \zeta(s) + H_{\text{vac}}(s), \quad H_{\text{vac}} \in \mathcal{O}(\operatorname{Re} s > 0).$$

In $\operatorname{Re} s > 1$, with logarithms defined by absolutely convergent Euler series,

$$Z_{\text{vac}}^\times(s) = \zeta(s)^\delta G_{\text{vac}}(s), \quad \log G_{\text{vac}}(s) = \sum_p (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centered partial sums satisfy

$$A_N := \sum_{n=1}^N (z_n - \delta) = \lfloor (N+1)\delta \rfloor - N\delta, \quad |A_N| < 1.$$

Abel summation represents the centered term by $s \int_1^\infty A_{\lfloor x \rfloor} x^{-s-1} dx$. The integral converges locally uniformly in $\operatorname{Re} s > 0$, proving holomorphy. The second identity follows by separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. $\square$

The decomposition reveals what the reading on primes adds: it examines the same vacancy sequence on a subset determined by factorization. Controlling discrepancy over all integers gives the first holomorphic continuation; the second observable preserves its own discrepancy on primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation explains their relevance to the arithmetic kernel without transferring the proof of the treatise's terminal results to this article.

7 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

7.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lceil su_c(j) \rceil - 1 = \lfloor sv_c \rfloor. \quad (7.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 7.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lceil su_c(j) \rceil - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (7.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 7.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (7.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (7.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lceil su_c(j) \rceil - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (7.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 7.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (7.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (7.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (7.5) and its residual bound. $\square$

7.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (7.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (7.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 7.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (7.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 7.2. For the second, apply (7.4) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (7.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (7.8) determines its compatibility with every subsequent extension exactly.

Example 7.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (7.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

7.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (7.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 7.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$\begin{aligned} L_0 &= \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, & L_1 &= \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \end{aligned}$$

Both matrices are invertible.

Proof. The matrices

$$\begin{aligned} A_0 &= \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 7.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

7.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (7.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 7.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (7.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (7.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (7.13)$$

Lemma 7.8 (Integer criterion for intersection). *The cylinders in (7.10) have a nonempty intersection if and only if (7.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (7.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (7.13). $\square$

Definition 7.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N + u, D),$$

$$T_{u,d}(n, k, N, D) = (n+1, k+1, 729N + u, 1000D + d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 7.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (7.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (7.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. $\square$

Theorem 7.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)} s_c(n, k) &= s_c(n+1, k), \\ T_{u_c(n), d_c(1000, k)} s_c(n, k) &= s_c(n+1, k+1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n+a) - P_c(729, n) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 7.2 places v_c simultaneously in both intervals in (7.10); the intersection criterion gives compatibility. The update identities are (7.5) in each base. Recovery follows from (7.4). Thus the defect recurrence and prefix recovery are consequences of the same update. $\square$

Example 7.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

7.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 7.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$\lfloor s\ell_c(j) \rfloor \leq 729P_c(729, n) + u \leq \lceil su_c(j) \rceil - 1. \quad (7.16)$$

Theorem 7.14 (Uniqueness of the prolongation compatible with the reader). *Condition (7.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (7.16) coincide with $P_c(729, n + 1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n + 1)$. By (7.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

7.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (7.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (7.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (7.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (7.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 7.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (7.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (7.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (7.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (7.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (7.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (7.23). $\square$

Example 7.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^W &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 7.17 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (7.24)$$

Proposition 7.18 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (7.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

7.7 Composition theorem for the block, cylinder, and signature

Theorem 7.19 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (7.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (7.14) and (7.15);*
2. *its six trits produce the signature $\mathcal{S}(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (7.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 7.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 7.11 to be applied. The deterministic application of (7.6) produces the bands used by (7.17)–(7.20). The recovery identities are those of Theorem 7.15. Finally, Theorem 7.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

7.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \tag{7.26}$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl}, \text{pr}, \text{au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \tag{7.27}$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (7.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

8 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

8.1 Regional bands, encoding, and integer signatures

8.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (8.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (8.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (8.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

8.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (8.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (8.4)$$

$$a_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (8.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (8.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 8.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

8.2 Conversion clock and first zero advance

8.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (8.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 8.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (8.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 8.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

8.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (8.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

8.3 Oriented construction of the twenty-four-trit descriptor

8.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative-positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

8.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are (5, 7, 7, 7, 7, 7); this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (8.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase-charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (8.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

8.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\Omega_{24} = (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \quad (8.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 8.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\ 222111\ 211201\ 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (8.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (8.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

8.4 Functional panel and terminal repertoire

8.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (8.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

8.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (8.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its 8! orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (8.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

8.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_{j,i}, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 6: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 6,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 7: Complete phase–charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Proposition 8.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 7, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing

$100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

8.5 Comparative readers, affine reader, and temporal incidence

8.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (8.14)$$

Lemma 8.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (8.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

8.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (8.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

8.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \text{ in } \mathbb{F}_3^6. \quad (8.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

8.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu, o, s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (8.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (8.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 8.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v, u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v, u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (8.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

8.6 Enumeration of cylinders and decimal candidates

8.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \quad (8.19)$$

8.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 8.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lfloor \frac{pD}{Q} \right\rfloor \leq N < \left\lfloor \frac{(p+1)D}{Q} \right\rfloor \right\}. \quad (8.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q)$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lfloor pD/Q \rfloor \leq N$; the second is equivalent to $N < \lfloor (p+1)D/Q \rfloor$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 8.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 8.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (8.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

8.7 Exceptional incidence and heterotypy condition

8.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (8.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (8.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (8.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

8.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (8.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (8.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (8.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

8.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{array}{lll} 1 \rightarrow 4, & 1 \rightarrow 5, & 2 \rightarrow 6, \\ 3 \rightarrow 7, & 4 \rightarrow 8, & 5 \rightarrow 9, \\ 6 \rightarrow 10, & 7 \rightarrow 11, & 8 \rightarrow 12. \end{array}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

8.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) = & [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ & \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \quad (8.26)$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 8.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_9$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (8.27)$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (8.22); the destination bands T_8, T_{12} and residues $R_{4,8}, R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (8.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (8.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

8.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_{72} = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

$$\begin{array}{cccc} 020022 & 222111 & 211201 & 021101 \\ 002001 & 020111 & 200110 & 121012 \\ 010122 & 001001 & 221110 & 222110 \\ 121020 & & & \end{array}$$

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

8.8 Regional composition and continuity of the realizations

8.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{8.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (8.13).

The register is then formed by the twelve Euclidean divisions in (8.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

8.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

8.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

8.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

8.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbitalSelection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (8.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean’s native evaluation; their reports record `Lean.ofReduceBool`, in addition to the library’s customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

9 Dodecaphasic extraction of the arithmetic–geometric coupling constant

Definition 9.1 (Holographic arithmetic–geometric coupling constant). This term denotes the ordered finite register produced by the oriented extraction of the APP–TRIT–TPK state, with its declared phase origin, normalization, and readers, whose positional representation enters the arithmetic closure and whose incidence and geometric realizations act on the same register. The symbol K denotes its canonical integral representation; κ_{ag} denotes its periodic rational readout in the specified chart.

Generation of the regional coordinates does not exhaust the information in the APP–TRIT–TPK state. One Archimedean readout determines a value through compatible cylinders; another preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphasic register belongs to this second readout before entering the closure of alpha. It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A word of twelve catalogue blocks, a list of twelve oriented events, a signed integer vector, and a base-one-thousand dodecaphasic register are distinct objects. Their comparison requires the maps that relate them. Equality of length does not provide these maps, and loss of signs is not merely a notational simplification.

This chapter retains the causal boundary established in the preceding chapters: the nine APP positions on the horizontal and vertical axes produce the arithmetic sheets; TRIT specifies regime and orientation; TPK selects, transports, updates, and prolongs, preserving residue, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite calendar. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ reversed: the last map is a reduced memory readout, not the generator of the signed register.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence between three coordinatizations of a register previously produced by APP–TRIT–TPK. The Hadamard transformation relates the signed register to the integer register; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12}/4} K \xrightleftharpoons[(D_3, D_4, Q)]{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. Once the base 1000, the 12 positions and the phase origin are fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplication by $1000^{12} - 1$ followed by Euclidean divisions recovers the complete register. The proofs in §9.8, §9.9 and §9.11 establish these equivalences in their respective domains.

Two compositions then act on that register. The positional composition combines its components with the closure, propagation and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and obtains

$$K \longmapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The complete register allows both compositions to be reconstructed. The support B_K specifically preserves membership in the class selected by the threshold. The identity $B_K = H_e \cap P_\pi$ in (11.12) links this integer extraction with the incidence of the regional codes. The negative support of its balance with K completes the flag $B_K \subset H_\alpha^-$ developed in the exceptional extension. Thus, the role of K is defined through an invertible transformation and precise subsequent maps; the additional information on depth and transport remains in the state from which the register arises.

9.1 Cyclic extension and memory grading

The observable calendar contains one hundred and eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \dots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the cyclic translation law. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphasic sector from the internal nonadic position. Neither preserves the total number of turns on its own.

Proposition 9.2 (Calendar carry). *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$, the defect of additivity is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and coincides with the image of ι . Reduction is surjective, and multiplication by nine is injective on the stated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{lcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the stated carry and verifies the section identity. $\square$

The result explains a distinction that will recur for alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts the nonadic turns, the calendar preserves only $\beta = [q_{\text{mem}}]_{12}$. After one hundred and eight steps, β returns, but q_{mem} increases by twelve. The register also preserves the cylinder, the incidence register, and the vertical memory. Closure of a finite chart does not turn this history into a cycle without memory.

9.2 The oriented register and the integral $1 + 5 + 6$ chart

For a history produced by TPK, let τ_m be the subroute over E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence entry. The register is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection to the calendar forgets precisely some of the data needed by the signed reader.

An initial chart of the register can be written in terms of oriented events. Oriented extraction specifies the set of event codes $\{2, 4, 6, 8\}$ and the sector orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the direction code at step t , we obtain

$$e_i = \Sigma_{12}(i+1) \# \{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event count separate from the orientation. It does not identify a cardinal direction with a TRIT sign. The vector e , whose components are signed local counts, is not yet the vector U that will appear below.

Opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

we define the integral chart

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The decomposition $1 + 5 + 6$ is not an arbitrary allocation of coordinates: one coordinate preserves charge, five compare the pairs relative to a reference pair, and six preserve the opposition of the half-crowns.

Lemma 9.3 (Inversion of the integral chart). *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, which gives the division by six. Once the p_i have been reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the stated solution. This solution is integral exactly under the parity congruence. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not justify omitting the remaining material. The complete register has a larger domain than this chart of counts. Extraction of the channels feeding U uses that complete register and cannot be deduced from the list (e_i) by identifying names.

9.3 Incidence evaluation preceding the dodecaphasic register

Recovering a register from its cyclic differences and total charge is a different question from evaluating those coordinates on the incidence register. The first problem is solved by the transverse extractor of Proposition 9.13. For the second, a historical presentation of the register provides three integer counts in each window. It is useful to make their composition explicit and determine exactly the information required for their evaluation.

Let $\mathcal{L}$ be a previously generated ledger of visits and oriented traces. Each visit retains the path, APP position, sheet, time, multiplicity, and boundary incidence. Arithmetic evaluation produces the integer and its decomposition

$$n_t = r_t + 9q_t, \quad 1 \leq r_t \leq 9.$$

The integer n_t is computed from the sum or product sheet; its residue is not received as an independent value. The distinction between corona incidence and interior incidence remains explicit for visits with residue 9.

In window E_m , the count-based presentation considers

$$A_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r(v)), \quad (9.1)$$

$$C_m = \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \quad (9.2)$$

$$V_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{9\}}(r(v)) \epsilon_\delta(v). \quad (9.3)$$

Here $w(v)$ is the incidence multiplicity, and $\epsilon_\delta(v) = 1$ on the corona, -1 in the interior. The traces counted by $\nu^\pm$ must be enumerated by the corresponding rule in the article. The integer sum requires finite support or an alternative construction determining its meaning. The number of chronological incidences and the reduced length $120(1 + 3j)$ have different types; relating them requires the reader's unit of length.

Define $[x]_{10} \in \{0, \dots, 9\}$ as the Euclidean representative, retaining the quotients

$$x = [x]_{10} + 10\lfloor x/10 \rfloor.$$

The incidence block is

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}. \quad (9.4)$$

Composition yields the map

$$\mathcal{E}_{\text{cnt}}(\mathcal{L}) = \left((z_m - z_{m+3})_{m=1}^{12}, (z_m - z_{m+4})_{m=1}^{12}, \sum_{m=1}^{12} z_m \right). \quad (9.5)$$

Indices are read cyclically modulo 12. In this subsection, $(D_j z)_m = z_m - z_{m+j}$, for $j = 3, 4$, and $Q(z) = \sum_{m=1}^{12} z_m$; hence the preceding composition is $(D_3 z, D_4 z, Q(z))$. This convention is retained in §9.9, where complete integral reconstruction is proved. This formula produces its coordinates from the counts. It does not use K , U , α , or the expected transverse coordinates as arguments. Identifying it with $\mathcal{E}_{108}^{90,120}$ requires composing it with the actual collection and incidences of the terminal state; this does not follow solely from invertibility of the transverse system.

Proposition 9.4 (Necessary and sufficient arithmetic information). *Let $N, N' \in \mathbb{Z}^{12 \times 3}$ be ordered by the counts (A, C, V) . For the map defined by (9.4)–(9.5),*

$$\mathcal{E}_{\text{cnt}}(N) = \mathcal{E}_{\text{cnt}}(N') \iff N - N' \in 10\mathbb{Z}^{36}.$$

In particular, the 36 sectorial residues determine exactly the output of 25 coordinates. Retention of the quotients and provenance constitutes additional information in the register.

Proof. Equality modulo 10 produces the same blocks and hence the same transverse coordinates. Conversely, if the transverse coordinates agree, $D_3(z - z') = D_4(z - z') = 0$. The steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of total charges makes that constant zero. Finally, the decimal representation of an integer between 0 and 999 has three unique digits. This recovers the three residues of each window. The statement is an algebraic identity for every N, N' , not an inference drawn from finite tests. $\square$

9.4 Conjugation of counts and boundary terms

Let $\rho(m) = 13 - m$. Consider an incidence conjugation that reverses the order of the windows, preserves the arithmetic and boundary classes of visits, and exchanges the trace channels. Then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

If $c_m = [C_m]_{10}$, decimal reduction produces the identity

$$z_m^\dagger = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \quad (9.6)$$

Indeed, $[-C]_{10} = 0$ when $[C]_{10} = 0$, and $[-C]_{10} = 10 - [C]_{10}$ otherwise. The formula retains exactly that difference. Negating the channel is not equivalent to negating the entire decimal block.

Applying this identity to a concrete TPK involution requires checking its action on incidences. For an occupancy reader evaluated before the advance, reversal of a path $\gamma = (x_0, \dots, x_n)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0).$$

The endpoint terms are incorporated before reduction modulo 10. They vanish on closed paths; on open windows they may alter the residues. This correction, already present in the bidirectional development of transport, specifies what the connection between the path and the counts must preserve.

Proposition 9.5 (Necessity of the register's nonperiodic information). *Suppose that the observable of a fixed collection of paths satisfies $o(t + 54) = o(t)$, that multiplicity is preserved under this return, and that the same local reader acts on the windows of 9 instants. Then its counts and blocks satisfy*

$$(A, C, V)_{m+6} = (A, C, V)_m, \quad z_{m+6} = z_m.$$

Proof. The translation $t \mapsto t + 54$ maps E_m bijectively onto E_{m+6} , preserving each counted incidence and its multiplicity. The equality passes to the sum and to decimal reduction. $\square$

Thus a presentation with 12 distinct blocks requires the reader to retain some information that does not factor through that observable of period 54: incidence selection, effective orientation, endpoints, multiplicity, or memory. The proposition identifies an inadmissible reduction of the state; it does not assert that the complete dynamics has period 54, nor does it determine its terminal reader by itself.

9.5 Complete census of cylinders and selection of the terminal boundary

The terminal readout jointly uses two kinds of information: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This section constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 6, not values introduced to define the APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent readout of these outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram products, weights, distances, and marginals is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what this profile selects on the specified catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

9.5.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readouts. A length of one thousand is sufficient for the integers that follow, as verified from the two bounds of each interval; no infinite depth is assumed. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (9.7)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphasic register K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 9.6 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1)-1}{h} \right\rfloor. \quad (9.8)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer value that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, which equals the second side because the numerator and denominator are integers. Equality of the endpoints of the range of values proves the assertion. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of the regional transport:

Channel	Thirty-trit prefix	P_χ
π	010211012222010211002111110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the prolongation already constructed; the decimal readout is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (9.9)$$

The catalogue includes all tails for which these two cylinders have a nonempty intersection.

Proposition 9.7 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max \left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r \right), \quad (9.10)$$

$$b_\chi = \min \left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r \right). \quad (9.11)$$

For the specified regional cylinders, no clipping applies and we obtain

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(9.12)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (9.9) intersect exactly when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D/Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1)/Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (9.8) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint can also be obtained directly from $\lfloor T_\chi D/Q \rfloor \pmod{729}$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 denotes the ternary representation of length six, including its leading zeros. By the table, each catalogue contains fewer than 729 consecutive elements, so this reduction is injective on it. Thus, the three boundary catalogues are completely described, without choosing the final triples:

$$\begin{aligned} \mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}. \end{aligned} \tag{9.13}$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would impose a different condition: it would lose the first cylinder that intersects the boundary and would give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

9.5.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions take integer values; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned} (g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3). \end{aligned} \tag{9.14}$$

The words are ordered as in (9.13). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} that of $[n(u) = 0][w(u) = 3]$.

Theorem 9.8 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (9.14) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331. \tag{9.15}$$

These four counts are, respectively,

$$\begin{aligned} N_g &= \text{tr}(A_{\pi e} A_{e \varphi} A_{\varphi \pi}), \\ N_{gn} &= \text{tr}(D_\pi^n A_{\pi e} D_e^n A_{e \varphi} D_\varphi^n A_{\varphi \pi}), \\ N_{gnw} &= \text{tr}(D_\pi^{nw} A_{\pi e} D_e^{nw} A_{e \varphi} D_\varphi^{nw} A_{\varphi \pi}), \\ N_{gnwh} &= \text{tr}(D_\pi^{nw} A_{\pi e}^h D_e^{nw} A_{e \varphi}^h D_\varphi^{nw} A_{\varphi \pi}^h). \end{aligned} \tag{9.16}$$

All products and traces in this formula are calculated over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace in (9.16) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the cardinality is summed. The final sum runs over $0 \leq i < 196, 0 \leq j < 197$. Substitution of the words in (9.13) and the profile (9.14) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the complete list of 331 triples; its program evaluates these sums with exact binary indicators and checks each survivor by scalar evaluation. No sample or tolerance is involved. $\square$

9.5.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_{\partial}(B) = (1, 2, 2, 1, 2, 0), \quad (9.17)$$

where w_{col} counts the nonzero elements in each column and q_{∂} sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Denote by $\mathcal{F}$ the set of 331 triples from the preceding theorem.

Proposition 9.9 (Reduction of the census to the terminal pair). *The sums of indicators over $\mathcal{F}$, successively imposing rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (9.18)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For each triple in the already enumerated set, the rank is computed by elimination over $\mathbb{F}_3$, the nonzero elements of the six columns are counted, and their entries are summed modulo three. The products of the indicators in (9.17) give (9.18). Substituting the two final indices into (9.13) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing these two matrices as the domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are $(0, 2, 0)$ and $(2, 2, 1)$; only the latter belongs to this line. This recovers the last step of the selector in Section 9.6, now preceded by the construction and enumeration of its complete domain. The stable readout of $[3^{1080}x_{\chi}]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same ranks (129, 197, 148). This subsequent comparison does not define the profile or enter the sums of indicators.

9.5.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 preserves the identity of each candidate here because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled marginals and the charge supply additional effective conditions.

Thus, a complete finite enumeration and its selector have been recovered on output cylinders, with an explicit readout profile. Construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a claim about alpha or about the complete continuum.

9.6 Two readouts of the terminal state

Denote by x_{term} the state produced by the composition of selection, transport, and update up to the terminal cut. In operator notation,

$$x_{\text{term}} = (\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1)(x_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readouts are

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary readout; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The terminal selector operates on the three regional catalogues of sizes 196, 197, and 196, constructed in Section 9.5. The local signatures reduce their $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331, and the column marginal $q_{\partial} = (1, 2, 2, 1, 2, 0)$ leaves two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the latter belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; therefore $B_{\text{term}} = B_1$. This check proves the last step of the selector from the declared census. It does not turn the column marginal into a consequence of the row charge or replace the preceding enumeration with inspection of these two matrices.

The signed reader has the typed composition

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (9.19)$$

The channel register determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}})) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding register, not as parameters fitted using alpha. The reconstruction proved in the following sections starts precisely from this typed output. The documentary trace of the preceding extractor and the scope of its self-contained transcription are preserved in the technical provenance note; the subsequent arithmetic is not presented as a substitute for that trace.

9.7 From the channel register to the signed vector

The harmonic reader Π_H can be written without using either K or alpha. To avoid confusing its orbital sums with the alpha blocks, we denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (9.20)$$

Division by three belongs to the reader of the three orbits. Its integrality is checked on compatible register outputs; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (9.21)$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and hence

$$U^{(1)} = (2378, -452, -668, -322),$$

$$U^{(2)} = (1406, 998, -204, -28),$$

$$U^{(3)} = (2479, -551, -371, -997).$$

Interleaving by columns gives

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (9.22)$$

The presence of negative values and components greater than 999 shows that U is not a word of base-one-thousand digits. Nor is it $U_6 \parallel U_6$, a catalogue concatenation of period six. The distinction concerns the domain and the information preserved, not merely the values.

The constructive order up to this point is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The following inversion produces the dodecaphasic register. Being able subsequently to recover the channels from the dodecaphasic register provides a reversibility check; it does not reverse this causal order.

9.8 Orbitwise Hadamard transform and integer reconstruction

On the cycle of twelve positions, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

Separation into four sectors on each orbit admits, once origin and orientation are fixed, the angular readout of ninety degrees. The transformation does not act on three contiguous quadruples in the natural order.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders according to the three indicated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 9.10 (Hadamard inversion and integrality). *We have*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integer elimination, or expansion of the determinant, gives the negative sign and the value -16 . The integrality condition is exactly divisibility by four of the four coordinates of $H_4 u$. $\square$

Definition 9.11 (Dodecaphasic integral transformation). The sublattice of signed registers and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical register K is the image of the signed register generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines precisely the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves the two inverse compositions. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 9.12 (Canonical integral register). *The inversion $K^{(r)} = H_4 U^{(r)}/4$ of the blocks in (9.21) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural order, the dodecaphasic register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (9.23)$$

It is the unique rational preimage of U , and it belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$\begin{aligned} H_4 U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4 U^{(2)} &= (2172, 2636, 232, 584), \\ H_4 U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (9.23). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in terms of lattices. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the minors of order two are even, and at least one has absolute value two; those of order three are multiples of four, and at least one has absolute value four. It follows that

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus, not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by the inversion. The factor $1/4$ is forced by the matrix and does not have the status of a free coefficient.

9.9 Cyclic differences and the uniform component

With cyclic indices modulo twelve, let

$$(D_3 z)_m = z_m - z_{m+3}, \quad (D_4 z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second, by four. The channel names 90 and 120 refer here to these separations in the oriented cycle, not to an identification with a physical operator from another domain.

Proposition 9.13 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphasic register.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. The total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral assertion, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every minor of order twelve must contain the row Q . Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in the row Q . All maximal minors are divisible by twelve, and the tree produces one of absolute value exactly twelve. The last factor is therefore twelve. $\square$

Checking (9.23) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulae (9.20) and (9.21) in turn reconstruct U from these differences. This closes a triangle of exact representations of the same object: transverse register, signed vector, and dodecaphasic register. Equality of results does not merge their codomains into one; it proves that the declared maps recover the relevant information.

9.10 Integral compatibility and determination of the transverse register

The domain of this construction is the representation of the register of an APP–TRIT–TPK history. We distinguish evaluation of the register's events, compatibility of its channels, and inversion of its coordinates. The following result gives a complete integral criterion for the second level and an explicit factorization connecting the first to it. The proof uses the register operators and establishes their relations for every element of the stated integer domains.

9.10.1 Characterization by adjacent increments

Indices belong to $\mathbb{Z}/12\mathbb{Z}$. Let

$$(Sz)_i = z_{i+1}, \quad D_3 = I - S^3, \quad D_4 = I - S^4, \quad Q(z) = \sum_{i=0}^{11} z_i.$$

We retain the transverse extractor

$$\mathcal{D} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}, \quad k \longmapsto (D_3 k, D_4 k, Q(k)).$$

For a triple of integer coordinates (b, c, q) , write

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j \quad (1 \leq i \leq 12), \quad T(w) = \sum_{i=0}^{11} P_i. \quad (9.24)$$

Theorem 9.14 (Integral image of the transverse extractor). *A triple $(b, c, q) \in \mathbb{Z}^{25}$ belongs to $\text{im } \mathcal{D}$ if and only if it satisfies*

$$b_i = w_i + w_{i+1} + w_{i+2} \quad (0 \leq i < 12), \quad (9.25)$$

$$\sum_{i=0}^{11} w_i = 0, \quad (9.26)$$

$$q + T(w) \equiv 0 \pmod{12}. \quad (9.27)$$

Its unique integer preimage is

$$k_i = \frac{q + T(w)}{12} - P_i, \quad 0 \leq i < 12. \quad (9.28)$$

The twelve equations in (9.25) and equation (9.26) are linearly independent over $\mathbb{Q}$.

Proof. For an output of $\mathcal{D}$, we have

$$c_i - b_{i+1} = (k_i - k_{i+4}) - (k_{i+1} - k_{i+4}) = k_i - k_{i+1}.$$

The sum of three increments gives b_i ; the cyclic sum of all of them is zero. Moreover, $P_i = k_0 - k_i$, so $q = 12k_0 - T(w)$. This yields all the conditions and the inverse formula.

Conversely, (9.27) makes formula (9.28) integer-valued; (9.26) allows its cyclic extension. Its adjacent differences are w_i . Therefore, $D_3 k = b$ and

$$(D_4 k)_i = w_i + w_{i+1} + w_{i+2} + w_{i+3} = w_i + b_{i+1} = c_i.$$

Summing the inverse gives $Q(k) = q$. The same formula proves uniqueness. For independence, the invertible integer change of variables $(b, c) \mapsto (b, w = c - Sb)$ converts the first twelve relations into $b = (I + S + S^2)w$. Each solves for a different coordinate of b . The condition $\sum w_i = 0$ is an additional independent relation. $\square$

The rational image has dimension twelve. Its integral structure adds a congruence of order twelve: this is the constructive manifestation of the last Smith factor of $\mathcal{D}$. The eleven increments $w_0, \dots, w_{10}$ and the charge q are sufficient coordinates, with $w_{11} = -\sum_{i=0}^{10} w_i$ and congruence (9.27). The original twenty-five coordinates retain these same twelve degrees of freedom through thirteen relations.

9.10.2 Commutativity with the harmonic readout

Let $\mathcal{H}_{12}$ be the Hadamard transformation of the register, with orbits $(r, r+3, r+6, r+9)$ for $r = 0, 1, 2$. Its block is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4.$$

For compatible (b, c, q) , define

$$B_r = \sum_{j=0}^3 c_{r+3j}, \quad A_0 = \frac{q + 2B_0 + B_1}{3}, \quad A_1 = A_0 - B_0, \quad A_2 = A_1 - B_1.$$

The reader Π_H in (9.21) has blocks

$$(\Pi_H(b, c, q))^{(r)} = (A_r, b_{r+3} - b_{r+9}, b_r + b_{r+6}, b_r - b_{r+6}).$$

Proposition 9.15 (Agreement of the two reconstructions). *On $\text{im } \mathcal{D}$, we have*

$$\Pi_H \mathcal{D} = \mathcal{H}_{12}, \quad \mathcal{D}^{-1} = \frac{1}{4} \mathcal{H}_{12} \Pi_H.$$

The image of Π_H restricted to that domain is exactly

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12} u \equiv 0 \pmod{4}\}.$$

Proof. For $k = \mathcal{D}^{-1}(b, c, q)$, let $a_r = \sum_{j=0}^3 k_{r+3j}$. The shift by four positions takes each orbit to the next, so $B_r = a_r - a_{r+1}$. Together with $q = a_0 + a_1 + a_2$, these identities give $A_r = a_r$. On one orbit, write (x_0, x_1, x_2, x_3) for the coordinates of k and $d_j = x_j - x_{j+1}$. Then

$$d_1 - d_3 = x_0 + x_1 - x_2 - x_3, \quad d_0 + d_2 = x_0 - x_1 + x_2 - x_3, \quad d_0 - d_2 = x_0 - x_1 - x_2 + x_3.$$

These are the three remaining rows of $H_4 k^{(r)}$. Inversion and the integral characterization follow from $\mathcal{H}_{12}^2 = 4I$. $\square$

The condition $\Pi_H(b, c, q) \in \mathcal{L}_H$ controls only the harmonic output. Membership of the twenty-five coordinates in $\text{im } \mathcal{D}$ additionally requires (9.25)–(9.27). For example, adding to c the vector $e_0 - e_3$ preserves the three orbital sums and leaves Π_H unchanged. Starting from $(0, 0, 0)$ thus gives a triple with $\Pi_H = 0$ that violates (9.25). This is an exact negative control for channel compatibility, independent of any terminal evaluation.

9.10.3 Determination and transport on register orbits

Theorem 9.14 determines k once its channels and charge have been produced. As a condition on outputs not yet individualized, the same image is a group isomorphic to $\mathbb{Z}^{12}$. With only a charge q fixed, the solutions form an affine coset of

$$L_0 = \{v \in \mathbb{Z}^{12} : \sum v_i = 0\} = \bigoplus_{i=0}^{10} \mathbb{Z}(e_i - e_{11}).$$

For example, $100\mathbf{1} + e_0$ and $100\mathbf{1} + e_1$ are two distinct registers of charge 1201, with components in $\{0, \dots, 999\}$. Each satisfies all image conditions and all Hadamard congruences through its own channels. These witnesses compare the compatibility equations; they are not assigned the status of terminal HMT histories.

Proposition 9.16 (Cyclic covariance and the register stabilizer). *Let ρ be the window translation on a domain of registers $\mathcal{R}$, and let S be the preceding translation on $\mathbb{Z}^{12}$. On the orbit of $R \in \mathcal{R}$, a covariant reader is determined by a vector*

$$k \in \text{Fix}(S^p),$$

where $p \mid 12$ is the least period of R under ρ . The charge condition $Q(k) = q$ admits integer solutions if and only if $12/p$ divides q ; when they exist, they form an affine coset of rank $p - 1$. In particular, on a free orbit, covariance and charge leave eleven degrees of freedom.

Proof. The rule $F(\rho^j R) = S^j k$ is well defined exactly when $S^p k = k$. Such vectors repeat $12/p$ times a word of length p . Their charge is $\frac{12}{p} \sum_{i=0}^{p-1} k_i$. Divisibility and rank follow from this formula. Channel covariance follows from $D_a S = S D_a$ and $Q S = Q$. On $\mathcal{L}_H$, the corresponding action is $\hat{S}_H = \mathcal{H}_{12} S \mathcal{H}_{12}/4$; by Proposition 9.15, both reconstructions intertwine these actions. $\square$

If the domain carries a marked origin, one must use the action that also transports that mark. The proposition concerns this declared cyclic action; it does not presuppose periodicity of the enriched history. Naturality with respect to truncations beyond the cut at 108 is obtained, for any already defined reader F_{108} , through $F_N = F_{108} \circ \tau_{N,108}$. This naturality preserves the initial evaluation rule; it does not by itself choose among the possible rules on the first one hundred and eight events.

9.10.4 Eventwise evaluation and composition of contributions

The preceding characterization provides a factorization of extraction through two effective readouts of the register:

$$\omega : \mathcal{R} \longrightarrow \{w \in \mathbb{Z}^{12} : \sum w_i = 0\}, \quad q_{\text{reg}} : \mathcal{R} \longrightarrow \mathbb{Z}, \quad q_{\text{reg}}(R) + T(\omega(R)) \equiv 0 \pmod{12}. \quad (9.29)$$

These maps must be evaluated on the events, orientation, incidence, and memory produced by APP-TRIT-TPK. Once their actions on those data are given, the composition

$$R \longmapsto (w, q) \longmapsto ((I + S + S^2)w, (I + S + S^2 + S^3)w, q) \xrightarrow{\Pi_H} u \xrightarrow{\mathcal{H}_{12}/4} k$$

is determined and satisfies all inverse checks. The formula $w = c - Sb$ also connects a producer that already supplies both channels: it is a check on its output, preceding any comparison with the terminal witness.

Proposition 9.17 (Accumulation of contributions and prospective uniqueness). *Suppose that the initial state and each event a_i of the register produce, through rules fixed beforehand, contributions $(\delta w_t, \delta q_t)$ with*

$$\sum_i \delta w_{t,i} = 0, \quad \delta q_t + T(\delta w_t) \equiv 0 \pmod{12}.$$

With compatible initial conditions $(w^{(0)}, q^{(0)})$, the update

$$w^{(t+1)} = w^{(t)} + \delta w_t, \quad q^{(t+1)} = q^{(t)} + \delta q_t$$

produces a unique register at each stage. Its increment is

$$\delta k_{t,i} = \frac{\delta q_t + T(\delta w_t)}{12} - P_i(\delta w_t).$$

The construction respects concatenation of registers, and the readout of a prefix remains unchanged under subsequent prolongations.

Proof. The compatibility conditions are preserved under addition. Formula (9.28) is additive in (w, q) and produces the stated increment. Induction fixes each state successively. The sum over two segments is the concatenated sum, and a prefix receives exactly the same summands before and after the history is prolonged. $\square$

The proposition provides a sufficient criterion, not a requirement of linearity for every TPK reader. If events are transported by nontrivial actions, their contributions can be expressed in the terminal chart through the corresponding affine cocycle. The cocycle equation then preserves the same independence from the partition of the trajectory. In both cases, the event rules and the initial state constitute the constructive data that must come from the article: choosing them to reproduce an expected vector would replace that provenance with a different task.

The result of this development is the exact determination condition (9.29), its proven inverse, and its transport identities. Its scope is the transverse register; the other HMT constructions retain their own domains and proofs.

9.11 Two rational readouts: κ_{36} and κ_{per}

Once K has been constructed, the base-one-thousand chart allows two distinct operations. The first reads a single turn; the second prolongs the dodecaphasic register periodically. Let $B = 1000$, and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (9.30)$$

The leading zeros of a block, such as 058, are part of its three-digit representation. They do not change its integer value, but they allow the twelve blocks to be concatenated unambiguously.

Definition 9.18 (Finite and periodic readouts of the dodecaphasic register). The single-turn readout is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 9.19 (Exact value and period of the prolonged dodecaphasic register). *We have*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12} \kappa_{\text{per}}. \quad (9.31)$$

The infinite word of the dodecaphasic register has least period twelve in base one thousand and least period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series gives $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; the least block period is therefore twelve. The decimal expansion is canonical, since the dodecaphasic register has no tail of blocks 999. If its least decimal period were d , it would divide 36. Grouping digits in threes would produce a block period $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve under this operation. Hence the least decimal period is 36. $\square$

Both readouts are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their proximity does not allow one to replace the other in an exact identity. In particular, the finite closure of alpha uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

9.12 Reversibility of the rational constant and phase change

The rational constant κ_{per} completely determines the register K when the base, length, and readout origin are preserved. Its numerical and structural functions are related through an exact inversion.

Proposition 9.20 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1) \kappa_{\text{per}}.$$

The integer expansion of $I(K)$ into twelve base-1000 blocks recovers all the K_j , including the leading zeros of each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, we have

$$\kappa(\rho K) = 1000 \kappa(K) - K_1. \quad (9.32)$$

Proof. The first identity inverts the definition of the periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. Furthermore,

$$I(\rho K) = 1000 I(K) - K_1 (1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ gives (9.32). $\square$

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the already constructed transformations $D_3 K, D_4 K, Q$. The rational evaluation preserves the signed register and the cyclic differences that depend on it. The total depth and transport operators of a history remain in their own coordinates: the preceding inversion has the integral dodecaphasic register as its domain.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical readout, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

The subsequent incidence also admits an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because this orbit first recovers the integer components. The signed support associated with α additionally uses its regional registers and boundary normalization. The relation between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the function of K : it provides a reversible integral coordinatization of the oriented register, whose periodic evaluation and incidence functions participate in subsequent constructions. The value, transformation, and domain are preserved jointly.

9.13 Periodicity and the cokernel of the transport operator

The periodicity just proved belongs to the dodecaphasic register. It does not imply periodicity of the regional coordinates, the carries of the closure, or alpha. It is useful to make this distinction precise through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written as

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \mapsto K + \partial_b h, \quad C \mapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it does not authorize modifying the dodecaphasic register already selected by the register.

Proposition 9.21 (Periodic carry quotient). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module has the presentation $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ amounts to imposing $t = b$, leaving the relation $b^{12} - 1 = 0$. Evaluation of the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $b^{12} - 1$ of κ_{per} and this quotient arise from the same cyclic length, but their objects remain distinct: one is a rational evaluation and the other a quotient of a carry module. The finite closure of alpha will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

The next chapter is thus prepared. The dodecaphasic register has been fixed before solving the alpha recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the register; and its two rational evaluations have separate domains and uses. None of these facts uses alpha as an input, and none yet determines the arithmetic type of the output of the complete closure.

10 Derivation of the fine-structure constant

In this development, the fine-structure constant receives an arithmetic determination combining the three regional coordinates and the integer dodecaphasic register. The operation combines their blocks with a fixed orientation and transports the quotients of positional normalization. The resulting parameter expresses the compatibility of that oriented difference with the extension of the cylinders. Its value and the incidence data used in its construction will remain related throughout the exceptional development.

The internal order matters. The coordinates π , e and φ , already constructed in the preceding section, are prior coordinates of the same arithmetic state with trajectory memory, and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select α . The correlated generation of the four coordinates shares an origin and coinductive compatibility, without requiring all four components to be evaluated in a single step.

The two routes express different mathematical functions of the same parameter. The first performs an integer closure by Euclidean division in base 1000. The second studies the analytic equilibrium of vacancies through the resolvent $E_{21/22}$ and its graded extensions. One route determines blocks and quotients; the other determines a root through an equation and a domain of uniqueness. Their identification is then formulated in the common cylinders at every depth. The exposition distinguishes these three results: positional construction, analytic determination and compatibility of the two.

10.1 First route: positional determination by integer closure

10.1.1 Domain and orientation of the integer closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and extend the dodecaphasic register by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphasic register is subtracted according to the dodecaphasic closure already typed. For $B = 1000$, the local operation distinguishes the signed balance

$$Z_j := P_j + E_j - \Phi_j - K_j$$

from the sum incorporating the incoming quotient. Normalization is

$$s_j = Z_j + c_{j+1}, \quad A_j = s_j - B \left\lfloor \frac{s_j}{B} \right\rfloor, \quad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \quad (10.1)$$

Thus, $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted because the table is printed from left to right.

Definition 10.1 (Finite-depth closure). For a depth N and an integer boundary t , the map $\mathcal{C}_N$ receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (10.1) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division leaves no residual choice of sign: the remainder is fixed in $\{0, \dots, 999\}$ and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of the closure are not anomalies; they are the integer information required to maintain the oriented equality.

10.1.2 Dodecaphasic window and carry conditions

The first twelve triads obtained are

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117). \end{aligned}$$

In the closure channel, the five regions determining the coordinate P retain their distinct antecedents. Sharing this reading does not identify them as regions or as histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. It shows each input, the incoming carry, the total before normalization, the resulting block and the outgoing carry. The table is a reproducible integer calculation, not a list of blocks selected for their resemblance to a target.

j	P_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 8: Initial closure with a zero right boundary. The dependence propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the reading of the window is the rational number

$$\alpha_{12}^{(0)} = \frac{7297352569283800997285105472380663}{10^{36}} = 0.007297352569283800997285105472380663. \quad (10.2)$$

The superscript recalls the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readings becomes visible when the carry from the remote tail is transported.

10.1.3 Telescoping and reversibility in a boundary fiber

The local form of the equality is

$$P_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \quad (10.3)$$

The carry term links adjacent positions. Removing it coordinate by coordinate yields a false equality. Only an evaluation preserving its endpoints allows it to telescope.

Proposition 10.2 (Integer identity at arbitrary depth). *For $\iota_N(z) = \sum_{j=1}^N z_j B^{N-j}$, every execution of $\mathcal{C}_N$ satisfies*

$$\iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(K) - \iota_N(A) = B^N c_1 - c_{N+1}. \quad (10.4)$$

Proof. Multiplying (10.3) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^N (Bc_j - c_{j+1}) B^{N-j} = B^N c_1 - c_{N+1}.$$

Each interior carry appears twice with opposite signs; only the two endpoints survive. $\square$

For Table 8, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphasic register constant appearing here is κ_{36} , not κ_{per} . The domain of this identity contains truncations and a finite boundary; its form does not authorize removing both conditions when passing to the complete section.

The local inversion is also exact:

$$K_j = P_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ , the blocks A and the complete carries, it recovers the dodecaphasic register. Concatenation also recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not be turned into a new causal recipe that starts from alpha and then manufactures the register. The direction of construction remains signed register, dodecaphasic register, closure and output.

10.1.4 Extension and canonical normalization of the complete section

The extension preserves the periodic dodecaphasic register and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} P_j B^{-j}, \quad e_0 = \sum_{j \geq 1} E_j B^{-j}, \quad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2 and 1, so that $p = \pi - 3$, $e_0 = e - 2$ and $f = \varphi - 1$.

Proposition 10.3 (Existence and uniqueness of the complete normalization). *The series*

$$y = p + e_0 - f - \kappa_{\text{per}} = \sum_{j \geq 1} (P_j + E_j - \Phi_j - K_j) B^{-j} \quad (10.5)$$

converges absolutely. In the domain of the blocks obtained, $0 < y < 10^{-2}$. There exists a unique canonical expansion (A_j) of y , with no tail eventually constant at 999, and a unique bounded integer sequence (c_j) satisfying (10.3) with $c_1 = 0$. These carries belong to the invariant set $\mathcal{C} = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B-1)$, so a geometric series dominates it. If $Z_j = P_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r \geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; therefore $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y . Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying the equality of the two series by B^{j-1} and separating their first $j - 1$ terms gives

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Moreover, $R_j(A) \in [0, 1)$, hence $-3 < c_j < 2$, which places all carries in $\mathcal{C}$. The shift identities for the tails give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y . The canonical expansion is unique and, once fixed, the tail formula determines all carries. Finally, if a finite execution is applied with $c_{j+1} \in \mathcal{C}$, then $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to $\mathcal{C}$. $\square$

The proposition makes the normalization of the subsequent reader explicit: it does not replace the generation of the three sequences with a conventional series introduced as a datum. Once those sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite procedure preserved in the corpus propagates the five boundaries of $\mathcal{C}$ and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. A boundary is not chosen because it yields favorable digits, nor is compatibility at every depth deduced from observing a long run.

The initial comparison shows why this precaution matters. The stabilized extension gives $c_{13}^\infty = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first turn. The first eleven triads do not change, but the extension begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

An occurrence of the block 999 does not violate the canonical convention: what is excluded is a tail eventually constant at 999. The window with a zero boundary ends in 663; the truncation of the complete section ends in 662. Both statements describe different and exact operations. They must not be merged into a single projection notation.

10.1.5 Operational definition and complete additive equation

Definition 10.4 (Alpha coordinate of the closure). The integer route determines the coordinate

$$\alpha_A = \sum_{j \geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\text{per}}, \quad (10.6)$$

where (A_j, c_j) is the compatible canonical normalization of (10.1). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}.} \quad (10.7)$$

The integer four is not calibrated: it is $3 + 2 - 1$, the combination of the integer parts of the three coordinates. The base one thousand comes from the cubic chart, and the period twelve from the dodecaphasic register. The signs are those of the oriented channel and the closure, and the remote boundary is the one imposed by the compatible section. These are the effective origins of the coefficients of the equation.

At this level, alpha may be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphasic return, calculated with compatible carry. The term *defect* denotes a determined algebraic difference, not an arbitrary loss or an imperfection of the system.

This description does not suppress the genealogy: without the regions, the signed register, the orientation and the carry law, the scalar equality would not be a sufficient exposition of the construction.

Three objects that alpha notation may accompany in subsequent readers must also remain separate: A is a word of blocks; the negative pre-carry support is obtained from the sign of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None of them is interchangeable with the real value α_A .

Signed balance and sum with an incoming quotient. Positional division consists of two consecutive stages. With $K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}$, which denotes the periodic extension also denoted by K_j in (10.1), the balance before quotient transport is

$$Z_j := P_j + E_j - \Phi_j - K_j^{\text{per}}.$$

The variable s_j in (10.1) denotes the sum that already includes the incoming contribution, so that

$$s_j = Z_j + c_{j+1}, \quad A_j = Z_j + c_{j+1} - 1000c_j.$$

The incidence configuration H_{α}^- uses the signs of the balance Z_j in the initial window; determining the digits also uses the linking quotients c_{j+1} and c_j . This distinction jointly preserves the signed support and the normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and the dodecaphasic register arise from the common APP-TRIT-TPK generation. Once produced, their positional composition has an explicit dependence on the reading depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}, \quad y_N = \sum_{j=1}^N Z_j B^{-j}, \quad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (10.5) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \quad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window with that of its extension, both expressed in the same coordinates.

Proposition 10.5 (Boundary transport and canonical cylinder). *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in \mathcal{C} = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^N A_j^{(N,t)} B^{-j}$. For the registers of the domain under consideration, we have*

$$\alpha_A - a_N^{(t)} = B^{-N}(R_{N+1}(Z) - t), \quad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \quad (10.8)$$

The quotient of the complete extension is $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. With this boundary,

$$0 \leq \alpha_A - a_N^{(c_{N+1}^{\infty})} < B^{-N},$$

so the resulting window is the canonical prefix of order N .

Proof. Telescoping (10.3) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, each incoming state belongs to $\mathcal{C}$ and the first dividend lies between 5 and 9, we have $c_1 = 0$. Substituting this equality into the decomposition of the series proves (10.8). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in \mathcal{C}$.

For the complete canonical normalization, $c_{N+1}^{\infty} = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. The integrality of the quotient implies $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c_{N+1}^{\infty})} = B^{-N} R_{N+1}(A)$, proving membership in the canonical cylinder. $\square$

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed at the scale of the next position. The calculation over the five boundaries preserves this information as a finite collection of possibilities until the blocks to the left stabilize. The quantitative contraction arises from depth and tail transport; its object is the normalization of the coordinates already generated.

Graded domain of the analytic resolvent. The analytic composition displayed in §10.2.2 acts on $\text{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation produces exactly $T_{7;9}$; degree 9 is the last component of that domain. The identity $N^3 = 0$ explains why the resolvent is expressed through three summands. To describe an extension to higher degrees, one must also specify the transport between successive graded domains and its action on the initial component of each domain. Local nilpotence and global compatibility thus play distinct mathematical roles.

10.1.6 Constructive determination and role of the dodecaphasic register

The preceding definition clarifies what it means to determine α from the structure. The domain of the operation contains four registers of known provenance: the three regional sequences and the integer phase register. Normalization also incorporates a boundary condition compatible with extension. Once these data are fixed, Euclidean division determines each remainder and its quotient. Proposition 10.3 extends that determination to the complete system. Independence from a target physical value is therefore a property of the order of construction: the value compared at the end does not participate in selecting any of the arguments of this map.

This causal independence must be distinguished from algebraic independence between numbers. The former is established by identifying the domain, the operations and the provenance of the coefficients; the latter would be a statement about polynomial relations. The result used here is the former. The construction determines a coordinate through registers produced by APP-TRIT-TPK and allows the relations it satisfies to be checked subsequently. This order makes it possible to use a coordinate already generated in a later operation without turning it into an external parameter.

The register K has a more precise role than that of a numerical correction added to a formula. Its twelve positions arise from the reversible transformation of the signed register. The equation $U = \mathcal{H}_{12}K$ preserves the possibility of returning to that register, while the transverse differences and the total sum allow another reconstruction. In the periodic evaluation, the order of the positions determines the positional weight of each component. The same set of components, arranged in a different order, generally produces a different rational coordinate. Thus, the phase origin and orientation form part of the definition of the number entering (10.7).

The normalization of α composes this information with the regions. Its local identity distinguishes two successive operations. First the oriented balance of blocks is formed; then that balance is represented through a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and allows a support to be extracted subsequently. The second allows the prefixes of the real coordinate to be formed. Both readings remain related because the state preserves the data before and after normalization. The signed support and the positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. The interior quotients cancel in pairs when a prefix is evaluated, while their endpoints remain in the equality. This cancellation is a property of the weighted sum; it neither removes the quotients from the state nor allows them to be suppressed before the sum is taken. The right boundary of a window receives the contribution of the continuation. For this reason, evaluating an isolated window and restricting an extended section are distinct operations. The change of the final block from 663 to 662, proved above, provides an explicit realization of this difference without altering the blocks already stabilized to its left.

In this sense, bidirectionality has a concrete arithmetic formulation. Generation provides compatible extensions; normalization transmits the integer information of their boundary from positions of lower weight towards those of higher weight. The equation preserves the contribution of both directions through its indices and its boundary conditions. Reversibility in the fixed fiber is checked by reconstructing K from the other registers and the complete quotients. This inversion certifies the consistency of the original operation; the genealogical direction remains the one that produces the register first and the α coordinate afterwards.

The second determination has complementary content. In it, the immediate object is a balance of vacancies, and the parameter appears as its root in an isolating domain. The positional route provides a canonical normalization of registers; the analytic route studies the vanishing of an operator. Presenting both requires precisely this difference of functions to be preserved. Their relations with the common state must be expressed through the refinement maps and the comparison of their complete realizations, as developed in the corresponding subsection. Agreement of a finite approximation constitutes a localized check; identification of the complete constructions has the domain and scope of its own statement.

Finally, this organization explains the position of α in the article. Its equation appears after regional generation and the construction of K , because it uses both results. The electron section then uses that coordinate as a factor in its composition. The exceptional extension receives the registers and their supports, where incidence information is preserved. These two continuations share the same provenance and develop distinct mathematical functions: evaluation of a central section and construction of a boundary configuration. Developing each one allows the structure to be followed through the equation, instead of restricting the comparison to its scalar value.

10.2 Second route: analytic determination through vacancies

10.2.1 Transport invariants and graded resolvent

The analytic route receives invariants of the oriented state, not the output α_A . Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \quad (10.9)$$

and the vacancy condition uses

$$\delta = \log_{10} \frac{10}{9}.$$

The ratio $10/9$ compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification reading; it is not a target value of α .

The genealogy of the coefficients must be preserved in full, but with its exact logical strength. Recent developments bring together a marked signature of the state. The synchronization matrix and the associated census are

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \quad N_9 = 43.$$

Its determinant yields

$$a = -\frac{\det D_{\text{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \quad k_* = mb - 1 = 20.$$

The vacancy reading in the base-one-thousand chart gives

$$v_- = \lfloor 1000\delta \rfloor = 45, \quad v_+ = \lceil 1000\delta \rceil = 46.$$

These integers are typed antecedents of the evaluation of the truncation. They are not obtained from its root or from a metrological approximation to it.

The lower branch also coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45.$$

The transverse rank of the register in the preceding chapter gives $r_{\text{dod}} = 11$, and the phase half-corona provides $f_{\text{ph}} = 8\binom{6}{2} = 120$. In particular, $45r_{\text{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{I}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{G}_9(\mathcal{I}_9)(x) = & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_- r_{\text{dod}}}. \end{aligned} \quad (10.10)$$

Substitution recovers the coefficients of P_6 and its ninth-order extension. The pair 45/46 crosses the division between the two: the upper branch occupies degree six, and the lower branch reappears in degrees eight and nine.

The factorization in (10.10) is formulated for a state preserving origin, channel order and orientation. Naturality with respect to morphisms preserving those data is not equivalent to uniqueness of every reader after they have been forgotten. The domain of this factorization is determined by the signature (10.10); the extension to all degrees is constructed in Section 10.4, with its recurrence class and its dependence on the state's pre-coordinate explicitly defined.

10.2.2 Nilpotent resolvent in degrees 7, 8 and 9

The first extension admits a particularly explicit finite operator construction. On the space with basis e_7, e_8, e_9 , let

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7.$$

The symbol N here denotes a nilpotent operator, not a truncation depth. Its domain and grading are fixed before the polynomial is evaluated at a number.

Lemma 10.6 (Exact nilpotent extension). *We have $N^3 = 0$ and*

$$(I - N)^{-1}v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

The evaluation $e_n \mapsto x^n$ produces

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (10.11)$$

Proof. The operator raises the degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \quad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. $\square$

Here the denominators 120, 45 and 495 are not decimal adjustments: they arise from the half-corona, the APP determinant and its product with the dodecaphasic rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only the symmetry of the cycle and replacing this operator with another symmetric operator does not necessarily preserve the coefficients.

10.2.3 Root of multiplicity one of the ninth-order truncation

Define

$$P_9 = P_6 + T_{7:9}, \quad E_{21/22}^{(9)}(x) = P_9(x) - \delta, \quad I_\alpha = (0, 10^{-2}).$$

This finite operator has its own existence and uniqueness result.

Proposition 10.7 (Root of multiplicity one of the truncation). *The operator $E_{21/22}^{(9)}$ has a unique root ξ_9 in I_α , and that root is simple. The following rational isolation is obtained:*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \quad (10.12)$$

Proof. The bounds on the internal coordinates $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ suffice for existence. At zero, $E_{21/22}^{(9)}(0) = -\delta < 0$. At 10^{-2} , the linear contribution minus the negative terms is greater than 0.0626, so the operator's value is positive.

For $0 \leq x \leq 10^{-2}$, discarding the positive terms of the derivative gives

$$(E_{21/22}^{(9)})'(x) \geq 2\pi - \frac{7}{2} \cdot 10^{-2} - \frac{10}{21} \cdot 10^{-8} - \frac{6}{46} \cdot 10^{-10} - \frac{7}{120} \cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. The narrower interval (10.12) results from a directed rational evaluation. Let x_- and x_+ be its endpoints. To compute $\delta = \log(10/9)/\log 10$, we use, for $z > 1$,

$$L_N(z) = 2 \sum_{j=0}^{N-1} \frac{y^{2j+1}}{2j+1}, \quad y = \frac{z-1}{z+1}, \quad 0 < \log z - L_N(z) < \frac{2y^{2N+1}}{(2N+1)(1-y^2)}.$$

The remainder is bounded by the geometric series of positive terms. This formula is applied to $z = 10/9$ and $z = 3$, with $N = 520$, using $\log 10 = \log(10/9) + 2 \log 3$. Evaluation of π uses the generated thousand-digit regional cylinder, including its width 10^{-1000} . Interval arithmetic in (10.10) gives

$$-\frac{4559}{10^{23}} < E_{21/22}^{(9)}(x_-) < -\frac{4558}{10^{23}}, \quad \frac{1698}{10^{23}} < E_{21/22}^{(9)}(x_+) < \frac{1699}{10^{23}}.$$

These signs and the proved monotonicity establish the isolation. The endpoints are results of evaluating the operator; its coefficients arise from the signature constructed beforehand. $\square$

The result concerns ξ_9 . It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. The root of a truncation may share a prefix with the root of the extended operator and differ at a later position. Simplicity allows the motion of the root under perturbations to be controlled; it does not eliminate those perturbations.

10.3 Compatibility of the two determinations at every depth

Remark 10.8 (Truncation, extension and compatibility). The operations in degrees 6 to 9 determine a finite jet. Positional normalization determines the complete coordinate through its compatible sequence of carries. The next proposition expresses the general criterion for identification through cylinders. Section 10.4 then constructs a correlated analytic chart of all degrees, with a recurrence, uniform convergence, uniqueness of the root and projective compatibility. The construction receives the shared coordinate of the state and the declared class of extensions; it thus preserves the distinction between an analytic representation of that coordinate and an independent selection.

The complete analytic route is organized into a collection of compatible graded readers

$$E_{21/22}^{(6)} \longleftarrow E_{21/22}^{(9)} \longleftarrow E_{21/22}^{(12)} \longleftarrow \cdots.$$

The symbol

$$E_{21/22} = \lim_{\longleftarrow m} E_{21/22}^{(6+3m)}$$

denotes that complete reader, with its transports and truncations, not a formal series freely chosen to make it pass through α_A . The second route determines its unique positive root in I_α , denoted by α_B , with the uniqueness and compatibility of the complete reader proved in Section 10.4.

The proofs in Section 10.4 establish that, at each depth, both systems determine the same surviving cylinder C_N^α , and that their diameters tend to zero. It is useful to state the mathematical consequence of this compatibility separately.

Proposition 10.9 (Identification of the compatible limits). *If the values of the integer and analytic routes belong to the same cylinders C_N^α for every N , and $\text{diam } C_N^\alpha \rightarrow 0$, then*

$$\alpha_A = \alpha_B = \alpha.$$

Proof. For every N , the distance between the two values is bounded by $\text{diam } C_N^\alpha$. Letting N tend to infinity gives zero distance. The existence of the compatible sections ensures that this is not the empty intersection of a system of approximations. $\square$

This proof identifies the consequence of the compatibility squares. Their construction at arbitrary depth appears in Section 10.4. Finite evaluations check realizations of that construction; the recurrence and its uniform bounds establish the passage to the limit.

The difference can be expressed algebraically. Let $F = \mathbb{Q}(\pi, \delta)$ and let $j_9 : F[[x]] \rightarrow F[x]/(x^{10})$ be the truncation. Then

$$\ker j_9 = x^{10}F[[x]].$$

All the operators

$$E_h(x) = E_{21/22}^{(9)}(x) + x^{10}h(x)$$

have the same ninth-order truncation. However, taking $h = 0$ and $h = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0,$$

so they do not share that root. The observation does not allow a tail foreign to HMT transport to be chosen; it demonstrates precisely why the generated tail and its compatibility must be preserved. Nor does an arbitrary formal series suffice to speak of analytic roots without its domain of convergence or the realization of the cylinder system.

10.4 Compatibility between the dodecaphasic and analytic representations of alpha

Compatibility is established at every depth. The reversible dodecaphasic representation and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal representations with distinct codomains, produced from the same enriched APP-TRIT-TPK state. The jet $E_{21/22}^{(9)}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below is not deduced from the bare jet: it receives the shared coordinate of the state and expresses its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This separation preserves the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \mathcal{X}_\alpha^{\text{enr}} \longrightarrow (P, E, \Phi, U_{12}^{\text{sgn}}, K) \longrightarrow \alpha_{\text{HMT}}.$$

The coordinates P, E, Φ and the register K are earlier outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition, not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\widehat{\mathcal{X}}_r^{\text{enr}, \alpha}$ be the sector of truncated states that extend the internal alpha seed fixed at R_{36} . Write

$$x_r^\alpha = (\tilde{x}_r, \chi_\alpha, \varepsilon_r^\alpha, N_r^\alpha),$$

where χ_α is the internal character of the sector, not the expressed scalar value. Each state preserves remainder, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, register, memory, survival, provenance and depth. If

$$d_r^\alpha = \text{Dig}_{729}(\varepsilon_r^\alpha), \quad \varepsilon_{r+6}^\alpha = 729\varepsilon_r^\alpha - d_r^\alpha, \quad N_{r+6}^\alpha = 729N_r^\alpha + d_r^\alpha,$$

the effective transition is

$$u_r^\alpha = \text{Upd}_r^\alpha \circ \text{Tra}_r^\alpha \circ \text{Sel}_r^\alpha, \quad \text{Ext}_{r+6,r}^{\alpha,\rightarrow} = \text{graph}(u_r^\alpha).$$

The single-valuedness of these extensions, with the seed R_{36} fixed, produces a unique compatible section: the phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}_\alpha^{\text{enr}} := \widehat{\Omega}_\alpha := \varprojlim_{r \geq 36} (\widehat{\mathcal{X}}_r^{\text{enr},\alpha}, \text{Ext}_{r+6,r}^{\alpha,\rightarrow}). \quad (10.13)$$

In this section, κ_{per} is the periodic reading κ_{ag} of the register already constructed; both notations denote the same fraction in the fixed base-one-thousand chart.

10.4.1 The dodecaphasic closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \dots, 999\}^N, \quad \mathsf{T}_{\alpha,N}(\mathbf{s}) := (A_1, \dots, A_N), \quad \mathbf{s} \in \mathcal{X}_\alpha^{\text{enr}},$$

where the blocks A_j are obtained through (10.1) with the compatible boundary $c_{N+1}^\infty = [R_{N+1}(Z)]$. If $N' \geq N$, let $\rho_{N',N} : \mathcal{W}_{N'} \rightarrow \mathcal{W}_N$ be the restriction to the first N blocks.

Theorem 10.10 (Complete dodecaphasic output and projective compatibility). *The collection $(\mathsf{T}_{\alpha,N})_{N \geq 1}$ is compatible:*

$$\rho_{N',N} \circ \mathsf{T}_{\alpha,N'} = \mathsf{T}_{\alpha,N} \quad (N' \geq N). \quad (10.14)$$

Its canonical cylinders

$$C_N^A(\mathbf{s}) := \left[\sum_{j=1}^N A_j 1000^{-j}, \sum_{j=1}^N A_j 1000^{-j} + 1000^{-N} \right) \cap I_\alpha \quad (10.15)$$

are nested, have diameter at most 1000^{-N} and satisfy

$$\bigcap_{N \geq 1} C_N^A(\mathbf{s}) = \{\alpha_A(\mathbf{s})\}, \quad \alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}. \quad (10.16)$$

Consequently, the first route expresses a unique infinite section, which we denote by $\alpha_{\text{HMT}} := \alpha_A$.

Proof. Proposition 10.3 provides the unique canonical expansion and the unique bounded sequence of carries. The local identity (10.3) and its telescoping (10.4) preserve the remote boundary; Proposition 10.5 proves that restricting an extended execution recovers exactly the preceding prefix. This gives (10.14). Membership of the full sum in each cylinder follows from the tail identity, and $\text{diam } C_N^A \leq 1000^{-N} \rightarrow 0$; the intersection contains a single point. Finally, (10.7) identifies that point with the expression in (10.16). All its coefficients and signs are fixed before the representation of α . $\square$

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^\infty = -1$. In particular,

$$\rho_{36,12}(\alpha_{A,36}) = \alpha_{A,12} \quad (10.17)$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

10.4.2 Analytic representation of the alpha coordinate

The analytic extension is specified through all its coefficients. Let $s \in \mathcal{X}_\alpha^{\text{enr}}$. Before decimal normalization, the same state already expresses the pre-coordinate

$$a(s) := p(s) + e_0(s) - f(s) - \kappa_{\text{per}}(s) \in I_\alpha, \quad (10.18)$$

with $p = \pi_{\text{HMT}} - 3$, $e_0 = e_{\text{HMT}} - 2$ and $f = \varphi_{\text{HMT}} - 1$. Equivalently, $a = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}$. No conventional α is introduced here: p, e_0, f are the three regional characters already generated, and κ_{per} is the rational reading of the terminal register. The integer closure applies to (10.18) the normalization in (10.1); the chart below expresses a compatible analytic representation on the same pre-coordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\text{per}})$ directly: it does not contain an edge $\alpha_A \rightarrow \lambda$, although the equality proved below allows both representation names to be identified subsequently.

Write $q = 1/729$, let F_s be the ordered field generated by the coordinates of s , and set $E_{9,s}(X) = E_{21/22}^{(9)}(s; X)$. We define the linking coefficient

$$\lambda(s) := -\frac{E_{9,s}(a(s))(1 - qa(s)^3)}{a(s)^{10}}. \quad (10.19)$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (10.19) must be read with its exact type: once the first representation of the state has produced $a(s)$, it fixes the unique extension within the geometric class defined below. It is not a second independent selection of $a(s)$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_s$ set

$$E_{s,\mu}(X) := E_{9,s}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(s) < 10^{-2}$ and $qa(s)^3 < 1$, we have

$$E_{s,\mu}(a(s)) = 0 \iff \mu = \lambda(s). \quad (10.20)$$

Thus, the coefficient seed contains no freedom once the state and the class of q -geometric extensions have been fixed. This uniqueness is relative to the declared class; it does not assert that the ninth-order jet by itself selects a tail among all possible analytic series.

The local action on blocks of three coefficients is

$$\mathcal{C}_\alpha : F_s^3 \longrightarrow F_s^3, \quad \mathcal{C}_\alpha(u, v, w) = q(u, v, w), \quad c^{(0)} = (\lambda(s), 0, 0). \quad (10.21)$$

Therefore, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(s), \quad b_{11+3n} = b_{12+3n} = 0. \quad (10.22)$$

This is the rule that computes each coefficient beyond degree nine. No free parameter remains at each extension.

For $M \geq 0$, let

$$E_{s,M}(X) := E_{9,s}(X) + \lambda(s) \sum_{n=0}^M q^n X^{10+3n}. \quad (10.23)$$

The complete function is

$$E_{s,\infty}(X) := E_{9,s}(X) + \lambda(s) \frac{X^{10}}{1 - X^3/729}. \quad (10.24)$$

To formulate the projective system without implicit types, we fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{F}_M := \coprod_{s \in \mathcal{X}_\alpha^{\text{enr}}} \{s\} \times F_s[X]/(X^{d_M+1}). \quad (10.25)$$

If $M' \geq M$, the restriction is

$$\tau_{M',M}(s, [P]_{d_{M'}}) := (s, [P]_{d_M}). \quad (10.26)$$

Proposition 10.11 (Naturality of the analytic recurrence). *Let*

$$\Gamma_M(s) = (s, [E_{s,M}]_{d_M}).$$

If $M' \geq M$, the truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_M = \Gamma_{M'}, \quad \Gamma_{E21}(s) = \varprojlim_M \Gamma_M(s). \quad (10.27)$$

The projection onto the three new degrees is precisely $\mathcal{C}_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (10.22) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ adds only the next block; truncating it recovers the preceding polynomial literally. Compatibility for $M' \geq M$ is obtained by composition. Thus, the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under $\mathcal{C}_\alpha$. $\square$

10.4.3 Uniform convergence, simple root and rational intervals

The regional cylinders and the periodic register provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}\},$$

let P_c be its certified thousand-digit nonadic prefix. The datum used is not the rational number P_c treated as though it were the complete value, but the half-open cylinder and its outer closure

$$\mathcal{C}_c^{(1000)} = [P_c, P_c + \varepsilon), \quad \overline{\mathcal{C}}_c^{(1000)} = [P_c, P_c + \varepsilon], \quad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively with these closures. If P_π, P_e, P_φ denote the three prefixes, we obtain

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\text{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\text{per}}, \\ A_{1000} &:= [a_-, a_+], \quad a(s) \in [a_-, a_+] = A_{1000}, \quad \text{diam } A_{1000} = 3\varepsilon. \end{aligned} \quad (10.28)$$

The natural interval extension of (10.19) then transports A_{1000} , the cylinder of π_{HMT} and the directed rational bound for δ to a closed interval $\Lambda_{1000} \ni \lambda(s)$; none of the three tails is frozen. In particular,

$$\frac{7297352569283800997285105472380662}{10^{36}} < a(s) < \frac{7297352569283800997285105472380663}{10^{36}}, \quad (10.29)$$

$$-8.711 \times 10^{-6} < \lambda(s) < -8.709 \times 10^{-6}.$$

These inequalities use certified prefixes with their rational tail bound; they do not receive a table of alpha values. The package verifier reproduces them from the three nonadic outputs and from the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \leq x \leq 10^{-2}$, the tail beyond M satisfies

$$|E_{s,\infty}(x) - E_{s,M}(x)| \leq 8.711 \times 10^{-26} \frac{r^{M+1}}{1-r}. \quad (10.30)$$

Moreover,

$$\left| \frac{d}{dx} \left(\lambda \frac{x^{10}}{1 - x^3/729} \right) \right| \leq 8.711 \times 10^{-6} \left(\frac{10 \cdot 10^{-18}}{1-r} + \frac{3 \cdot 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \quad (10.31)$$

The tail of the derivatives beyond block M also admits the depth-dependent estimate

$$\sup_{0 \leq x \leq 10^{-2}} |E'_{s,\infty}(x) - E'_{s,M}(x)| \leq 8.711 \times 10^{-24} r^{M+1} \left(\frac{13 + 3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow{M \rightarrow \infty} 0. \quad (10.32)$$

Theorem 10.12 (Limit function and unique simple root of the correlated chart). *The sequence $E_{s,M}$ converges uniformly, together with its derivatives, to $E_{s,\infty}$ on $\bar{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{s,\infty}(a(s)) = 0, \quad \inf_{x \in \bar{I}_\alpha} E'_{s,\infty}(x) > 6.239. \quad (10.33)$$

Consequently, $a(s)$ is the unique simple root of the complete function in I_α .

Proof. The ratio of two consecutive terms of the tail is bounded by $r < 1$, which gives (10.30); term-by-term differentiation and summation of the two geometric series give (10.31) and (10.32). The proof for the jet establishes $E'_{0,s} > 6.24$ on $\bar{I}_\alpha$; hence, the complete perturbation leaves the derivative above 6.239. Finally, substituting (10.19) into (10.24), the two summands evaluated at $a(s)$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with that pre-coordinate. $\square$

To make the identification effective at every depth, directed endpoint evaluations first certify

$$E_{s,\infty}(0) < 0 < E_{s,\infty}(10^{-2}), \quad E'_{s,\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{on } [0, 10^{-2}].$$

By continuity there is a root in the interval and, by the second inequality, it is unique. Only after this existence, monotonicity and the invariant that the bracket contains the root have been established is the mean-value-theorem fallback applied. We start from $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{s,\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}, \delta$ and λ , with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4} (s_j - r_j). \quad (10.34)$$

When $F_j^+ < 0$, the right half is retained; when $F_j^- > 0$, the left half is retained. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{s,\infty} \geq c = 6$ and the mean value theorem imply that any root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4} \right]. \quad (10.35)$$

Indeed, let ξ_j be the root and let $y_j = E_{s,\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (10.34), we have $|y_j| \leq cw_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, proving (10.35). The argument includes the case $y_j = 0$: the root is then at the midpoint, but the algorithm does not need to recognize this equality in order to retain it.

If the available enclosure does not satisfy (10.34), the source cylinders are first refined and the evaluation is repeated; no sign is declared and no insufficient contraction is accepted. In each of the three branches the new interval retains the root and has diameter at most $w_j/2$. For each finite target depth, directed refinement allows the width bound to be met without an uncertified comparison of a real number with zero. Monotonicity proves inductively that

$$a(s) \in D_{j+1} \subseteq D_j, \quad \text{diam } D_{j+1} \leq \frac{1}{2} \text{diam } D_j, \quad \text{diam } D_j \longrightarrow 0, \quad (10.36)$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule to twenty-four base-one-thousand levels and checks that the complete outer closure A_{1000} , not only its lower endpoint, lies inside each expressed cylinder. Its negative tests reject an excessively wide enclosure, an incorrectly ordered enclosure, a point other than the midpoint, a domain without a sign change, and a degenerate bracket. Coinductive extension allows one thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$ the verifier computes

$$j_N^- := \lfloor S_N a_- \rfloor, \quad j_N^+ := \lfloor S_N a_+ \rfloor \quad (10.37)$$

and expresses the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and avoids assigning to the left cell an endpoint lying exactly on its right boundary.

Let $C_N^A(s)$ be the integer cylinder in (10.15). We recursively choose $m(N) > m(N-1)$ so that $\text{diam } D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(s). \quad (10.38)$$

Define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is nonempty because both factors contain $a(s)$. Since the cylinder is half-open, u_N may not belong to $I_{B,N}$. Evaluation at that endpoint is understood through the continuous extension of $E_{s,\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(s), \quad \text{diam } I_{B,N} \longrightarrow 0, \quad E_{s,\infty}(\ell_N) \leq 0 \leq E_{s,\infty}(u_N). \quad (10.39)$$

The bound (10.30) allows a truncation to be chosen that preserves the sign at each endpoint where the complete function does not vanish. If an endpoint coincides with the root, the identity of the complete function and the membership already proved are used; that same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{s,\infty}(x) - E_{s,M}(x) = \lambda(s)x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

In the positive domain under consideration, where $\lambda(s) < 0$, we obtain at the root

$$E_{s,M}(a(s)) = -\lambda(s)a(s)^{10} \frac{(qa(s)^3)^{M+1}}{1 - qa(s)^3} > 0 \quad (M < \infty).$$

Therefore, the certification of the weak signs in (10.39) refers to $E_{s,\infty}$. Its evaluation uses the complete function or the truncation accompanied by the signed remainder. This covers both the included infimum and the supremum that may be excluded from the cylinder.

Proposition 10.13 (Compatibility squares and common cylinders). *The restrictions of words and of analytic representations commute:*

$$\begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\tau_{\alpha,N'}} & \mathcal{W}_{N'} \\ \parallel & & \downarrow \rho_{N',N} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\tau_{\alpha,N}} & \mathcal{W}_N \end{array} \quad \begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{F}_{M'} \\ \parallel & & \downarrow \tau_{M',M} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{F}_M. \end{array}$$

The collection $I_{B,N}$ provides the effective connection between the two squares and satisfies concretely $I_{B,N} \subseteq C_N^A$ for every N .

Proof. The first square is the compatibility of integer normalization; the second is Proposition 10.11. The inclusion follows from the definition (10.38), and nonemptiness and the signs follow from Theorem 10.12. No step identifies the root of a truncation with that of an arbitrary tail. $\square$

We now define, without leaving the prefix reader implicit,

$$\alpha_B(s) := \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x), \quad \text{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \quad \alpha_{B,N} := \text{pref}_N(\alpha_B). \quad (10.40)$$

Theorem 10.14 (Projective equality of the two correlated representations). *For every N , the integer and analytic routes express the same prefix:*

$$\boxed{\alpha_{A,N} = \text{pref}_N(\alpha_A) = \text{pref}_N(\alpha_B) = \alpha_{B,N}.} \quad (10.41)$$

In the limit,

$$\boxed{\alpha_A = \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x) = \alpha_B = \alpha_{\text{HMT}}.} \quad (10.42)$$

Proof. Integer normalization expresses exactly $a(s)$. Theorem 10.12 proves that the analytic chart represents that same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix, and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Therefore, their prefixes coincide for each N ; nesting and diameters tending to zero give equality of the limits. Metrological recognition occurs afterwards. $\square$

The analytic function constitutes a correlated representation of the same state, not a second derivation causally independent of the integer closure. This precision does not weaken the equality proved: it identifies its exact mechanism, prevents attributing to the jet a tail that it does not determine, and avoids promoting a conventional value to a generator.

11 Exceptional prolongation: from dodecaphasic incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves the two sheets and division with remainder and quotient; the TRIT determines the local regime and its orientation; the TPK composes selection, transport, carry, memory, and return. The Archimedean reading contracts compatible cylinders. The reading now under study instead preserves the incidence of positions and their transport between frames. Both start from the same arithmetic state with trajectory memory. An exceptional geometry is not reconstructed from the final decimals of π , φ , e , or α .

This point clarifies the meaning of the prolongation $\alpha/K \rightarrow$ exceptional incidence. The arrow does not assert that an isolated real number determines a code, a lattice, or a group. It transports the dodecaphasic record and the signed data that entered the determination of α , together with the words and frames arising from APP-TRIT-TPK. In particular, the ordered record K can be recovered from its rational constant κ under the previously fixed base, length, and origin conventions. Projection onto a support, by contrast, preserves only the selected incidences. For its part, K_{ph} denotes a different, reduced-memory reading, subsequent to the holonomy Γ_9 .

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; and subsequently recognizing the resulting objects through the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech, and Monster designate those recognitions and the explicit realizations that make them verifiable. They are not retrospective selectors of the seeds or coefficients of the generation.

The argument begins with compatibility of prefixes and frames, constructs the ternary code, and determines the flag through K and the signed record of α . Marked incidence then selects the lattice origin and the Leech neighbor. The final step realizes the vertex algebra and its automorphisms. At each stage, the structure preserved will be specified: linear combinations, supports, discriminant classes, bilinear form, or graded product. Explanations of the function of K , normalization, and the origin of norm 54 accompany their respective constructions in Sections 11.3 and 11.9.

11.1 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\text{cal}} \rightarrow X_m^{\text{cal}}$ for $m \leq n$. The superscript indicates that the declared initial frame and its transport are preserved, not that a parameter is fitted against an experimental value. We write (w_j, f_j) for the visible six-trit word and the frame at step j . The frames satisfy a causal law

$$f_{j+1} = g_j(x)f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this reading. The return of the phase resets neither the frame nor the memory of the state. Likewise, the census of 468 decimal emissions, the 243 visible six-trit words, and the ambient space $\mathbb{F}_3^6$, with 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a six-trit block b , the Archimedean reading takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \quad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class belongs first to the internal completion $\mathbb{R}_{\text{HMT}}$; its subsequent identification with the usual real line does not enter the choice of blocks.

For the incidence reading we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad w \longmapsto (w, wA). \quad (11.1)$$

Words are written as row vectors. If $A^f = fA f^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{A^f}(w f^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (11.2)$$

The calibrated codomain retains f together with the encoded word. We therefore define

$$Q_n(x) = ((f_j, \gamma_{f_j A_w f_j^{-1}}(w_j)))_{0 \leq j < n}. \quad (11.3)$$

Proposition 11.1 (Compatibility and prolongation of the incidence reading). *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Thus the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves the visible word exactly, since $\text{pr}_1 \gamma_A = \text{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators through index $m - 1$. By induction, they have the same frames at those indices. The first m components of (11.3) agree, proving the square. The universal property of the inverse limit gives Q_∞ . The recovery claim is the projection onto the first six coordinates of each graph. $\square$

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ does not by itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are also transported. The support calculations below are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

11.2 The finite realization and the ternary code

The elliptic operator of the TRIT admits the following finite realization in the six-position chart. These positions coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, with the order $\infty, 0, 1, 2, 3, 4$ declared, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$, and $C_{a,b} = \chi(b - a)$ at the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$, and $\chi(0) = 0$. This is a coordinate realization of the quadratic structure already transported; identifying the six positions with the projective line is chart data, not a new numerical input into the generator.

The quadratic-sum identities give $C^2 = 5I_6$. Reducing modulo three yields

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6. \quad (11.4)$$

The quadratic equality realizes the elliptic regime; it does not identify the field $\mathbb{F}_3$ with an Archimedean realization of the imaginary unit. They are distinct realizations of a common operator relation, with their corresponding domains.

Definition 11.2 (Code of the dodecaphasic chart). The ternary code associated with (11.4) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}.$$

Proposition 11.3 (Self-duality, distance, and the Witt design). *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\text{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (11.5)$$

The set $\mathcal{H}$ has 132 elements and constitutes an $S(5, 6, 12)$ system: every set of five positions is contained in a unique hexad.

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by the symmetry of A_W , satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^T = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (11.5); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two weight-six words have the same support, one of the signs $1, -1$ occurs at least three times among the six ratios of their nonzero coordinates. Subtracting the corresponding multiple would produce a nonzero word of weight at most three, which is impossible. Hence a support corresponds to exactly the pair $\{c, -c\}$, and there are $264/2 = 132$ supports. If two distinct supports shared five positions, the same argument would cancel at least three coordinates in a union of size seven, producing weight at most four. Thus a five-element set belongs to at most one hexad. Finally,

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. □

The subsequent recognition is the extended ternary Golay code and the small Witt design [28]. The preceding proof explains which object is recognized. It does not replace the APP-TRIT-TPK genealogy with a search among known codes.

11.3 Dodecaphasic flag associated with K and α

Articulation of regional coordinates and closure incidence. The function of this prolongation is understood through a distinction among three operations: preserving a record, evaluating a coordinate, and extracting an incidence relation. All three act on information produced by APP-TRIT-TPK. Their articulation makes explicit the article's structural thesis: a numerical realization has an operator provenance, and certain data of that provenance additionally admit a combinatorial and lattice realization. The equations make it possible to follow both routes with their domains and the information remaining in each.

The starting point is the terminal state with its compatible prolongations. It preserves the residues and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation, and self-scaling regions are determined in that domain before their readers establish the coordinates π, e, φ . At the same time, the oriented record of the twelve windows determines the signed vector U . The subsequent appearance of K arises from the integral transformation proved in §9.8:

$$U = \mathcal{H}_{12}K, \quad K = \frac{1}{4}\mathcal{H}_{12}U, \quad \mathcal{H}_{12}^2 = 4I_{12}. \quad (11.6)$$

The congruence defining $\mathcal{L}_H$ guarantees integrality of the inverse. Thus, K constitutes a reversible coordinatization of the signed record in the pertinent sublattice. Its positions preserve phase order; its differences D_3K, D_4K preserve transversal relations; the sum $Q(K)$ determines the uniform component. That structure explains why the record enters both numerical composition and selection of a boundary configuration.

The periodic evaluation

$$K \mapsto \kappa_{\text{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable when base, length, and phase origin are fixed. Extracting a support performs a different operation: it preserves the membership of each position in a class defined by a threshold and groups records having the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete record from which they arise.

In the frame inherited from the dodecaphasic record, the data are

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), \\ w_\pi &= 010211, & w_e &= 201101, \\ w_\varphi &= 121200, & w_{\text{APP}} &= 022020. \end{aligned} \quad (11.7)$$

Applying γ_{A_w} yields the following incidences:

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first is the support of a weight-nine word; the other three are hexads. Through its high-block reading, the record supplies

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi. \quad (11.8)$$

Here the integer 729 marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in (11.2). In this particular record, all integer thresholds between 660 and 729 produce the same support: the support recovers neither the threshold nor the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$Z_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \quad (11.9)$$

Its negative support is

$$H_{\alpha}^{-} = \{4, 5, 6, 7, 9, 10\}. \quad (11.10)$$

These signs are not extracted from the decimal expansion of the scalar α ; they are retained in the state preceding its numerical expression.

Normalization of alpha and its signed configuration. The coordinate of α is determined along the positional route by composing regional blocks and the dodecaphase record. Before positional division, the oriented combination has components $Z_j = B_j + E_j - \Phi_j - K_j^{\text{per}}$, and the balance with incoming quotient is $s_j = Z_j + c_{j+1}$. Normalization satisfies

$$A_j = s_j - 1000c_j = Z_j + c_{j+1} - 1000c_j, \quad 0 \leq A_j < 1000. \quad (11.11)$$

This identity simultaneously preserves the normalized block and the contribution of the transported quotient. The limit of compatible prefixes determines the scalar closure coordinate. The signed configuration, for its part, preserves which positions in the combination before normalization exhibit negative defect. In the initial window, that configuration is the vector Z_0 of (11.9), and its negative support is H_{α}^{-} .

Two contents of the same operation are thus defined. Positional normalization determines a real coordinate through its compatible remainders and quotients; negative support determines a configuration of positions in the twelve-component cycle. The sign refers to the balance of records before normalization. Its incidence reading remains available because the state preserves that balance alongside the normalized result. This articulation corresponds to the proved positional route; comparison with the analytic vacancy operators preserves the precise scope established in the chapter on α .

Proposition 11.4 (Agreement of the two flag selectors). *Within $\mathcal{H}$, the support (11.10) is the unique hexad H satisfying*

$$B_K \subset H \subset P_{\pi}, \quad (|H \cap H_e|, |H \cap H_{\varphi}|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Thus the signed reading and the incidence reading select the same flag $B_K \subset H_{\alpha}^{-}$.

Proof. The four hexads containing B_K are obtained by adjoining, respectively, the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third pairs are contained in P_{π} . For the second, the intersection with H_{φ} has size three, and that with H_{APP} has size three; it does not meet the last condition. For the third, the three intersections have sizes 4, 3, 2. This last pair produces precisely the negative support of (11.9). $\square$

It is useful to retain both the flag and its marked frame:

$$\mathcal{J}_* = (B_K \subset H_{\alpha}^{-}), \quad \tilde{\mathcal{J}}_* = (\mathcal{J}_*; P_{\pi}, H_e, H_{\varphi}, H_{\text{APP}}).$$

Union, intersection, and difference are equivariant under a simultaneous relabeling of the design. The marked class, not the position numbers taken in isolation, is the transported object.

Lemma 11.5 (Origin selected by the marked incidence). *For the preceding frame,*

$$o_* = (H_e \cap H_{\varphi}) \setminus (H_{\alpha}^{-} \cup H_{\text{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\tilde{\mathcal{J}}_$.*

Proof. The first intersection is $\{1, 3, 4, 9\}$. The subtracted union contains 3, 4, 9 and does not contain 1. Equivariance follows because a bijection commutes with set operations. $\square$

This origin will select a direction of the lattice neighbor once the A_2 metric and the positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

From ternary coding to the marked flag. The regions also provide the codes w_π, w_e, w_φ and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \mapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and adds its transformation by A_W . The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \text{supp}(c)$, preserves the nonzero positions. For weight-six words, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. In this way, the quotient by sign produces the hexads, while the complete code remains available for operations requiring its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \quad (11.12)$$

The left-hand side arises from an integer comparison of the components of K ; the right-hand side, from the intersection of the coded supports of propagation and closure. The threshold belongs to the declared block chart and remains explicit in the reader's definition. The complete record preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes that tetrad to a hexad. The membership condition in support P_π , together with intersection sizes $(4, 3, 2)$ relative to $H_e, H_\varphi, H_{\text{APP}}$, selects a unique hexad. The following proofs bring together the two determinations: that obtained through the sign of Z_0 and that obtained through incidence. The result is organized as

$$\tilde{J}_* = (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\text{APP}}). \quad (11.13)$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π here recover their internal differentiation. They share w_π as visible ternary code, but retain distinct emissions, signatures, and multiplicities. In the binary realization on a marked dodecad, the transversal region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to lifts of the four hexads of its star. The negative region is distinguished by H_α^- , and chart order individualizes the three positive regions. This correspondence preserves regional roles and the distribution $432 + 4 \cdot 144 = 1008$, in addition to realizing the incidence partition $1 + 4$.

11.4 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p. \quad (11.14)$$

Let $V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}$. The two Euclidean realizations are subsequent languages for the same finite design here.

Proposition 11.6 (Isometry of the dodecaphasic screen). *We have*

$$I^\top I = 36I_{12} + 30J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6}I|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\text{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the automorphisms of the design.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^\top I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I\rho_{12}(g) = \rho_{\mathcal{H}}(g)I$. $\square$

The incidence information also has a concrete spectral reading. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 11.7 (Action of the intersection kernel). *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

Thus G acts by the scalar 30 on E_{inc} .

Proof. The (H, p) entry of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, there are ten four-element sets T containing p , each contained in four hexads, and five not containing it, each with a unique extension of the five-element set $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen four-element subsets of H each give one extension: the total is 15. This yields the identity. The rectangular term vanishes on V_{11} . $\square$

There is indeed a material intertwiner here: its domain, codomain, entries, and normalization are determined. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc} , with $U_W P = Q U_W$. This statement does not identify all the operators acting on these spaces solely by their trace.

11.5 The direction selected by the dodecaphasic register

The register K has a geometric role complementing its rational evaluation and support readout. Once the representation of the twelve positions has been constructed, its components determine a direction in the space of zero-mean variations. The reduction $12 \rightarrow 11$ removes the uniform mode; the next reduction $11 \rightarrow 10$ removes the direction selected by the register itself. The two operations therefore have different antecedents.

To specify the chart used completely, consider the group A_5 of even permutations of five objects and the subgroup $C_5 = \langle (1\ 2\ 3\ 4\ 5) \rangle$. The twelve left cosets $r_p C_5$ are ordered by the following representatives, written as lists of images:

p	r_p	p	r_p
1	(4, 3, 5, 2, 1)	7	(5, 4, 3, 2, 1)
2	(4, 2, 1, 3, 5)	8	(1, 3, 4, 2, 5)
3	(1, 2, 4, 5, 3)	9	(4, 2, 3, 5, 1)
4	(2, 5, 3, 4, 1)	10	(3, 2, 5, 4, 1)
5	(4, 3, 1, 5, 2)	11	(4, 5, 2, 3, 1)
6	(5, 1, 2, 3, 4)	12	(5, 3, 2, 4, 1).

The action by left multiplication defines an orthogonal representation $\rho : A_5 \rightarrow O(\mathbb{R}^{12})$: $\rho(a)e_p = e_q$ when $ar_p C_5 = r_q C_5$. This representation chart acts on the register already obtained; the group characters are not used to choose its twelve values.

Let χ_3 have the values of the three-dimensional character

class	1	2^2	3	5_a	5_b
χ_3	3	-1	0	$(1 + \sqrt{5})/2$	$(1 - \sqrt{5})/2$.

The class 5_a contains the displayed generator of C_5 , and 5_b contains its square. This convention distinguishes the two conjugate three-dimensional sectors of the representation. Define

$$P_3 = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \rho(a), \quad P_{11} = I_{12} - \frac{1}{12} J_{12}. \quad (11.15)$$

Both matrices are determined by finite sums in $\mathbb{Q}(\sqrt{5})$. The appearance of this quadratic field belongs to the realization of the representation, downstream of the regional generation of φ .

Lemma 11.8 (Projection of the register onto the three-dimensional sector). *We have*

$$P_3 = P_3^\top = P_3^2, \quad P_3 \mathbf{1} = 0, \quad \text{tr } P_3 = 3, \quad P_3 P_{11} = P_{11} P_3 = P_3.$$

For the register (11.7), the vector

$$u_K = P_3 P_{11} K \quad (11.16)$$

is nonzero. Its exact norm is

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0. \quad (11.17)$$

Proof. The character sum in (11.15) is the isotypic projector. The real character and the substitution $a \mapsto a^{-1}$ give symmetry. The multiplicity of the sector in the action on A_5/C_5 is the dimension of its subspace fixed by C_5 ,

$$\frac{1}{5} \left(3 + 2 \frac{1 + \sqrt{5}}{2} + 2 \frac{1 - \sqrt{5}}{2} \right) = 1.$$

Hence the rank is three. Orthogonality to the trivial character gives $P_3 \mathbf{1} = 0$ and the two compositions with P_{11} . These identities can also be checked by multiplying the matrices in the finite sum.

Finally, $\sum K_p = 6263$, and substituting the twelve components into that same sum gives

$$\|u_K\|^2 = K^\top P_3 K = \frac{1}{20} \sum_{a \in A_5} \chi_3(a^{-1}) K^\top \rho(a) K = \frac{6638585 + 2275584\sqrt{5}}{20}.$$

This is an exact evaluation of sixty specified terms, reproduced by the accompanying certificate using quadratic-field arithmetic with rational coefficients. The two positive coefficients prove nonvanishing. $\square$

Definition 11.9 (Dodecaphasic direction and transverse complement). The direction selected by K in the declared chart is $\ell_K = \mathbb{R}u_K$. Its transverse complement within $V_{11} = \mathbf{1}^\perp$ is

$$V_{10}(K) = V_{11} \cap u_K^\perp.$$

Proposition 11.10 (Orthogonal reduction determined by K). *The operator*

$$P_{10}(K) = P_{11} - \frac{u_K u_K^\top}{\|u_K\|^2} \quad (11.18)$$

is the rank-ten orthogonal projector onto $V_{10}(K)$. We have

$$V_{11} = \ell_K \oplus^\perp V_{10}(K), \quad P_{10} P_{11} = P_{10}, \quad P_{10} u_K = 0.$$

Proof. Let $E_K = u_K u_K^\top / \|u_K\|^2$. The proven nonvanishing ensures that it is defined. It is symmetric, $E_K^2 = E_K$, has rank one, and, since $u_K \in V_{11}$, $P_{11} E_K = E_K P_{11} = E_K$. Hence

$$(P_{11} - E_K)^2 = P_{11} - 2E_K + E_K = P_{11} - E_K.$$

Its image is $V_{11} \cap u_K^\perp$, and its trace is $11 - 1 = 10$. The decomposition and the remaining identities follow by projection onto the two orthogonal summands. $\square$

This result makes precise a role of K not exhausted by its designation as a register: besides reversibly representing oriented data and selecting the tetrad B_K , it determines a linear polarization. The direction depends on the complete register through $P_3 K$, not merely on the four points of its support. The order of register, representation, direction, and dimensional reduction has thus been retained (Proposition 11.10).

The transformation has a precise covariance law. If the chart is changed simultaneously by an orthogonal permutation matrix S , $K' = SK$ and $P'_3 = SP_3 S^{-1}$, then

$$u'_{K'} = Su_K, \quad P'_{10}(K') = SP_{10}(K)S^{-1}.$$

On the other hand, replacing u_K by au_K , with $a \neq 0$, does not change E_K . The direction retains a polarization and does not by itself allow recovery of either the amplitude of u_K or the entire register. Reversibility of $\kappa \leftrightarrow K$ remains within its own domain.

The geometric realization established here is a reduction of inner-product spaces. Its subsequent use as a compact direction requires transporting this line to the corresponding physical realization. Subsection 11.17 presents the algebraic map that keeps this reduction and the lattice screen coordinated.

11.6 The Witt star and the generation of M_{12}

Definition 11.11 (Star on a four-element set). For $B \in \binom{[12]}{4}$, the Witt star is the collection of the four hexads containing B . Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and exchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

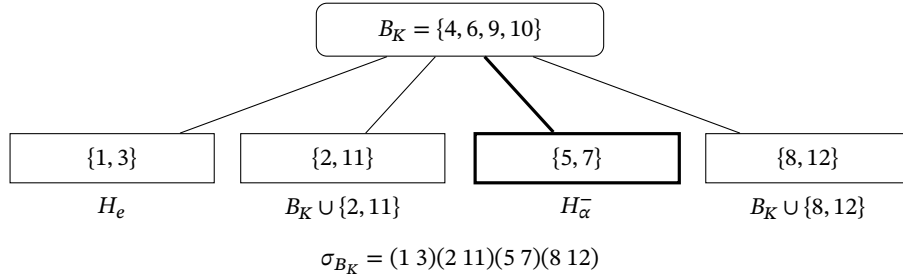


Figure 8: The concrete star of the dodecaphasic flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α . The four pairs exhaust the eight outer positions.

Proposition 11.12 (The collection of stars). *The $\binom{12}{4} = 495$ permutations σ_B preserve $\mathcal{H}$. The group they generate has order 95040 and is the Mathieu group M_{12} in its action on the small Witt design.*

Proof. The preservation claim is a finite check on the 132 hexads of (11.2): for each four-element set the four petals are constructed and one verifies $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$. Closure under composition of these permutations produces 95040 elements. In the lexicographic enumeration, it suffices to add successively six stars not yet contained in the group; the intermediate orders are 2, 6, 18, 360, 7920, 95040, and all 495 stars belong to the final group. The procedure and the six generators are made explicit in the technical provenance note. Finally, one applies the recognition of the automorphism group of $S(5, 6, 12)$ as M_{12} , whose order is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$. $\square$

An individual star generates a cyclic group of order two. It does not by itself generate M_{12} , still less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 11.13 (Equality of traces does not identify operators). On V_{11} , a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (11.14) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

11.7 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. Its weights are 0, 8, 12, 16, 24, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D , the support of a weight-twelve word. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π .

Proposition 11.14 (Residue of a dodecad and lifting of hexads). *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad, } |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D . Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^\perp$ with $O_B^\perp \cap D = B$.

Proof. A five-element set $A \subset D$ belongs to a unique octad O . If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the permitted weights, with $0 \leq j \leq 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \geq 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift by selecting five of its points. Moreover,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \quad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads on B give four distinct octads. The fifth cannot meet D in six points, since it would be another residual hexad; because it contains B , its intersection has size four and equals B . $\square$

A hexad-preserving chart $\ell : [12] \rightarrow D$ identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is the recognition of the design already constructed as the small Witt design; a concrete choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \quad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. The domain of this injectivity is the hexads, not the record K .

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	(501, 614, 498, 169, 272, 272)	555555
$++_1$	144	(810, 923, 870, 169, 272, 272)	339555
$++_2$	144	(810, 983, 810, 169, 272, 272)	393555
$++_3$	144	(870, 923, 810, 169, 272, 272)	933555
$--$	144	(870, 923, 810, 418, 674, 521)	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible tritic word $w_\pi = 010211$, but not the same regional genealogy. The

three positive regions preserve the tail 169|272|272; the negative region changes it to 418|674|521; the transverse region has a constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O_{\ell(B_K)}^\perp$, the negative region to $\iota_D(H_\alpha^-)$, and the three positive regions to the other three lifts of the star. This realizes the $4 + 1$ partition of the five octads on $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart order; incidence does not determine it without that datum. The precise claim is therefore a marked correspondence of regions, compatible with their roles and with the flag, not a canonical bijection inferred from the cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also, $1008 = 36 \cdot 28$, and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalog checks, not substitutes for the map ι_D or the regional chart.

11.7.1 Pentadic representation of the microfiber and its incidence

The arrangement of the five regions makes it possible to visualize simultaneously the unity of the ternary quotient and the differentiation preserved by the catalog. The relevant map is the restriction

$$\Psi|_{\mathcal{R}_\pi^{\text{micro}}} : \mathcal{R}_\pi^{\text{micro}} \longrightarrow \{010211\}, \quad \Psi(U) = (-U) \bmod 3.$$

Its five antecedents are the vectors U_6 in the preceding table. Figure 9 recovers their radial organization, preserving chart order, signatures, and multiplicities of the catalog (equation (6.1)).

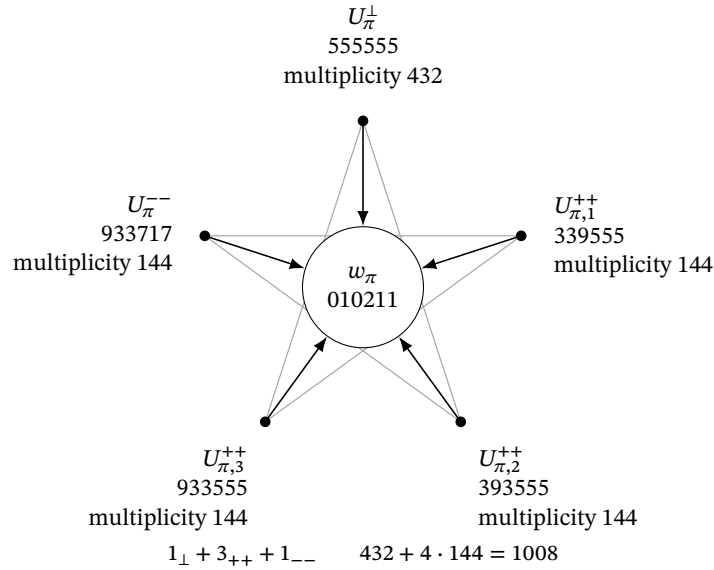


Figure 9: The five regions of π in a pentadic arrangement. Each position preserves its multiplicative signature and multiplicity; the radial arrows represent the common projection $\Psi(U) = 010211$. The star-shaped outline organizes the five positions of the chart. Their realization in the five Witt octads is performed through regional alignment and the $4 + 1$ decomposition in the text.

Projection onto the center identifies the five ternary images; the record of exterior positions retains the regional information making incidence realization possible. The distribution $1_\perp + 3_{++} + 1_{--}$ specifies one transversal region, three positive regions, and one negative region. Their normalized weights are $3/7$ and $1/7$ for each of the remaining four: transversal multiplicity is triple, even though all five images under Ψ coincide.

The link with the Witt star is formulated through the flag $B_K \subset H_\alpha^-$, the alignment ℓ , and the lift ι_D already constructed. The five regional positions are realized in the five octads over $\ell(B_K)$:

$$\begin{aligned} U_\pi^\perp &\longmapsto O_{\ell(B_K)}^\perp, \\ U_\pi^{--} &\longmapsto \iota_D(H_\alpha^-), \\ \{U_{\pi,1}^{++}, U_{\pi,2}^{++}, U_{\pi,3}^{++}\} &\longmapsto \{\iota_D(B_K \cup P) : P \in \mathcal{P}_+\}, \\ \mathcal{P}_+ &= \{\{1, 3\}, \{2, 11\}, \{8, 12\}\}. \end{aligned}$$

Assignment of the three positive positions to the three pairs is performed in an ordered chart. The transversal and negative rows are distinguished by their regional predicates and by the flag.

The two figures are thus articulated: Figure 8 shows the four hexads of the small Witt design over B_K ; Figure 9 shows the five regions that, under alignment, occupy the four residual octads and the transversal octad of the binary design. The law $4 + 1$, proved in Proposition 11.14, specifies the change in incidence. In this composition, K determines the marked tetrad, and the signature of α distinguishes one of its hexads. In addition to their common ternary value, the regions preserve the structure required for that transport toward the exceptional realization.

11.8 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four need not first go through the binary code. It is performed directly from C_W by discriminant gluing. Two compatible prolongations thus remain distinct: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ with Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

Represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with this discriminant. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \quad (11.19)$$

Theorem 11.15 (Ternary gluing). *The object (11.19) is a positive, even, unimodular lattice of rank 24. Its root system is exactly A_2^{12} ; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes stable under addition. Self-orthogonality annihilates the discriminant pairing: the latter is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a glue class is $2 \text{ wt}(c)/3$ modulo $2\mathbb{Z}$. The weights in (11.5) are multiples of three, so the lattice is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(3^6)^2} = 1.$$

Positivity follows from the orthogonal sum of the twelve A_2 metrics. In a nonzero class of A_2^*/A_2 , the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Hence a vector in a nontrivial glue class has norm at least $6 \cdot 2/3 = 4$. No norm-two roots occur outside A_2^{12} . Within it, the roots are precisely those of its twelve components. $\square$

11.9 Incidence selection of the rootless unimodular neighbor

From the incidence origin to lattice modification. The marked flag determines a position through an equivariant set operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Its function is to distinguish one component among the twelve entering the lattice realization. The complete ternary code acts as a glue code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and self-duality, weights, and distance of the code translate, respectively, into properties of integrality and determinant, parity, and norm bounds for the glue classes. Here the symbolic information of the code performs a function no longer performed by the isolated support.

The A_2 metric and positive radial chamber are fixed in the lattice development. Within that chamber, the congruence required for an even index-three neighbor has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the declared chamber. The subscript α records the incidence provenance of the selection: its content is the frame that entered closure and now individualizes a lattice direction.

Pairing with v_α determines the index-three sublattice $N_{\alpha,0}$, and adding the class $v_\alpha/3$ produces the neighbor N_α of (11.21). This modification preserves rank and recovers unimodularity and parity; congruence selection eliminates the previous roots, and local bounds exclude roots in the new classes. Identification with Leech appears after these verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the second receives the ternary code directly as glue data.

The label $o_* = 1$ of the flag is now realized as one of the twelve A_2 components. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \quad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

this congruence is equivalent to $\sum_i b_i \equiv 1 \pmod{3}$. The least norm in the chamber is obtained with a single $b_i = 1$ and all others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 54. \quad (11.20)$$

The word “least” refers to this positive chamber. It does not assert minimality throughout the residual class: the vector $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not an omissible conclusion.

Let $N = N(C_W)$. Define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3}. \quad (11.21)$$

Theorem 11.16 (Leech neighbor determined by the marked flag). *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank 24, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24} .*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \bmod 3$ is nonzero: a simple root of a component pairs with $a_i \rho_i$ to give $a_i \equiv 1 \pmod{3}$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$ belonged to N , the integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. It follows that

$$[N_\alpha : N_{\alpha,0}] = 3, \quad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

Hence it is even.

It remains to exclude roots, the decisive step. The roots of N belong to A_2^{12} . Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24} . $\square$

The argument does not recognize Leech from a coincidence of dimension. It constructs the gluing, proves integrality, evenness, and the determinant, and eliminates roots through a local bound. Only then does it apply the classification theorem. The dependence on α is preserved in the flag selecting o_* and in the vector (11.20); the scalar value of the constant is not converted into a length of the lattice.

11.10 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the rank-twenty-four Heisenberg oscillator space; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the central extension of the lattice. The cocycle and the lift of the isometries are part of the construction: a bare vector-space sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ . With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\text{HMT}}^\natural = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \quad (11.22)$$

The sign $+$ denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 11.17 (Prolongation to the Moonshine module). *The algebra (11.22), constructed on the neighbor (11.21), is a realization of the Moonshine algebra $V^{\natural}$. It has central charge 24, zero weight-one component, and automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\mathrm{Tr}_{V_{\mathrm{HMT}}^{\natural}} q^{L_0-1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \quad q = e^{2\pi i\tau}. \quad (11.23)$$

Proof. Theorem 11.16 verifies the lattice hypotheses: positivity, evenness, unimodularity, rank twenty-four, and absence of roots. The Frenkel–Lepowsky–Meurman construction and theorem for the Leech lattice then apply. They identify the orbifold with $V^{\natural}$ and establish its character and its automorphism group. Composing the flag, the neighbor, and the VOA construction functor gives the realization stated here. The group is not inferred from the coefficient 196884. $\square$

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof of the FLM construction in this article [26]. Moreover, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It does not introduce π or e as inputs into the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes an alternating sign. Hence

$$\mathrm{Tr}_{V_{\Lambda}} \theta q^{L_0-1} = t_2(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character of $V^{\natural}$ without insertion. Nor does it by itself identify a Monster conjugacy class: the space, insertion, and lift must be specified before functions are compared.

Grading, unity, and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_{α} once its properties have been verified. It incorporates the oscillator space, the central extension of the lattice, the vertex product, and the sector twisted by the lift of negation. The result is the algebra $V_{\mathrm{HMT}}^{\natural}$ of (11.22), whose automorphism group is isomorphic to the Monster. Its action domain is thus fixed by an algebraic structure with grading and product. Borchers’s Moonshine theorem subsequently determines the graded traces with insertion of those automorphisms [26, 27].

Unity acquires a precise realization at this level: the weight-zero component is the vacuum line and contributes the coefficient 1 of q^{-1} in the graded character. At the cylinder level, refinement maps preserve the constant function through precomposition. Every conservation statement therefore has its object and map. The development of the proofs makes visible this continuity of construction, from arithmetic operations and memory to incidence, metric, and vertex products.

This articulation places K and the closure of α in the main argument. K reversibly preserves an integral coordinate of the history; normalization of α composes regional records with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. The following proofs develop these maps, distinguishing reversible equivalences, information quotients, and realizations adding metric or algebraic structure. A demonstrable relation between structure, equation, and value is thus expressed, preserving the attribution of each construction and the scope of its theorems.

11.11 State-based construction of the twelve cyclotomic fibres

On the APP marks, multiplication $a : r \mapsto 4r$ in $\mathbb{Z}/9\mathbb{Z}$ satisfies $a^3 = 1$. The sheets satisfy

$$S(ax, ay) = aS(x, y), \quad P(ax, ay) = a^{-1}P(x, y),$$

since $16 \equiv 7 \equiv 4^{-1} \pmod{9}$. The ordered unit orbits are $\mathcal{O}_0 = (1, 4, 7)$, $\mathcal{O}_1 = (2, 8, 5)$. In each, the augmentation module $A(\mathcal{O}) = \ker(\mathbb{Z}[\mathcal{O}] \xrightarrow{\Sigma} \mathbb{Z})$ inherits the inner product making the e_r orthonormal. In the basis $b_1 = e_r - e_{4r}$, $b_2 = e_{4r} - e_{7r}$, its matrices are

$$G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^\top G_{A_2} C = G_{A_2}.$$

The lattice form therefore originates in the orbits of the marks. For the pairs $(1, 2), (2, 3), (3, 1)$ and sheets $\mu \in \{\Sigma, \Pi\}$, we obtain

$$W_{\text{APP}} = \bigoplus_{(i,j)} \bigoplus_{\mu} \bigoplus_{\varepsilon=0}^1 A(\mathcal{O}_\varepsilon), \quad \ell_{ij}^\Sigma(x) = x_i + x_j, \quad \ell_{ij}^\Pi(x) = x_i x_j \pmod{9}.$$

Each summand preserves pair, sheet, and orbit. The state map is

$$\Psi : (\mathbb{Z}/9\mathbb{Z})^3 \longrightarrow W_{\text{APP}}, \quad \Psi(x)_{ij,\mu,\varepsilon} = \beta_\varepsilon(\ell_{ij}^\mu(x)), \quad \beta_\varepsilon(r) = \begin{cases} e_r - e_{4r}, & r \in \mathcal{O}_\varepsilon, \\ 0, & r \notin \mathcal{O}_\varepsilon. \end{cases}$$

Proposition 11.18 (Equivariance and rank of the state module). *The action of C on the six additive fibres and of C^{-1} on the six multiplicative fibres satisfies $\Psi(ax) = \rho(a)\Psi(x)$. The images span $W_{\text{APP}} \otimes \mathbb{Q}$, of dimension twenty-four.*

Proof. The sheet identities and $\beta(4r) = C\beta(r)$ prove equivariance. For rank, order the coordinates by the three stated pairs, by Σ, Π , by $\mathcal{O}_0, \mathcal{O}_1$, and by b_1, b_2 . In each orbit, β takes successive values $(1, 0), (0, 1), (-1, -1)$. The matrix of the images of the following states, read row by row, has determinant 9:

$$\begin{array}{cccc} (0, 0, 1) & (0, 0, 2) & (0, 0, 4) & (0, 0, 5) \\ (0, 1, 0) & (0, 1, 1) & (0, 1, 2) & (0, 1, 3) \\ (0, 1, 4) & (0, 1, 5) & (0, 1, 6) & (0, 2, 0) \\ (0, 2, 2) & (0, 2, 3) & (0, 4, 0) & (0, 5, 0) \\ (1, 0, 1) & (1, 0, 2) & (1, 0, 4) & (1, 0, 5) \\ (1, 1, 0) & (1, 2, 0) & (1, 4, 0) & (1, 5, 0). \end{array}$$

Successive elimination, reducing each row by its predecessors and normalizing its first nonzero coefficient, gives pivot columns numbered from zero

$$(8, 10, 9, 11, 0, 12, 14, 16, 13, 15, 17, 2, \\ 18, 19, 1, 3, 20, 22, 21, 23, 4, 6, 5, 7)$$

and pivots

$$(1, 1, 1, -1, 1, 1, 1, 1, -1, 3, 1, 1, 3, 1, -1, 1, 1, 1, -1, 1, 1, 1, -1).$$

Their product, multiplied by the sign of the column permutation, is 9. The invertible minor establishes full rank. $\square$

11.12 Dodecaphasic alignment with the Witt positions

The pair index is $i \in \mathbb{Z}/3\mathbb{Z}$, the sheet index is $\mu \in \{0, 1\}$, and the orbit index is $\varepsilon \in \{0, 1\}$. With the dodecaphasic origin fixed, $\lambda(i, \mu, \varepsilon) = 4i + 2\mu + \varepsilon \pmod{12}$ is bijective: the four values for each i form a block, and the three blocks are disjoint. Let $R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Multiplication proves $RCR = C^{-1}$ and $R^\top G_{A_2} R = G_{A_2}$. For the trivializations ι_b and frames P_b transported jointly with Witt incidence, set

$$g_b = \iota_b^{-1} P_b C P_b^{-1} \iota_b, \quad T_{i\mu\varepsilon} = \iota_{\lambda(i,\mu,\varepsilon)}^{-1} P_{\lambda(i,\mu,\varepsilon)} R^\mu.$$

Then

$$g_{\lambda(i,\mu,\varepsilon)} T_{i\mu\varepsilon} = T_{i\mu\varepsilon} C^{(-1)^\mu}.$$

The direct sum is a C_3 -equivariant isomorphism between the twelve state fibres and the twelve lattice positions. It preserves pair, sheet, orbit, and form under simultaneous transport of origin, order, and orientation. These data persist in the lattice gluing and passage to the neighbour constructed below.

11.13 Ternary orientation of the Leech neighbour

The order-two presentation constructed above uses negation of the Leech lattice. The same lattice, obtained from the incidence marked by K and the dodecaphasic support of α , admits a second presentation compatible with the ternary character of the construction. To make this continuity explicit, the gluing is retained, its order-three isometry is constructed, and preservation of the already selected neighbour is verified. This step requires no alteration of K , its rational reading, or the preceding definition of α . The order-three prolongation is developed here, including the lattice operations and the precise conditions of the classical theorems used.

We work in the basis of simple roots of each A_2 factor, with

$$B_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \rho = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (11.24)$$

Multiplication gives

$$C^3 = I_2, \quad C^2 + C + I_2 = 0, \quad C^T B_2 C = B_2, \quad R C R = C^{-1}, \quad R^T B_2 R = B_2.$$

These identities fix both the metric and the orientation of the action. In the discriminant $A_2^*/A_2 \cong \mathbb{F}_3$, we take the class of $\varpi = (2, 1)^T/3$ as generator. We have $C\varpi - \varpi = (-1, 0)^T$, so C acts trivially on the discriminant; the reflection R reverses its sign.

The Paley–Witt matrix already constructed produces a full-support element by the following operation:

$$\begin{aligned} q_* &= (1, 1, 1, 1, 1, 1), & q_* A_W &= (2, 1, 1, 1, 1, 1), \\ c_* &= (q_*, q_* A_W) = (1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1) \in C_W. \end{aligned} \quad (11.25)$$

Every coordinate of c_* is nonzero. For the twelve positions, define the frames and the isometry

$$R_b(c_*) = \begin{cases} R, & (c_*)_b = 1, \\ I_2, & (c_*)_b = 2, \end{cases} \quad g_c = \bigoplus_{b=1}^{12} R_b(c_*) C R_b(c_*)^{-1}. \quad (11.26)$$

The symbol $R_b(c_*)$ here denotes an invertible frame of an A_2 factor, not the regional projector P_3 .

Proposition 11.19 (Order-three isometry on the marked neighbour). *The operator g_c is an order-three isometry with no nonzero fixed vectors. It preserves the Niemeier lattice $N = N(C_W)$ and the neighbour N_α constructed in (11.21).*

Proof. Each block of g_c is C or C^{-1} ; its minimal polynomial is $X^2 + X + 1$. This proves its order and the absence of fixed vectors. Both blocks act trivially on the discriminant, so they preserve every gluing class and therefore the lattice N .

Write $v = v_\alpha = 4\rho_{o_*} + \sum_{b \neq o_*} \rho_b$, with the frames and incidence origin already fixed. Its radial coefficients are $a_{o_*} = 4$ and $a_b = 1$ for $b \neq o_*$. All satisfy $a_b \equiv 1 \pmod{3}$. In one factor,

$$(C - I_2)\rho = (-2, -1)^T, \quad (C^{-1} - I_2)\rho = (-1, -2)^T.$$

After division by three, their discriminant classes are 2 and 1, respectively. Consequently, the vectors

$$y = \frac{g_c v - v}{3}, \quad z = \frac{g_c^{-1} v - v}{3} \quad (11.27)$$

have discriminant words c_* and $-c_*$, both in C_W . Hence $y, z \in N$. The local product is

$$\left\langle \frac{(C^{\pm 1} - I_2)a_b \rho}{3}, a_b \rho \right\rangle = -a_b^2,$$

and summing gives $\langle y, v \rangle = \langle z, v \rangle = -(16 + 11) = -27$. Thus, $y, z \in N_0 = \{x \in N : \langle x, v \rangle \equiv 0 \pmod{3}\}$. For $x \in N_0$,

$$\langle g_c x, v \rangle = \langle x, g_c^{-1} v \rangle = \langle x, v \rangle + 3\langle x, z \rangle \equiv 0 \pmod{3}.$$

The argument applied to g_c^{-1} proves $g_c N_0 = N_0$. Finally, $g_c(v/3) = v/3 + y$ preserves the class generating $N_\alpha = N_0 + \mathbb{Z}(v/3)$. We obtain $g_c N_\alpha = N_\alpha$. $\square$

The proof preserves more than the lattice dimension: it uses the gluing code, its orientation, the flag origin, and the congruence of the radial vector. These data explain why the second presentation remains linked to the dodecaphasic record from which the construction began.

11.14 Ternary trace, twisted weight, and order-three extension

Let $\Lambda = N_\alpha$, and let $\hat{g}_c$ be the standard lift of g_c to the lattice algebra $V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda]$, with the central extension and cocycle already specified. The odd order allows this lift to be chosen of order three. Each complex A_2 factor contributes the eigenvalues ζ_3 and ζ_3^2 , where $\zeta_3^2 + \zeta_3 + 1 = 0$.

Proposition 11.20 (Graded trace and type of the lift). *For $j = 1, 2$,*

$$\text{Tr}_{V_\Lambda}(\hat{g}_c^j q^{L_0-1}) = t_3(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12}. \quad (11.28)$$

The lowest weight of the nontrivial twisted modules is $4/3$, and the lift is of type zero.

Proof. Only the zero vector contributes to the lattice part of the trace, because g_c and g_c^2 have no nonzero fixed vectors. At each oscillator degree, the trace of symmetric powers is the inverse of the determinant $I - g_c^j q^n$. Each A_2 factor contributes $1 + q^n + q^{2n}$, proving (11.28).

The twisted-vacuum weight formula for an order-three lattice automorphism, applied to the two twelve-dimensional eigenspaces, gives

$$\rho_{\hat{g}_c} = \frac{1}{4 \cdot 3^2} (1 \cdot 2 \cdot 12 + 2 \cdot 1 \cdot 12) = \frac{4}{3}.$$

The type is the class of $3^2 \rho_{\hat{g}_c} = 12$ modulo three, which is zero. The weight formula is a result of twisted-module theory; the multiplicities and isometry supplied to it have been constructed here. $\square$

Theorem 11.21 (Second presentation of the Moonshine algebra). *The type-zero cyclic extension*

$$V_{(3)} = \text{Orb}_{\mathbb{Z}/3}(V_\Lambda, \hat{g}_c) \quad (11.29)$$

is isomorphic to the vertex operator algebra $V^\natural$.

Proof. Theorem 11.16 provides a positive, even, unimodular, rootless lattice of rank twenty-four. Proposition 11.19 constructs an order-three fixed-point-free isometry on that same lattice; Proposition 11.20 verifies the lift and its type. The Chen–Lam–Shimakura theorem on the order-three orbifold of the Leech lattice algebra then applies [32, Theorem 1.1]. This identification is also covered by the uniform treatment of Abe–Lam–Yamada [33, Theorem 4.4 and Sections 2–4]. This application follows construction of the lattice data; equality of characters does not replace the hypotheses of the theorem used. $\square$

To specify the dual symmetry, write $V_{(3)} = W_0 \oplus W_1 \oplus W_2$, where W_i is the integral sector selected in the $\hat{g}_c^i$ -twisted module. The product satisfies $Y(W_i, z)W_j \subset W_{i+j \bmod 3}((z))$. Hence

$$g^\vee|_{W_i} = \zeta_3^i I \quad (11.30)$$

defines an order-three automorphism. Indeed, the factor on a product is $\zeta_3^{i+j} = \zeta_3^i \zeta_3^j$, and the vacuum and conformal vector belong to W_0 . The classical identification of this automorphism in the Monster is class $3B$, with series $T_{3B} = t_3 + 12$ [34, 27]. This normalization can be checked in the construction. The lattice character is $\text{ch } V_\Lambda = J + 24$: its initial pole is q^{-1} , its weight-one space has dimension 24, and the weight-zero modular character is determined by these data. If F is the character of the fixed subspace and H that of each integral twisted sector, projection onto invariants and conjugation of the two sectors give

$$F = \frac{J + 24 + 2t_3}{3}, \quad F + 2H = J, \quad \text{Tr } g^\vee q^{L_0-1} = F - H = t_3 + 12.$$

Its domain is the extended algebra, whereas $\hat{g}_c$ acts on the preceding lattice algebra.

11.15 Equivalence of the order-two and order-three presentations

Denote by

$$V_{(2)} = (V_\Lambda)^+ \oplus (V_\Lambda^T)^+$$

the FLM presentation in (11.22). Both presentations preserve the same Leech neighbour as their antecedent. Their extensions employ different automorphisms and vertex products determined by the respective lifting data.

Theorem 11.22 (Graded equivalence and transport of automorphisms). *Given vertex operator algebra isomorphisms*

$$\phi_2 : V_{(2)} \xrightarrow{\cong} V^\natural, \quad \phi_3 : V_{(3)} \xrightarrow{\cong} V^\natural,$$

the composition

$$\Phi_{23} = \phi_2^{-1} \phi_3 : V_{(3)} \xrightarrow{\cong} V_{(2)} \quad (11.31)$$

preserves the vacuum, conformal vector, grading, and vertex product. Conjugation by Φ_{23} identifies the automorphism groups and preserves their graded traces.

Proof. The composition of the two isomorphisms preserves the stated structures. In particular,

$$\Phi_{23} Y_{(3)}(a, z)b = Y_{(2)}(\Phi_{23}a, z)\Phi_{23}b, \quad \Phi_{23} L_0^{(3)} = L_0^{(2)} \Phi_{23}.$$

If h is an automorphism of $V_{(3)}$, then $h' = \Phi_{23} h \Phi_{23}^{-1}$ is an automorphism of $V_{(2)}$. Equality of traces on each finite-dimensional homogeneous component gives

$$\text{Tr}_{V_{(3)}} h q^{L_0-1} = \text{Tr}_{V_{(2)}} h' q^{L_0-1}.$$

□

The choice of ϕ_2 and ϕ_3 forms part of the statement; Φ_{23} is not presented as a canonical identification independent of these choices. Nor does it transform an order-three automorphism into the order-two involution: conjugation preserves order. Its content is that two different extensions realize the same graded structure, with explicit transport of their products, representations, and traces.

11.16 Eta functions, Fricke involution, and vacuum normalization

For $\text{Im } \tau > 0$, the traces of the two lattice automorphisms admit the expressions

$$\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n), \quad t_2 = \left(\frac{\eta(\tau)}{\eta(2\tau)} \right)^{24}, \quad t_3 = \left(\frac{\eta(\tau)}{\eta(3\tau)} \right)^{12}. \quad (11.32)$$

They follow directly from the factorizations $1 - q^{2n} = (1 - q^n)(1 + q^n)$ and $1 - q^{3n} = (1 - q^n)(1 + q^n + q^{2n})$. The order in q of the level-three quotient, before raising it to the twelfth power, is

$$\frac{1}{24} - \frac{3}{24} = -\frac{1}{12}; \quad 12 \left(-\frac{1}{12} \right) = -1.$$

Here $-1/12$ denotes the exact difference between the initial exponents of two eta functions. The operation is specified and does not require interpreting a divergent sum as a convergent one. Agreement with other normalized defects in the corpus preserves their respective realizations; it does not identify their operators through equality of the scalar.

Proposition 11.23 (Fricke involution on the order-two trace). *Let $w_2\tau = -1/(2\tau)$. Then*

$$t_2(w_2\tau) = \frac{2^{12}}{t_2(\tau)}, \quad T_{2A} = t_2 + 24 + \frac{2^{12}}{t_2} \quad (11.33)$$

and $T_{2A}(w_2\tau) = T_{2A}(\tau)$.

Proof. The transformation law $\eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau)$ gives

$$\frac{\eta(-1/(2\tau))}{\eta(-1/\tau)} = \sqrt{2} \frac{\eta(2\tau)}{\eta(\tau)}.$$

The twenty-fourth power yields the first identity. Substitution interchanges the two variable terms of T_{2A} and fixes the constant term. Recognition of this expression as the McKay–Thompson series of class 2A corresponds to the level-two symmetrization of Conway–Norton [34, §§4 and 6, Table 3] and the Moonshine theorem [27]. $\square$

The modular uniformizations of levels two and three, with the preceding normalizations, are

$$j = \frac{(t_2 + 256)^3}{t_2^2}, \quad (11.34)$$

$$J = j - 744 = t_3 + 12 + \frac{10 \cdot 3^9}{t_3} + \frac{4 \cdot 3^{14}}{t_3^2} + \frac{3^{18}}{t_3^3}. \quad (11.35)$$

They are used as classical modular identities on the traces already constructed, not as rules for selecting the code or the neighbour. The second can also be written $j = (t_3 + 27)(t_3 + 243)^3/t_3^3$. The direct product expansion of (11.28) begins with

$$t_3 = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + 1188q^4 + O(q^5).$$

In particular, $[q]J = 54 + 10 \cdot 3^9 = 196884$; this is a consequence of the traces and uniformization. The construction of $V^\natural$ retains its lattice and orbifold proof, independently of this check of the first coefficient.

The three operations occurring in this extension have different domains. The dual symmetry $g^\vee$ acts on sectors of the vertex algebra; Fricke acts on the modular parameter and its function; radius duality interchanges momentum and winding in the compact lattice sector. Their reciprocity formulas can be coordinated through the corresponding maps, but the name “duality” does not replace those maps. This distinction allows the construction to continue towards dimensional screens while preserving incidence, grading, and the origin of each operation.

11.17 Exceptional incidence and coordination of the screens

The direction selected by K and the Leech lattice arise from different representations of the constructed incidence. The former determines the orthogonal reduction $12 \rightarrow 11 \rightarrow 10$; the latter enters the isotropic reduction $26 \rightarrow 24$. Their coordination preserves the preceding space, its relations and the information conveyed by each projection.

The sections on T-duality and coordinated screens in [2] develop this coordination jointly. The former brings together the involution of integer coordinates, unitary conjugation with its domains and the change of residual section retaining the quotient memory. The latter constructs the morphism $\mathcal{R}_M^{\text{alg}}$ defined there and proves the isomorphism of the quotient presentations in the corresponding theorem on isomorphic screens. The graph preserves the coordinate in V_{11} together with its projection in V_{10} , while the isotropic quotient recovers the Leech lattice. The respective kernels and codomains are thereby preserved.

The action ellipse is constructed before duality is applied and fixes its dimensionless radius together with the reciprocal radius (the action-ellipse section of [2]). The compact sector then connects exceptional incidence with a determined quadratic form. The fields, action and operators of the subsequent physical realization are presented in their own domain, preserving this algebraic antecedent and its internal proofs.

11.18 Graded Moonshine traces and representation of the Monster

For $g \in \text{Aut}(V_{\text{HMT}}^{\mathfrak{h}})$, the trace with insertion is

$$T_g(\tau) = \text{Tr}_{V_{\text{HMT}}^{\mathfrak{h}}} gq^{L_0-1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [27, Theorem 1.1]. The character J , the group $\mathbb{M}$, and the collection $g \mapsto T_g$ are realizations of the algebra already constructed. They do not form a causal sequence in which one first guesses J and then infers the Monster.

The unit has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V_0^{\mathfrak{h}} = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one the dimension is zero, accounting for the absence of the constant term in $J = j - 744$. A morphism of vertex algebras preserves the vacuum vector. This property provides the technical meaning of preservation of the unit at this level; the unit of cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

appears. The first decomposition corresponds to the trivial conformal line and the irreducible component of dimension 196883 in the Monster representation. The other expressions have lattice-theoretic or arithmetic readings. Their equality does not automatically identify their summands with TPK regions, words, or routes. Doing so would require presenting the maps between those spaces and proving that they respect the relevant actions.

In particular, an isometry $N_\alpha \simeq \Lambda_{24}$ induces a realization of (11.22), and a choice of isomorphism with $V^{\mathfrak{h}}$ transports its automorphism group. This does not construct an action of the Monster on the digits of π , on the blocks of K , or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\text{TPK}}(g) = \rho_{V^{\mathfrak{h}}}(g)F,$$

including domain, codomain, and preserved information. The absence of that step from the present claim does not remove the incidence intertwiner U_W , the flag, or the neighbor already constructed; it only delimits a different promotion, namely an action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned} & \text{APP-TRIT-TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, Z_0) \\ & \longrightarrow (C_W, \mathcal{H}; B_K \subset H_{\alpha}^-, P_{\pi}, H_e, H_{\varphi}, H_{\text{APP}}) \\ & \longrightarrow (N(C_W), o_*, \text{metric and chamber}) \longrightarrow N_{\alpha} \longrightarrow V_{\text{HMT}}^{\natural}. \end{aligned}$$

At the incidence level, the stars generate M_{12} , and the residual chart realizes the regional $4 + 1$ structure in W_{24} . At the lattice and VOA level, the construction gives Leech and the Moonshine module. These are prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber, and lift.

The result of this section is therefore an assembled derivation: the finite and lattice proofs make the passage from dodecaphasic incidence to the hypotheses of the recognition theorems effective. The conclusion comprises the structures and transports specified in the preceding propositions, with their explicit domains and choices. Comparison of constants and experimental validation concern assertions distinct from this algebraic construction.

12 Hilbert-space realization of the nonadic phase and preservation of the unit

The construction of Triadic Modular Holography first provides the states, their transitions, and the restriction maps. The Hilbert-space realization adds a linear structure and an inner product to their discrete coordinates. This order determines what each operator represents: the basis labels arise from the nonadic construction; linear operations subsequently act on those labels. Genealogical memory remains in the TPK state and accompanies the projections used to represent it.

12.1 Phase fiber, labeled basis, and inner product

For an integer depth $d \geq 1$, let

$$\mathcal{C}_{9,d} = (\mathbb{Z}/9\mathbb{Z})^d, \quad \mathcal{H}_d = \ell^2(\mathcal{C}_{9,d}), \quad \mathcal{A}_d = B(\mathcal{H}_d). \quad (12.1)$$

The inner product is $\langle f, g \rangle = \sum_x \overline{f(x)}g(x)$. The functions δ_x form an orthonormal basis labeled by the phase coordinates. Thus $\dim \mathcal{H}_d = 9^d$ and $\mathcal{A}_d \cong M_{9^d}(\mathbb{C})$. The matrix algebra represents this finite fiber; routes, quotients, and memory omitted by a phase projection still belong to the starting state.

Coordinate factorization induces the unitary map

$$\mathcal{H}_{d+1} \longrightarrow \mathcal{H}_d \otimes \mathbb{C}^9, \quad \delta_{(x,r)} \longmapsto \delta_x \otimes \delta_r. \quad (12.2)$$

Define the normalized trace $\tau_d(a) = 9^{-d} \text{Tr}(a)$. Finite counting and normalization are distinguished: $\text{Tr}(I) = 9^d$, whereas $\tau_d(I) = 1$. Likewise, the group $(\mathbb{Z}/9\mathbb{Z})^3$ and the vector space $\mathbb{F}_3^6$ both have 729 elements, but retain different algebraic laws. Realization of a phase fiber does not replace the ternary encoding of APP emissions.

12.2 Translation and phase: finite Weyl representation

In the complex realization subsequent to the constructions of π and the structure $i^2 = -1$, fix $\omega = \exp(2\pi i/9)$. On $\mathcal{H}_1$, let

$$(Xf)(r) = f(r-1), \quad (Zf)(r) = \omega^r f(r). \quad (12.3)$$

The first operation shifts the phase coordinate; the second evaluates it through a character. Both are unitary and satisfy

$$X^9 = Z^9 = I, \quad ZX = \omega XZ. \quad (12.4)$$

Indeed, $(ZXf)(r) = \omega^r f(r-1)$, whereas $(XZf)(r) = \omega^{r-1} f(r-1)$. The character appears here as a realization of the phase already constructed and does not select the TPK numerical registers.

12.2.1 Oriented area and central extension of translations

The translation support is $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. The mixed APP curvature constructed in Section 4.4 assigns +1 to the oriented plaquette of the sum-product pair and -1 to the pair in the opposite order. Its bilinear extension to two displacements $u = (a, b)$ and $v = (c, d)$ is the alternating form

$$\sigma_{\text{ar}}(u, v) = ad - bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \quad (12.5)$$

Indeed, alternation makes the evaluation vanish on two displacements along the same axis. Values on the oriented basis are $\sigma_{\text{ar}}((1, 0), (0, 1)) = 1$ and $\sigma_{\text{ar}}((0, 1), (1, 0)) = -1$; distributing both arguments gives (12.5). The same formula is the sum of plaquettes of any oriented cellular chain realizing that parallelogram: interior edges cancel. Changing an integer representative by a multiple of nine preserves its representation in $\mathbb{Z}/9\mathbb{Z}$.

The order $X^a Z^b$ of the already defined operators fixes a section of the central extension. Denote its composition factor by

$$c_W(u, v) = bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \quad (12.6)$$

This cocycle preserves the order of the two operations, while σ_{ar} preserves the orientation of the parallelogram.

Proposition 12.1 (Central extension and commutator sign). *The set $\mathfrak{H}_9 = (\mathbb{Z}/9\mathbb{Z})^3$, with product*

$$(a, b, t)(c, d, s) = (a + c, b + d, t + s + bc), \quad (12.7)$$

is a group of order 729. The map

$$\varrho(a, b, t) = \omega^t X^a Z^b$$

is a faithful unitary representation on $\mathcal{H}_1$. With the convention $[g, h] = ghg^{-1}h^{-1}$, one has

$$[\varrho(a, b, 0), \varrho(c, d, 0)] = \omega^{bc-ad} I = \omega^{-\sigma_{\text{ar}}((a,b),(c,d))} I. \quad (12.8)$$

The center and the translation group form the exact sequence

$$0 \longrightarrow \mathbb{Z}/9\mathbb{Z} \xrightarrow{t \mapsto (0,0,t)} \mathfrak{H}_9 \xrightarrow{(a,b,t) \mapsto (a,b)} X_9 \longrightarrow 0.$$

Proof. For $w = (e, f)$, the cocycle identity reads

$$c_W(u, v) + c_W(u + v, w) = bc + (b + d)e = de + b(c + e) = c_W(v, w) + c_W(u, v + w).$$

It proves associativity of (12.7); the identity is $(0, 0, 0)$, and the inverse of (a, b, t) is $(-a, -b, -t + ab)$. The three residues are free, giving order $9^3 = 729$. From $ZX = \omega XZ$ it follows that

$$(X^a Z^b)(X^c Z^d) = \omega^{bc} X^{a+c} Z^{b+d},$$

which proves the representation property. If $\omega^t X^a Z^b = I$, its action on δ_r first forces $a = 0$. Evaluation at $r = 0$ gives $t = 0$, and evaluation at $r = 1$ gives $b = 0$, since ω has order nine. The representation is therefore faithful. Comparing the products in both orders gives the factor $bc - ad$. A central element must satisfy $bc - ad = 0$ for every (c, d) . Taking successively $(c, d) = (1, 0), (0, 1)$ yields $b = a = 0$. This identifies the center and proves exactness. $\square$

The Wilson phase for orientation (12.5) is $\omega^{\sigma_{\text{ar}}(u,v)}$; the commutator of the ordered section $X^a Z^b$ expresses its inverse. In particular, $c_W(u, v) - c_W(v, u) = -\sigma_{\text{ar}}(u, v)$. This identity relates two determined conventions without identifying the polarized cocycle with the alternating form. The extension belongs to the translation group of the APP support. Its representation acts on the phase fiber; the trajectory

register and carries retain their residence in the TPK state that produced that fiber. The group order 729 and the cardinality 729 of the ternary ambient space preserve the different types established in (12.1). The center $\mathbb{Z}/9\mathbb{Z}$ of this extension records the commutator phase. The memory C_{12} of the sequence $0 \rightarrow C_{12} \rightarrow C_{108} \rightarrow C_9 \rightarrow 0$, constructed in Proposition 9.2, belongs to the dodecaphasic calendar. Each extension preserves its own composition law and projection.

Proposition 12.2 (Algebra generated by phase and translation). *The 81 operators $X^a Z^b$, with $0 \leq a, b < 9$, form a basis of $\mathcal{A}_1 = M_9(\mathbb{C})$. In particular, the representation (12.3) is irreducible.*

Proof. If $a \neq 0 \pmod{9}$, the matrix $X^a Z^b$ has no diagonal entries. If $a = 0$, its trace is $\sum_{r=0}^8 \omega^{br}$, equal to 9 for $b = 0$ and 0 otherwise. By (12.4), the product $(X^a Z^b)^* X^{a'} Z^{b'}$ is a factor of modulus one times $X^{a'-a} Z^{b'-b}$. Therefore,

$$\tau_1((X^a Z^b)^* X^{a'} Z^{b'}) = \delta_{a,a'} \delta_{b,b'}.$$

These are 81 linearly independent matrices in a space of dimension 81. They generate the whole matrix algebra, whose commutant consists of scalars; irreducibility follows. $\square$

Theorem 12.3 (Uniqueness with fixed primitive central character). *Every irreducible unitary representation of the group presented by $X^9 = Z^9 = c^9 = I$, c central, and $ZX = cXZ$, in which c acts as ωI , is unitarily equivalent to (12.3).*

Proof. Let $v \neq 0$ be an eigenvector of Z , with eigenvalue λ . Since $Z^9 = I$, λ is a power of ω . The vectors $X^k v$, $0 \leq k < 9$, have eigenvalues $\omega^k \lambda$, distinct because the character is primitive. They are orthogonal, and their span is invariant under X , Z , and the center. Irreducibility forces it to be the entire space. Reordering the basis to begin with eigenvalue 1, Z is diagonal with entries $1, \omega, \dots, \omega^8$, and X is cyclic translation. Normalizing the basis produces the asserted unitary equivalence. $\square$

Primitivity of the character and irreducibility are part of the statement. At depth d , operators acting on different factors commute, and tensor products of the preceding bases generate $\mathcal{A}_d$. The phase structure is thus represented at each level without identifying phase return with return of the complete genealogical state.

12.3 Two lifts and two different normalizations

Factorization (12.2) permits preserving the entire new fiber or fixing one of its labels. For $p_j = |\delta_j\rangle\langle\delta_j|$, define

$$\iota_d(a) = a \otimes I_9, \quad J_{d,j}(a) = a \otimes p_j. \quad (12.9)$$

Proposition 12.4 (Preservation of unit and trace). *The inclusion ι_d is unital and preserves normalized trace. The inclusion $J_{d,j}$ is a corner embedding and satisfies*

$$\tau_{d+1}(\iota_d(a)) = \tau_d(a), \quad \iota_d(I) = I, \quad \tau_{d+1}(J_{d,j}(a)) = \frac{\tau_d(a)}{9}, \quad J_{d,j}(I) = I \otimes p_j.$$

Proof. The trace of a tensor product is the product of the traces. Since $\text{Tr}(I_9) = 9$ and $\text{Tr}(p_j) = 1$, dividing by 9^{d+1} produces both formulas. The images of the identity are read directly from (12.9). $\square$

The unital inclusion preserves the set of 9 phases and its global normalization. The corner preserves a marked continuation and its support projector. Both operations are legitimate, but transmit different data to the next depth.

12.4 Inductive closure and tracial realization

Theorem 12.5 (Nonadic uniformly hyperfinite algebra). *The norm closure of the union through the inclusions ι_d is*

$$\mathcal{A}_{9^\infty} = \varinjlim (\mathcal{A}_d, \iota_d) \cong \bigotimes_{n=1}^{\infty} M_9(\mathbb{C}).$$

It is a UHF algebra of type $9^\infty = 3^\infty$, with a unique normalized trace τ , whose restriction to $\mathcal{A}_d$ is τ_d .

Proof. The inclusions are injective and unital. The closure of the increasing union of matrix algebras is precisely the UHF construction. The compatibility in Proposition 12.4 defines τ on the union; positivity and norm one allow its extension to the closure. Every normalized trace on the limit restricts to the unique normalized trace on each matrix algebra. Since the union is dense, those restrictions determine the trace uniquely. $\square$

Let $\mathcal{H}_\tau$ be the completion of $\mathcal{A}_{9^\infty}$ for $\langle a, b \rangle_\tau = \tau(a^*b)$, and let π_τ be its representation by left multiplication. The vector defined by the identity is denoted Ω_τ .

Theorem 12.6 (Tracial factor of complete refinement). *The weak closure $\mathcal{R}_9 = \pi_\tau(\mathcal{A}_{9^\infty})''$ is a hyperfinite factor of type II_1 . The nonadic inclusions preserve its unit and normalized trace.*

Proof. The product trace extends to a faithful finite normal trace in the tracial representation. To see that the center is trivial, let E_d be the tracial expectation onto $\pi_\tau(\mathcal{A}_d)$. On a tensor of length $n > d$, it takes the trace of the last $n - d$ factors. These maps extend continuously to the corresponding orthogonal projections in L^2 . If z is central, $E_d(z)$ commutes with $\pi_\tau(\mathcal{A}_d)$, by bimodularity of E_d . It is therefore scalar and equal to $\tau(z)I$. Density of the matrix union implies $E_d(z) \rightarrow z$ in L^2 , hence $z = \tau(z)I$. The algebra is thus a finite factor. It contains unital matrix algebras of unbounded sizes 9^d , so it is not a finite matrix factor. It is of type II_1 . It is hyperfinite because the same union of matrices is weakly dense. Preservation of unit and trace follows from each finite inclusion and persists upon taking closures. $\square$

12.5 Integer ranks and continuous tracial dimension

In $\mathcal{A}_d$, a projection P has normalized dimension $\tau_d(P) = \text{rank}(P)/9^d$. Ranks remain integers at every finite depth. Their compatible normalizations allow the following passage to the limit.

Proposition 12.7 (Realization of each tracial dimension). *For each $t \in [0, 1]$, there exists a projection $P_t \in \mathcal{R}_9$ with $\tau(P_t) = t$.*

Proof. Let $m_d = \lfloor 9^d t \rfloor$. Then $0 \leq m_{d+1} - 9m_d \leq 8$. After choosing a diagonal projection of rank m_d , its inclusion has rank $9m_d$; add $m_{d+1} - 9m_d$ diagonal positions in its complement. This yields an increasing sequence of projections within $\mathcal{R}_9$. Its strong limit is a projection P_t , and normality of the trace gives

$$\tau(P_t) = \lim_{d \rightarrow \infty} \frac{\lfloor 9^d t \rfloor}{9^d} = t.$$

$\square$

Here t denotes the dimension coordinate represented in the closure, not an input to the APP generator. The assertion describes the scope of the resulting tracial realization. To compare it with a particular observable, it remains necessary to identify the projection and the genealogical information it preserves.

By contrast, the isometries $V_{d,j}\xi = \xi \otimes \delta_j$ realize $J_{d,j}(a) = V_{d,j}aV_{d,j}^*$. In the limit of a marked route, the finite spaces are dense. The corners approximate all finite-rank operators; their norm closure is $\mathcal{K}(\mathcal{H}_\infty)$, and their bicommutant is $B(\mathcal{H}_\infty)$, of type I_∞ . This difference preserves the operational content of the two choices in (12.9): selection of one continuation and unital prolongation of the entire fiber have different limits.

The Hilbert-space realization, its phase operators, and the tracial closure are thus constructed in this section, with their inclusion choices explicit. They constitute the operator-theoretic support to be used subsequently for coordinate interchange and the compact Hamiltonian.

Part II

Positive geometry, refinement and lattice duality

13 Tritic orientation, compatible prolongation and double projection

The generated character of the real coordinates is expressed in the relation between arithmetic depth and evaluation. The primitive datum is an admissible history of operations with its residues, quotients and orientations. An Archimedean reading associates increasingly narrow intervals with its prefixes; an incidence reading preserves boundary relations of the same state. The double projection of the corpus [1, 7] brings these two operations together without identifying a scalar evaluation with the entire history. The article uses this antecedent to study the subsequent realizations of the dodecaphasic record.

13.1 Positional cylinders and determination of the real coordinate

For a ternary block $b = (b_1, \dots, b_6)$, with representatives $b_j \in \{0, 1, 2\}$ of $\mathbb{F}_3$, the integer evaluation $\nu(b) = \sum_{j=1}^6 b_j 3^{6-j}$ belongs to $\{0, \dots, 728\}$. If b_{n+1} is the next admissible block, positional prolongation satisfies

$$N_{n+1} = 729N_n + \nu(b_{n+1}), \quad a_n = \frac{N_n}{729^n}, \quad b_n^+ = \frac{N_n + 1}{729^n}.$$

Here N_0 is the integer part determined by the regional reader. The symbol b_n^+ denotes a real endpoint and is distinct from the ternary block b . The representation cylinder is $[a_n, b_n^+]$; its closure is used in the argument establishing existence of the limit. Carry preserves the passage from a prefix to its predecessor through Euclidean division.

Proposition 13.1 (Limit of cylinder centres). *For a compatible prolongation, the cylinder closures are nested and determine a unique coordinate x . If $c_n = (a_n + b_n^+)/2$, then*

$$x = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n^+ = \lim_{n \rightarrow \infty} c_n, \quad |x - c_n| \leq \frac{1}{2 \cdot 729^n}.$$

Proof. The inequality $0 \leq \nu(b_{n+1}) < 729$ places the subcylinder in the closure of its predecessor. Its endpoints are rational, its width is 729^{-n} , and it remains inside the first bounded closed interval. Nesting and decreasing width produce a unique Cauchy class of endpoints. In the Archimedean realization, this class has the representative x ; the distance to the midpoint is at most half the width. The boundary convention assigns a representation when two positional expansions represent the same limit. $\square$

The arithmetic mean thus enters a precise evaluation: the centre of each interval bounding the coordinate. The limit preserves the exactness of the value as the width vanishes. The enriched representation additionally preserves the prefix sequence and its incidence. The numerical projection and the exceptional projection are two maps with different codomains; one preserves the limiting value and the other transports the relations enabling the lattice construction. The passage to $\mathbb{R}$ is thus situated in the realization of the reader, after the arithmetic generation of the data it evaluates.

To make compatibility explicit, let $p_{n+1,n}$ be the restrictions of the enriched state, with $x_n = p_{n+1,n}(x_{n+1})$. If $I_n(x_n)$ is the numerical cylinder and $\mathcal{B}_n(x_n)$ the boundary incidence, the two transports simultaneously preserve

$$\overline{I_{n+1}(x_{n+1})} \subseteq \overline{I_n(x_n)}, \quad r_{n+1,n} \mathcal{B}_{n+1}(x_{n+1}) = \mathcal{B}_n(x_n).$$

The second equality uses the incidence maps of the corpus; Proposition 27.1 makes their realization through transported frames and codes explicit. Its five consubstantial constructions—spectral, solenoidal, gauge, cohomological and determinantal—belong to that same compatible system. The subsequent geometric study preserves this unity of origin by specifying which of its observations each theorem uses.

13.2 Oriented trajectories and the temporality of update

Let $x \xrightarrow{\text{adm}} y$ be the TPK admissibility relation on the enriched space $\widehat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is

$$\gamma : \mathcal{T} \longrightarrow \widehat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+).$$

The compatibility relation determines the transitions; the parametrization allows them to be ordered. In the space $E = \mathcal{T} \times \widehat{X}$, with $p(t, x) = t$, the trajectory determines the section $\tilde{\gamma}(t) = (t, \gamma(t))$, which satisfies $p \circ \tilde{\gamma} = \text{id}_{\mathcal{T}}$. This distinction preserves the difference between a state, its trajectory and the coordinate used to describe its evolution.

The tritic signature $\tau \in \{+1, 0, -1\}$ types the generators through

$$J_\tau^2 = -\tau I.$$

In the temporal reader of the corpus, +1 represents return and memory; 0, threshold update; and −1, the opening of conjugate prolongations. Their interpretation as past, present and future refers to the functions of the preserved antecedent, the effective transition and the admissible continuation. The three tenses thus acquire an operator-theoretic content on histories.

If $\mathcal{P}_n(x)$ is the set of admissible continuations of length n , deleting the final step defines $r_n : \mathcal{P}_{n+1}(x) \rightarrow \mathcal{P}_n(x)$. An unbounded continuation is a collection (γ_n) with $r_n(\gamma_{n+1}) = \gamma_n$. Restriction reconstructs antecedents; the update U_t acts at the transition; prolongation maintains the prefix boundary. The existence of these collections is used in the surviving domains fixed by the prolongation theorem of the corpus.

On a finite signature $\sigma = (\tau_1, \dots, \tau_n)$, oriented inversion is

$$C_{\text{rev}}\sigma = (-\tau_n, \dots, -\tau_1), \quad C_{\text{rev}}^2\sigma = \sigma.$$

The conjugate trajectory also preserves the sheet, position and boundary transformations of its transport. Bidirectionality is therefore expressed in the full history, while this formula gives its signature. The phase return of Γ_0 admits a memory increment: states with identical phase may occupy different depths.

13.3 Quadratic algebras and activity of the parabolic threshold

In a fixed regime, conjugation of $aI + bJ_\tau$ changes the sign of b , and its algebraic norm is

$$(aI + bJ_\tau)(aI - bJ_\tau) = (a^2 + \tau b^2)I.$$

The elliptic, parabolic and hyperbolic types correspond to the three quadratic relations. Their local groups are represented by

$$e^{tJ_{+1}} = \cos t I + \sin t J_{+1}, \quad e^{tJ_0} = I + tJ_0, \quad e^{tJ_{-1}} = \cosh t I + \sinh t J_{-1}.$$

The proof separates even and odd terms in the exponential series; in the nilpotent type, only degrees zero and one remain. Each collection satisfies the composition law in its parameter and the inversion $e^{-tJ_\tau} = (e^{tJ_\tau})^{-1}$.

In particular, the zero tritic value permits a nonzero generator with square zero. For $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$,

$$e^{tN}(q, p) = (q + tp, p).$$

The threshold has an effective linear action on its fibre. Its parameter, its displacement and the memory of the transition remain determined even though the regime signature is zero. This is the precise content allowing the present to be developed narratively as an active update.

The positive elliptic form has circular level sets in orthonormal coordinates and elliptical ones after a linear transformation; the hyperbolic form has two isotropic directions and hyperbolic level sets; the parabolic type has a degenerate form and unipotent transport. Spherical geometry additionally requires the homogeneous space or metric realizing it. The preceding classification fixes the generators that may participate in that realization and distinguishes algebraic type from the curvature of a subsequent space.

The mode calendar makes the connection with self-scaling concrete. The rule $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$, $0 : m \mapsto m$ applied to the cyclic block $(+1, -1, 0)$ produces (Σ, Π, Π) . Counting its three transitions in the order (Σ, Π) determines

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (13.1)$$

Its positive eigenray, normalized as $(1, x)^\top$, satisfies $x = \lambda$, $1 + x = \lambda x$, and hence $x^2 - x - 1 = 0$. The positive root is the self-scaling coordinate obtained as φ . The matrix arises from the calendar of operations; the radical expression and the following exponential identities are its subsequent realizations.

13.4 Exponential realizations of the generated coordinates

The already determined regional coordinates allow the concrete realizations of the quadratic TRIT collection to be recognized. The order is substantive: construction of the regions precedes evaluation of an exponential at the parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not participate in selecting the prefixes that produce them.

13.4.1 Ternary cycle, nilpotent transport and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, formed by integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. In residue evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization thus preserves a contragredient orientation between sum and product. The quotients of the integer lift are preserved in the corresponding arithmetic record.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (13.1) gives

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the plane of nilpotent transport and the modal incidence plane. Having the same matrix dimension does not identify these spaces or their coordinates.

Proposition 13.2 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 13.3 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{\text{av}}^2. \quad (13.2)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotence removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

The exponential therefore equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the full turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic step. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use different parameters of a common exponential collection.

In that collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI) \exp(tI) = \exp((s + t)I)$. Its role is not restricted to the parabolic type. The coordinate e , previously obtained from its region, is recognized here as the unit-parameter evaluation of the exponential; this identity does not replace its coinductive genealogy.

13.4.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

Using Q distinguishes this operator square from the dodecaphasic record K .

Proposition 13.4 (Polynomial regime projectors). *The maps*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I and recover the three summands of the representation.

Proof. The three polynomials take the values $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross-products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

It follows that

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos t I + \sin t \mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh t I + \sinh t \mathbb{J}). \end{aligned} \quad (13.3)$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The full representation brings together the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation and memory. The exponential articulation allows their types to be recognized without replacing that dynamics by an identification of the fibres.

14 Transport from the modular block to the incidence quotient

The central APP block and face–vertex incidence provide two realizations of the same involution. Their relation is expressed through a rectangular intertwiner, before any metric interpretation is made. The adjacent diagonal congruence

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

selects $k = 4$ in the declared window. The corresponding block is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I_2 + 2R_c = 9P_+ + 5P_-, \quad R_c = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad P_{\pm} = \frac{I_2 \pm R_c}{2}.$$

The difference between the two eigenvalues is four. Incidence transport then determines how this difference is preserved when sector multiplicities change.

The cube faces are indexed by (i, ϵ) , with $i \in \{1, 2, 3\}$ and $\epsilon \in \{-1, 1\}$; its vertices are indexed by $v \in \{-1, 1\}^3$. Define

$$M_{(i,\epsilon),v} = \mathbf{1}_{\{v_i=\epsilon\}}.$$

Let R_F be face opposition and R_V vertex inversion $v \mapsto -v$. Entrywise, one verifies

$$\mathbf{1}_{\{v_i=-\epsilon\}} = \mathbf{1}_{\{(-v)_i=\epsilon\}}, \quad R_F M = M R_V.$$

The matrix M therefore transports the full involution.

Theorem 14.1 (Incidence descent of the central block). *Let $B_F = 7I_6 + 2R_F$ and $B_V = 7I_8 + 2R_V$. Then*

$$B_F M = M B_V.$$

If $u = (1, 1, 1, 1, 1, 1)^T$, $P_u = uu^T/6$ and $P_- = (I_6 - R_F)/2$, the quotient

$$Q_{\square} = \mathbb{R}^6 / \ker(MM^T) \simeq \mathbb{R}u \oplus E_-$$

has dimension 1 + 3, and the induced operator is

$$\bar{B}_F = 9P_u + 5P_-.$$

Proof. The first identity follows by multiplying $R_F M = M R_V$ by two and adding $7M$. Each face contains four vertices; two adjacent faces share two, and opposite faces share none. This count gives, entrywise,

$$MM^T = 12P_u + 4P_-.$$

The first projector has rank one and the second rank three; their images are orthogonal. The eigenvalues are 12, 4, 0, with multiplicities 1, 3, 2. Opposition acts as +1 on $\mathbb{R}u$ and as −1 on E_- , and preserves the kernel of the Gram matrix. Thus, B_F descends to the quotient and acts by nine and five on those two summands. Its characteristic polynomial is $(\lambda - 9)(\lambda - 5)^3$. $\square$

The temporal assignment to the axis $\mathbb{R}u$ fixes the signed form $J_L = P_- - P_u$. On the quotient, $I_Q = P_u + P_-$, so

$$J_L = \frac{7I_Q - \bar{B}_F}{2}.$$

This equality relates the form of signature $(3, 1)$, with the stated temporal convention, to the transported modular operator. The initial Gram matrix retains its positivity; the signature is introduced in the subsequent realization through the sign assigned to each sector. The dilations nine and five remain those of the block: the affine identity determines the form and does not turn the block into an isometry.

14.1 Arithmetic aggregation, lattice norm and route functional

Preservation of the integer before reduction becomes visible by summing the two sheets. The modular aggregation matrices are

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

Their sums are fifty-one and forty-five. Upon reconstructing the nine elements of each block, the integer and its carry satisfy

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129.$$

The residue aggregate and the quotient constitute two coordinates of this equality. In the subsequent lattice construction, the value fifty-four appears as the norm of the marked chamber $v_{\alpha} = 4\rho_1 + \rho_2 + \dots + \rho_{12}$, for orthogonal vectors $\rho_j^2 = 2$. In the material realization it appears as the evaluation of the route functional D_1 of the profile $(80, 54, 6)$. Each occurrence retains its originating map: aggregation with carry, lattice form and oriented route evaluation. The equality of their values acquires structural content through these maps, whose domains and proofs are developed in their respective sections.

15 Dodecaphasic extraction and isotypic projection of the direction

Before evaluating numbers, the readers distinguish dynamical regions. The mirror symmetry of the corpus acts on the block $B = (u^{\pi}, u^e, u^{\varphi}) \in (\mathbb{F}_3^6)^3$ by

$$cB = (u^e, u^{\pi}, u^{\varphi}).$$

With $q = u^{\pi} + u^e + u^{\varphi}$, $a = u^{\pi} + u^e - u^{\varphi}$ and $d = u^{\pi} - u^e$, the formulas

$$u^{\varphi} = 2(q - a), \quad s = q - u^{\varphi}, \quad u^{\pi} = 2(s + d), \quad u^e = 2(s - d)$$

are componentwise identities in $\mathbb{F}_3$. The mirror preserves the axis and the aggregate in a fibre and reverses the antisymmetric distribution. Nonadic transport evolves the fibre itself. The fifth phase has boundary datum 111111, residual signature 000, one local solution and one extension: the marking of φ at five identifies the regional axis of that operation. In phases six, seven and eight, the regional aggregates are 012100, 101001 and 112000, respectively. These data belong to the ternary domain; readers of compatible histories subsequently produce their real evaluations.

The twelve-window record and the regional blocks thus preserve their domains and their connection within the same state. The positive realization below uses the resulting record, with its order and scale. Transporting the regional mirror to a geometric deformation requires first composing its action on that record; a shared genealogy alone says less than commutativity of that square. This distinction preserves the full regional dynamics when studying one of its finite realizations.

The calendar divides the one hundred and eight positions into twelve nonadic windows. Terminal extraction has the composition

$$x_{\text{term}} \mapsto \mathcal{R}_{12}(x_{\text{term}}) \mapsto (b^{90}, b^{120}, Q) \mapsto U_{\text{sgn}} \mapsto K.$$

The first two arrows preserve the oriented history in the readers of the corpus. To display in full the finite reconstruction used in this article, their integer outputs are

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \quad Q = 6263. \end{aligned}$$

Fix the cyclic order of twelve positions. Let $B_r = \sum_{j=0}^3 b_{r+3j}^{120}$ and $d_{r,j} = b_{r+3j}^{90}$, for $r = 1, 2, 3$. Then

$$s_1 = \frac{Q + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2, \quad U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}).$$

Evaluation gives

$$U^{(1)} = (2378, -452, -668, -322), \quad U^{(2)} = (1406, 998, -204, -28), \quad U^{(3)} = (2479, -551, -371, -997).$$

The matrix

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4,$$

reconstructs each orbit through $H_4 U^{(r)}/4$. Restoring the order of the positions yields

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \quad Q = \sum_{j=1}^{12} K_j = 6263. \quad (15.1)$$

Every component is positive. Division by three comes from the three orbits and division by four from the Hadamard inverse. The additional direct check is $(I - S_3)K = b^{90}$, $(I - S_4)K = b^{120}$, where $(S_a K)_j = K_{j+a}$ with cyclic indices. These identities make the reconstruction reproducible without selecting coefficients through a target geometry.

The incidence chart of the corpus realizes the twelve positions as cosets of A_5/C_5 . If ρ is that representation and χ_3 its real three-dimensional character, its projector is

$$P_3 = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \rho(a), \quad P_{11} = I - \frac{1}{12} \mathbf{1}\mathbf{1}^\top.$$

The enumeration and orientation of cosets are preserved in the source of this chart. To make the geometric extension directly evaluable, its exact output $u = P_3 P_{11} K$ is written with $r = \sqrt{5}$:

$$\begin{aligned} u = & \left(-495/4 - 233r/10, \quad 403/4 + 169r/10, \right. \\ & -403/4 - 169r/10, \quad 495/4 + 233r/10, \\ & 601/4 + 1031r/10, \quad 203/4 + 211r/5, \\ & -203/4 - 211r/5, \quad -601/4 - 1031r/10, \\ & 313/4 + 569r/10, \quad 162 + 219r/20, \\ & \left. -162 - 219r/20, \quad -313/4 - 569r/10 \right). \end{aligned} \quad (15.2)$$

Addition and multiplication in $\mathbb{Q}(\sqrt{5})$ verify that

$$\mathbf{1}^\top u = 0, \quad \|u\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

In particular, its coordinates are nonconstant. The direction u , the full record K and the projector $P_{10} = P_{11} - uu^\top/\|u\|^2$ preserve different amounts of information. This distinction permits using the direction for a flow without substituting it for the record used by the action readers.

16 Reconstruction of positive geometry from nonadic memory

Dodecaphasic extraction preserves more information than a deformation direction. The direction selects a motion; the full record also determines the partition on which that motion acts. Recovering the record from its memories allows this distinction to be formulated as a reconstruction theorem. The result follows from the composition of the nonadic compressions of the corpus [2]; its geometric application preserves the uniform charge and the complements of both stages.

In this section, coordinates are numbered from zero to eleven. Let $S : \mathbb{R}^{12} \rightarrow \mathbb{R}^{12}$ be the shift $(Sx)_i = x_{i+1}$, with indices modulo twelve. In the Euclidean chart subsequent to the enriched state, define

$$T_j = \frac{8I + S^j}{9}, \quad D_j = \frac{\sqrt{8}}{9}(S^j - I), \quad j \in \{3, 4\}.$$

The coefficient eight counts the ordinary positions of the nonadic chart; the ninth position contains transport. Orthogonality of the shift implies, by direct multiplication,

$$T_j^* T_j + D_j^* D_j = I.$$

The first memory is the complement of the first compression; the second is taken on the state that has already passed through that compression. Thus,

$$q = \mathbf{1}^* K, \quad m_0 = D_3 K, \quad m_1 = D_4 T_3 K, \quad MK = (m_0, m_1). \quad (16.1)$$

The order of composition is part of the object. The map $M : \mathbb{R}^{12} \rightarrow \mathbb{R}^{24}$ preserves two correlated observations, produced by the same linear trajectory.

Theorem 16.1 (Exact inversion of the memory readout). *The map $K \mapsto (q, MK)$ is a linear isomorphism between $\mathbb{R}^{12}$ and $\mathbb{R} \times \text{im } M$. Its inverse is obtained through*

$$T_3^{-1} = \frac{512I - 64S^3 + 8S^6 - S^9}{455}, \quad (16.2)$$

$$b = -\frac{9}{\sqrt{8}}m_0, \quad c = -\frac{9}{\sqrt{8}}T_3^{-1}m_1, \quad w_i = c_i - b_{i+1},$$

$$B_0 = 0, \quad B_i = \sum_{j=0}^{i-1} w_j, \quad K_i = \frac{q + \sum_{j=0}^{11} B_j}{12} - B_i. \quad (16.3)$$

Proof. Writing $X = S^3$, one has $X^4 = I$ and $(8I + X)(512I - 64X + 8X^2 - X^3) = 4095I$. Dividing by nine proves (16.2). All operators used are polynomials in the same shift and commute. Hence,

$$b = (I - S^3)K, \quad c = (I - S^4)K, \quad w_i = K_i - K_{i+1}.$$

Telescoping gives $B_i = K_0 - K_i$; summing this equality over the twelve positions yields (16.3). Furthermore, $MK = 0$ is equivalent to invariance under S^3 and S^4 . Since $S = S^4(S^3)^{-1}$, the kernel is exactly $\mathbb{R}\mathbf{1}$. The charge recovers this component and proves injectivity. The decomposition $K = (q/12)\mathbf{1} + K^\perp$ proves surjectivity onto the declared codomain. $\square$

The geometric form is determined by information that a compressed reading appeared to have displaced. The vector complement contains the differences between positions; the charge contains the common component. Their combination reconstructs the twelve coordinates before normalizing them as lengths. This assertion relates effective operations: compression, recording of complements and inversion. Conservation of a norm expresses a distinct property, which by itself is weaker than this recoverability.

16.1 Compatible memory cone and stability of minors

Let L be the pseudoinverse of M , with image $\mathbf{1}^\perp$. Equivalently,

$$L = (M^*M|_{\mathbf{1}^\perp})^{-1} M^*, \quad LM = P_{11}, \quad K = \frac{q}{12}\mathbf{1} + Lm.$$

The restricted inverse notation is interpreted as an operator taking values in $\mathbf{1}^\perp$. Strict positivity of the record is equivalent to membership in the open cone

$$\mathcal{M}_+ = \left\{ (q, m) \in \mathbb{R} \times \text{im } M : \frac{q}{12}\mathbf{1} + Lm > 0 \right\}. \quad (16.4)$$

The inequalities are coordinatewise. The cone has dimension twelve; its section at a fixed positive charge has dimension eleven. In particular, the twenty-four entries of the two memories satisfy compatibility relations.

Proposition 16.2 (Optimal reconstruction constant in Euclidean norm). *One has $\|L\| = 9/4$. For arbitrary perturbations of the observations,*

$$\|\delta K\|^2 \leq \frac{|\delta q|^2}{12} + \frac{81}{16}(\|\delta m_0\|^2 + \|\delta m_1\|^2). \quad (16.5)$$

Proof. The Fourier basis diagonalizes S . If $z^{12} = 1$, the eigenvalue of M^*M in the corresponding mode is

$$\lambda(z) = \frac{8}{81}|z^3 - 1|^2 + \frac{8}{81}|z^4 - 1|^2 \left| \frac{8 + z^3}{9} \right|^2.$$

Evaluating all twelve modes gives

$$\text{Spec}(M^*M) = \{0^{[1]}, (16/81)^{[2]}, (8/27)^{[2]}, (32/81)^{[1]}, (952/2187)^{[4]}, (1256/2187)^{[2]}\}.$$

Bracketed exponents indicate multiplicity. The smallest positive eigenvalue is $16/81$, giving $\|L\| = 9/4$. The uniform component $(\delta q/12)\mathbf{1}$ is orthogonal to $L\delta m$; the Pythagorean identity and the norm of L yield (16.5). $\square$

To pass to interval geometry, return to numbering from one to twelve and set $d_j = K_j/q$, $t_0 = 0$, $t_j = \sum_{r=1}^j d_r$. The cone (16.4) determines exactly $0 = t_0 < t_1 < \dots < t_{12} = 1$, and therefore

$$\Delta_{ij} = t_j - t_i = \sum_{i < r \leq j} d_r > 0 \quad (i < j).$$

These are the minors of the path matrix constructed in Section 18. Their dependence on memory is explicit. The points (t_j, t_j^2) are likewise determined and produce the rank-one amplituhedral realization developed below.

If the square root of the right-hand side of (16.5) is smaller than $\min_j K_j$, the perturbed reconstruction remains in the positive cone. For the resulting record, $\min_j K_j = 58$; at fixed charge it suffices that

$$\sqrt{\|\delta m_0\|^2 + \|\delta m_1\|^2} < \frac{232}{9}.$$

The bound preserves vertex order and the sign of every ordered minor. It describes stability of this mathematical chart, with the stated normalization. For observations outside $\text{im } M$, reconstruction uses the projection MLm ; the residue $(I - ML)m$ is retained separately when recovery of the full observation is required.

16.2 Nodal recovery and boundary reduction

The resulting partition also has a quadratic realization:

$$\mathcal{E}_d(f) = \sum_{j=1}^{12} \frac{|f_j - f_{j-1}|^2}{d_j} = f^* L_d f.$$

The nodal Laplacian recovers each length through $d_j = -1/(L_d)_{j-1,j}$. If only the global endpoints are retained, minimization eliminates the interior nodes and instead produces the matrix $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, since $\sum_j d_j = 1$. The nodal reader distinguishes the interval form; the two-endpoint reader observes only its total length.

The proof of this elimination is given in full in the energy section: the minimizing affine prolongation Hf and the detail z , which vanishes at the old nodes, satisfy

$$\mathcal{E}_{\tilde{d}}(Hf + z) = \mathcal{E}_d(f) + \mathcal{E}_{\tilde{d}}(z).$$

The second term is nonnegative and vanishes exactly for zero detail. The effective reading preserves the minimum energy; the pair (f, z) also preserves the refined configuration. Thus, positivity of minors, energy positivity and memory recoverability are related through maps defined on the same partition, retaining their respective target spaces.

17 Complete reconstruction of the register and incidence isometry

The reconstruction in Section 16 preserves positive geometry, exceptional incidence, and the deformation direction within a single composition. The initial information for this composition is the register produced by the enriched APP-TRIT-TPK state: the two arithmetic sheets preserve their operations; the tritic orientation determines the regime; transport, updating, and nonadic prolongation preserve the differences between positions and the uniform component. The linear realizations below act on this already determined register. Their function is to transport information and make its invariants explicit.

The fundamental distinction concerns the preserved domain. The register K contains twelve ordered coordinates; the tetrad B_K preserves four positions selected by incidence; a geometric direction preserves a line in a representation space. Each operation has its own image and fiber. The following construction first preserves the twelve coordinates and then derives the reductions involved in the geometry. An exact reconstruction is thus available before extracting a support or performing a dimensional reduction.

17.1 Gram form of the dodecaphasic design

We number the positions from one to twelve and denote their set by $\Omega = \{1, \dots, 12\}$. The Paley-Witt encoding of the enriched state determines the set $\mathcal{H}$ of its 132 hexads. A hexad is a six-position support of a weight-six vector in the extended ternary code. The explicit construction of the code and the proof that every quintuple belongs to a unique hexad are developed in Theorem 27.3 and Proposition 27.5. We use precisely this incidence here, with the positions transported in the same order as the coordinates of the register.

Let $V = \mathbb{R}^{12}$ and $W = \mathbb{R}^{\mathcal{H}}$ be equipped with their Euclidean inner products. The incidence matrix defines the map

$$\mathcal{J} : V \longrightarrow W, \quad (\mathcal{J}k)_H = \sum_{p \in H} k_p, \quad \mathcal{J}_{H,p} = \mathbf{1}_{\{p \in H\}}. \quad (17.1)$$

The symbol $\mathbf{1} \in V$ denotes the uniform vector with twelve components. We define

$$\mu(k) = \frac{1}{12} \mathbf{1}^\top k, \quad P_0 = I_{12} - \frac{1}{12} \mathbf{1} \mathbf{1}^\top, \quad V_0 = \mathbf{1}^\perp.$$

The projector P_0 coincides with P_{11} from Section 16; the zero subscript indicates the zero-mean condition in this presentation. The orthogonal decomposition is $k = \mu(k) \mathbf{1} + P_0 k$.

Lemma 17.1 (Gram identity and incidence normalization). *The incidence satisfies*

$$\mathcal{J}^\top \mathcal{J} = 36I_{12} + 30\mathbf{1}\mathbf{1}^\top. \quad (17.2)$$

Consequently,

$$U_W = \frac{1}{6}\mathcal{J}|_{V_0} : V_0 \longrightarrow E_{\text{inc}} := \mathcal{J}(V_0)$$

is an isometry onto an eleven-dimensional subspace.

Proof. For a fixed point, the number of quintuples containing it is $\binom{11}{4}$. Each hexad containing that point includes $\binom{5}{4}$ of them. The uniqueness of the extension of each quintuple gives 66 hexads per point. For a fixed pair of points, the same argument gives $\binom{10}{3}/\binom{4}{3} = 30$ hexads per pair. The (p, r) entry of $\mathcal{J}^\top \mathcal{J}$ counts exactly the hexads containing both p and r . Its diagonal entries are therefore 66, and all other entries are 30, proving (17.2).

For $z \in V_0$, the uniform term vanishes and

$$\left\| \frac{1}{6}\mathcal{J}z \right\|^2 = \frac{1}{36}z^\top \mathcal{J}^\top \mathcal{J}z = \|z\|^2.$$

Injectivity and the dimension of the image follow from this equality. The factor six is thus fixed by the Gram form of the design. $\square$

The incidence count determines the scale of the isometry. Sums over hexads are linear observables of the positions; their inner products contain the multiplicity with which each position and each pair occur in the design. Removing the uniform mode separates these two counting levels and leaves the quadratic factor 36. This operation provides an explicit link between the twelve-position representation and its representation on hexads.

17.2 Preservation of the mean and inversion of the twelve coordinates

The uniform component is preserved through an additional scalar coordinate. We equip $\mathbb{R} \oplus E_{\text{inc}}$ with the inner product

$$\langle (\mu, y), (\nu, z) \rangle_* = 12\mu\nu + \langle y, z \rangle.$$

The weight twelve is the squared norm of $\mathbf{1}$, fixed by the number of positions. We define

$$F : V \longrightarrow \mathbb{R} \oplus E_{\text{inc}}, \quad F(k) = \left(\mu(k), \frac{1}{6}\mathcal{J}P_0k \right). \quad (17.3)$$

Theorem 17.2 (Isometric isomorphism of the complete register). *The map F is a linear isometric isomorphism. Its inverse is*

$$F^{-1}(\mu, y) = \mu\mathbf{1} + \frac{1}{6}P_0\mathcal{J}^\top y, \quad y \in E_{\text{inc}}. \quad (17.4)$$

In particular,

$$12|\mu(k)|^2 + \left\| \frac{1}{6}\mathcal{J}P_0k \right\|^2 = \|k\|^2. \quad (17.5)$$

Proof. The uniform-centered decomposition is orthogonal, and Lemma 17.1 preserves the norm of its second summand. This proves (17.5) and injectivity. Moreover,

$$\frac{1}{36}P_0\mathcal{J}^\top \mathcal{J}P_0 = P_0.$$

Substituting $F(k)$ into (17.4) gives $\mu(k)\mathbf{1} + P_0k = k$. Conversely, if $y \in E_{\text{inc}}$, there is a unique $z \in V_0$ with $y = \mathcal{J}z/6$. The proposed formula returns $\mu\mathbf{1} + z$, whose image is (μ, y) . This proves surjectivity onto the stated codomain and both inversion identities. $\square$

The restriction $y \in E_{\text{inc}}$ expresses an effective linear compatibility. The operator

$$\Pi_{\text{inc}} = \frac{1}{36} \mathcal{J} P_0 \mathcal{J}^\top : W \longrightarrow W \quad (17.6)$$

is the orthogonal projector onto E_{inc} . Indeed, it is symmetric, and its square reduces to itself by (17.2); it acts as the identity on the image of $\mathcal{J} P_0$. Compatibility is therefore checked through $\Pi_{\text{inc}} y = y$. For an arbitrary datum in W , reconstruction followed by incidence returns $\Pi_{\text{inc}} y$; the orthogonal component $(I_W - \Pi_{\text{inc}})y$ quantifies the information external to this representation of the register.

Proposition 17.3 (Covariance of reconstruction). *Let g be an automorphism of the design, and let $\rho(g)$ and $\sigma(g)$ be its permutation actions on positions and hexads. Then*

$$F\rho(g) = (1 \oplus \sigma(g))F.$$

The mean is preserved, and reconstruction commutes with the simultaneous transport of the positions and their hexads.

Proof. The equivalence $p \in H \iff g(p) \in g(H)$ implies $\mathcal{J}\rho(g) = \sigma(g)\mathcal{J}$. Permutations fix $\mathbf{1}$, preserve the mean, and commute with P_0 . Substituting these identities into (17.3) proves the claim. Covariance of the inverse follows by composition with F^{-1} . $\square$

Isometric normalization and arithmetic inversion serve different functions. In the Hadamard coordinatization of the register, $U = \mathcal{H}_{12}K$ and $K = \mathcal{H}_{12}U/4$, with $\mathcal{H}_{12}^\top = \mathcal{H}_{12}$ and $\mathcal{H}_{12}^2 = 4I_{12}$. The operator $J_H = \mathcal{H}_{12}/2$ is orthogonal. Hence FJ_H preserves the norm of the normalized signed register $U/2$ and represents it in the enlarged incidence space. Integer inversion through $\mathcal{H}_{12}/4$ also preserves the corresponding congruence sublattice. Both cases transport the generated register, with the phase order fixed before the transformation [2].

17.3 Exact composition from the two nonadic memories

We return to M , q , m_0 , and m_1 from (16.1), together with the reconstruction operator $L = (M^*M|_{V_0})^{-1}M^*$ from Section 16. The compatible domain is $\mathcal{D} = \mathbb{R} \times \text{im } M$, with $m = (m_0, m_1)$. Theorem 16.1 establishes the reconstruction

$$\mathcal{R}(q, m) = \frac{q}{12} \mathbf{1} + Lm, \quad LM = P_0.$$

Since L takes values in V_0 , composition with incidence has a direct expression:

$$\mathcal{C} = F\mathcal{R}, \quad \mathcal{C}(q, m) = \left(\frac{q}{12}, \frac{1}{6} \mathcal{J} Lm \right). \quad (17.7)$$

Corollary 17.4 (Equivalence between compatible memory and complete incidence). *The map $\mathcal{C}$ is an isomorphism between $\mathcal{D}$ and $\mathbb{R} \oplus E_{\text{inc}}$, with inverse*

$$\mathcal{C}^{-1}(\mu, y) = \left(12\mu, \frac{1}{6} M P_0 \mathcal{J}^\top y \right). \quad (17.8)$$

Its preserved quadratic form is

$$\|\mathcal{C}(q, m)\|_*^2 = \frac{|q|^2}{12} + \|Lm\|^2 = \|\mathcal{R}(q, m)\|^2.$$

Proof. Formula (17.7) follows from $P_0 \mathbf{1} = 0$ and $P_0 L = L$. The inverse is obtained by applying $k \mapsto (\mathbf{1}^\top k, Mk)$ to (17.4), using $M \mathbf{1} = 0$. The composition identities follow from $LM = P_0$ and $MLm = m$ for $m \in \text{im } M$. The norm equality is (17.5) applied to $\mathcal{R}(q, m)$. $\square$

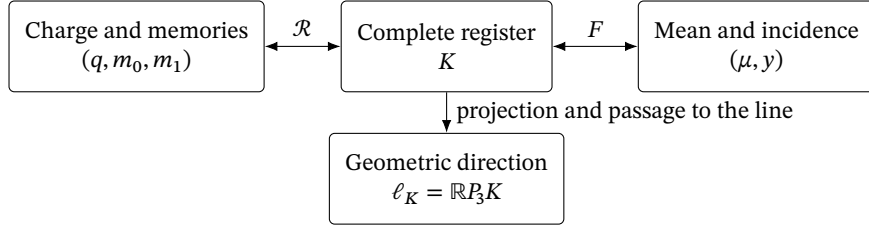


Figure 10: The charge and the two compatible memories reconstruct the twelve coordinates. The mean and the incidence component provide another reversible representation of the same register. The geometric direction is a subsequent reduction with positive-dimensional fibers.

This composition also determines the positive partition in 16. If $q > 0$, its lengths are recovered through

$$d_j = \frac{1}{12} + \frac{1}{6q}(P_0 \mathcal{J}^\top y)_j, \quad y = \frac{1}{6} \mathcal{J} P_0 K. \quad (17.9)$$

The twelve conditions $d_j > 0$ are the equations of the positive chart expressed in incidence coordinates. With $t_j = \sum_{r \leq j} d_r$, the ordered minors $t_j - t_i = \sum_{i < r \leq j} d_r$ are reconstructed from (μ, y) or from (q, m_0, m_1) . Thus the positivity of the minors and the isometric preservation of incidence are related by explicit maps. The former is an inequality in the ordered chart; the latter is an identity of inner products.

17.4 Recovery of polarization and scope of the reductions

The realization of the twelve positions on the cosets of A_5/C_5 determines the three-dimensional projector

$$P_3 = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \varrho(a).$$

Here ϱ is the permutation representation and χ_3 the three-dimensional character chosen in the marked chart. Its representatives, character values, and effective matrix are fixed in (28.2); Lemma 28.1 proves $P_3 = P_3^\top = P_3^2$, $\text{rank } P_3 = 3$ and $P_3 \mathbf{1} = 0$. In this chart, the register direction is obtained through

$$u_K = P_3 P_0 K = P_3 L m = \frac{1}{6} P_3 P_0 \mathcal{J}^\top y. \quad (17.10)$$

The equalities follow from the two inverses already proved. The same direction is therefore recovered from the memories or from the incidence component; the mean is preserved to reconstruct K , even though the three-dimensional projector annihilates it.

For the register in (28.1), the exact evaluation (28.3) gives

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

The line $\ell_K = \mathbb{R}u_K$ and the transverse complement $V_{10}(K) = V_0 \cap u_K^\perp$ are thus determined. Its projector is

$$P_{10}(K) = P_0 - \frac{u_K u_K^\top}{\|u_K\|^2}. \quad (17.11)$$

Indeed, the second summand is a rank-one orthogonal projector contained in V_0 ; its product with P_0 , in either order, equals itself. Squaring (17.11) recovers the same operator, and its trace is $11 - 1 = 10$. We obtain $V_0 = \ell_K \oplus^\perp V_{10}(K)$. This is the reduction that subsequently enters the polarization of the twelve lattice planes.

Incidence also transports these operators. On E_{inc} , the operator $\hat{P}_3 = U_W P_3 U_W^*$ satisfies $\hat{P}_3 U_W = U_W P_3$; the identity follows from $U_W^* U_W = I_{V_0}$. Transporting $P_{10}(K)$ in the same way preserves its rank and inner product. The correspondence between the two representations is therefore established on vectors and operators, as well as on their dimensions.

The need to preserve the different components admits precise checks. The centered observation of k coincides with that of $k + c\mathbf{1}$ for every c , whereas their charges differ by $12c$. In particular, $k = 0$ and $k = \mathbf{1}$ have the same centered incidence component and different means. The coordinate μ is exactly the information separating these two registers. Likewise, $u_{ak+b\mathbf{1}} = au_k$ for $a \neq 0$; the resulting line coincides with ℓ_k , even though the amplitude and mean change. These identities describe the fibers of the maps on the stated ambient space.

The extraction $k \mapsto \{j : k_j \geq 729\}$ has other fibers. For the specific register K , both K and $K + e_1$ have support $\{4, 6, 9, 10\}$, where e_1 is the first standard basis vector. These are two different ambient vectors with the same tetrad. This check characterizes the information loss of the support map; the terminal selection of the register remains fixed by its prior construction. The flag $B_K \subset H_{\alpha}^-$ also preserves the hexad determined by the signs before the carry, whose agreement with the incidence selection is proved in Proposition 27.6.

The common result of these compositions is an explicit mathematical continuity. Compatible memory reconstructs the register; its uniform-centered decomposition determines an isometric incidence representation; reconstruction of its lengths determines the positive chart; and three-dimensional projection determines the deformation direction. The tetrad and flag preserve the incidences used by the exceptional constructions, whereas F preserves the entire dodecaphasic register. The enriched history, in turn, retains the orientation, path, and prolongation that place this register within the joint discrete structure of the continuum.

18 Planar boundary measurement, Plücker coordinates, and positive mutations

The dodecaphasic representation of the enriched state admits a positive realization whose network, coordinates, and chart changes can be written out completely. The starting composition is

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U_{\text{sgn}} \longrightarrow K.$$

This reading acts on the enriched state preserving the two sheets, orientation, carry, memory, and boundary. The already generated register is

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \quad Q = \sum_{r=1}^{12} K_r = 6263.$$

Its extraction is recovered exactly from the two channels and total charge:

$$b^{90} = (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461),$$

$$b^{120} = (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128).$$

On the position orbits $(r, r+3, r+6, r+9)$, for $r = 1, 2, 3$, let $B_r = \sum_{j=0}^3 b_{r+3j}^{120}$ and $d_{r,j} = b_{r+3j}^{90}$. We obtain

$$s_1 = (Q + 2B_1 + B_2)/3, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2,$$

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}), \quad K|_r = \frac{1}{4}H_4 U^{(r)},$$

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4.$$

The network constructed below realizes the positive values produced by this extraction. Its combinatorial form is a subsequent formalization of the register, with stated origin and orientation.

18.1 Directed acyclic network and determinants of nonintersecting paths

For a positive register of length L , set

$$d_r = K_r/Q > 0, \quad t_0 = 0, \quad t_j = \sum_{r=1}^j d_r \quad (1 \leq j \leq L).$$

Thus $t_L = 1$. The L separations produce $L + 1$ marked points; in the dodecaphasic case there are thirteen points. The matrix realizing them is

$$C(d) = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ t_0 & t_1 & \cdots & t_L \end{pmatrix}. \quad (18.1)$$

The Grassmannian $\text{Gr}(2, L + 1)$ identifies rank-two matrices under invertible changes of the two rows. Its positive part consists of classes admitting all ordered minors of order two strictly positive.

Construct a directed network with vertices a_j, b_j , for $0 \leq j \leq L$, and sinks q_j . There are horizontal edges $a_{j-1} \rightarrow a_j$ and $b_{j-1} \rightarrow b_j$, of weight one; vertical edges $a_j \rightarrow b_j$, of weight d_j , for $j \geq 1$; and edges $b_j \rightarrow q_j$, of weight one. The ordered sources are b_0, a_0 . The two rows are drawn horizontally, the vertical edges descend, and the sinks exit below the lower row. This drawing is planar, and the orientation is acyclic. The boundary measurement M_{ij} is the sum of the products of weights of all paths from source i to sink q_j . The path network has two sources and $L + 1$ sinks, hence $L + 3$ external connections. The matrix it realizes has $L + 1$ columns. The plabic seed introduced below has precisely $L + 1$ boundary vertices and constitutes another presentation of its coordinates; these two boundary counts are kept explicit.

Theorem 18.1 (Boundary measurement and register minors). *The boundary measurement of this network is $M = C(d)$. For $0 \leq i < j \leq L$,*

$$\Delta_{ij}(C(d)) = t_j - t_i = \sum_{r=i+1}^j d_r > 0. \quad (18.2)$$

The same number is the sum of weights of the vertex-disjoint path pairs $b_0 \rightarrow q_i, a_0 \rightarrow q_j$.

Proof. The lower source has a unique path to each sink, of weight one. A path from a_0 to q_j descends exactly once, along an edge of index $1 \leq r \leq j$; its weight is d_r . This proves the matrix formula. The lower path to q_i occupies all vertices $b_0, \dots, b_i$. A path from the other source to q_j is therefore disjoint from it precisely when it descends at $i < r \leq j$. Its sum of weights is the right-hand side of (18.2). The pair with exchanged destinations always shares a vertex. Expanding the determinant cancels all intersecting pairs and retains exactly the disjoint ones. Here the determinantal mechanism of nonintersecting paths has been verified directly for the specific network. $\square$

Identification of planar-network measurements with cells of the nonnegative Grassmannian provides their subsequent geometric recognition; see A. Postnikov, *Total positivity, Grassmannians, and networks*. The preceding proof also gives a finite formula for every minor without using a metrological value.

The collection $d_r > 0, \sum_r d_r = 1$, has dimension $L - 1$. Its map to the Grassmannian is injective: if the rows of $C(d)$ and $C(d')$ generate the same plane, the second rows differ by an affine transformation; the conditions $t_0 = t'_0 = 0$ and $t_L = t'_L = 1$ fix that transformation. In particular, for $L = 12$ one obtains an eleven-dimensional collection inside the twenty-two-dimensional positive cell. The point lies in the top cell; the collection retains its own dimension. The thirteen points and twelve separations thus have different mathematical roles.

Allowing $d_r \geq 0$ preserves nonnegativity. When $d_r = 0$, consecutive columns $r - 1$ and r coincide and form parallel elements of the matroid. The rank remains two because $t_0 = 0$ and $t_L = 1$. The bases are exactly the pairs $i < j$ crossing some positive separation. This degeneration produces strata of consecutive incidence. A zero column, which would give a loop, corresponds to a different operation from the coincidence of two nonzero columns.

18.2 Triangulation seed and exchange relation

Let $N = L + 1$, and consider the vertices $0, \dots, L$ of a convex polygon in cyclic order. To each side or diagonal (i, j) , assign the coordinate $A_{ij} = \Delta_{ij}(C(d))$, with ordered indices. The fan triangulation from vertex zero has diagonals $(0, j)$, $2 \leq j \leq L - 1$. Its sides are frozen variables, and its diagonals are mutable variables. The quiver is obtained by cyclically orienting the three sides of each triangle and canceling opposite arrow pairs. This is a seed of type A_{N-3} ; for the dodecaphasic register, its type is A_{10} .

The seed has an explicit plabic realization. Place a black vertex inside each triangle; at the polygon vertices, place a white vertex if it is incident to a diagonal, and a black one otherwise. Join the centers to the three corners of their triangles, remove the original sides, and contract edges between same-colored vertices. Add a boundary leaf at each marked vertex. This construction follows the triangulation algorithm of Kodama and Williams, *KP solitons and total positivity for the Grassmannian*, Algorithm 12.1. The coordinate labels of this seed are evaluated at the minors already produced by the two-row network. For a mutable face, the $\mathcal{X}$ -type coordinate is the ratio of products of incident $\mathcal{A}$ coordinates prescribed by its quiver column; in a quadrilateral it will be written explicitly as a cross-ratio. Changing the graph changes the chart of these minors while preserving their provenance.

Theorem 18.2 (Positive exchange in the register realization). *For four vertices $a < b < c < d$,*

$$A_{ac}A_{bd} = A_{ab}A_{cd} + A_{ad}A_{bc}. \quad (18.3)$$

Consequently, the diagonal change $(a, c) \leftrightarrow (b, d)$ is evaluated through

$$A_{bd} = \frac{A_{ab}A_{cd} + A_{ad}A_{bc}}{A_{ac}} > 0.$$

All triangulations provide compatible positive charts on the same realization $C(d)$.

Proof. Substituting $A_{ij} = t_j - t_i$ reduces the equality to

$$(t_c - t_a)(t_d - t_b) = (t_b - t_a)(t_d - t_c) + (t_d - t_a)(t_c - t_b).$$

Expanding both sides proves the identity. The differences are positive; therefore each flip preserves a positive evaluation, and the quotient is defined. Connectivity of triangulations by flips allows passage between them through these exchanges. The cluster interpretation of the Plücker relation is the subsequent recognition developed by J. Scott in *Grassmannians and Cluster Algebras*. The algebraic proof of the specific evaluation is already contained in the displayed identity. $\square$

18.3 Refinement of separations and naturality of exchanges

Let $B \geq 2$. At depth n , each separation d_r is divided into B^n equal consecutive separations:

$$d_{(r-1)B^n+c}^{(n)} = d_r/B^n, \quad 1 \leq c \leq B^n.$$

This defines $LB^n + 1$ points and a matrix C_n . If $t_n(j) = Bj$ identifies a point with its same-position descendant, then

$$C_{n+1}|_{t_n(\{0, \dots, LB^n\})} = C_n, \quad \Delta_{t_n(i), t_n(j)}(C_{n+1}) = \Delta_{ij}(C_n). \quad (18.4)$$

The refined network replaces each vertical edge of weight d by a sequence of B descent positions, with weights d/B . The path sum between inherited endpoints remains d . Composition of refinements preserves this equality because finite sums are grouped associatively.

Proposition 18.3 (Naturality on inherited endpoints). *Every exchange (18.3) among four inherited endpoints commutes exactly with refinement. Every old triangulation extends to one of the refined polygon while preserving its interior triangles: the polygons added between consecutive endpoints are triangulated outside those triangles. Flips between old diagonals then preserve their exchange relations.*

Proof. The first conclusion follows by substituting the equality in (18.4) for each minor. For the second, preserve the chords joining consecutive old endpoints and triangulate each outer pocket formed by the new marks. These regions have disjoint interiors. The two triangles incident to an old diagonal remain unchanged, and so do their quadrilateral and Ptolemy identity. $\square$

This naturality specifies the endpoints on which it acts. An old side becomes a diagonal of the larger polygon when new vertices are added along its boundary arc. A formerly frozen variable can therefore be mutable in the enlarged problem. The preceding compatibility is exact compatibility of the evaluation system and inherited flips. An inclusion of seed patterns with the same coefficient regime is obtained by keeping these chords and the new auxiliary diagonals frozen. This states the operation that preserves the seed; increasing depth and performing a mutation remain different operations.

18.4 Projective diagonal action and invariance of cross-ratios

If $\lambda_j > 0$, the column action $C \mapsto C \operatorname{diag}(\lambda_0, \dots, \lambda_L)$ gives

$$A_{ij} \mapsto \lambda_i \lambda_j A_{ij}.$$

It preserves positivity and all strata defined by vanishing minors. The cross-ratio associated with a quadrilateral is

$$X_{abcd} = \frac{A_{ab}A_{cd}}{A_{ad}A_{bc}} > 0. \quad (18.5)$$

The four factors $\lambda_a \lambda_b \lambda_c \lambda_d$ cancel exactly. Consequently, every positive diagonal column action fixes X_{abcd} , even if it depends on a parameter and has determinant one. Changing column weights preserves the projective point of each column; their cross-ratios express this preservation. For the register to change the projective shape, it must act on the ordered separations.

18.5 Register-directed positive deformation of separations

The corpus fixes the representation of the twelve positions and its orthogonal projector P_3 onto the three-dimensional sector. With $P_{11} = I_{12} - J_{12}/12$, the recovered direction is

$$u = P_3 P_{11} K = P_3 K \in \mathbf{1}^\perp.$$

In the stated chart, setting $r = \sqrt{5}$, its exact evaluation is

$$\begin{aligned} u = (&-495/4 - 233r/10, 403/4 + 169r/10, -403/4 - 169r/10, \\ &495/4 + 233r/10, 601/4 + 1031r/10, 203/4 + 211r/5, \\ &-203/4 - 211r/5, -601/4 - 1031r/10, 313/4 + 569r/10, \\ &162 + 219r/20, -162 - 219r/20, -313/4 - 569r/10). \end{aligned}$$

We retain $\sum_r u_r = 0$ and $\|u\|^2 = (6638585 + 2275584\sqrt{5})/20 > 0$.

For $s \in \mathbb{R}$, define the deformation of the weights by

$$d_r(s) = \frac{K_r e^{su_r}}{\sum_{\ell=1}^{12} K_\ell e^{su_\ell}}, \quad t_j(s) = \sum_{r=1}^j d_r(s), \quad C(s) = C(d(s)). \quad (18.6)$$

The formula uses the already generated register and its direction. It is a new analytic realization of the reader; the parameter s measures its deformation. When the internal output $\rho_A = \pi_{\text{HMT}}/\varphi_{\text{HMT}}^2$ is

used, writing $s = \tau \log \rho_A$ expresses exactly the same collection. This reparametrization acts after the generation of the constants. Identifying s or τ with a physical time requires the temporal law of the corresponding reader.

Theorem 18.4 (Shape change, positivity, and compatibility with depth). *The collection (18.6) belongs to $\text{Gr}_{>0}(2, 13)$ for every real s and preserves $t_0 = 0, t_{12} = 1$. For $i < j$, let*

$$\mu_{ij}(s) = \frac{\sum_{r=i+1}^j K_r u_r e^{s u_r}}{\sum_{r=i+1}^j K_r e^{s u_r}}, \quad \mu(s) = \frac{\sum_r K_r u_r e^{s u_r}}{\sum_r K_r e^{s u_r}}.$$

Then

$$\frac{d}{ds} \log A_{ij}(s) = \mu_{ij}(s) - \mu(s), \quad \frac{d}{ds} \log X_{abcd}(s) = \mu_{ab}(s) + \mu_{cd}(s) - \mu_{ad}(s) - \mu_{bc}(s). \quad (18.7)$$

The deformation commutes with every refinement assigning the successors of separation r the weight d_r/B and the same velocity u_r .

Proof. All exponential factors and all K_r are positive. Normalization makes the separations sum to one; the minors are sums of positive separations by (18.2). Differentiating the logarithm of

$$A_{ij}(s) = \frac{\sum_{r=i+1}^j K_r e^{s u_r}}{\sum_r K_r e^{s u_r}}$$

gives the first identity. The combination of four logarithms in (18.5) cancels the four global means with their signs and gives the second. Finally, replacing K_r by B copies of K_r/B with identical u_r preserves numerator and denominator when the successors are grouped. The equality holds for all parameters, at any depth, and for every composition of refinements. $\square$

There is an exact witness that this collection changes shape. For the first four endpoints,

$$X_{0123}(0) = \frac{1560}{23711}, \quad \left. \frac{d}{ds} \log X_{0123}(s) \right|_{s=0} = -\frac{309899}{917} - \frac{268596}{4585} \sqrt{5} < 0. \quad (18.8)$$

The calculation uses only the first three separations (234, 543, 140) and their already recovered velocities. The strict inequality proves an effective projective motion. The Ptolemy identities remain exact during this motion, since they apply to the columns $(1, t_j(s))$ at every parameter value. The register thus controls a positive shape deformation and, simultaneously, a system of charts that remains compatible with it.

The proved scope brings together a rank-two planar network, its explicit measurement, a triangulation plabic seed, all its mutations by flips, naturality of inherited endpoints, and a deformation of cross-ratios directed by K . Each operation has a verifiable domain and result. Comparison of this realization with enriched temporal dynamics and the material branch uses their respective reading maps; no coordinate identity replaces those maps.

19 Rank-one amplituhedral geometries and residue compatibility

The dodecaphasic register allows positive data, Grassmannian domain, kinematic image, covering, degree on cells, and canonical form to be assembled explicitly. The first result concerns the two-dimensional amplituhedron with parameters $(n, k, m) = (13, 1, 2)$; the construction then extends to rank-one cyclic polytopes of higher dimension. This is a subsequent geometric realization of the register produced by APP-TRIT-TPK. The stated additional operation is lifting an ordered rational partition by powers. This choice fixes a particular realization; subsequent physical correspondences retain their own readers and conditions.

The provenance of the register preserves the composition

$$x_{\text{term}} \mapsto \mathcal{R}_{12}(x_{\text{term}}) \mapsto (b^{90}, b^{120}, Q_{\text{TPK}}) \mapsto U_{\text{sgn}} \mapsto K.$$

APP preserves both sheets and their remainders and quotients; TRIT orients the regime of each transition; TPK transports and updates the state with carry, boundary, route, and memory. The representation K is read in the same system of enriched states that jointly generates the continuum. Its signed extraction and Hadamard inversion produce

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \quad S_K = \sum_{j=1}^{12} K_j = 6263. \quad (19.1)$$

The geometric proof uses the positivity of these twelve coordinates, their order, and their total. Its coefficients come from that extraction, not from retrospective selection of a physical value.

19.1 Rational partition and positive lift

Define thirteen rational partition points:

$$t_0 = 0, \quad t_i = \frac{K_1 + \cdots + K_i}{S_K} \quad (1 \leq i \leq 12), \quad Z_i = (1, t_i, t_i^2)^\top, \quad Z = (Z_0 \cdots Z_{12}). \quad (19.2)$$

Their numerators, with common denominator 6263, are

$$(0, 234, 777, 917, 1646, 2305, 3129, 3750, 3808, 4722, 5516, 5662, 6263).$$

Thus $0 = t_0 < t_1 < \cdots < t_{12} = 1$. The labeled partition and the total preserve the register through the exact inverse

$$K_i = S_K(t_i - t_{i-1}). \quad (19.3)$$

The partition preserves the proportions; retaining S_K also preserves the integer scale.

Proposition 19.1 (Strictly positive kinematic data). *The matrix Z has rank three, and all its ordered minors of size three are positive. For $0 \leq a < b < c \leq 12$,*

$$\langle abc \rangle := \det(Z_a, Z_b, Z_c) = (t_b - t_a)(t_c - t_a)(t_c - t_b) > 0. \quad (19.4)$$

Proof. Subtract the first column from the last two and expand along the first row. Each difference $t_j^2 - t_a^2$ factors as $(t_j - t_a)(t_j + t_a)$, giving the stated product. The strictly increasing partial sums prove positivity. Any of these minors establishes rank three. $\square$

19.2 Image, fibers, and degree-one cells

The Grassmannian $\text{Gr}_{\geq 0}(1, 13)$ is the projective space of nonnegative nonzero vectors $c = (c_0, \dots, c_{12})$, modulo positive scaling. The chart $\sum c_j = 1$ identifies it with the closed twelve-dimensional simplex. Define

$$\Phi_Z : \text{Gr}_{\geq 0}(1, 13) \longrightarrow \mathbb{P}^2, \quad [c] \longmapsto \left[\sum_{j=0}^{12} c_j Z_j \right]. \quad (19.5)$$

In the chart $Y_0 = 1$, let $v_j = (t_j, t_j^2)$ and $P_K = \text{conv}(v_0, \dots, v_{12})$.

Theorem 19.2 (Polygonal amplituhedron of the register). *The closed image of Φ_Z is exactly the convex polygon $P_K = \mathcal{A}_{13,1,2}(Z)$, of dimension two with thirteen vertices. The image of the strictly positive Grassmannian is the interior of P_K . The eleven cells*

$$\mathcal{S}_i = \{[c] : c_0, c_i, c_{i+1} > 0, c_j = 0 \text{ if } j \notin \{0, i, i+1\}\}, \quad 1 \leq i \leq 11, \quad (19.6)$$

map bijectively, with degree one and positive orientation, onto the interiors of $T_i = [v_0, v_i, v_{i+1}]$. Their closures cover P_K , and their interiors are pairwise disjoint.

Proof. Under the normalization $\sum c_j = 1$, the map is $c \mapsto \sum c_j v_j$, whose image is the convex hull. The minors in (19.4) show that every ordered triple has positive orientation. Each consecutive side leaves all other vertices strictly to its left; so does the final side from v_{12} to v_0 . All points are therefore extreme vertices of a convex polygon.

Every combination with $c_j > 0$ is interior, since for any supporting line some vertex lies strictly in the interior half-plane. Conversely, let y be interior and $b = 13^{-1} \sum v_j$. For sufficiently small $\varepsilon > 0$, $y' = (y - \varepsilon b)/(1 - \varepsilon)$ remains in the polygon. Writing $y' = \sum a_j v_j$, $a_j \geq 0$, $\sum a_j = 1$, gives $y = \sum((1 - \varepsilon)a_j + \varepsilon/13)v_j$, with all coefficients positive.

The diagonals from v_0 divide the polygon into the eleven triangles. On each triple support, the barycentric coordinates are unique because $\langle 0, i, i+1 \rangle > 0$; their explicit inverse appears in (19.8). Each support defines a rank-one positroid cell with coordinates $[1 : a : b]$, $a, b > 0$, at those positions. A directed planar network with one source and three paths of weights 1, a , b to the three destinations realizes these coordinates by boundary measurement. The positive determinant proves orientation and degree one. $\square$

The generic fiber of the full map has dimension ten: $\sum c_j = 1$, $\sum c_j t_j = x$, and $\sum c_j t_j^2 = y$ are three independent equations in thirteen coordinates. For interior y , the fiber contains a strictly positive point, so its intersection with the simplex has dimension $13 - 3 = 10$. Degree one concerns the triangular cells; the two-dimensional form is obtained through these cells.

19.3 Canonical form and oriented residues

For $Y = (1, x, y)^T$, define

$$\ell_{ab}(Y) = \det(Z_a, Z_b, Y) = (t_b - t_a)\{y - (t_a + t_b)x + t_a t_b\}. \quad (19.7)$$

The expression also holds when $b < a$. For $a < b < c$, let $D_{abc} = \langle abc \rangle$ and

$$\lambda_a = \frac{\ell_{bc}}{D_{abc}}, \quad \lambda_b = \frac{\ell_{ca}}{D_{abc}}, \quad \lambda_c = \frac{\ell_{ab}}{D_{abc}}, \quad \lambda_a + \lambda_b + \lambda_c = 1. \quad (19.8)$$

The affine orientation is $dx \wedge dy$. Fix the boundary convention $\text{Res}^\partial(\eta \wedge du/u) = \eta|_{u=0}$: the normal differential occupies the last position. In dimension two, the Poincaré residue placing du/u first has the opposite sign. The following cancellations uniformly use the stated convention.

Theorem 19.3 (Rational form and cancellation of diagonals). *The canonical form of the oriented triangle $[v_a, v_b, v_c]$ is*

$$\Omega_{abc} = \frac{D_{abc}^2 dx \wedge dy}{\ell_{ab}(Y)\ell_{bc}(Y)\ell_{ca}(Y)}. \quad (19.9)$$

The canonical form of the polygon is

$$\Omega_{P_K}(x, y) = \sum_{i=1}^{11} \frac{t_i t_{i+1} (t_{i+1} - t_i) dx \wedge dy}{(y - t_i x) \{y - (t_i + t_{i+1})x + t_i t_{i+1}\} (t_{i+1} x - y)}. \quad (19.10)$$

The poles of interior diagonals cancel. The only polar divisors are the thirteen lines of the exterior sides. Parametrizing each side from its initial to its final vertex by $s \in [0, 1]$, its boundary residue is $ds/[s(1-s)]$.

Proof. The form of the barycentric simplex is $d\lambda_b \wedge d\lambda_c / (\lambda_a \lambda_b \lambda_c)$. The Jacobian satisfies $dx \wedge dy = D_{abc} d\lambda_b \wedge d\lambda_c$. Substituting (19.8) gives (19.9). For $a = 0$, canceling the three scalar factors of the lines gives each summand of (19.10).

On side ab , set $\lambda_c = u$, $\lambda_b = (1-u)s$, $\lambda_a = (1-u)(1-s)$. Then

$$\Omega_{abc} = \frac{ds}{s(1-s)} \wedge \frac{du}{u(1-u)}.$$

The residue is the form of the oriented interval. Two adjacent triangles induce opposite orientations on their diagonal and produce opposite residues. Since all poles are simple, a vanishing residue removes the diagonal pole as a divisor. Each exterior side appears once. The homogeneous formula has degree zero as a projective form and has no other polar divisors. If two rational projective forms had the same residues and no other poles, their difference would be a holomorphic two-form on $\mathbb{P}^2$, necessarily zero. This proves characterization and uniqueness. $\square$

We use the conventional notion of positive geometry through logarithmic poles and canonical boundary residues, developed by N. Arkani-Hamed, Y. Bai, and T. Lam, *Positive Geometries and Canonical Forms*, JHEP 11 (2017), 039, <https://arxiv.org/abs/1703.04541>, Sections 2, 3, 5.2, 6.1, and 7.2. The selection of the coefficients of the present polygon precedes this recognition and comes from K .

19.4 Carry refinement on the fixed geometry

In a positively oriented triangle $T = [A, B, C]$, write

$$Y(u, r) = (1 - u)A + u\{(1 - r)B + rC\}, \quad 0 < u, r < 1. \quad (19.11)$$

Vertex A corresponds to $u = 0$, and the opposite side to $u = 1$. The form becomes

$$\Omega_T = \frac{du}{u(1 - u)} \wedge \frac{dr}{r(1 - r)}. \quad (19.12)$$

Let $B_0 \geq 2$ be a stated integer base, distinct from vertex B . The coordinate r admits the remainder-and-carry update

$$c = \lfloor B_0 r \rfloor, \quad r' = B_0 r - c, \quad r = \frac{c + r'}{B_0}. \quad (19.13)$$

The endpoints retain their boundary marks. The digit c remains as memory and allows the update to be inverted. The charts of the two affine transports should be specified. The subcell zoom acts on $r \in [c/B_0, (c+1)/B_0]$ through $r' = B_0 r - c$. By contrast, transport of the lifted pair $L_c(x, y) = (B_0 x - c, B_0 y + c)$, with $x + y = M$, changes the total mass to $B_0 M$ and expresses the coordinate

$$\frac{B_0 x - c}{B_0 M} = \frac{x}{M} - \frac{c}{B_0 M}.$$

These maps have different domains and normalizations. Both preserve the carry in the enriched state and admit recovery with their scale data. The following proof uses the subcell zoom and the geometric partition it identifies.

Theorem 19.4 (Exact invariance under subdivision with memory). *Let $s_j = j/B_0$ and $W_j = (1 - s_j)B + s_j C$. The triangles $T_j = [A, W_j, W_{j+1}]$, $0 \leq j < B_0$, subdivide T and satisfy*

$$\sum_{j=0}^{B_0-1} \Omega_{T_j} = \Omega_T. \quad (19.14)$$

The equality remains valid at every finite depth and for any ordered rational partition of the side. Subdividing the eleven triangles of P_K leaves Ω_{P_K} exactly invariant.

Proof. The regions $r \in [s_j, s_{j+1}]$ cover T with disjoint interiors. In the common coordinates,

$$\Omega_{T_j} = \frac{du}{u(1 - u)} \wedge \frac{(s_{j+1} - s_j) dr}{(r - s_j)(s_{j+1} - r)}.$$

The rational identity

$$\frac{s_{j+1} - s_j}{(r - s_j)(s_{j+1} - r)} = \frac{1}{r - s_j} + \frac{1}{s_{j+1} - r}$$

telescopes: each interior term cancels the one from the adjacent subinterval. This leaves $1/r + 1/(1-r) = 1/[r(1-r)]$. This proves equality of rational forms; iteration gives every finite depth. The proof uses only $s_0 = 0 < s_1 < \dots < s_N = 1$, so it covers nonuniform partitions. The change $r' = (r - s_j)/(s_{j+1} - s_j)$ locally returns the unit form. The carry word identifies the subcell, and its inverse recovers the previous coordinate. The new interior poles cancel through the same telescoping identity. $\square$

The preceding sum assembles the complete geometric partition. When a survival reader selects only some histories, its image is the union of the selected subcells and its form is the corresponding sum; it agrees with the form of the full polygon precisely when that union covers it with oriented multiplicity one. The covering criterion for a specific selection of histories is therefore expressed by a checkable condition on its cell chain.

This refinement preserves the fixed polygon. Inserting a new point (t, t^2) on the parabola between consecutive parameters $a < t < b$ produces another geometry: the chord height minus t^2 is $(t-a)(b-t) > 0$, and the new point lies outside the previous polygon. The proved invariance concerns rectilinear subdivision of cells, whose exterior boundary remains fixed.

19.5 Triangulation changes and positive reparametrization

For four ordered vertices A, B, C, D , we have

$$\Omega_{ABC} + \Omega_{ACD} = \Omega_{ABD} + \Omega_{BCD}. \quad (19.15)$$

Both sides have the same exterior residues and cancel their interior diagonal; their equality follows from the proved uniqueness. It is also checked by multiplying through by their linear denominators. Every change between triangulations of a convex polygon decomposes into these diagonal changes, so the form is independent of triangulation. Identification with cluster coordinates must additionally preserve the variety and its change relations.

If $D = \text{diag}(d_0, \dots, d_{12})$, $d_j > 0$, then

$$\mathcal{A}_{13,1,2}(ZD) = \mathcal{A}_{13,1,2}(Z). \quad (19.16)$$

The automorphism $[c] \mapsto [Dc]$ reparametrizes the kinematic map. In the affine chart, c_j changes to $c_j d_j / \sum_e c_e d_e$, which again ranges over the simplex. The image and its canonical form remain unchanged. Displacement of a parametrized history and change of the entire geometric set are thus distinguishable operations.

19.6 Rank-one extension to higher dimensions

For $2 \leq m \leq 12$, define

$$Z_i^{(m)} = (1, t_i, t_i^2, \dots, t_i^m)^T, \quad P_K^{(m)} = \text{conv}\{(t_i, t_i^2, \dots, t_i^m) : 0 \leq i \leq 12\}. \quad (19.17)$$

The ordered maximal minors are the Vandermonde products $\prod_{a < b} (t_{i_b} - t_{i_a}) > 0$. The matrix has rank $m + 1$. The image of $\text{Gr}_{\geq 0}(1, 13)$ is the cyclic polytope $P_K^{(m)} = \mathcal{A}_{13,1,m}(Z^{(m)})$, of dimension m ; its generic fiber has dimension $12 - m$. Here we use the mathematical extension of the amplituhedral definition to any m ; customary applications to amplitudes impose their own values of m .

The triangulation is determined by the fixed label order. Choose the least vertex of a polytope, recursively triangulate all facets not containing it, and take cones from that vertex. Induction begins with points. A line from the initial vertex to an interior point meets the opposite boundary in a facet; this proves covering. Uniqueness of the intersection point and compatibility of the restrictions induced by the same order prove that interiors are disjoint and intersections are common faces. Dimension decreases at each step, so the procedure terminates. This is a pulling triangulation determined by the labels.

For an oriented simplex $T = [p_0, \dots, p_m]$ of this triangulation, let

$$A_T = (p_1 - p_0 \ \cdots \ p_m - p_0), \quad (\lambda_1, \dots, \lambda_m)^\top = A_T^{-1}(x - p_0), \quad \lambda_0 = 1 - \sum_{j=1}^m \lambda_j.$$

Orient its vertices so that $\det A_T > 0$. Its canonical form is

$$\Omega_T(x) = \frac{dx_1 \wedge \cdots \wedge dx_m}{\det(A_T) \prod_{j=0}^m \lambda_j(x)}, \quad \Omega_{P_K^{(m)}} = \sum_T \Omega_T. \quad (19.18)$$

The barycentric change proves the formula and that the kinematic restriction to the support of T has degree one. The residues are the face forms with signs induced by orientation. On an interior face, the two orientations are opposite and the total residue vanishes. On an exterior face, the sum of the forms in that face's triangulation remains. Induction on dimension proves that this sum is its canonical form. Repeating the residue along a flag of faces reaches the vertices, where the normalization is ± 1 . This gives residue correspondence in every codimension for this rank-one collection.

The same proof covers any compatible simplicial subdivision of the fixed polytope: interior faces cancel and exterior faces preserve the residues. Nonexistence of top-degree holomorphic forms on $\mathbb{P}^m$ ensures uniqueness of the rational form. In particular, refinement stability, covering, and residues are settled in this collection of dimensions $2, \dots, 12$. Grassmannian degrees $k > 1$ retain their specific maps and proof obligations.

20 Moment matrices and higher-rank amplituhedral realizations

The partition of the dodecaphasic record extends the rank-one construction to explicit higher-rank points. The same thirteen ordered nodes simultaneously provide positive Plücker minors and a positive-definite moment form. These two positivities are obtained from a single list of intervals; the second proves that the composition preserves the required rank.

Let $d_j = K_j/Q$, where $Q = \sum_{j=1}^{12} K_j = 6263$, and define

$$t_0 = 0, \quad t_j = \sum_{r=1}^j d_r.$$

Positivity of the twelve blocks gives $0 = t_0 < t_1 < \cdots < t_{12} = 1$. For each $2 \leq k \leq 9$, set

$$C_{ai}^{(k)} = t_i^a \quad (0 \leq a < k), \quad Z_{bi}^{(k)} = t_i^b \quad (0 \leq b < k+4), \quad 0 \leq i \leq 12.$$

The convention $t_i^0 = 1$ includes the zero node. Thus, $C^{(k)}$ is a $k \times 13$ matrix, while $Z^{(k)}$ has size $(k+4) \times 13$.

Theorem 20.1 (Positivity and rank of the moment realization). *The matrix $C^{(k)}$ represents a point of $\text{Gr}_{>0}(k, 13)$, and all ordered maximal minors of $Z^{(k)}$ are positive. The matrix*

$$Y^{(k)} = C^{(k)}(Z^{(k)})^\top, \quad Y_{ab}^{(k)} = \sum_{i=0}^{12} t_i^{a+b},$$

has rank k . In particular, for the positive external datum $Z^{(k)}$ already constructed from the record,

$$[Y^{(k)}] \in \mathcal{A}_{13,k,4}(Z^{(k)}), \quad 2 \leq k \leq 9,$$

where

$$\mathcal{A}_{13,k,4}(Z^{(k)}) = \{[C(Z^{(k)})^\top] : [C] \in \text{Gr}_{\geq 0}(k, 13)\}.$$

Proof. A maximal minor of either initial matrix has the form

$$\det(t_{ij}^a)_{a=0}^{r-1} = \prod_{p < q} (t_{i_q} - t_{i_p}) > 0,$$

for increasing indices. The inequality proves both Plücker assertions. The initial $k \times k$ block of $Y^{(k)}$ is the Hankel matrix $H^{(k)}$. For $x \neq 0$,

$$x^T H^{(k)} x = \sum_{i=0}^{12} \left(\sum_{a=0}^{k-1} x_a t_i^a \right)^2 > 0.$$

Indeed, a nonzero polynomial of degree less than $k \leq 9$ cannot vanish at thirteen distinct points. Hence $H^{(k)}$ is positive definite, its determinant is strictly positive and $Y^{(k)}$ has rank k . Amplituhedral membership is then the composition of the two preceding positive matrices, with rank already verified. $\square$

In rank two, the proof takes the particularly explicit form

$$\det H^{(2)} = 13 \sum_i t_i^2 - \left(\sum_i t_i \right)^2 = \sum_{i < j} (t_j - t_i)^2 > 0.$$

Node separation, produced by positivity of the record, is thus converted into coercivity of a quadratic form. Each fixed record corresponds to a specified point for each rank. The complete coverage, triangulations and residues proved in the preceding section retain their rank-one domain; the present theorem adds the explicit collection of rank-two to rank-nine points and their moment matrices. This distinction allows each proof to be reused within its precise scope.

21 Nonadic refinement, positive flags, and compatibility of moments

The length of a finite representation determines how many minors can be formed from its columns. Refinement preserves inherited columns and adds others with known provenance. Thus the dimension of a matrix realization can grow without replacing the register that fixes its separations. The following construction combines this growth with two precise notions of compatibility: restriction of evaluations to the previous marks and averaging of observables over descendant cells. The former preserves polynomial flags; the latter preserves a measure and separates the information increment into orthogonal details.

The starting datum is the dodecaphasic output already produced by APP-TRIT-TPK. The two APP sheets preserve their complete divisions; TRIT orients the regime, and the operator $U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t$ transports the enriched state. The prolongation $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ preserves phase return and memory advance. The joint discrete structure of the continuum correlatively preserves the representations of that state, with their sheets, orientation, memory, and boundaries. The regional representations and the signed register come from this same state; inversion of the latter gives

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \quad Q = \sum_{j=1}^{12} K_j = 6263.$$

This output is used with its marks and orientation; the moment matrices are subsequent realizations. The refinement index introduced below counts subdivisions of the geometric reader. Identifying it with a physical duration or a number of TPK microtransitions requires preserving the clock of that other realization.

21.1 Marked partitions and cofinal growth

Write $d_j = K_j/Q$, $a_0 = 0$, and $a_j = \sum_{h=1}^j d_h$. For $r \geq 0$, define

$$L_r = 12 \cdot 9^r, \quad N_r = L_r + 1, \quad t_{r,(j-1)9^r+c} = a_{j-1} + \frac{c}{9^r} d_j, \quad 0 \leq c < 9^r.$$

Shared endpoints are counted only once. The L_r cells are indexed by (j, c) , with $1 \leq j \leq 12$ and $0 \leq c < 9^r$, and have length $\ell_{r,j,c} = d_j/9^r$. The expression of c in r nonadic digits, including leading zeros, is also preserved. A successor has index $9c + h$, with $0 \leq h \leq 8$; Euclidean division recovers its predecessor and last digit. Shared endpoints retain their boundary mark; the half-open interval convention only organizes the cells and removes no mark.

Proposition 21.1 (Cofinality and recovery of the register). *The marks satisfy*

$$0 = t_{r,0} < t_{r,1} < \cdots < t_{r,L_r} = 1, \quad t_{r+1,9i} = t_{r,i}.$$

The maximum length is $h_r = (\max_j K_j)/(Q9^r)$, which tends to zero. For every pair of integers $k \geq 1$, $m \geq 1$, there is an r_0 such that $k + m \leq N_r$ for $r \geq r_0$. The register is recovered at any level through

$$K_j = Q \sum_{c=0}^{9^r-1} \ell_{r,j,c}.$$

Proof. Each consecutive difference is $d_j/9^r > 0$. Replacing c by $9c$ in the next-level formula gives exactly the previous mark. The nine successor lengths sum to the predecessor length; iteration proves recovery. Finally, $N_r = 12 \cdot 9^r + 1$ grows without bound. An explicit choice is any $r_0 \geq 0$ with $9^{r_0} \geq (k + m - 1)/12$. The entire argument preserves Q , the block index, and the descent digits. $\square$

Normalization determines the ratios K_j/Q . Recovery of the integers also uses the charge Q preserved by the register. This distinguishes the normalized geometry from the scale information accompanying its representation.

21.2 Evaluation flags and realizations in every finite dimension

For $1 \leq q \leq N_r$, let

$$V_r^{(q)} = (t_{r,i}^a)_{0 \leq a < q, 0 \leq i < N_r}, \quad F_r^q = \text{row } V_r^{(q)}.$$

The zeroth power is one at the zero node as well.

Theorem 21.2 (Positive flag and arbitrary-rank composition). *The spaces F_r^q form a flag $F_r^1 \subset F_r^2 \subset \cdots \subset F_r^{N_r}$, with $\dim F_r^q = q$, and every matrix $V_r^{(q)}$ has all its ordered maximal minors strictly positive. For $k \geq 1$, $m \geq 1$, $k + m \leq N_r$, the matrices*

$$C_r = V_r^{(k)}, \quad Z_r = V_r^{(k+m)}, \quad Y_r = C_r Z_r^\top$$

satisfy

$$(Y_r)_{ab} = \sum_{i=0}^{L_r} t_{r,i}^{a+b}, \quad \text{rank } Y_r = k, \quad [Y_r] \in \mathcal{A}_{N_r,k,m}(Z_r).$$

The construction reaches every finite pair (k, m) through a sufficiently deep level of the same marked partition.

Proof. For $i_1 < \cdots < i_q$, the corresponding minor is

$$\det(t_{r,i_b}^{a-1})_{a,b=1}^q = \prod_{a < b} (t_{r,i_b} - t_{r,i_a}) > 0.$$

This proves rank, positivity, and the flag inclusions. The initial $k \times k$ block of Y_r is the Gram matrix of the evaluations of $1, t, \dots, t^{k-1}$. For $x \neq 0$,

$$x^\top(Y_r)_{[0,k-1],[0,k-1]}x = \sum_i \left(\sum_{a=0}^{k-1} x_a t_{r,i}^a \right)^2 > 0,$$

because a nonzero polynomial of degree less than k cannot vanish at the $N_r \geq k + m$ distinct nodes. The block is invertible, and Y_r , which has k rows, has rank k . The stated membership is the application of the positive external matrix Z_r to the positive point represented by C_r . $\square$

We use the mathematical convention

$$\mathcal{A}_{N,k,m}(Z) = \{[CZ^\top] : [C] \in \text{Gr}_{\geq 0}(k, N)\}.$$

The definition for general m can be found in S. N. Karp, L. K. Williams, and Y. X. Zhang, *Decompositions of amplituhedra*, Definition 1.1. Here the external datum already comes from the partition expressed by HMT. The conclusion constructs a determined collection of points and flags; the dimensions of the ambient Grassmannian and of an atlas of its image are objects additional to the ranks proved.

Let $E_r : \mathbb{R}^{N_{r+1}} \rightarrow \mathbb{R}^{N_r}$ be restriction to the positions $9i$. For every $q \leq N_r$,

$$E_r((V_{r+1}^{(q)})^\top x) = (V_r^{(q)})^\top x.$$

Thus $E_r : F_{r+1}^q \rightarrow F_r^q$ is an isomorphism on these evaluation spaces: its two realizations identify the same polynomial of degree less than q . Composing restrictions agrees with direct restriction. Minors on inherited indices preserve their value exactly.

Proposition 21.3 (Recovery from a calibrated flag). *If the column order, endpoint marks, and calibration $t_{r,0} = 0, t_{r,L_r} = 1$ are preserved, the plane F_r^2 determines the nodes and separations. Together with Q and the descent marks, it determines K .*

Proof. The plane contains the constant vector and the node vector. Any other coordinate that, together with the constant vector, generates the same plane is $a + bt$, with $b \neq 0$. The two calibrations fix $a = 0, b = 1$. Consecutive differences recover the lengths, and Proposition 21.1 recovers the register. $\square$

21.3 Rank-one convex bodies and growth of the domain

For $m \geq 1$ and $m + 1 \leq N_r$, define

$$P_r^{(m)} = \text{conv}\{(t_{r,i}, t_{r,i}^2, \dots, t_{r,i}^m) : 0 \leq i \leq L_r\}.$$

Normalizing a positive row of C as barycentric coefficients proves $\mathcal{A}_{N_r,1,m}(Z_r) = P_r^{(m)}$. The Vandermonde determinant provides $m + 1$ affinely independent points, so the body has dimension m .

Proposition 21.4 (Simplicial covering in finite dimension and convex limit). *Every $P_r^{(m)}$ admits a triangulation using its vertices, with degree one over the interior of every simplicial cell. The sum of their canonical forms has canceled interior residues. Moreover,*

$$P_r^{(m)} \subseteq P_{r+1}^{(m)}, \quad \bigcup_{r \geq r_0} P_r^{(m)} = \text{conv}\{(t, t^2, \dots, t^m) : 0 \leq t \leq 1\}.$$

Proof. Order the vertices by their marks. In each face, choose its least vertex; recursively triangulate the facets not containing it and form the corresponding cones. Induction on dimension proves covering and disjoint interiors, and the same rule on shared faces guarantees compatibility. In each oriented simplex

$[v_0, \dots, v_m]$, with matrix $A = (v_1 - v_0, \dots, v_m - v_0)$ of positive determinant and barycentric coordinates $\lambda_0, \dots, \lambda_m$, the barycentric map is affine and invertible. Its form is

$$\Omega_{[v_0, \dots, v_m]} = \frac{dx_1 \wedge \dots \wedge dx_m}{\det A \prod_{j=0}^m \lambda_j}.$$

The residual restriction to each face is its oriented form. Two adjacent simplices induce opposite orientations on their common face, so the interior residues cancel. This is the same triangulation and residue operation applied in the previous rank-one realization, with the number of vertices now increased by refinement.

Inclusion of inherited nodes proves inclusion of the bodies. The mesh tends to zero, so the sampled curves are dense in the compact moment curve. Every finite convex combination of curve points is approximated by combinations using nodes from a common level. Conversely, all the bodies lie in the compact convex hull of the curve. Both inclusions prove the equality. $\square$

Cancellation of residues acts on a triangulation or subdivision of a fixed body. Passing from $P_r^{(m)}$ to $P_{r+1}^{(m)}$ adds vertices and can enlarge the body. Their forms belong to their respective domains. The preceding convex limit specifies convergence of the bodies without imposing an identity between their forms.

21.4 Calendar measure and convergence of nodal moments

Let $F_K : [0, 1] \rightarrow [0, 1]$ be the increasing piecewise affine function taking $(j-1)/12$ to a_{j-1} and $j/12$ to a_j . On the corresponding interval, its slope is $12d_j > 0$. The transported calendar measure is

$$\mu_K = (F_K)_*(ds), \quad \frac{d\mu_K}{dt} = \frac{1}{12d_j} \quad (a_{j-1} < t < a_j).$$

Each initial interval has mass $1/12$, and each level- r cell has mass $1/L_r$. This choice assigns equal weight to the calendar positions before realizing their lengths. The length measure dt is another reader; the two coincide when all initial separations are equal.

Its moments are rational and determined by K :

$$M_p = \int_0^1 t^p d\mu_K(t) = \frac{1}{12(p+1)} \sum_{j=1}^{12} \frac{a_j^{p+1} - a_{j-1}^{p+1}}{d_j}, \quad p \geq 0. \quad (21.1)$$

Proposition 21.5 (Recovery through moments and quantiles). *The full sequence $(M_p)_{p \geq 0}$ determines μ_K . If $H_K(t) = \mu_K([0, t])$ is its distribution function, then*

$$a_j = H_K^{-1}(j/12), \quad K_j = Q(H_K^{-1}(j/12) - H_K^{-1}((j-1)/12)).$$

Consequently, the complete moments, the charge Q , and the calendar order recover the initial register.

Proof. Two probability measures on $[0, 1]$ with the same moments have the same integrals for all polynomials. Uniform approximation of continuous functions by polynomials and finite mass extend this equality to all continuous functions, which determine the Borel measure. The density of μ_K is positive and constant on each initial interval. Its distribution is therefore continuous and strictly increasing, and satisfies $H_K(a_j) = j/12$. Inverting this equality and taking the difference of endpoints prove the formulas. $\square$

The nodal probability measure corresponding to Theorem 21.2 is $\nu_r = N_r^{-1} \sum_{i=0}^{L_r} \delta_{t_{r,i}}$.

Theorem 21.6 (Limit of the nodal representation). *We have $\nu_r \rightarrow \mu_K$. If f is continuous,*

$$\left| \int f d\nu_r - \int f d\mu_K \right| \leq \omega_f(h_r) + \frac{\text{osc}(f)}{N_r},$$

where ω_f is its modulus of continuity. For $p \geq 1$, the error of the moment of order p is bounded by $ph_r + 1/N_r$. For every fixed pair (k, m) ,

$$\frac{1}{N_r} Y_r \longrightarrow (M_{a+b})_{0 \leq a < k, 0 \leq b < k+m} =: Y_\infty^{(k,m)}, \quad \text{rank } Y_\infty^{(k,m)} = k.$$

Proof. The measure assigning mass $1/L_r$ to each left cell endpoint approximates its integral with error at most $\omega_f(h_r)$. Indeed, each cell has that mass and diameter no greater than h_r . The measure ν_r is L_r/N_r times this measure plus δ_1/N_r , adding at most $\text{osc}(f)/N_r$. For $f(t) = t^p$, the Lipschitz constant is p . Convergence of each entry follows from these bounds. The initial block of the limit matrix is positive definite: the integral of the square of a nonzero polynomial is positive, because μ_K has strictly positive density on every open interval of $[0, 1]$. Thus the limit preserves rank k . $\square$

Nodal moments change exactly through the addition of nodes. If $\mathcal{J}_r$ is the set of new positions and $z_q(t) = (1, t, \dots, t^{q-1})^\top$, then

$$V_{r+1}^{(q)}(V_{r+1}^{(q)})^\top = V_r^{(q)}(V_r^{(q)})^\top + \sum_{i \in \mathcal{J}_r} z_q(t_{r+1,i}) z_q(t_{r+1,i})^\top.$$

Each summand is positive semidefinite. The same update holds for the rectangular matrices Y_r , using $z_k z_{k+m}^\top$. The equality expresses the change in counting measure; normalization and the preceding theorem describe its limit. The rank- k limit object has coordinates in a finite Grassmannian, whereas the external data Z_r belong to the different representations at each level.

21.5 Cell averages, compatible operators, and orthogonal detail

For a cell $I_{r,i} = [b_{r,i}, b_{r,i} + \ell_{r,i}]$, set

$$a_{r,i}^{(p)} = \frac{(b_{r,i} + \ell_{r,i})^{p+1} - b_{r,i}^{p+1}}{(p+1)\ell_{r,i}}, \quad A_r^{(q)} = (a_{r,i}^{(p)})_{0 \leq i < L_r, 0 \leq p < q}.$$

These are the monomial means with respect to the conditional probability of μ_K on each cell. This probability is uniform on the cell, although μ_K is not the global length measure.

Let $\mathcal{H}_r = \mathbb{R}^{L_r}$ with inner product $\langle x, y \rangle_r = L_r^{-1} \sum_i x_i y_i$. The map $J_r : \mathcal{H}_r \rightarrow \mathcal{H}_{r+1}$ repeats a predecessor's value on its nine successors. Its adjoint $R_r = J_r^*$ averages those nine values. We have $R_r J_r = I$.

Theorem 21.7 (Exact compatibility and moment balance). *For every $p \geq 0$,*

$$R_r a_{r+1}^{(p)} = a_r^{(p)}, \quad \frac{1}{L_r} \sum_i a_{r,i}^{(p)} = M_p.$$

Define

$$D_r^{(q)} = A_{r+1}^{(q)} - J_r A_r^{(q)}, \quad G_r^{(q)} = \frac{1}{L_r} (A_r^{(q)})^\top A_r^{(q)}.$$

Then

$$R_r D_r^{(q)} = 0, \quad G_{r+1}^{(q)} = G_r^{(q)} + \frac{1}{L_{r+1}} (D_r^{(q)})^\top D_r^{(q)},$$

and

$$G_r^{(q)} \longrightarrow H_K^{(q)} = (M_{a+b})_{0 \leq a, b < q}.$$

The details and initial level reconstruct all levels. For $q \leq L_r$, the matrix $A_r^{(q)}$ has rank q , and all ordered minors of order q are positive.

Proof. The nine successors have equal masses, and their intervals partition the predecessor interval. Additivity of the integral gives the first identity; summing over all cells gives the second. The decomposition $A_{r+1} = J_r A_r + D_r$ is orthogonal in each column because $R_r D_r = 0$. Expanding its Gram matrix cancels the cross terms, and J_r is an isometry. This proves the matrix balance.

The cellwise constant observable taking the value $a_{r,i}^{(p)}$ uniformly approximates t^p , with error no greater than ph_r for $p \geq 1$. It therefore converges in $L^2(\mu_K)$, and the inner products converge to M_{a+b} . Reconstruction is obtained by iterating $A_{r+1} = J_r A_r + D_r$.

To prove the minors, choose q ordered cells. By multilinearity and Fubini, the determinant of their averages is the mean over the product of these cells of $\prod_{a < b} (x_b - x_a)$. This integrand is positive in the interior of the product because the cells are disjoint and ordered. Their boundaries have measure zero. The mean is strictly positive. $\square$

The identity extends to the degree pair k and $k + m$:

$$\bar{Y}_r = \frac{1}{L_r} (A_r^{(k)})^\top A_r^{(k+m)}, \quad \bar{Y}_{r+1} - \bar{Y}_r = \frac{1}{L_{r+1}} (D_r^{(k)})^\top D_r^{(k+m)}.$$

For $k + m \leq L_r$, the averaging matrices define a second positive amplituhedral realization, with L_r columns, and $\bar{Y}_r$ has rank k . These are cell averages, whereas Y_r uses evaluations at $N_r = L_r + 1$ nodes. Both realizations converge to $Y_\infty^{(k,m)}$; their agreement in the limit follows from the preceding proofs.

The operator version is equally explicit. For a continuous function f , let $\Psi_r(f)$ be the diagonal operator of its conditional means on $\mathcal{H}_r$. It is linear, positive, and unital, and

$$R_r \Psi_{r+1}(f) J_r = \Psi_r(f).$$

This equality is proved by applying both sides to each basis vector. For $X(t) = t$, the first two operators determine, cell by cell,

$$\Psi_r(X)_i = b_{r,i} + \frac{\ell_{r,i}}{2}, \quad \Psi_r(X^2)_i - \Psi_r(X)_i^2 = \frac{\ell_{r,i}^2}{12}.$$

If $v_{r,i}$ denotes this variance, the formula $\ell_{r,i} = \sqrt{12v_{r,i}}$ recovers the length; the first identity recovers the initial endpoint. With orientation, marks, and Q , one again recovers K . Averaging is a positive compression: the displayed variance quantifies why $\Psi_r(X^2)$ and $\Psi_r(X)^2$ are different operations.

21.6 Transport of the measure and scope of dimensional growth

Two positive lists of separations d and d' , with the same marks, determine an increasing piecewise affine homeomorphism $F = F_{K'} \circ F_K^{-1}$. It takes each cell to its corresponding cell and satisfies $F_* \mu_K = \mu_{K'}$. Hence

$$\mathcal{U}_F : L^2(\mu_K) \longrightarrow L^2(\mu_{K'}), \quad \mathcal{U}_F f = f \circ F^{-1}$$

is a surjective isometry. If P_r and $P_{r'}$ are the projections onto cellwise constant functions, then $P_{r'} \mathcal{U}_F = \mathcal{U}_F P_r$: conditional means are transported because the masses of corresponding cells agree. This applies in particular to the positive separation flow directed by K , preserving each block's direction in its successors. The moments of the spatial coordinate may change; the isometry transports the observable together with the measure and its projections.

The growth proved is cofinal in every finite external dimension. It preserves a system of positive flags, an explicit collection of amplituhedral points, complete convex bodies in rank one, and a measure whose averaging operators are exactly compatible. Memory of the details allows inversion of compression between levels. The rank-one covering and residue proofs preserve their domain; for higher ranks, the present result fixes matrices, positivity, recovery, compatibility, and limit with explicit types. This separation locates the scope of each construction within the same development without confusing dimension growth, parametrization of a point, and classification of all cells of an image.

22 Genealogical moments, Jacobi operators, and spectral residues

The calendar-induced measure provides a link between cell geometry and a positive spectral realization. This link uses the entire marked partition: each of its twelve intervals preserves the same chronological mass, whereas its length comes from the corresponding register coordinate. The density exactly compensates for that length. Consequently, the spatial distribution of the measure retains the anisotropy of the register even though the distribution among windows is uniform.

Let $K_j > 0$, $Q = \sum_{j=1}^{12} K_j$, $d_j = K_j/Q$, and $t_j = \sum_{i \leq j} d_i$, with $t_0 = 0$. The probability measure used in this section is

$$d\mu_K(t) = \sum_{j=1}^{12} \frac{\mathbf{1}_{[t_{j-1}, t_j]}(t)}{12d_j} dt.$$

Shared endpoints have measure zero. Its moments are rational for every integer register:

$$m_r = \int_0^1 t^r d\mu_K(t) = \frac{1}{12(r+1)} \sum_{j=1}^{12} \frac{t_j^{r+1} - t_{j-1}^{r+1}}{d_j}, \quad r \geq 0. \quad (22.1)$$

The equality $m_0 = 1$ expresses conservation of total mass. For each $n \geq 1$, the Hankel matrix $H_n = (m_{a+b})_{0 \leq a, b < n}$ is positive definite: its quadratic form is the integral of the square of a nonzero polynomial over an interval where the density is positive. This observation allows passage from moments to an operator without choosing a target spectrum.

Theorem 22.1 (Tridiagonal recurrence determined by the partition). *There is a unique sequence of monic polynomials p_n , of degree n , orthogonal for μ_K . If*

$$h_n = \int p_n^2 d\mu_K > 0, \quad a_n = \frac{\int t p_n^2 d\mu_K}{h_n}, \quad \beta_n = \frac{h_n}{h_{n-1}} > 0 \quad (n \geq 1),$$

then $p_0 = 1$, $p_1 = t - a_0$, and

$$p_{n+1}(t) = (t - a_n)p_n(t) - \beta_n p_{n-1}(t), \quad n \geq 1. \quad (22.2)$$

All coefficients a_n, β_n are rational for the stated integer register. In the orthonormal basis $\psi_n = p_n/\sqrt{h_n}$, the compression of multiplication by t to polynomials of degree less than n has the symmetric matrix

$$J_n = \begin{pmatrix} a_0 & \sqrt{\beta_1} & & 0 \\ \sqrt{\beta_1} & a_1 & \ddots & \\ & \ddots & \ddots & \sqrt{\beta_{n-1}} \\ 0 & & \sqrt{\beta_{n-1}} & a_{n-1} \end{pmatrix}.$$

Proof. Positivity of H_n guarantees that the linear system imposing orthogonality on a monic polynomial has a unique solution. Since its coefficients are rational moments, the solution is rational. Expand $t p_n$ in the monic basis through degree $n+1$. For $j \leq n-2$, symmetry of the inner product gives $\langle t p_n, p_j \rangle = \langle p_n, t p_j \rangle = 0$. Only the terms of degrees $n+1, n, n-1$ remain. Their coefficients are, respectively, $1, a_n, h_n/h_{n-1}$, proving the recurrence. Normalization by $\sqrt{h_n}$ turns the two adjacent coefficients into $\sqrt{\beta_n}$. $\square$

For the register expressed in Section 15, the first diagonal entry and the square of the first coupling are

$$a_0 = m_1 = \frac{71195}{150312}, \quad \beta_1 = m_2 - m_1^2 = \frac{2206226167}{22593697344}.$$

The first equality calculates the center of mass of the chronological partition; the second calculates its dispersion and determines the first link of the tridiagonal matrix. Both numbers are obtained by integrating the generated lengths, and their structural meaning precedes any decimal evaluation.

Theorem 22.2 (Simple spectrum, quadrature, and positive residues). *The eigenvalues $\lambda_1 < \dots < \lambda_n$ of J_n are simple and belong to $(0, 1)$. There are unique weights $w_i > 0$, summing to one, such that*

$$\int f(t) d\mu_K(t) = \sum_{i=1}^n w_i f(\lambda_i) \quad (\deg f \leq 2n - 1). \quad (22.3)$$

The resolvent observed at the unit constant e_0 satisfies

$$g_n(z) = \langle e_0, (zI - J_n)^{-1} e_0 \rangle = \sum_{i=1}^n \frac{w_i}{z - \lambda_i} = \frac{1}{z - a_0 - \frac{\beta_1}{z - a_1 - \frac{\beta_2}{\ddots - \frac{\beta_{n-1}}{z - a_{n-1}}}}}. \quad (22.4)$$

In particular, its poles and residues are obtained from the register through the operations of taking moments, orthogonalization, and compression.

Proof. For any nonzero polynomial p of degree less than n ,

$$0 < \int_0^1 t |p(t)|^2 d\mu_K(t) < \int_0^1 |p(t)|^2 d\mu_K(t).$$

The variational characterization places the spectrum of J_n in $(0, 1)$. Each subdiagonal entry is positive. The eigenvector equations recursively determine all its coordinates from the first; if that coordinate vanished, the entire vector would vanish. Every eigenspace therefore has dimension one. Symmetry of the matrix gives orthogonal diagonalization and simplicity of the spectrum. For unit eigenvectors v_i , set $w_i = |\langle e_0, v_i \rangle|^2 > 0$. Completeness of the eigenbasis gives $\sum_i w_i = 1$ and the first expression for the resolvent.

By expansion of the determinant, the tridiagonal recurrence shows that $\det(tI - J_n) = p_n(t)$. For $\deg f \leq 2n - 1$, Euclidean division writes $f = qp_n + r$, with $\deg q, \deg r < n$. Orthogonality makes the integral of qp_n vanish, and its evaluation at the λ_i also vanishes. For r of degree less than n , successive multiplications of the constant by t remain within the polynomial space up to the required degree; hence $\int r d\mu_K = \langle e_0, r(J_n)e_0 \rangle$. The spectral resolution proves the quadrature formula. Its uniqueness follows from invertibility of the Vandermonde matrix evaluated at the distinct roots. Finally, the Schur complement of the first row and column of $zI - J_n$, iterated over the trailing submatrices, produces the continued fraction. $\square$

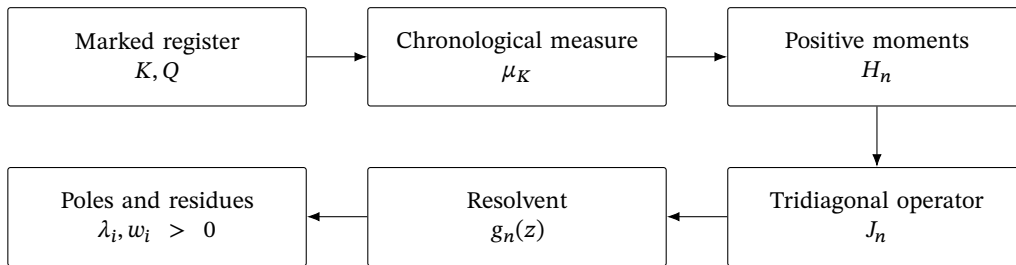


Figure 11: Spectral realization of the HMT partition. Cell geometry determines the measure; positivity of its moments produces the operator and the positive residues of its resolvent.

The resolvent formula provides a spectral reading of spatial memory. The positive weights describe how much the constant coordinate observes of each mode; the poles are the algebraic frequencies of the compressed multiplication operator. Moments of increasing order recover successive details of the measure, and the continued-fraction coefficients preserve this information through a local three-term

recurrence. The classical appearance of a Jacobi matrix is recognized after its coefficients have been constructed from the HMT cells. The causal chain is fixed by

$$K \mapsto (d, t) \mapsto \mu_K \mapsto (m_r)_{r \geq 0} \mapsto ((a_n)_{n \geq 0}, (\beta_n)_{n \geq 1}) \mapsto (J_n, g_n).$$

Proposition 22.3 (Dimension and refinement compatibility). *The matrices J_n are compatible principal compressions of the same bounded multiplication by t on $L^2(\mu_K)$. Their spectral measures $\sum_i w_i \delta_{\lambda_i}$ converge weakly to μ_K . If each nonadic subinterval is represented by its midpoint and its mass $1/(12 \cdot 9^r)$, the resulting atomic measure $\mu_{K,r}$ approximates the moments of μ_K . For a fixed degree n , construction from these moments recovers the coefficients a_j, β_j of the same matrix as refinement tends to infinite depth.*

Proof. Orthogonalization of the complete sequence uses the same formulas as n increases, so each principal block remains fixed. Quadrature exactly reproduces any polynomial once $2n - 1$ exceeds its degree. Uniform approximation of continuous functions on $[0, 1]$ by polynomials, together with total mass one, proves weak convergence. The midpoint sums converge on each of the twelve intervals with its constant density, and thus approximate all moments. In fixed dimension, the orthogonalization coefficients are rational functions of finitely many moments, with denominators formed from positive Gram determinants. Convergence of these moments and positivity of the limit ensure convergence of the coefficients. $\square$

Falsification criterion 22.4. *A zero length invalidates the stated density and can destroy the rank of the moments. Replacing the fixed measure by unnormalized counts on different meshes changes the operator. The residues of g_n are spectral weights of the resolvent, whereas the residues of a canonical form are defined on geometric divisors: identifying these notions requires specifying the form and the corresponding map. Likewise, J_n is dimensionless; identifying it with a physical Hamiltonian requires the representation and the action and clock scale used by that sector.*

Theorem 22.5 (Faithfulness of the marked spectral realization). *The infinite Jacobi operator J , together with its unit cyclic vector e_0 and charge Q , determines the register K . Two realizations of this type are unitarily equivalent through a map preserving the marked vector and charge if and only if their registers agree. The spectrum as a set, by contrast, is $[0, 1]$ for every positive register in the domain.*

Proof. Polynomials are dense in $L^2(\mu_K)$: first approximate continuous functions and then use their density for the finite measure. The constructed orthonormal basis defines a unitary map taking e_n to ψ_n . Under this map, J is multiplication by t , a bounded self-adjoint operator, and e_0 represents the constant function one. Therefore,

$$\langle e_0, J^r e_0 \rangle = m_r \quad (r \geq 0).$$

These moments determine the measure: two measures with the same moments agree on polynomials, then on continuous functions by uniform approximation, and finally as Borel measures. Its distribution $F(t) = \mu_K([0, t])$ is continuous and strictly increasing. Each initial interval has mass $1/12$, so

$$t_j = F^{-1}(j/12), \quad K_j = Q(F^{-1}(j/12) - F^{-1}((j-1)/12)).$$

A unitary equivalence preserving e_0 preserves the moments and therefore the register. The converse follows from the unique construction.

For the last statement, the density of μ_K is positive on every subinterval. Each $\lambda \in [0, 1]$ admits unit-norm functions concentrated on progressively smaller intervals around λ , on which the norm of $(J - \lambda I)f$ tends to zero. Thus the entire interval belongs to the spectrum. Outside it, multiplication by $1/(t - \lambda)$ is bounded and defines the resolvent. This proves equality of the spectrum. $\square$

The theorem distinguishes the extent of the spectrum from the information it preserves relative to a marked observation. The same spectral interval admits the different distributions of the collection; the measure observed from the unit recovers the original separations. The link with the Grassmannian

representation is now reversible: marked minors reconstruct K , the register constructs (J, e_0, Q) , and its moments and quantiles recover the same register. Every reader factoring through (K, ξ) therefore has a marked spectral expression once the remaining coordinates ξ are preserved. The spectral realization contains structure, in addition to pole positions or eigenvalues.

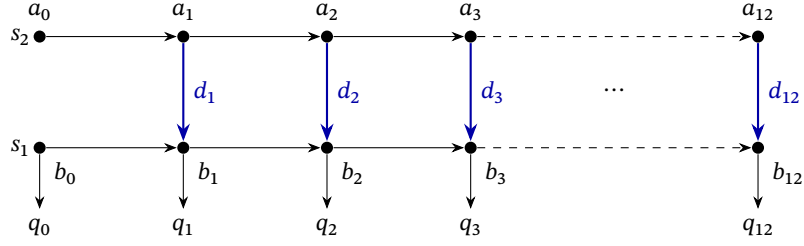


Figure 12: Two-row planar network. Horizontal and outgoing edges have weight one. Each path from the upper source descends exactly once. For two destinations $i < j$, disjoint pairs descend after i and before j ; their total weight is $\Delta_{ij} = d_{i+1} + \dots + d_j$. Dashed lines retain the eight intermediate positions omitted only from the drawing.

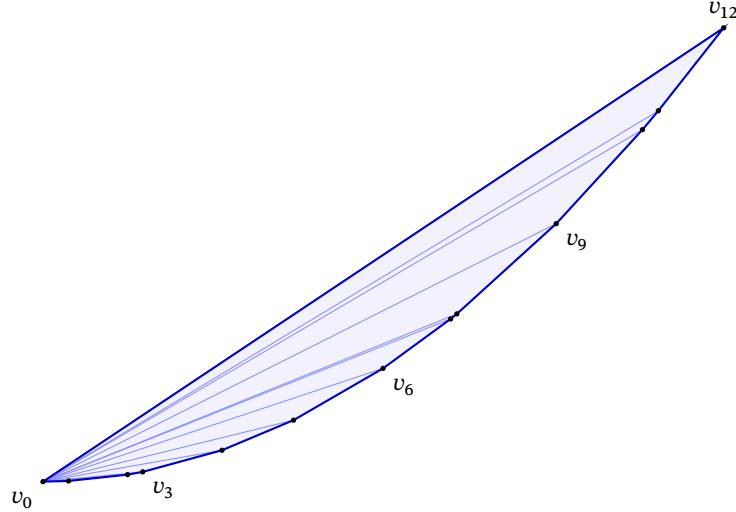


Figure 13: Rational polygon determined by the thirteen parameters t_j in (19.2), plotted with its actual coordinates (t_j, t_j^2) . The triangulation from v_0 has eleven cells. Each interior diagonal receives two opposite residues. Five vertices are labelled for legibility; all thirteen are represented.

23 Weight flow, information geometry and Legendre duality

Interval deformation admits an exact variational description. Let $d_j = K_j/Q > 0$, $\sum_j d_j = 1$, and let u_j be the coordinates of the direction (15.2). For $s \in \mathbb{R}$, define

$$Z(s) = \sum_{j=1}^{12} d_j e^{s u_j}, \quad \psi(s) = \log Z(s), \quad d_j(s) = \frac{d_j e^{s u_j}}{Z(s)}. \quad (23.1)$$

All exponential arguments are dimensionless. The parameter s is the coordinate of this geometric deformation. Assigning a physical duration belongs to a declared temporal reader.

Theorem 23.1 (Strict convexity and dual coordinate). *The function ψ is analytic and strictly convex on $\mathbb{R}$. If $u_- = \min_j u_j$ and $u_+ = \max_j u_j$, the map $v = \psi'(s)$ is a diffeomorphism from $\mathbb{R}$ onto (u_-, u_+) . For each v in that interval there is a unique coordinate $s(v)$, and*

$$\psi^*(v) = s(v)v - \psi(s(v)), \quad (\psi^*)'(v) = s(v), \quad (\psi^*)''(v) = \frac{1}{\psi''(s(v))} > 0.$$

Proof. The finite sums allow termwise differentiation. This gives

$$\begin{aligned} \psi'(s) &= \sum_j d_j(s) u_j =: \bar{u}(s), \\ \psi''(s) &= \sum_j d_j(s) (u_j - \bar{u}(s))^2. \end{aligned} \quad (23.2)$$

The weights are strictly positive and the u_j are not all equal; the variance is therefore strictly positive. As s tends to $+\infty$, dividing numerator and denominator of ψ' by e^{su_+} shows that only indices with $u_j = u_+$ survive. Thus $\psi'(s) \rightarrow u_+$. The argument with u_- gives the opposite limit. Monotonicity and the inverse function theorem provide smooth bijectivity. The function $sv - \psi(s)$ has its unique maximum at $s(v)$; differentiating the maximum value proves the two dual identities. $\square$

The evolution of each interval is

$$\dot{d}_j(s) = d_j(s)(u_j - \bar{u}(s)). \quad (23.3)$$

The derivatives sum to zero. Total length therefore remains one while the internal lengths are re-distributed. This conservation of global scale allows the change in cross-ratios to be attributed to a deformation of positions rather than a common dilation.

On the open simplex of weights, the metric $g_d(\xi, \zeta) = \sum_j \xi_j \zeta_j / d_j$, on tangent vectors of sum zero, is positive. The linear function $F(d) = \sum_j d_j u_j$ has, with respect to this metric, exactly the field in (23.3) as its gradient, since

$$g_d(d_j(u_j - \bar{u}), \xi) = \sum_j (u_j - \bar{u}) \xi_j = \sum_j u_j \xi_j = dF(\xi).$$

Consequently $dF/ds = \psi''(s) > 0$ along the flow. The ascending character corresponds to this specific function and metric.

Theorem 23.2 (Variational invariance under inherited subdivision). *For each interval j , let $r_{j,c} > 0$, $\sum_c r_{j,c} = 1$, be the relative weights of a fixed subdivision. Assign to the subintervals*

$$d_{j,c} = r_{j,c} d_j, \quad u_{j,c} = u_j.$$

The partition function, its logarithm, all its derivatives and the Legendre transform agree exactly with those of the predecessor partition. Furthermore, the flow commutes with summation over subintervals:

$$\sum_c d_{j,c}(s) = d_j(s), \quad d_{j,c}(s) = r_{j,c} d_j(s).$$

Proof. Grouping the finite sum gives

$$Z_{\text{ref}}(s) = \sum_{j,c} r_{j,c} d_j e^{su_j} = \sum_j d_j e^{su_j} = Z(s).$$

All conclusions follow from this identity and from dividing each weight by the same $Z(s)$. The same argument holds after any finite number of subdivisions. For a countable subdivision, the sum of positive terms converges to the same value when grouped by predecessor intervals. $\square$

The inherited-direction hypothesis has verifiable content. If an interval with direction a is divided into two halves with directions $a + b$ and $a - b$, its contribution changes from de^{sa} to $de^{sa} \cosh(sb)$. For $b \neq 0$, equality of partition functions fails even though the total weight at $s = 0$ is the same. Geometric and variational conservation thus requires preserving the relevant directional information as

well. The proved identity applies to inherited subdivision; its transport to a TPK update that changes those directions is determined by composing the reader of that update.

There is a regular Hamiltonian formulation of the dual coordinate. In the cotangent plane with coordinates (q, p) , take $H(q, p) = \psi(p)$. The equations are $\dot{q} = \psi'(p)$, $\dot{p} = 0$. For velocities $\dot{q} \in (u_-, u_+)$, the Legendre transform gives

$$L(q, \dot{q}) = \psi^*(\dot{q}), \quad \frac{\partial^2 L}{\partial \dot{q}^2} > 0.$$

This system variationally realizes the potential of the interval flow. Its equation $\dot{p} = 0$ shows that Hamiltonian time differs from the parameter s traversing the collection (23.1). Evaluation of H and L is preserved under the subdivision of the theorem, because ψ and ψ^* remain identical.

In turn, any flow $\dot{d} = F(d)$ admits, in a local chart of the simplex, a cotangent lift with $\mathcal{H}(d, p) = p \cdot F(d)$. This second construction reproduces the flow (23.3) itself on its base, but its Hessian with respect to p is zero. The two realizations answer different questions: one produces a regular convex duality and the other transports dynamics on a given base. This distinction preserves the meaning of Hamiltonian and Lagrangian when comparing this geometry with the physical readers of the corpus.

24 Dirichlet forms and variational equivalence of refinement

The positive representation of the dodecaphasic register defines an ordered partition through

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \quad d_j = K_j/6263, \quad t_i = \sum_{j=1}^i d_j.$$

The register is received from the composition $x_{\text{term}} \mapsto \mathcal{R}_{12} \mapsto (b^{90}, b^{120}, Q_{\text{TPK}}) \mapsto U_{\text{sgn}} \mapsto K$, subsequent to APP-TRIT-TPK and its enriched state, in accordance with the readers reconstructed in [2]. This section applies a quadratic reader to the resulting partition. Depths will be compared through restrictions to inherited endpoints, so that every energy coefficient retains its origin in an already produced positive separation.

The corpus contains a weighted difference energy for the graph of the two memories and a Schur reduction of the complete forms. Here these operations are brought together on the interval network of the register. The specialization to conductances $1/d_j$, the interpolation operator, and its compatibility with deformation of the separations constitute the formalization developed in this section.

24.1 Reconstruction of separations from the stiffness matrix

Let $d = (d_1, \dots, d_m)$, with $d_j > 0$, and let $f = (f_0, \dots, f_m) \in \mathbb{C}^{m+1}$ be the values of an observable at the partition endpoints. Define

$$\mathcal{E}_d(f) = \sum_{j=1}^m \frac{|f_j - f_{j-1}|^2}{d_j}, \quad L_d = D^* \text{diag}(d_1^{-1}, \dots, d_m^{-1}) D, \quad (24.1)$$

where $D : \mathbb{C}^{m+1} \rightarrow \mathbb{C}^m$ is the oriented incidence operator, $(Df)_j = f_j - f_{j-1}$, and the asterisk denotes the Hermitian adjoint. Thus $\mathcal{E}_d(f) = f^* L_d f$. The stiffness matrix has entries

$$(L_d)_{00} = d_1^{-1}, \quad (L_d)_{mm} = d_m^{-1}, \quad (L_d)_{ii} = d_i^{-1} + d_{i+1}^{-1} \quad (1 \leq i < m),$$

$$(L_d)_{i-1,i} = (L_d)_{i,i-1} = -d_i^{-1},$$

and all other entries vanish. It is Hermitian positive semidefinite, its kernel consists of the constant functions, and its form is positive definite on the quotient by that kernel. The complete matrix recovers the separations through

$$d_j = -\frac{1}{(L_d)_{j-1,j}}. \quad (24.2)$$

Proposition 24.1 (Equivalence between separations and the complete nodal reader). *The map $d \mapsto L_d$ is a bijection between $(0, \infty)^m$ and the set of real symmetric tridiagonal matrices whose entries immediately adjacent to the diagonal are negative and whose row sums are zero. Its inverse is (24.2). The normalization $\sum_j d_j = 1$ is equivalent to $\sum_{j=1}^m -1/(L_d)_{j-1,j} = 1$.*

Proof. The explicit formulas for L_d satisfy all the conditions. Conversely, the negative adjacent entries determine unique positive separations through (24.2). The zero sum of each row determines its diagonal entries as the sum of incident conductances. Tridiagonality makes all remaining entries zero. The matrix in (24.1) is recovered exactly; its positive semidefiniteness then follows from the factorization $D^* \text{diag}(1/d_j)D$. $\square$

The weighted discrete derivative $J_j = (f_j - f_{j-1})/d_j$ is the oriented flow of edge j . At interior vertices, $(L_d f)_i = J_i - J_{i+1}$; at the endpoints, $(L_d f)_0 = -J_1$ and $(L_d f)_m = J_m$. The interior equation $L_d f = 0$ expresses that the flow has the same value on every edge. This defines a quadratic observable of the interval reader; its realization as a physical quantity requires the relevant dimensional chart.

24.2 Harmonic interpolation, Schur complement, and detail reconstruction

Refine each separation as $d_j = \sum_{c=1}^{r_j} \delta_{j,c}$, with $\delta_{j,c} > 0$. Let $\tilde{d}$ be the concatenation of these separations, R restriction to the old endpoints, and H the interpolation defined by

$$(Hf)_{j,c} = (1 - \theta_{j,c})f_{j-1} + \theta_{j,c}f_j, \quad \theta_{j,c} = \frac{\sum_{a=1}^c \delta_{j,a}}{d_j}. \quad (24.3)$$

Shared endpoints are counted only once. We have $RH = I$.

Theorem 24.2 (Isometric refinement of the energy). *For every vector of old values f ,*

$$H^* L_{\tilde{d}} H = L_d, \quad L_{\tilde{d}} H = R^* L_d, \quad \min_{Rg=f} \mathcal{E}_{\tilde{d}}(g) = \mathcal{E}_d(f). \quad (24.4)$$

The minimizer is unique and equals Hf . Every g admits the unique decomposition

$$g = H(Rg) + z, \quad Rz = 0, \quad \mathcal{E}_{\tilde{d}}(g) = \mathcal{E}_d(Rg) + \mathcal{E}_{\tilde{d}}(z). \quad (24.5)$$

Proof. On an old edge, write $h_c = g_{j,c} - g_{j,c-1}$, so that $\sum_c h_c = f_j - f_{j-1} = a$. If $h_c = \delta_{j,c}a/d_j + r_c$, then $\sum_c r_c = 0$ and

$$\sum_c \frac{|h_c|^2}{\delta_{j,c}} = \frac{|a|^2}{d_j} + \sum_c \frac{|r_c|^2}{\delta_{j,c}}.$$

The equality follows by expansion; the cross term is proportional to $\sum_c r_c$ and vanishes. The minimum is attained exactly when all r_c vanish, the condition determining (24.3). Summing over the edges gives the minimum and the energy decomposition. The flow of Hf is constant within every old edge, so its interior components of $L_{\tilde{d}} Hf$ vanish and its inherited components agree with $L_d f$. This proves the second matrix identity; multiplying by H^* , using $RH = I$, gives the first. $\square$

The term $\mathcal{E}_{\tilde{d}}(z)$ is nonnegative and vanishes exactly for $z = 0$: a zero-energy detail would be constant and, vanishing at the inherited endpoints, would vanish throughout the network. The pair (Rg, z) preserves the complete refined vector and allows its recovery through $g = H(Rg) + z$. Minimization selects its harmonic component; storing the detail z retains the information that the reduced form alone does not distinguish. Conservation of the observed energy and conservation of the refined state are thus expressed by different explicit maps.

Ordering vertices as inherited endpoints E and interior vertices I , write

$$L_{\tilde{d}} = \begin{pmatrix} A & B^* \\ B & C \end{pmatrix}.$$

For a refinement with interior vertices, the block C is positive definite: a function vanishing at the endpoints and having zero energy is constant on every subinterval and therefore identically zero. We obtain

$$H = \begin{pmatrix} I \\ -C^{-1}B \end{pmatrix}, \quad \text{Schur}_I(L_{\tilde{d}}) = A - B^*C^{-1}B = L_d. \quad (24.6)$$

If refinement preserves all vertices and introduces none, the same statement reduces to $H = I$ and $L_{\tilde{d}} = L_d$.

Proposition 24.3 (Naturality of prolongations and elimination). *For three nested partitions, harmonic interpolations and Schur complements satisfy*

$$H_{n+2 \leftarrow n} = H_{n+2 \leftarrow n+1} H_{n+1 \leftarrow n}, \quad \text{Schur}_{n+1 \rightarrow n}(\text{Schur}_{n+2 \rightarrow n+1}(L_{n+2})) = L_n. \quad (24.7)$$

Proof. Interpolation of affine values on an interval remains the same affine function after any subdivision of that interval. This proves the first identity. Minimizing first over vertices added at the last level and then over those at the intermediate level amounts to minimizing over all of them, with the same inherited endpoints fixed. The variational identity in (24.4) and uniqueness of the minimizer prove the second identity. Restriction of the minimum is therefore performed through the same composition as harmonic prolongation of the data. $\square$

24.3 Dirichlet-to-Neumann response and energy defect of an extension

If only the initial and final endpoints are preserved, let $D_0 = \sum_{j=1}^m d_j$. For values (a, b) , the harmonic solution is

$$f_i = a + (b - a) \frac{\sum_{j=1}^i d_j}{D_0}, \quad J_j = (b - a)/D_0.$$

Its Dirichlet-to-Neumann operator, taking boundary values to outgoing flows, is

$$\Lambda_d = \frac{1}{D_0} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad \min \mathcal{E}_d = \frac{|b - a|^2}{D_0}. \quad (24.8)$$

When values are prescribed at all inherited vertices, the same argument gives the multipoint response operator $\Lambda = L_d$.

An approximate extension $g = (f, y)$, where y collects the values at interior vertices, has interior residual $r = Bf + Cy$. The exact identity

$$\mathcal{E}_{\tilde{d}}(g) - f^* \Lambda f = r^* C^{-1} r \geq 0 \quad (24.9)$$

follows from $y + C^{-1}Bf = C^{-1}r$ and the decomposition

$$g^* L_{\tilde{d}} g = f^* (A - B^* C^{-1} B) f + (y + C^{-1} B f)^* C (y + C^{-1} B f).$$

Thus the effective value and the detail retain a complete Hermitian decomposition. In particular, if a linear extension is $\tilde{H} = \begin{pmatrix} I \\ Z \end{pmatrix}$, its residual matrix is $\mathcal{R} = B + CZ$, and

$$\tilde{H}^* L_{\tilde{d}} \tilde{H} - \Lambda = \mathcal{R}^* C^{-1} \mathcal{R} \geq 0.$$

A zero residual characterizes exact harmonic interpolation. This energy residual lives in the space of interior vertices; cancellation of residues of meromorphic forms belongs to another boundary map and retains its own definition.

24.4 Commutation of refinement and the separation flow

Let $u = P_3 P_{11} K$ be the algebraic direction already recovered from the register readers [2]. For the real parameter s , define

$$d_j(s) = \frac{K_j e^{su_j}}{\sum_{\ell} K_{\ell} e^{su_{\ell}}}.$$

Fix positive fractions $\vartheta_{j,c}$ summing to one for each j . The inherited refinement is

$$\delta_{j,c}(s) = \vartheta_{j,c} d_j(s), \quad u_{j,c} = u_j.$$

First producing the refined weights $K_j \vartheta_{j,c}$ and then deforming them gives the same collection: the successors have identical exponential factors, and their coefficients sum to K_j . For every s ,

$$\text{Schur}_I(L_{\tilde{d}(s)}) = L_{d(s)}, \quad H^* L_{\tilde{d}(s)} H = L_{d(s)}. \quad (24.10)$$

In this parent–children comparison, H is constant: its coefficients are the partial sums of $\vartheta_{j,c}$. By contrast, interpolation from the two global endpoints uses $t_i(s) = \sum_{j \leq i} d_j(s)$ and changes with the parameter. Both statements correspond to different restrictions of the same network.

The normalization $\sum_j d_j(s) = 1$ makes the global two-endpoint operator remain equal to $\begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$. The all-node matrix $L_{d(s)}$ does preserve the deformation and recovers it through (24.2). The information loss in the two-endpoint reader is identified exactly: it retains the total length and its flows, whereas the nodal reader retains all separations. The detail archive provides the complement needed to reconstruct the refined observable.

25 Analytic reconstruction of energy and refinement memory

The positive separations constructed from the dodecaphasic register determine an ordered partition and, by the preceding section, a Dirichlet form on its nodal values. Its refinement has two complementary operations: harmonic interpolation preserves the energy of inherited data, whereas each detail records the variation added between those data. Passage to unlimited depth consists in assembling this compatible collection through a norm controlling the sum of its details. The result concerns the one-dimensional analytic realization of energy; it retains as its antecedent the HMT construction of the register and the joint discrete structure of the continuum.

25.1 Nested partitions and interpolation spaces

Fix the initial partition $P_0 = \{0 = t_0 < t_1 < \dots < t_{12} = 1\}$, whose lengths are $d_j = t_j - t_{j-1} = K_j / \sum_i K_i > 0$. Let $(P_n)_{n \geq 0}$ be a sequence of nested finite partitions of $[0, 1]$, obtained by subdividing the preceding intervals. We write

$$|P_n| = \max\{b - a : [a, b] \text{ is an interval of } P_n\}, \quad \lim_{n \rightarrow \infty} |P_n| = 0.$$

Repeated subdivision of each interval into a fixed number of equal successors, at least two, satisfies this condition. For other rules, mesh contraction is an explicit hypothesis of this reader. Successor labels and their carry data accompany the subdivision and preserve the provenance of the refinement.

Let V_n be the complex space of continuous functions on $[0, 1]$ that are affine on each interval of P_n . The energy and energy norm are defined by

$$\mathcal{E}_n(f) = \sum_{[a,b] \in P_n} \frac{|f(b) - f(a)|^2}{b - a} = \int_0^1 |f'(x)|^2 dx, \quad \|f\|_{\mathcal{E}}^2 = |f(0)|^2 + \mathcal{E}_n(f). \quad (25.1)$$

The integral equality follows because the derivative of an affine function is constant on each interval. The energy thus agrees exactly with the nodal form already constructed. The inclusion $V_n \subset V_{n+1}$ interprets

the same affine function on a finer partition; it preserves both $f(0)$ and the energy and is isometric for $\|\cdot\|_{\mathcal{E}}$.

The initial value has a precise function: it fixes the constant component identified by the Dirichlet form as its kernel. Energy controls differences, and the pair consisting of the initial value and those differences determines the observable. In the Sobolev space $H^1([0, 1]; \mathbb{C})$, whose elements have an absolutely continuous representative with weak derivative in L^2 , this norm is equivalent to the usual norm. Indeed, $f(x) = f(0) + \int_0^x f'(t) dt$ controls the norm of f in terms of $|f(0)|$ and $\|f'\|_{L^2}$; integration of the same identity in turn controls $|f(0)|$ through $\|f\|_{L^2} + \|f'\|_{L^2}$.

25.2 Reconstruction of a function from its compatible values

Theorem 25.1 (Energy reconstruction and completion of interpolations). *Let $f_n \in V_n$ be a collection compatible under restriction to inherited nodes:*

$$f_{n+1}(a) = f_n(a) \quad \text{for every node } a \text{ of } P_n.$$

If $\sup_n \mathcal{E}_n(f_n) < \infty$, there is a unique function $f \in H^1([0, 1]; \mathbb{C})$, considered in its continuous representative, whose values at all these nodes are the prescribed ones. Moreover, $f_n \rightarrow f$ uniformly and in H^1 , and

$$\int_0^1 |f'(x)|^2 dx = \lim_{n \rightarrow \infty} \mathcal{E}_n(f_n). \quad (25.2)$$

Conversely, nodal interpolation of any element of H^1 produces a collection of this kind. Thus the completion of $\bigcup_n V_n$ in the energy norm is $H^1([0, 1]; \mathbb{C})$.

Proof. Set $g_n = f'_n \in L^2([0, 1]; \mathbb{C})$. For $m \geq n$, compatibility of the values at the endpoints of each interval $[a, b]$ of P_n gives

$$\frac{1}{b-a} \int_a^b g_m(x) dx = \frac{f_m(b) - f_m(a)}{b-a} = \frac{f_n(b) - f_n(a)}{b-a} = g_n|_{(a,b)}.$$

Let W_n be the space of L^2 functions constant on each interval of P_n , and let $Q_n : L^2 \rightarrow W_n$ be the orthogonal projection given by averages over these intervals. The preceding identity states $Q_n g_m = g_n$. Consequently, $g_m - g_n$ is orthogonal to g_n , and

$$\|g_m - g_n\|_{L^2}^2 = \mathcal{E}_m(f_m) - \mathcal{E}_n(f_n). \quad (25.3)$$

The energies are nondecreasing and bounded. The Pythagorean identity proves that (g_n) is a Cauchy sequence in L^2 , with limit g . Define

$$f(x) = f_0(0) + \int_0^x g(t) dt.$$

This function belongs to H^1 , has weak derivative g , and, by the Cauchy–Schwarz inequality, satisfies

$$\sup_{x \in [0,1]} |f_n(x) - f(x)| \leq \|g_n - g\|_{L^2} \rightarrow 0. \quad (25.4)$$

Uniform convergence preserves every prescribed nodal value; together with convergence of the derivatives in L^2 , it gives convergence in H^1 . Convergence of the derivative norms proves (25.2). The union of the nodes is dense because $|P_n| \rightarrow 0$; two continuous representatives with the same values on that union agree. This gives uniqueness.

For the converse, take $f \in H^1$ in its continuous representative, and let f_n be its interpolation at the nodes of P_n . Its derivative is $g_n = Q_n f'$. The projections Q_n are contractive on L^2 . For a continuous function v , the error $|Q_n v - v|$ is bounded by its modulus of continuity evaluated at $|P_n|$, which tends to

zero. Density of continuous functions in L^2 and contractivity imply $Q_n g \rightarrow g$ for every $g \in L^2$: given a continuous approximant v , use

$$\|Q_n g - g\|_{L^2} \leq 2\|g - v\|_{L^2} + \|Q_n v - v\|_{L^2}.$$

Applying this convergence to $g = f'$ gives convergence of the derivatives and, through (25.4), of the functions. Contractivity bounds their energies by $\|f'\|_{L^2}^2$, and convergence of the norms again proves the limit identity. Every function in H^1 is therefore an energy limit of elements of $\bigcup_n V_n$. Equivalence of norms and completeness of H^1 identify the stated completion. $\square$

The argument describes concretely what is preserved when passing from the network to the function. Each finite derivative records the average of all subsequent derivatives on its interval; refinement adds orthogonal components that retain this average. Bounded energy controls the total sum of these additions. The continuous value is then recovered by integrating the limit derivative and preserving the initial value. Through this explicit route, the analytic realization assembles the compatible information of the finite levels.

25.3 Orthogonal decomposition of memory by scales

Let H_n be harmonic interpolation of the values on P_n to the nodes of P_{n+1} , interpreted as a function in V_{n+1} , and define the detail

$$z_{n+1} = f_{n+1} - H_n f_n.$$

By (24.5), the detail vanishes at inherited nodes. Its derivative has zero average on each old interval and is orthogonal to W_n . For different levels, the derivative of an earlier detail belongs to that space of constant functions; the derivatives of the details are therefore mutually orthogonal. The same orthogonality holds against f'_0 .

Proposition 25.2 (Energy identity and recovery of the details). *For every finite depth N ,*

$$\mathcal{E}_N(f_N) = \mathcal{E}_0(f_0) + \sum_{n=0}^{N-1} \mathcal{E}_{n+1}(z_{n+1}). \quad (25.5)$$

If the collection satisfies the hypotheses of Theorem 25.1, the series on the right converges and its sum is $\int_0^1 |f'|^2$. The data f_0 and all the details determine the complete compatible collection and its analytic realization.

Proof. The finite identity follows by summing $\mathcal{E}_{n+1}(f_{n+1}) = \mathcal{E}_n(f_n) + \mathcal{E}_{n+1}(z_{n+1})$, the orthogonal decomposition of the preceding section. The summands are nonnegative, and the partial sums converge by (25.2). Finite reconstruction is performed recursively through $f_{n+1} = H_n f_n + z_{n+1}$; the preceding theorem then determines the limit function. Conversely, a function in H^1 determines its interpolations and, by taking differences, all these details. Each arrow thus preserves its own data. $\square$

Variational elimination of interior nodes selects the zero detail at the eliminated level. Preserving the details allows recovery of any compatible finite-energy configuration. The effective response and the complete configuration thus have a precise relationship: the former records the minimum over the interior variables; the latter adds the orthogonal components and their energy.

25.4 Transport of the analytic realization under the positive flow

The direction $u = P_3 P_1 K$, recovered before energy evaluation [2], determines the separation collection in (24.10). Its parameter $s \in \mathbb{R}$ describes this analytic deformation. Write

$$d_j(s) = \frac{K_j e^{su_j}}{\sum_{\ell} K_{\ell} e^{su_{\ell}}}, \quad t_i(s) = \sum_{j=1}^i d_j(s), \quad a_j(s) = \frac{d_j(s)}{d_j(0)} > 0.$$

Let $F_s : [0, 1] \rightarrow [0, 1]$ be the map sending each $t_i(0)$ to $t_i(s)$ and affine on the initial intervals. It is an increasing homeomorphism, with slope $a_j(s)$ on the interval $[t_{j-1}(0), t_j(0)]$. Its inverse is also piecewise affine, and both maps are Lipschitz.

Suppose each subdivision inherits the length fractions and direction of its predecessor interval, as in the preceding section. All refined nodes are then transported through the same F_s . Denote their partitions by $P_n(s) = F_s(P_n(0))$ and their interpolation spaces by $V_n(s)$. The map

$$U_s f = f \circ F_s^{-1}$$

transports $V_n(0)$ to $V_n(s)$, preserves corresponding nodal values, and extends this action to H^1 .

Theorem 25.3 (Energy equivalence and commutation with refinement). *For every $f \in H^1([0, 1]; \mathbb{C})$,*

$$\int_0^1 |(U_s f)'(y)|^2 dy = \sum_{j=1}^{12} \frac{1}{a_j(s)} \int_{t_{j-1}(0)}^{t_j(0)} |f'(x)|^2 dx. \quad (25.6)$$

For s in any compact interval, the energy norms before and after transport are uniformly equivalent, and $|P_n(s)| \rightarrow 0$ uniformly on that compact interval. Harmonic interpolation and restriction to inherited nodes commute with U_s ; reconstruction and minimization of the details are transported both at finite depth and in the H^1 realization.

Proof. On initial interval j , the chain rule and change of variable $y = F_s(x)$ give $(U_s f)'(F_s(x)) = f'(x)/a_j(s)$ and $dy = a_j(s) dx$. Integration and summation over the intervals prove (25.6). The formula holds for absolutely continuous representatives, and its right-hand side is finite.

Fix a compact parameter interval. Continuity and positivity of the twelve factors give constants $0 < a_- \leq a_+ < \infty$ such that $a_- \leq a_j(s) \leq a_+$ for every j and every s in that compact interval. Therefore,

$$\frac{1}{a_+} \mathcal{E}(f) \leq \mathcal{E}(U_s f) \leq \frac{1}{a_-} \mathcal{E}(f), \quad |P_n(s)| \leq a_+ |P_n(0)|.$$

Since $(U_s f)(0) = f(0)$, we also obtain

$$\min\{1, a_+^{-1}\} \|f\|_{\mathcal{E}}^2 \leq \|U_s f\|_{\mathcal{E}}^2 \leq \max\{1, a_-^{-1}\} \|f\|_{\mathcal{E}}^2.$$

These bounds prove uniform equivalence and preservation of the decreasing-mesh condition.

On each predecessor edge, the interpolation coefficients are the partial sums of the inherited fractions, independent of s . If $H_n(s)$ denotes its interpolation, then $U_s H_n(0) = H_n(s) U_s$ on the corresponding spaces. Restriction commutes because the nodes are transported with their values. The decomposition into harmonic function and detail is preserved, and the Schur identity in (24.10) preserves its variational characterization. Finally, bounded continuity of U_s allows passage to the H^1 limit. In particular, a collection reconstructed before transport subsequently converges to $U_s f$, with the same transported nodal values. $\square$

The law (25.6) identifies the exact metric factors of the deformation: the length change of each interval inversely modifies its energy contribution. Equivalence of norms preserves finite-energy configurations, and the detail archive preserves their reconstruction. This gives a complete analytic realization of this reader of the positive register. Its compatibility with canonical forms is articulated on subdivisions of a fixed domain: summing forms preserves the meromorphic object, whereas variational reduction and the details preserve the energy response and refined observable. Identification with higher-rank amplituhedral forms retains the specific conditions of covering, degree, and residue correspondence established for that realization.

26 Nonadic transports and memory preservation

The reconstruction in Section 16 provides an effective link between transport memory and the geometry of the dodecaphasic record. The record generated by APP-TRIT-TPK determines its incidence readings; the two compressions preserve, in separate components, the terminal representation and its complements; combining the memories with the charge restores the twelve coordinates. The geometric direction and dimensional projector are obtained after this restoration. The following development establishes the operator balance underlying this chain and its behaviour under refinement, spectral realization, and change of metric [6, 2].

26.1 Nine-phase calendar and orthogonal decomposition

Let $\mathcal{H}$ be a complex Hilbert space and let $C \in \mathcal{B}(\mathcal{H})$. The orthogonal sum of nine copies, $\mathcal{H}^{\oplus 9}$, separately preserves the nine positions of the calendar. We define

$$\begin{aligned} S_C(v_0, \dots, v_8) &= (Cv_8, v_0, \dots, v_7), \\ J : \mathcal{H} &\longrightarrow \mathcal{H}^{\oplus 9}, \quad Jv = \frac{1}{3}(v, \dots, v). \end{aligned} \quad (26.1)$$

The normalization $1/3 = 9^{-1/2}$ gives $J^*J = I$. The operator $P = JJ^*$ is the orthogonal projection onto the uniform section; its action replaces the nine components by their mean. Each component crosses the junction containing C once during one cycle, so that

$$S_C^9 = I_9 \otimes C. \quad (26.2)$$

The return of the calendar position thus realizes an action of C on the memory fibre.

Definition 26.1. The uniform compression of the transport and its complement are

$$T_C = J^*S_CJ = \frac{8I + C}{9}, \quad \eta_C = (I - P)S_CJ. \quad (26.3)$$

The first map has codomain $\mathcal{H}$; the second has codomain $J(\mathcal{H})^\perp \subset \mathcal{H}^{\oplus 9}$.

For $z = (C - I)v$, the complement is written

$$\eta_C v = \frac{1}{27}(8z, -z, -z, -z, -z, -z, -z, -z, -z). \quad (26.4)$$

The sum of its components is zero. The coefficients of the decomposition therefore arise from averaging eight identical continuations and one transported continuation.

Proposition 26.2 (Balance of compression and of the memory junction). *For every $C \in \mathcal{B}(\mathcal{H})$,*

$$\eta_C^* \eta_C = \frac{8}{81}(I - C)^*(I - C), \quad (26.5)$$

$$I - T_C^* T_C = \eta_C^* \eta_C + \frac{1}{9}(I - C^* C). \quad (26.6)$$

*If $C^*C = I$, the map $v \mapsto (T_C v, \eta_C v)$ is isometric.*

Proof. Expression (26.4) gives

$$\|\eta_C v\|^2 = \frac{8^2 + 8}{27^2} \|(C - I)v\|^2 = \frac{8}{81} \|(C - I)v\|^2.$$

Polarization proves (26.5). On the other hand,

$$T_C^* T_C = \frac{64I + 8C + 8C^* + C^* C}{81}.$$

Adding (26.5) yields

$$T_C^* T_C + \eta_C^* \eta_C = \frac{8I + C^* C}{9}.$$

Subtracting this result from I gives (26.6). When $C^* C = I$, the sum of the two Gram products is the identity, which preserves all inner products. $\square$

The term $\eta_C^* \eta_C$ measures dispersion among the nine components. The term $(I - C^* C)/9$ separately records the change in norm at the memory junction. The unitary realization places all visible variation in the first term and preserves the norm in the full orthogonal sum.

Proposition 26.3 (Preservation of an observable). *Let C be unitary and $A = A^* \in \mathcal{B}(\mathcal{H})$ satisfy $C^* A C = A$. Then*

$$A = T_C^* A T_C + \eta_C^* (I_9 \otimes A) \eta_C = T_C^* A T_C + \frac{8}{81} (I - C)^* A (I - C). \quad (26.7)$$

For $A \geq 0$, both terms in the sum are positive.

Proof. The complement formula, applied to the sesquilinear product defined by A , proves the second equality. Expanding the two terms gives

$$\frac{1}{81} (64A + 8AC + 8C^* A + C^* A C + 8A - 8AC - 8C^* A + 8C^* A C) = \frac{8A + C^* A C}{9} = A.$$

Positivity is preserved under a map of the form $B \mapsto L^* B L$. $\square$

26.2 Iterating compression and preserving the fixed sector

In this subsection we fix a unitary operator C and write $T = T_C$, $\eta = \eta_C$. The transport S_C^N retains all phases for N steps. The power T^N , by contrast, successively applies uniform compression; at each stage $\eta T^j v$ preserves the complement of the previous representation. The following operators express this distinction.

Proposition 26.4 (Isometric record at a finite horizon). *For every integer $N \geq 1$, the map*

$$V_N v = (T^N v, \eta v, \eta T v, \dots, \eta T^{N-1} v) \quad (26.8)$$

from $\mathcal{H}$ to $\mathcal{H} \oplus (\mathcal{H}^{\oplus 9})^{\oplus N}$ is isometric. If A satisfies the hypotheses of (26.7), then

$$A = T^{*N} A T^N + \sum_{j=0}^{N-1} T^{*j} \eta^* (I_9 \otimes A) \eta T^j. \quad (26.9)$$

Proof. Multiplying (26.7) by T^{*j} and T^j gives

$$T^{*j} A T^j - T^{*(j+1)} A T^{j+1} = T^{*j} \eta^* (I_9 \otimes A) \eta T^j.$$

The telescoping sum proves (26.9). For $A = I$, the right-hand side is the Gram product of V_N . The step from N to $N + 1$ replaces only $T^N v$ by $(T^{N+1} v, \eta T^N v)$, preserving all memory components previously produced. $\square$

Theorem 26.5 (Compression limit and unlimited memory). *Let P_{fix} be the orthogonal projection onto $\ker(I - C)$. Then*

$$\lim_{N \rightarrow \infty} T^N = P_{\text{fix}}, \quad (26.10)$$

$$I = P_{\text{fix}} + \sum_{j=0}^{\infty} T^{*j} \eta^* \eta T^j. \quad (26.11)$$

The series converges strongly. For every bounded self-adjoint observable satisfying $C^ A C = A$, the following is also preserved:*

$$A = P_{\text{fix}} A P_{\text{fix}} + \sum_{j=0}^{\infty} T^{*j} \eta^* (I_9 \otimes A) \eta T^j. \quad (26.12)$$

Proof. The spectral theorem for the unitary operator realizes C as the coordinate function z on the unit circle. Compression corresponds to $t(z) = (8 + z)/9$, and

$$1 - |t(z)|^2 = \frac{8}{81} |1 - z|^2, \quad |z| = 1. \quad (26.13)$$

Thus $t(z)^N$ converges to zero at every $z \neq 1$ and retains the value one at $z = 1$. The bound $|t(z)^N| \leq 1$ allows dominated convergence to be applied to each vector spectral measure. This yields (26.10) and, by the same argument, $T^{*N} \rightarrow P_{\text{fix}}$ strongly. The boundedness of A , together with $\|T^N\| \leq 1$, gives $T^{*N}AT^N \rightarrow P_{\text{fix}}AP_{\text{fix}}$. The limit of (26.9) proves both identities. $\square$

The record

$$V_\infty v = (P_{\text{fix}}v, \eta v, \eta T v, \eta T^2 v, \dots)$$

is therefore an isometry. The fixed sector preserves the part that persists in the terminal representation; the other components remain in the memory of the complements.

For the bilateral shift $Ce_n = e_{n+1}$ on $\ell^2(\mathbb{Z})$, a fixed sequence is constant. Square summability forces it to vanish, so $P_{\text{fix}} = 0$. The entire norm is then represented in the complements. The spectrum of this shift is the full circle, and the spectral maximum of $|t(z)|^N$ is one; hence $\|T^N\| = 1$ for every N . Strong convergence retains this non-uniform behaviour.

For a cyclic permutation of d positions, the fixed sector is the line of uniform vectors. If the permutation has several orbits, each orbit contributes its own uniform direction. This dimension determines the terminal part that must be preserved.

Proposition 26.6 (Compression of an order-three transport). *Let C be unitary with $C^3 = I$. Then*

$$P_{\text{fix}} = \frac{I + C + C^2}{3}, \quad T_C^* T_C = P_{\text{fix}} + \frac{19}{27}(I - P_{\text{fix}}). \quad (26.14)$$

Proof. The average of the three powers is the orthogonal projector onto the fixed vectors. Since $C^* = C^2$,

$$T_C^* T_C = \frac{65I + 8(C + C^2)}{81} = \frac{57I + 24P_{\text{fix}}}{81},$$

which agrees with (26.14). $\square$

In the twelve-position realization on $\mathbb{R}^{12}$, $(Sx)_i = x_{i+1}$ with indices modulo twelve, S^3 has order four and S^4 has order three. The four orbits of S^4 give rank $P_{\text{fix}} = 4$. The exceptional realization g_c of Article X satisfies $g_c^3 = I$ and $\ker(I - g_c) = 0$ on its own carrier; there (26.14) reduces to $T_{g_c}^* T_{g_c} = 19I/27$ [2]. The coefficient and the fixed sector are transported together with the domain.

The radial contraction $M = qC$, $0 < q < 1$, has another exact record:

$$L_N v = (\sqrt{1 - q^2} v, \sqrt{1 - q^2} qCv, \dots, \sqrt{1 - q^2} q^{N-1} C^{N-1} v, q^N C^N v). \quad (26.15)$$

Its Gram product is the identity because $(1 - q^2) \sum_{j=0}^{N-1} q^{2j} + q^{2N} = 1$. The terminal has norm $q^N \|v\|$, and the infinite record preserves the inner product. The factor $q = 5/9$ of the central differential mode in Article VIII applies to a full cycle; its uniform distribution over nine steps uses $q^{1/9} S_C$, whose ninth power is $q(I_9 \otimes C)$ [6].

26.3 Record reduction and trace preservation

The fixed direction of the complement in the nine components allows a minimal realization in two summands. Define

$$D_C = \frac{\sqrt{8}}{9}(I - C).$$

The preceding identities give $T_C^* T_C + D_C^* D_C = I$ for unitary C . The change of sign relative to the expression for η_C corresponds to an orientation of the complementary component and preserves its Gram product.

Proposition 26.7 (Positive channel of compression with memory). *For unitary C , the map*

$$\mathcal{Q}_C(X) = T_C X T_C^* + D_C X D_C^* = \frac{8X + CXC^*}{9} \quad (26.16)$$

is completely positive on $\mathcal{B}(\mathcal{H})$ and preserves the identity. Its restriction to trace-class operators preserves the trace.

Proof. Expanding the two terms in the first expression cancels the products CX and XC^* . The remaining coefficients are $72/81 = 8/9$ for X and $9/81 = 1/9$ for CXC^* , proving (26.16). For any integer $m \geq 1$, the amplification to matrices of order m has the form

$$X \mapsto (I_m \otimes T_C)X(I_m \otimes T_C)^* + (I_m \otimes D_C)X(I_m \otimes D_C)^*.$$

Each summand preserves positivity; hence the map is completely positive. The second expression in (26.16) gives $\mathcal{Q}_C(I) = I$. If X is trace class, invariance of the trace under unitary conjugation implies $\text{Tr } \mathcal{Q}_C(X) = \text{Tr } X$. $\square$

The map (26.16) is also obtained by preparing the isometry $v \mapsto T_C v \otimes e_0 + D_C v \otimes e_1$ and taking the partial trace over $\mathbb{C}^2$. Preservation of the full vector belongs to the isometry; preservation of the trace of the reduced state belongs to $\mathcal{Q}_C$. Both statements follow from the same orthogonal balance.

A complementary preservation appears in the nonadic prolongation of finite algebras. For $\mathcal{A}_d = M_{9^d}(\mathbb{C})$, with $\tau_d(a) = 9^{-d} \text{Tr } a$, the maps

$$\iota_d(a) = a \otimes I_9, \quad J_{d,j}(a) = a \otimes p_j, \quad p_j = |e_j\rangle\langle e_j|$$

satisfy

$$\tau_{d+1}(\iota_d(a)) = \tau_d(a), \quad \iota_d(I) = I, \quad (26.17)$$

$$\tau_{d+1}(J_{d,j}(a)) = \frac{\tau_d(a)}{9}, \quad J_{d,j}(I) = I \otimes p_j. \quad (26.18)$$

The proof uses $\text{Tr}(a \otimes b) = \text{Tr}(a) \text{Tr}(b)$, $\text{Tr } I_9 = 9$ and $\text{Tr } p_j = 1$. The first inclusion preserves the entire fibre and the unit; the second preserves a marked continuation and its support.

This distinction also appears in the tree of enriched histories. Let $\mathcal{H}_n$ be the surviving prefixes of length n , and let $d(h) \geq 1$ be the number of retained outgoing emissions of h . Each emission with multiplicity is represented by distinct marked successors; the following sum runs over this set of $d(h)$ successors. The weight law

$$\mu_{n+1}(h\epsilon) = \mu_n(h) p_h(\epsilon), \quad p_h(\epsilon) = d(h)^{-1}$$

preserves $\sum_{\epsilon} \mu_{n+1}(h\epsilon) = \mu_n(h)$. Therefore, on $\mathcal{L}_n = L^2(\mathcal{H}_n, \mu_n)$, the pullback

$$R_n f(h\epsilon) = f(h)$$

is isometric and preserves the unit function. Its adjoint is

$$E_n g(h) = \sum_{\epsilon} p_h(\epsilon) g(h\epsilon), \quad E_n R_n = I, \quad E_n 1 = 1.$$

Indeed, grouping the sum for $\|R_n f\|^2$ by predecessor yields $\sum_h \mu_n(h) |f(h)|^2$; the same grouping in the inner product determines E_n . Memory and multiplicities remain present in the domain of histories. These maps constitute the functional realization of the refinement described in Article X [2].

26.4 Spectral chart and radius reciprocity

The preceding transport admits spectral realizations with their own domains. The Möbius chart of VIII acts, in finite coordinates, by

$$\tau(z) = \frac{z-1}{z}, \quad z \in \mathbb{C} \setminus \{0\}. \quad (26.19)$$

The inverse is $z = (1 - \tau)^{-1}$, for $\tau \neq 1$. The extension to the Riemann sphere sends 0 to ∞ and ∞ to 1.

Proposition 26.8 (Spectral mirror and reciprocal radius). *Let $z^\sharp = 1 - \bar{z}$. For $z \notin \{0, 1\}$,*

$$\tau(z^\sharp) = \frac{1}{\tau(z)}. \quad (26.20)$$

If $z = \sigma + it$, then

$$|\tau(z)|^2 = \frac{(\sigma-1)^2 + t^2}{\sigma^2 + t^2}; \quad |\tau(z)| = 1 \iff \sigma = \frac{1}{2}. \quad (26.21)$$

Proof. Direct substitution gives

$$\tau(1 - \bar{z}) = \frac{-\bar{z}}{1 - \bar{z}} = \frac{\bar{z}}{\bar{z} - 1} = \frac{1}{(z-1)/z}.$$

The ratio of squared moduli yields (26.21); equality between numerator and denominator is equivalent to $-2\sigma + 1 = 0$. $\square$

Writing $\tau(z) = re^{i\theta}$, with $r > 0$, the mirror preserves θ and transforms r into r^{-1} . Its fixed set in this chart is the unit circle. Conjugation $z \mapsto \bar{z}$, in turn, produces $\tau \mapsto \bar{\tau}$ and reverses the angular orientation. The two involutions preserve, respectively, the angular coordinate and the radial coordinate.

On $z = 1/2 + i\gamma$, with $\gamma \in \mathbb{R}$,

$$\tau_\gamma = \frac{\gamma + i/2}{\gamma - i/2}, \quad \gamma = \frac{i}{2} \frac{\tau_\gamma + 1}{\tau_\gamma - 1} = \frac{1}{2} \cot \frac{\theta}{2}, \quad \tau_\gamma = e^{i\theta}. \quad (26.22)$$

Compactification preserves the height through the inverse; the point $\tau = 1$ corresponds to infinity.

Let $\mathcal{D} = C_c^\infty(\mathbb{R}; \mathbb{C})$, with its usual test-function topology. The positive realization of the Weil form explicitly uses its positive semidefiniteness, its translation invariance, and its continuity in that topology, on the domain proved by the corresponding spectral reader. If $\mathcal{W}(f, g) = \langle Ef, Eg \rangle$, its kernel is $\mathcal{N} = \{f \in \mathcal{D} : \mathcal{W}(f, f) = 0\}$, and we define

$$\mathcal{H}_\mathcal{W} = \overline{\mathcal{D}/\mathcal{N}}.$$

Translations induce a strongly continuous unitary group $U_t[f] = [f(\cdot - t)]$: invariance preserves the norm, and continuity of translations in $\mathcal{D}$ and of the form proves strong continuity on a dense subspace and then on the entire completion. With the convention $U_t = e^{itH}$, its self-adjoint generator acts on test-function classes by $H[f] = [if']$. The Cayley transform of this realization is the bounded operator

$$C_\mathcal{W} = (H + iI/2)(H - iI/2)^{-1} = I + i(H - iI/2)^{-1}. \quad (26.23)$$

Functional calculus makes it unitary, since the quotient in (26.22) has modulus one for real γ . The compression identity applies to $C_\mathcal{W}$ on $\mathcal{H}_\mathcal{W}$. The space, the domain of the generator, and the positivity that permits its construction are the premises of this spectral realization [6].

Bilateral memory, by contrast, acts on $\ell^2(\mathbb{Z})$, and the permutations S^3, S^4 act on $\mathbb{R}^{12}$. Each realization determines its spectrum, its fixed sectors, and the action of one cycle. The respective transport maps specify the compositions between these spaces.

26.5 Cayley on test functions and preservation of the arithmetic form

The same arithmetic representation admits a Cayley reader defined directly on test functions. Fix $a > b > 1/2$ and

$$\mathcal{E}_b = \left\{ f \in C^\infty(\mathbb{R}) : \sup_x e^{b|x|} |f^{(j)}(x)| < \infty \text{ for every } j \geq 0 \right\}.$$

Smooth compactly supported functions belong to $\mathcal{E}_b$. We define

$$(R_a^+ f)(x) = \int_0^\infty e^{-at} f(x+t) dt, \quad (R_a^- f)(x) = \int_0^\infty e^{-at} f(x-t) dt, \quad (26.24)$$

$$C_a f = f - 2aR_a^+ f, \quad C_a^{-1} f = f - 2aR_a^- f. \quad (26.25)$$

Each derivative of $R_a^\pm f$ satisfies the bound for the corresponding seminorm of f , multiplied by $(a-b)^{-1}$. This follows from $e^{b|x|} |f(x \pm t)| \leq M_0 e^{bt}$. Both maps preserve $\mathcal{E}_b$.

With the transform $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-i\xi x} dx$, Fubini's theorem gives

$$\widehat{C_a f}(\xi) = c_a(\xi) \widehat{f}(\xi), \quad c_a(\xi) = \frac{\xi - ia}{\xi + ia}. \quad (26.26)$$

The multiplier of the second formula in (26.25) is c_a^{-1} , proving the inverse. For $\xi \in \mathbb{R}$, $|c_a(\xi)| = 1$; Plancherel provides its unitary extension to $L^2(\mathbb{R})$.

To specify arithmetic preservation, let

$$\mathcal{C}_{f,g}(u) = \int_{\mathbb{R}} f(x) \overline{g(x-u)} dx, \quad P_f^\pm = \int_{\mathbb{R}} f(x) e^{\pm x/2} dx.$$

The lengths $\ell_{p,k} = k \log p$ and weights $w_{p,k} = (\log p) p^{-k/2}$ are the prime-power representations of the arithmetic reader. The complete form of Article VIII is written

$$\begin{aligned} \mathcal{W}(f, g) &= c_\Gamma \mathcal{C}_{f,g}(0) \\ &+ \int_0^\infty \frac{e^{-u/2}}{1 - e^{-2u}} [2\mathcal{C}_{f,g}(0) - \mathcal{C}_{f,g}(u) - \mathcal{C}_{f,g}(-u)] du \\ &- \sum_{p,k} w_{p,k} [\mathcal{C}_{f,g}(\ell_{p,k}) + \mathcal{C}_{f,g}(-\ell_{p,k})] + P_f^+ \overline{P_g^-} + P_f^- \overline{P_g^+}, \end{aligned} \quad (26.27)$$

where $c_\Gamma = \psi(1/4) - \log \pi$, and $\psi = \Gamma'/\Gamma$ is the logarithmic derivative of the gamma function. These functions and constants enter the subsequent analytic realization of the arithmetic reader of the corpus.

The bound

$$|\mathcal{C}_{f,g}(u)| \leq M_f M_g (|u| + b^{-1}) e^{-b|u|}$$

controls the prime sum by a convergent series of the type $\sum_{n \geq 2} (\log n)(1 + \log n) n^{-b-1/2}$. The symmetric difference of correlations is $O(u^2)$ at the origin, while the gamma weight is $O(u^{-1})$. At infinity that weight decays exponentially. The margin $b > 1/2$ also ensures convergence of the two polar readers. This fixes the domain of all terms in (26.27).

Proposition 26.9 (Arithmetic invariance and nonadic balance). *For $f, g \in \mathcal{E}_b$,*

$$\mathcal{W}(C_a f, C_a g) = \mathcal{W}(f, g). \quad (26.28)$$

With $\mathcal{T}_a = (8I + C_a)/9$, we have

$$\mathcal{W}(f, g) = \mathcal{W}(\mathcal{T}_a f, \mathcal{T}_a g) + \frac{8}{81} \mathcal{W}((I - C_a)f, (I - C_a)g). \quad (26.29)$$

Proof. The multiplier c_a is unitary on the real line and commutes with translations. Consequently, $\mathcal{C}_{C_a f, C_a g}(u) = \mathcal{C}_{f,g}(u)$ for every u . By Fubini, the polar readers transform according to

$$P_{C_a f}^+ = -\frac{a-1/2}{a+1/2} P_f^+, \quad P_{C_a f}^- = -\frac{a+1/2}{a-1/2} P_f^-.$$

Their factors are real and reciprocal, so that each cross polar product remains unchanged. Combining the terms of (26.27) proves (26.28). The sesquilinear expansion of the right-hand side of (26.29) cancels the mixed terms and yields $(8\mathcal{W}(f, g) + \mathcal{W}(C_a f, C_a g))/9$, equal to $\mathcal{W}(f, g)$. $\square$

Finite iteration preserves the complete form:

$$\mathcal{W}(f, g) = \mathcal{W}(\mathcal{T}_a^N f, \mathcal{T}_a^N g) + \frac{8}{81} \sum_{j=0}^{N-1} \mathcal{W}((I - C_a)\mathcal{T}_a^j f, (I - C_a)\mathcal{T}_a^j g).$$

This sesquilinear identity simultaneously preserves gamma, primes, and polar terms. Its evaluation in a positive realization allows an interpretation as an energy distribution; its algebraic validity precedes that interpretation. The margin $a > b > 1/2$ belongs to the exponential test-function domain of this construction. The Cayley transform (26.23) with parameter $1/2$ belongs to the Hilbert-space domain of its generator.

26.6 Metric covariance of the dodecaphasic flow

The reconstruction in Section 16 determines $u_K \in \mathbb{R}^{12}$, with $\mathbf{1}^\top u_K = 0$ and $u_K \neq 0$. We write $A_K = \text{diag}(u_K)$, with the direction and normalization declared in (15.2), and

$$g_s = e^{sA_K} \in \text{GL}(12, \mathbb{R}).$$

The zero trace of A_K implies $\det g_s = 1$. Each g_s transports the weights and geometric coordinates according to the corresponding realization. Preserving the memory balance between charts also requires transporting the inner product.

Proposition 26.10 (Metric transport of compression and complement). *Let V be a finite-dimensional Hermitian space, C_0 unitary, and $g : V \rightarrow V$ invertible. We write $g^{-*} = (g^{-1})^*$ and define*

$$C_g = gC_0g^{-1}, \quad G_g = g^{-*}g^{-1}, \quad \langle v, w \rangle_g = \langle G_g v, w \rangle. \quad (26.30)$$

Then $C_g^* G_g C_g = G_g$, and

$$T_{C_g} g = g T_{C_0}, \quad (26.31)$$

$$\eta_{C_g} g = g^{\oplus 9} \eta_{C_0}, \quad (26.32)$$

$$G_g = T_{C_g}^* G_g T_{C_g} + \frac{8}{81} (I - C_g)^* G_g (I - C_g). \quad (26.33)$$

Proof. The positivity of G_g follows from $\langle G_g v, v \rangle = \|g^{-1}v\|^2$. Substituting (26.30),

$$C_g^* G_g C_g = g^{-*} C_0^* g^* g^{-*} g^{-1} g C_0 g^{-1} = g^{-*} C_0^* C_0 g^{-1} = G_g.$$

The identity $C_g g = g C_0$ gives (26.31) upon adding $8g$ and dividing by nine. It also gives $(C_g - I)g = g(C_0 - I)$; substituting it into (26.4) proves (26.32). Finally, the expansion of (26.7), with $A = G_g$ and $C_g^* G_g C_g = G_g$, proves (26.33). $\square$

Applying the proposition to $g = g_s$ and to the operators $C_0 = S^3, S^4$ of the dodecaphasic realization transports the compressions, the memories, and their inner products through explicit commutative squares. The identification $g_s : (V, \langle \cdot, \cdot \rangle) \rightarrow (V, \langle \cdot, \cdot \rangle_{g_s})$ is isometric. The formula therefore also transports the fixed sectors and the records at every depth:

$$P_{\text{fix},s} = g_s P_{\text{fix},0} g_s^{-1}.$$

The projector on the right-hand side is orthogonal with respect to G_{g_s} .

The volume condition and the metric condition have distinct effects. For the rational matrix

$$g = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \quad \det g = 1,$$

direct substitution into the compression gives

$$\frac{8I + g}{9} = \begin{pmatrix} 10/9 & 0 \\ 0 & 17/18 \end{pmatrix}, \quad T_g^* T_g + \frac{8}{81}(I - g)^*(I - g) = \frac{8I + g^* g}{9}.$$

The first coordinate expands in the Euclidean metric. The balance (26.33), by contrast, preserves the metric unit when a unitary operator C_0 is transported together with its inner product. The determinant controls volume; G_g controls norms and cross products.

The flow satisfies $g_{-s} = g_s^{-1}$. The radius reciprocity in (26.20) belongs to the spectral chart, and inversion of g_s belongs to the dimensional realization of the record. The constructive link used here preserves the explicit maps from memory to K and from K to its geometric direction, followed by the metric transport of each realization. This composition maintains, within a single development, exact reconstruction, positive refinement, boundary recording, and preservation of inner products.

27 Ternary incidence, discriminant gluing, and the Leech construction

The lattice construction uses the incidence information that accompanies the dodecaphasic record. Its domain preserves the ternary words, the frame in which they are represented, the pre-carry orientation, and the flag distinguishing one of the twelve positions. The passage to a lattice transmits these data through successive operations: linear encoding, incidence selection, discriminant gluing, and formation of a neighbour. Lattice classification enters after proving positivity, evenness, self-duality, and the absence of roots in the resulting construction.

These operations continue the same APP-TRIT-TPK generation and the same compatible refinement of the joint continuum that precede numerical reading [5]. The finite matrix, code, and metric defined below are realizations with precise domains. The lattice proof receives the marked state; the final scalar of a constant plays the role of neither a glue code nor a length.

27.1 Elliptic chart and incidence transport

The six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport, admit a six-position chart. We identify it with $\mathbb{P}^1(\mathbb{F}_5) = \{\infty, 0, 1, 2, 3, 4\}$, in that order. This identification fixes coordinates for realizing the quadratic relation of the elliptic regime of TRIT.

Let $\chi : \mathbb{F}_5 \rightarrow \{0, 1, -1\}$ be the quadratic character: $\chi(0) = 0$, $\chi(1) = \chi(4) = 1$, and $\chi(2) = \chi(3) = -1$. We define the integer matrix C_P with zero diagonal, $(C_P)_{\infty, a} = (C_P)_{a, \infty} = 1$, and finite entries $(C_P)_{a, b} = \chi(b - a)$. Its square is $5I_6$. Indeed, each diagonal entry of the square sums five units. The off-diagonal entries involving ∞ vanish because the sum of χ is zero. For two distinct finite indices, the contribution of one from position ∞ cancels with $\sum_c \chi(c - a)\chi(c - b) = -1$. This last identity is verified directly on the five residues, or through the affine substitution taking (a, b) to $(0, 1)$.

Reduction modulo three produces the symmetric matrix

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6 \quad \text{in } \mathbb{F}_3. \quad (27.1)$$

The equality preserves the elliptic type in this finite codomain. The finite field and the subsequent Archimedean realization of the imaginary unit have different domains.

For a matrix $A \in M_6(\mathbb{F}_3)$ and a row word $w \in \mathbb{F}_3^6$, we write

$$\gamma_A(w) = (w, wA) \in \mathbb{F}_3^{12}.$$

If $f \in \text{GL}_6(\mathbb{F}_3)$, then

$$\gamma_{fAf^{-1}}(wf^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (27.2)$$

The first half is wf^{-1} ; the second is $wf^{-1}fAf^{-1} = wAf^{-1}$, proving the identity.

Let X_n^{mar} be the space of enriched prefixes of length n , with a declared initial frame, and let $p_{n,m} : X_n^{\text{mar}} \rightarrow X_m^{\text{mar}}$ be its truncations. At step j , the prefix contains a visible word w_j and a frame f_j . The frame update has the form $f_{j+1} = g_j(x)f_j$, where $g_j(x)$ depends only on information already contained in the prefix. A compatible prolongation therefore preserves all previous frames.

Proposition 27.1 (Naturality of incidence transport). *The map*

$$\mathcal{Q}_n(x) = ((f_j, \gamma_{f_j A_W f_j^{-1}}(w_j)))_{0 \leq j < n} \quad (27.3)$$

satisfies $r_{n,m}\mathcal{Q}_n = \mathcal{Q}_m p_{n,m}$, where $r_{n,m}$ truncates to the first m encoded components. It therefore induces a map between the inverse limits of the prefix and marked-incidence systems.

Proof. Two compatible prefixes have the same words and the same transport operators up to step $m-1$. Starting from the same initial frame, the recursive relation gives by induction the same f_j up to that step. The first m components of (27.3) therefore agree. The universal property of the inverse limit produces the prolongation map. $\square$

In each frame, the first projection of γ_A exactly recovers the visible word. This scope is local: deep memory is preserved in the enriched state, and its recovery requires the corresponding coordinates. Passing from words to supports retains even less information. Likewise, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ also transports the Hamming metric and the incidence of the chart. Monomial changes preserve them directly. The supports used in this section are computed in the explicit chart (27.1).

The preceding naturality makes explicit the incidence reading that accompanies the numerical cylinders. Cylinder nesting and restriction of (27.3) act on the same compatible prefix, preserving respectively its positional evaluation and its marked relations. The ambient space $\mathbb{F}_3^6$, of 729 words, provides the linear domain of the encoding; it remains distinct from the 243 visible words and the 468 emissions in the admissible APP-TRIT-TPK census.

27.2 Self-dual ternary code and enumeration of its weights

Definition 27.2 (Code of the dodecaphasic chart). The linear code associated with (27.1) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

The weight $\text{wt}(c)$ counts the nonzero coordinates of c , and $\text{supp}(c) \subset \{1, \dots, 12\}$ is its set of nonzero positions. A hexad is the support of a word of weight six; their collection is denoted by $\mathcal{H}$.

Theorem 27.3 (Self-duality, minimum distance, and enumerator). *The code C_W is self-dual with parameters $[12, 6, 6]_3$ and has enumerator*

$$W_{C_W}(z) = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (27.4)$$

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six. By the symmetry of A_W and (27.1),

$$[I_6 \mid A_W][I_6 \mid A_W]^T = I_6 + A_W^2 = 0.$$

Thus the code is self-orthogonal of dimension half its length and is consequently self-dual.

The remaining enumeration is a finite integer operation on the explicit matrix. For $r \in \{0, \dots, 6\}$, we group the words by $\text{wt}(w) = r$. The following table gives the number of words of each total weight:

r	0	6	9	12	total
0	1	0	0	0	1
1	0	12	0	0	12
2	0	60	0	0	60
3	0	120	40	0	160
4	0	60	180	0	240
5	0	12	180	0	192
6	0	0	40	24	64
total	1	264	440	24	729

Each row contains $\binom{6}{r}2^r$ words. To verify all its entries, let q be an integer from 0 to 728, and define its six ternary digits

$$d_i(q) = \left\lfloor \frac{q}{3^{6-i}} \right\rfloor \bmod 3, \quad e_j(q) = \left(\sum_{i=1}^6 d_i(q)(A_W)_{ij} \right) \bmod 3.$$

The row and column of the contribution of q are, respectively,

$$r(q) = \sum_i \mathbf{1}_{\{d_i(q) \neq 0\}}, \quad h(q) = r(q) + \sum_j \mathbf{1}_{\{e_j(q) \neq 0\}}.$$

The accumulation recurrence starts with $T_{r,h}^{(0)} = 0$ and applies

$$T_{r,h}^{(q+1)} = T_{r,h}^{(q)} + \mathbf{1}_{\{r=r(q)\}} \mathbf{1}_{\{h=h(q)\}} \quad (0 \leq q < 729).$$

It produces exactly the table shown. Ternary expansion gives a bijection between these 729 integers and all words in $\mathbb{F}_3^6$, so the operation exhausts the code. Summing its rows proves (27.4); the first nonzero weight is six. $\square$

Remark 27.4 (Reproducible evaluation of the finite recurrence). The following loop literally implements the sums in the proof. Its only data are the integers of the matrix (27.1); it computes the enumerator before comparing it with the equality obtained.

```

from collections import Counter
from itertools import product
A = ((0,1,1,1,1,1), (1,0,1,2,2,1),
      (1,1,0,1,2,2), (1,2,1,0,1,2),
      (1,2,2,1,0,1), (1,1,2,2,1,0))
rows = [Counter() for _ in range(7)]
for w in product(range(3), repeat=6):
    v = tuple(sum(w[i]*A[i][j] for i in range(6)) % 3
              for j in range(6))
    r = sum(x != 0 for x in w)
    h = r + sum(x != 0 for x in v)
    rows[r][h] += 1
total = sum(rows, Counter())
if dict(total) != {0:1, 6:264, 9:440, 12:24}:
    raise ArithmeticError("Incompatible enumerator")

```

Proposition 27.5 (Hexads and the Witt design). *The collection $\mathcal{H}$ contains 132 hexads, and every five-element set of positions belongs to a unique hexad.*

Proof. Two words of weight six with the same support differ by a scalar. Indeed, among their six ratios of nonzero coordinates, one of the signs 1, -1 occurs at least three times. Subtracting the corresponding multiple, if the words were not proportional, would yield a nonzero word of weight at most three, contradicting Theorem 27.3. Each support thus corresponds to the pair $\{c, -c\}$, and there are $264/2 = 132$ supports.

If two distinct supports shared five positions, one of the two ratios would occur at least three times in that intersection. The linear combination cancelling those coordinates would be supported on a union of size seven and have weight at most four. The minimum distance rules this out. A five-element set therefore belongs to at most one hexad. Since

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

it belongs to exactly one. □

The constructed object is the extended ternary Golay code with its small Witt design. Recognition occurs after encoding, self-duality, and the distance computation. The lattice construction requires only the properties already proved and the marked flag obtained next.

27.3 Dodecaphasic flag and origin selection

The frame of the generated record preserves the words

$$w_\pi = 010211, \quad w_e = 201101, \quad w_\varphi = 121200, \quad w_{\text{APP}} = 022020.$$

The subscripts π, e, φ identify the corresponding reading regions of the constants; here w_e preserves the Euler region. Applying γ_{A_w} and taking supports gives

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first support has weight nine; the remaining three are hexads. The record

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$$

selects

$$B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi.$$

The threshold 729 belongs to the block format of the declared reading. In this particular record, integer thresholds from 660 to 729 produce the same support, so the support preserves the selected incidence, while the complete record also preserves the blocks and the threshold of its reading.

The signed pre-carry record is

$$\begin{aligned} Z_0 &= (7, 297, 353, -430, -715, -1199, \\ &\quad -3, 286, -894, -528, 380, 663). \end{aligned}$$

Its negative support is the hexad

$$H_{\bar{\alpha}} = \{4, 5, 6, 7, 9, 10\}.$$

The signs are received from the pre-carry state and are not extracted from the expansion of the final scalar.

Proposition 27.6 (Agreement of signed and incidence selection). *The hexad H_α^- is the unique $H \in \mathcal{H}$ satisfying*

$$B_K \subset H \subset P_\pi, \\ (|H \cap H_e|, |H \cap H_\varphi|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Proof. Encoding the words of weight six, or applying the preceding finite recurrence to their supports, gives the four hexads over B_K . Their complementary pairs are

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third are contained in P_π . The hexad of the second has intersections of size three with H_φ and with H_{APP} , and therefore fails the last condition. That of the third has intersections 4, 3, 2, and agrees with H_α^- . $\square$

Lemma 27.7 (Origin of the marked flag). *The origin selected by incidence is*

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Selection commutes with every bijection applied simultaneously to the supports.

Proof. The intersection $H_e \cap H_\varphi$ is $\{1, 3, 4, 9\}$. The subtracted union contains 3, 4, 9 and excludes 1. Equivariance follows because bijections commute with union, intersection, and difference. $\square$

The flag distinguishes one component among twelve before the metric step. The radial vector is determined afterwards, within a declared positive chamber. Combinatorial selection and the lattice operation are thereby kept distinct and connected.

27.4 Ternary gluing of the twelve components

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$, with bilinear form of matrix

$$B_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

For $x = ua_1 + va_2$,

$$x^2 = 2(u^2 - uv + v^2).$$

The identity $4(u^2 - uv + v^2) = (2u - v)^2 + 3v^2$ shows positivity. The determinant is three. The three classes of A_2^*/A_2 are represented by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

Pairing with a_1, a_2 verifies that these vectors belong to A_2^* , and $2\varpi_1 - \varpi_2 = a_1$. Hence $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with the discriminant.

For $c \in C_W$, we define $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and

$$N = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \tag{27.5}$$

Theorem 27.8 (Even unimodular glue lattice). *The set N is a positive, even, unimodular lattice of rank 24. Its norm-two vectors are exactly the roots of A_2^{12} .*

Proof. Linearity of the code and additivity of the discriminant make the union a subgroup of $(A_2^*)^{12}$ containing A_2^{12} . It is therefore a lattice of rank 24.

The pairing of discriminant classes is $2cd/3$ modulo $\mathbb{Z}$ in each component. Self-orthogonality of C_W makes their sum vanish and proves integrality of N . The norm of a class represented by $\phi(c)$ is $2 \text{wt}(c)/3$ modulo $2\mathbb{Z}$. All weights in (27.4) are multiples of three, so N is even.

Its index over A_2^{12} is $|C_W| = 3^6$. Using the determinant of the Gram matrix,

$$\det N = \frac{\det(A_2^{12})}{[N : A_2^{12}]^2} = \frac{3^{12}}{(3^6)^2} = 1.$$

The ambient form is the orthogonal sum of twelve positive forms; this proves positivity. Integrality and determinant one imply $N^* = N$.

In a nonzero class of the local discriminant, a vector has the form $(ua_1 + va_2)/3$, with residues $(u, v) \equiv (2, 1) \text{ or } (1, 2) \pmod{3}$. The integer $u^2 - uv + v^2$ is positive and divisible by three, so it is at least three. The representatives ϖ_1, ϖ_2 attain that value: the minimum norm of each nonzero class is $2/3$. A nonzero codeword has at least six nonzero coordinates. Every vector in a nontrivial glue class therefore has norm at least $6(2/3) = 4$.

In A_2^{12} , norm two can only come from a single component, because each component has minimum positive even norm two. The solutions of $u^2 - uv + v^2 = 1$ are $(\pm 1, 0), (0, \pm 1), (1, 1), (-1, -1)$. These are the six roots of A_2 , and complete the census of roots of N . $\square$

The preceding properties identify N as the Niemeier lattice with root system A_2^{12} . The description of all its roots is needed to control which ones survive passage to the neighbour.

27.5 Radial chamber and rootless neighbour

We fix the radial chamber

$$v(a) = \sum_{j=1}^{12} a_j \rho_j, \quad a_j = 1 + 3b_j, \quad b_j \in \mathbb{Z}_{\geq 0},$$

where ρ_j is the copy of ρ in component j . Evenness of an index-three neighbour requires $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_j (1 + 3b_j)^2 = 24 + 12 \sum_j b_j + 18 \sum_j b_j^2,$$

the congruence is equivalent to $\sum_j b_j \equiv 1 \pmod{3}$. The minimum norm within the chamber is attained with a single $b_j = 1$ and all others zero, and equals 54. The origin $o_* = 1$ of Lemma 27.7 selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 54. \quad (27.6)$$

The qualification “minimum” refers to the declared chamber. In the full residue class, $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. This distinction preserves the condition determining the coefficient four.

We define

$$\begin{aligned} N_0 &= \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \\ \Lambda &= N_0 + \mathbb{Z} v_\alpha/3. \end{aligned} \quad (27.7)$$

Theorem 27.9 (Even self-dual rootless neighbour). *The construction (27.7) is a positive, even, self-dual lattice of rank 24 with no norm-two vectors. Consequently, it is isometric to the Leech lattice.*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \pmod 3$ is nonzero. A simple root of component j has pairing $a_j \equiv 1 \pmod 3$ with $a_j \rho_j$. Hence $[N : N_0] = 3$, whence $\det N_0 = 9$.

For $x \in N_0$, the pairing $\langle x, v_\alpha/3 \rangle$ is an integer, so $v_\alpha/3 \in N_0^*$. Moreover, $\langle v_\alpha, v_\alpha \rangle = 54$ implies $v_\alpha \in N_0$. If $v_\alpha/3 \in N$, integrality of N would force all pairings of N with v_α to be divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ over N_0 therefore has order exactly three. It follows that

$$[\Lambda : N_0] = 3, \quad \det \Lambda = \frac{9}{3^2} = 1.$$

The pairings of N_0 with the new generator are integers, and the norm of this generator is six. The extension is integral. For $x \in N_0$ and $m \in \mathbb{Z}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

Thus Λ is even; its integrality and determinant one prove self-duality. Positivity and rank are inherited from the ambient Euclidean space.

To exclude roots, we first consider the class N_0 . All roots of N belong to A_2^{12} by Theorem 27.8. The pairings of its six local roots with $a_j \rho_j$ are $\pm a_j, \pm 2a_j$, congruent to $\pm 1, \pm 2 \pmod 3$. No root belongs to N_0 .

The other two classes of Λ/N_0 can be represented by $\pm v_\alpha/3$. In each component, their vectors belong to one of the classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

The distinguished component has the same description because $4\rho/3 \equiv \rho/3 \pmod{A_2}$. For the numerators (u, v) in $(ua_1 + va_2)/3$, the six residues are

$$(1, 1), (0, 2), (2, 0), (2, 2), (1, 0), (0, 1) \pmod 3.$$

All are nonzero. Positivity of $u^2 - uv + v^2$ forces it to be at least one, and the representatives $(1, 1), (0, -1), (-1, 0), (-1, -1), (1, 0), (0, 1)$ attain that minimum in all six classes. Each component has norm at least $2/9$. A vector in either of the two new classes has norm at least

$$12 \cdot \frac{2}{9} = \frac{8}{3} > 2.$$

Roots have been excluded in all three classes. The classification theorem for positive even unimodular lattices of rank twenty-four identifies the unique rootless case with the Leech lattice [9]. $\square$

The final identification uses an external classification on an object whose properties have already been proved. Selection of the distinguished component preserves the flag arising from the signed record; the subscript α expresses that provenance. The norm and coefficients of the radial vector are obtained from the metric and the chamber through the stated congruence.

The lattice Λ , its bilinear form, its twelve ambient planes, and the vector $v_\alpha/3$ are thereby determined with all the properties used by the subsequent deformation. The code also contains the word

$$c_* = (1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1),$$

since $q_* = (1, 1, 1, 1, 1, 1)$ satisfies $q_* A_W = (2, 1, 1, 1, 1, 1)$. This is the full-support datum that allows us to construct and prove stability of the order-three isometry in Section 28.

28 Dodecaphasic polarization, lattice duality, and theta reciprocity

The exceptional incidence and the polarization of the dodecaphasic record arise from two realizations of the same enriched state. The first preserves the boundary relations, the glue code, and the marked origin determining the lattice; the second lets a representation act on the twelve integer coordinates already obtained by the readers of the Topological Phase Kernel. Their composition allows us to study a metric deformation whose parameter changes the relative lengths of twelve planes while simultaneously preserving the total volume and an explicit ternary symmetry.

The causal starting point remains the APP-TRIT-TPK construction of the enriched state and the joint continuum. The record K is received after the dodecaphasic record and its incidence and orientation operations. The representation used acts on this output and preserves its twelve values as already determined data. The lattice realization is received with its bilinear form, its glue classes, and its marked vector. These structures, developed respectively in [2, 5], are retained when composed. The collection constructed below is a geometric realization of these data with a common twelve-position chart.

28.1 Polarization of the record and marking of the twelve planes

The dodecaphasic record and its sum are

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \quad (28.1)$$

$$\sum_{j=1}^{12} K_j = 6263.$$

Let A_5 be the group of even permutations of five elements and let $C_5 = \langle (1\ 2\ 3\ 4\ 5) \rangle$. Its twelve left cosets determine a permutation representation $\varrho : A_5 \rightarrow O(\mathbb{R}^{12})$. The marked chart is fixed by the following representatives, written as lists of images:

j	r_j	j	r_j
1	(4, 3, 5, 2, 1)	7	(5, 4, 3, 2, 1)
2	(4, 2, 1, 3, 5)	8	(1, 3, 4, 2, 5)
3	(1, 2, 4, 5, 3)	9	(4, 2, 3, 5, 1)
4	(2, 5, 3, 4, 1)	10	(3, 2, 5, 4, 1)
5	(4, 3, 1, 5, 2)	11	(4, 5, 2, 3, 1)
6	(5, 1, 2, 3, 4)	12	(5, 3, 2, 4, 1).

Specifically, $\varrho(a)e_j = e_\ell$ if $ar_jC_5 = r_\ell C_5$. The three-dimensional character χ_3 takes the values

class	1	2^2	3	5_a	5_b
χ_3	3	-1	0	$(1 + \sqrt{5})/2$	$(1 - \sqrt{5})/2$.

The class 5_a contains the stated generator of C_5 , and 5_b contains its square. This choice distinguishes the two conjugate three-dimensional characters. If I_{12} is the identity and $\mathbf{1}$ is the vector of twelve ones, define

$$P_3 = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \varrho(a), \quad P_{11} = I_{12} - \frac{1}{12} \mathbf{1} \mathbf{1}^\top. \quad (28.2)$$

The field $\mathbb{Q}(\sqrt{5})$ appears here in the realization of the representation acting on the generated record.

Lemma 28.1 (Zero-sum polarization). *The operators in (28.2) satisfy*

$$P_3 = P_3^\top = P_3^2, \quad \text{rank } P_3 = 3, \quad P_3 \mathbf{1} = 0, \quad P_3 P_{11} = P_{11} P_3 = P_3.$$

Consequently,

$$u = P_3 P_{11} K = P_3 K, \quad \sum_{j=1}^{12} u_j = 0, \quad \|u\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0. \quad (28.3)$$

Proof. The character sum is the isotypic projector of the representation. The reality of the character and the substitution $a \mapsto a^{-1}$ give its symmetry. The multiplicity of the three-dimensional character in the representation of A_5/C_5 is the dimension of its C_5 -invariants, which equals

$$\frac{1}{5} \left(3 + 2 \frac{1 + \sqrt{5}}{2} + 2 \frac{1 - \sqrt{5}}{2} \right) = 1.$$

The projector therefore has rank three. Its orthogonality to the trivial character implies $P_3 \mathbf{1} = 0$ and the identities with P_{11} . Applying the finite sum to the record (28.1) produces the twelve coordinates

$$\begin{aligned} u = & (-495/4 - 233\sqrt{5}/10, \quad 403/4 + 169\sqrt{5}/10, \\ & -403/4 - 169\sqrt{5}/10, \quad 495/4 + 233\sqrt{5}/10, \\ & 601/4 + 1031\sqrt{5}/10, \quad 203/4 + 211\sqrt{5}/5, \\ & -203/4 - 211\sqrt{5}/5, \quad -601/4 - 1031\sqrt{5}/10, \\ & 313/4 + 569\sqrt{5}/10, \quad 162 + 219\sqrt{5}/20, \\ & -162 - 219\sqrt{5}/20, \quad -313/4 - 569\sqrt{5}/10). \end{aligned} \quad (28.4)$$

Summing them and squaring them in $\mathbb{Q}(\sqrt{5})$ gives (28.3). Equivalently,

$$\|u\|^2 = K^\top P_3 K = \frac{1}{20} \sum_{a \in A_5} \chi_3(a^{-1}) K^\top \varrho(a) K.$$

The positive coefficients of its evaluation prove that $u \neq 0$. □

The zero sum has a concrete geometric function: expansion of some planes can compensate for contraction of the others in the total determinant. Non-vanishing of u , by contrast, ensures that this compensation contains an actual deformation.

The ambient Euclidean space of the lattice realization is

$$E = \bigoplus_{j=1}^{12} E_j, \quad B = \text{diag}(B_2, \dots, B_2), \quad B_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \quad (28.5)$$

Each E_j is the real plane spanned by the root lattice A_2 , expressed in its simple-root basis. The orthogonal decomposition belongs to the ambient space. The Leech lattice is obtained by gluing and passage to a neighbour, operations relating the twelve factors.

Let $C_W \subset \mathbb{F}_3^{12}$ be the six-dimensional ternary code of the Paley–Witt construction. Identifying the discriminant of each A_2 factor with $\mathbb{F}_3$, the gluing is

$$N = N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)),$$

where $\phi(c)$ chooses representatives of the discriminant classes. Self-orthogonality and self-duality of the code transmit evenness and integrality to N ; its determinant is

$$\det N = \frac{3^{12}}{(36)^2} = 1.$$

The exceptional construction fixes the marked origin $j = 1$ and the vector

$$v_\alpha = 4\rho_1 + \rho_2 + \dots + \rho_{12}, \quad \rho_j = (1, 1)_j, \quad \|v_\alpha\|_B^2 = 54. \quad (28.6)$$

Using it, we form

$$\begin{aligned} N_{\alpha,0} &= \{x \in N : \langle x, v_\alpha \rangle_B \equiv 0 \pmod{3}\}, \\ \Lambda &= N_{\alpha,0} + \mathbb{Z} v_\alpha/3. \end{aligned} \quad (28.7)$$

The exceptional realization establishes that Λ is a positive, even, unimodular, rootless lattice of rank 24, so it is isometric to the Leech lattice [5]. In particular, $\Lambda^* = \Lambda$, with dual taken with respect to B . These properties and the membership $v_\alpha/3 \in \Lambda$ are the precise lattice hypotheses used in the following theorems.

28.2 Ternary isometry and stability of the neighbour

We define the two-dimensional matrices

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \varpi = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Direct multiplication verifies

$$C^3 = I_2, \quad C^2 + C + I_2 = 0, \quad C^\top B_2 C = B_2, \quad RCR = C^{-1}, \quad R^\top B_2 R = B_2.$$

Moreover, $C\varpi - \varpi = (-1, 0)^\top$, so C acts trivially on A_2^*/A_2 . The reflection R reverses the sign of the discriminant class. The code C_W contains the full-support word

$$c_* = (1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1).$$

In the presentation $C_W = \{(q, qA_W) : q \in \mathbb{F}_3^6\}$, this word arises from $q_* = (1, 1, 1, 1, 1, 1)$ and $q_*A_W = (2, 1, 1, 1, 1, 1)$.

For each plane we fix the frame

$$Q_j = \begin{cases} R, & (c_*)_j = 1, \\ I_2, & (c_*)_j = 2, \end{cases} \quad g_c = \bigoplus_{j=1}^{12} Q_j C Q_j^{-1}. \quad (28.8)$$

The symbols Q_j denote plane frames; the symbol P_3 is reserved for the projector onto the three-dimensional sector.

Proposition 28.2 (Order-three symmetry of the marked lattice). *The transformation g_c is an order-three isometry of E , has zero as its only fixed vector, and satisfies*

$$g_c N = N, \quad g_c N_{\alpha,0} = N_{\alpha,0}, \quad g_c \Lambda = \Lambda.$$

Proof. Each block is C or C^{-1} , whose minimal polynomial is $X^2 + X + 1$. This proves the order and the claim about fixed vectors, as well as preservation of B . Each block acts trivially on the discriminant and therefore preserves all glue classes of N .

Write $v = v_\alpha = \sum_j a_j \rho_j$, with $a_1 = 4$ and $a_j = 1$ for $j > 1$. All a_j are congruent to one modulo three. The identities

$$(C - I_2)\rho = (-2, -1)^\top, \quad (C^{-1} - I_2)\rho = (-1, -2)^\top$$

show that

$$y = \frac{g_c v - v}{3}, \quad z = \frac{g_c^{-1} v - v}{3}$$

have discriminant classes c_* and $-c_*$, respectively. Both belong to C_W , so $y, z \in N$. In each factor,

$$\left\langle \frac{(C^{\pm 1} - I_2)a_j \rho}{3}, a_j \rho \right\rangle_{B_2} = -a_j^2.$$

Summing gives $\langle y, v \rangle_B = \langle z, v \rangle_B = -27$; hence $y, z \in N_{\alpha,0}$. If $x \in N_{\alpha,0}$, then

$$\langle g_c x, v \rangle_B = \langle x, g_c^{-1} v \rangle_B = \langle x, v \rangle_B + 3\langle x, z \rangle_B \equiv 0 \pmod{3}.$$

The same operation with g_c^{-1} proves the equality $g_c N_{\alpha,0} = N_{\alpha,0}$. Finally, $g_c(v/3) = v/3 + y$ preserves the class generating $\Lambda = N_{\alpha,0} + \mathbb{Z}(v/3)$. $\square$

The ternary symmetry therefore acts on the complete glued lattice. Its preservation uses the codeword and the congruence of the marked vector, in addition to the rotation of each plane. This precision allows the symmetry to be transported to the flow without losing the arithmetic information that made its construction possible.

28.3 Anisotropic flow and covolume preservation

Definition 28.3 (Deformation determined by polarization). For a dimensionless parameter $s \in \mathbb{R}$, define

$$H_u = \bigoplus_{j=1}^{12} u_j I_2, \quad D_s = \exp(sH_u) = \bigoplus_{j=1}^{12} e^{su_j} I_2, \quad \Lambda_s = D_s \Lambda. \quad (28.9)$$

The coordinate u_j is assigned to the plane E_j of the same marked chart.

The positive factor e^{su_j} preserves the orientation of each plane. The twelve labels transmit the correspondence between polarization and the lattice realization. This correspondence is part of the starting structure; a permutation of the labels is handled by transporting it simultaneously in both objects.

Theorem 28.4 (Covolume and symmetry of the flow). *For all $s, t \in \mathbb{R}$, we have*

$$\begin{aligned} D_{s+t} &= D_s D_t, & D_0 &= I, & D_s^{-1} &= D_{-s}, \\ D_s^\dagger &= D_s, & D_s g_c &= g_c D_s, & \det D_s &= 1, \end{aligned}$$

where $A^\dagger = B^{-1} A^\top B$. Each Λ_s is a discrete lattice of rank 24 and covolume one, preserved by the order-three isometry g_c .

Proof. The group identities hold in each exponential block. The scalar blocks are self-adjoint for B_2 , and commute with the blocks of g_c . The zero sum in (28.3) gives

$$\det D_s = \prod_{j=1}^{12} e^{2su_j} = \exp\left(2s \sum_{j=1}^{12} u_j\right) = 1.$$

The image of a lattice under an invertible operator is a lattice of the same rank; its covolume is multiplied by the absolute value of the determinant. The covolume of Λ is one, so that of Λ_s is also one. Commutation and Proposition 28.2 give

$$g_c \Lambda_s = g_c D_s \Lambda = D_s g_c \Lambda = D_s \Lambda.$$

The order and the fixed-point space of g_c remain unchanged. □

Remark 28.5 (Parameter and preserved symmetry). The direction u is determined by the record; s ranges over a collection of realizations. Replacing u by $u/\|u\|$ reparametrizes the same collection. A physical interpretation selecting a particular value of s adds its realization rule. The theorem preserves the explicit symmetry g_c ; transporting other automorphisms requires their corresponding relations with H_u .

28.4 Euclidean dual and actual deformation

Definition 28.6 (Euclidean dual). For a lattice $L \subset E$, its dual with respect to B is

$$L^* = \{y \in E : \langle y, x \rangle_B \in \mathbb{Z} \text{ for every } x \in L\}.$$

Theorem 28.7 (Dual reciprocity). *The collection (28.9) satisfies*

$$\Lambda_s^* = \Lambda_{-s} \quad (s \in \mathbb{R}). \quad (28.10)$$

Its self-duality at a parameter s is equivalent to $D_{2s} \Lambda = \Lambda$.

Proof. For an invertible operator A ,

$$(AL)^* = A^{-\dagger} L^*.$$

Indeed, $\langle y, Ax \rangle_B = \langle A^\dagger y, x \rangle_B$ is an integer for every $x \in L$ precisely when $A^\dagger y \in L^*$. Taking $A = D_s$, self-adjointness, the identity $D_s^{-1} = D_{-s}$, and $\Lambda^* = \Lambda$ prove (28.10). The equality $D_s \Lambda = D_{-s} \Lambda$, upon multiplication by D_s , is equivalent to $D_{2s} \Lambda = \Lambda$. □

Covolume controls the volume of a fundamental domain; self-duality controls all integer pairings with the lattice. The first property is continuous along this collection. The second imposes the exact arithmetic condition of the preceding theorem. We shall therefore use the expression *covolume one* for the Euclidean flow and specify integrality whenever arithmetic unimodularity is involved.

Proposition 28.8 (Witness to local loss of integrality). *There exists $\delta > 0$ such that, if $0 < |s| < \delta$, the lattice Λ_s is nonintegral and is not isometric to the Leech lattice.*

Proof. The vector $v_\alpha/3$ belongs to Λ . Its transported norm is

$$f(s) = \left\| D_s \frac{v_\alpha}{3} \right\|_B^2 = \frac{2}{9} \left(16e^{2su_1} + \sum_{j=2}^{12} e^{2su_j} \right). \quad (28.11)$$

By $\sum_j u_j = 0$ and the first coordinate in (28.4),

$$\begin{aligned} f(0) &= 6, \\ f'(0) &= \frac{4}{9} \left(16u_1 + \sum_{j=2}^{12} u_j \right) = \frac{20}{3} u_1 = -825 - \frac{466}{3} \sqrt{5} \neq 0. \end{aligned} \quad (28.12)$$

Continuity of f' allows us to choose an interval $(-\delta, \delta)$ on which f is strictly monotone. Shrinking δ also gives $11/2 < f(s) < 13/2$. For $0 < |s| < \delta$, we obtain $f(s) \neq 6$; six is the only integer in that interval. Thus Λ_s contains a vector of noninteger norm. An integral lattice has integer norm on all its vectors. Since Leech is integral and isometries preserve norms, the second statement is also proved. $\square$

The deformation preserves volume and ternary symmetry while actually changing the arithmetic metric structure. Opposite pairs of coordinates of u define a permutation of the ambient space interchanging dilations and contractions. Promoting it to an automorphism of the glued lattice would require preservation of the code and the neighbour. The dual reciprocity already proved uses only self-adjointness of the flow and initial self-duality.

28.5 Convergence and reciprocity of theta functions

For $s \in \mathbb{R}$ and $t > 0$, define

$$\Theta_s(t) = \sum_{\lambda \in \Lambda} \exp(-\pi t \|D_s \lambda\|_B^2). \quad (28.13)$$

This is the Gaussian sum of all lengths of the deformed lattice, with their multiplicities. If $M = \max_j |u_j|$, we have

$$e^{-2|s|M} \|x\|_B^2 \leq \|D_s x\|_B^2 \leq e^{2|s|M} \|x\|_B^2. \quad (28.14)$$

On a compact subset of $\mathbb{R} \times (0, \infty)$, the lower bound provides a dominating Gaussian on the fixed lattice. The series converges absolutely and uniformly. Derivatives of finite order add polynomial factors in the coordinates of λ ; those factors are dominated by another Gaussian. Term-by-term differentiation is therefore legitimate on those compact sets.

Theorem 28.9 (Gaussian reciprocity). *With the Euclidean volume associated with B ,*

$$\Theta_s(t) = t^{-12} \Theta_{-s}(t^{-1}), \quad s \in \mathbb{R}, \quad t > 0. \quad (28.15)$$

In particular, $\Theta_s(1) = \Theta_{-s}(1)$.

Proof. We fix the Fourier transform

$$\hat{f}(y) = \int_E f(x) e^{-2\pi i \langle x, y \rangle_B} dx.$$

For a lattice $L \subset E$ and $f_t(x) = e^{-\pi t \|x\|_B^2}$, we periodize

$$F_t(x) = \sum_{\ell \in L} f_t(x + \ell).$$

Normal convergence of the series and its derivatives gives a smooth function on E/L . If $\xi \in L^*$, its Fourier coefficient is

$$\begin{aligned} c_\xi &= \frac{1}{\text{covol}(L)} \int_{E/L} F_t(x) e^{-2\pi i \langle x, \xi \rangle_B} dx \\ &= \frac{\hat{f}_t(\xi)}{\text{covol}(L)}. \end{aligned}$$

The second equality decomposes E into translates of a fundamental domain; the character has value one on L . The product of the one-dimensional Gaussian integrals, in a basis orthonormal for B , gives

$$\hat{f}_t(\xi) = t^{-12} \exp(-\pi \|\xi\|_B^2/t).$$

The Fourier series is absolutely convergent. Evaluating it at $x = 0$, we obtain Poisson summation

$$\sum_{\ell \in L} e^{-\pi t \|\ell\|_B^2} = \frac{t^{-12}}{\text{covol}(L)} \sum_{\xi \in L^*} e^{-\pi \|\xi\|_B^2/t}.$$

Now $L = \Lambda_s$ has covolume one and $L^* = \Lambda_{-s}$ by Theorem 28.7. Substitution proves (28.15). The case $t = 1$ is immediate. $\square$

The preceding periodization is the Gaussian proof of Poisson summation; it can be compared with [8, Theorem 2, pp. 10–11]. The positive-sign Fourier convention leads to the same Gaussian transform, by its evenness. The theta equality concerns the complete series and preserves the analytic domain of convergence.

Proposition 28.10 (Holomorphic extension). *For τ in the upper half-plane $\mathbb{H} = \{\tau \in \mathbb{C} : \text{Im } \tau > 0\}$, the series*

$$\theta_s(\tau) = \sum_{\lambda \in \Lambda} e^{\pi i \tau \|D_s \lambda\|_B^2} \quad (28.16)$$

converges normally and satisfies

$$\theta_s(-1/\tau) = (-i\tau)^{12} \theta_{-s}(\tau). \quad (28.17)$$

Proof. The absolute value of each summand is $e^{-\pi \text{Im } \tau \|D_s \lambda\|_B^2}$. On compact subsets of the upper half-plane, the imaginary part has a positive lower bound, so (28.14) proves normal convergence and holomorphy. For $\tau = it$, with $t > 0$, (28.15) applied at t^{-1} gives $\theta_s(it) = t^{12} \theta_{-s}(it)$. Both sides of (28.17) are holomorphic on $\mathbb{H}$; the identity theorem extends the equality from the positive imaginary axis. $\square$

Modular inversion relates the parameters s and $-s$. Additional periodicity under $\tau \mapsto \tau + 1$ uses evenness of the norms. Proposition 28.8 exhibits parameters arbitrarily close to zero for which Euclidean integrality is lost. The collection therefore preserves the proved analytic reciprocity, while a realization through scalar modular forms or lattice vertex algebras additionally incorporates the corresponding arithmetic hypotheses. In the initial exceptional realization, those hypotheses support continuation towards vertex algebras and Monstrous Moonshine [5].

28.6 Split form and self-duality in dimension forty-eight

The loss of integrality of one Euclidean projection leads us to consider both projections simultaneously. On the space $E \oplus E$, introduce the bilinear form

$$\mathcal{B}((x, p), (x', p')) = \langle x, p' \rangle_B + \langle p, x' \rangle_B. \quad (28.18)$$

Its products pair the two coordinates. Define

$$\begin{aligned}\mathcal{D}_s &= D_s \oplus D_{-s}, \\ \Gamma_s &= \mathcal{D}_s(\Lambda \oplus \Lambda) \\ &= \{(D_s \lambda, D_{-s} \mu) : \lambda, \mu \in \Lambda\}.\end{aligned}\tag{28.19}$$

Theorem 28.11 (Integral self-duality for the split form). *The form $\mathcal{B}$ has signature $(24, 24)$. For every $s \in \mathbb{R}$, $\mathcal{D}_s$ is an isometry of $\mathcal{B}$, and Γ_s is an integral, even, self-dual lattice with respect to this form.*

Proof. Introducing

$$p_L = \frac{x+p}{\sqrt{2}}, \quad p_R = \frac{x-p}{\sqrt{2}},$$

yields

$$\mathcal{B}((x, p), (x, p)) = \|p_L\|_B^2 - \|p_R\|_B^2.$$

Since B is positive in dimension 24, the signature is $(24, 24)$. Self-adjointness and reciprocity of the blocks give

$$\langle D_s x, D_{-s} p' \rangle_B = \langle x, p' \rangle_B,$$

and similarly for the second cross product. Hence $\mathcal{B}(\mathcal{D}_s \xi, \mathcal{D}_s \eta) = \mathcal{B}(\xi, \eta)$.

On $\Gamma_0 = \Lambda \oplus \Lambda$, all cross products are integers, and

$$\mathcal{B}((\lambda, \mu), (\lambda, \mu)) = 2\langle \lambda, \mu \rangle_B \in 2\mathbb{Z}.$$

This proves integrality and evenness. If (x, p) belongs to the dual of Γ_0 for $\mathcal{B}$, pairing with $(\lambda, 0)$ and $(0, \mu)$ requires respectively $p, x \in \Lambda^* = \Lambda$. The dual is therefore Γ_0 . The isometry $\mathcal{D}_s$ transports these three properties to Γ_s . $\square$

The metric specifies the content of the conclusion. The collection Γ_s preserves integrality for the cross products of $\mathcal{B}$, even when Λ_s loses integrality for B . As abstract lattices with split form, the Γ_s are isometric. Their two Euclidean projections, considered with their respective positive metrics, vary with the parameter.

Proposition 28.12 (Exchange of reciprocal projections). *The involution $\mathcal{J}(x, p) = (p, x)$ satisfies*

$$\mathcal{J}\Gamma_s = \Gamma_{-s}, \quad \mathcal{J}\mathcal{D}_s\mathcal{J} = \mathcal{D}_{-s}.$$

In the coordinates (p_L, p_R) , it fixes p_L and reverses the sign of p_R .

Proof. The exchange turns $(D_s \lambda, D_{-s} \mu)$ into $(D_{-s} \mu, D_s \lambda)$, which ranges over Γ_{-s} . Conjugation of the two blocks gives the second identity. Finally, $x + p$ is preserved and $x - p$ reverses sign. $\square$

The involution expresses a duality of the complete lattice construction. Any eventual physical representation preserves, in addition to the split form, the operators and identification data proper to that realization.

28.7 Lattice energy and unitary equivalence

The reciprocal-radius energy $m^2/R^2 + w^2R^2$ interchanges its two terms under $(m, w, R) \mapsto (w, m, R^{-1})$, with a unitary conjugation preserving the operator domains [5]. The realization on the twelve marked planes uses

$$\mathcal{E}_s(\lambda, \mu) = \|D_s \lambda\|_B^2 + \|D_{-s} \mu\|_B^2, \quad \lambda, \mu \in \Lambda.\tag{28.20}$$

The energy is positive and satisfies $\mathcal{E}_s(\lambda, \mu) = \mathcal{E}_{-s}(\mu, \lambda)$.

Theorem 28.13 (Positive operator and conjugation of domains). *On the Hilbert space $\ell^2(\Lambda \oplus \Lambda)$, let*

$$(H_s \psi)(\lambda, \mu) = \mathcal{E}_s(\lambda, \mu) \psi(\lambda, \mu)$$

with domain

$$\mathcal{D}(H_s) = \left\{ \psi \in \ell^2(\Lambda \oplus \Lambda) : \sum_{\lambda, \mu} \mathcal{E}_s(\lambda, \mu)^2 |\psi(\lambda, \mu)|^2 < \infty \right\}. \quad (28.21)$$

Then H_s is positive and self-adjoint with compact resolvent. The map $U\psi(\lambda, \mu) = \psi(\mu, \lambda)$ is unitary, and

$$U\mathcal{D}(H_s) = \mathcal{D}(H_{-s}), \quad UH_s U^{-1} = H_{-s}. \quad (28.22)$$

Proof. Multiplication by a real function on a discrete set, with the maximal domain (28.21), is self-adjoint. This can be checked directly using the orthonormal basis of singleton-supported vectors: membership in the adjoint domain requires the sequence obtained by multiplying by $\mathcal{E}_s$ to be square summable, exactly the same condition. Moreover,

$$\langle \psi, H_s \psi \rangle = \sum_{\lambda, \mu} \mathcal{E}_s(\lambda, \mu) |\psi(\lambda, \mu)|^2 \geq 0.$$

By (28.14),

$$\mathcal{E}_s(\lambda, \mu) \geq e^{-2|s|M} (\|\lambda\|_B^2 + \|\mu\|_B^2).$$

A lattice has finitely many points in any bounded set; hence every sublevel set of $\mathcal{E}_s$ is finite. The diagonal coefficients $(1 + \mathcal{E}_s)^{-1}$ tend to zero outside finite sets, so $(H_s + I)^{-1}$ is a norm limit of finite-rank operators and is compact.

Exchanging the two indices preserves the norm of $\ell^2(\Lambda \oplus \Lambda)$, so U is unitary and $U^{-1} = U$. Equality of energies upon reversing s transforms the domain condition of H_s into that of H_{-s} , and evaluation on each pair (λ, μ) proves (28.22). $\square$

Three preservation laws with distinct domains are thus connected. On the Euclidean planes, the flow maintains covolume and ternary symmetry and relates each lattice to the dual at the opposite parameter. In the doubled space, the cross products preserve an integral, even, self-dual lattice. In the operator realization, the reciprocal exchange preserves energy through a unitary equivalence of operators together with their domains. The theta formulas express the same reciprocity in the convergent sum of lengths of the entire lattice.

29 Positive congruences, action readers and Hamiltonian dynamics

The relation between the dodecaphasic record and the General Mass Law has an effective composition. APP produces the two sheets with their residues and quotients; TRIT orients the transition and determines its regime; TPK composes selection, transport and updating while preserving route, carry, boundary, survival and memory. The regional readouts and the twelve-position record arise jointly from the enriched state and its compatible prolongation. Numerical closure and the material reading share this origin and retain their specific operations.

With the phase origin, orientation and base one thousand declared, the record is

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601).$$

The positional reader uses

$$\kappa_{\text{ag}}(K) = \frac{\sum_{j=1}^{12} K_j 1000^{12-j}}{1000^{12} - 1}, \quad \alpha = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{ag}}(K). \quad (29.1)$$

The values on the right are internal readouts of the same coinductive process. The formula expresses the compatible infinite-closure reader, with the carry boundary that defines it.

Action receives these readouts through the readers

$$\begin{aligned} H_5 &= \frac{1}{2}\sqrt{1000\alpha/\varphi} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \\ D_A &= \exp(-100\pi\alpha/9), \quad \mathcal{C}_\pi = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \\ \eta_{\text{ret}} &= H_5 - \frac{90}{\pi}\mathcal{C}_\pi, \quad R_{\text{act}} = \frac{H_5}{\eta_{\text{ret}}} > 0. \end{aligned} \quad (29.2)$$

Here and in the following formulae, π, φ, e abbreviate the HMT readouts already constructed. The local determinantal norm $\Sigma_H = \varphi\alpha^{16}$ fixes the decade -34 , and the section $\hbar_{\text{ret}} = \eta_{\text{ret}}10^{-34}\mathcal{U}_S$ belongs to the declared action line. The ratio R_{act} preserves the relation between the two action sections when their common dimensional basis changes.

The central electron route provides, through its displacement and autocorrelation readers, the triple $(80, 54, 6)$. With

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729}\right), \quad b_e^{\text{alg}} = \frac{\sqrt{3}}{4} [\exp(\varphi/\pi^2) - 22\alpha^3](1 + 15\Delta_4),$$

its material readout contains

$$\kappa_e = 80 - 54\alpha + 6\Delta_4, \quad \varepsilon_e^\beta = b_e^{\text{alg}} R_{\text{act}}^{\kappa_e}. \quad (29.3)$$

The composition (29.1)–(29.3) identifies the specific role of K : its closure reading contributes to α , and this output subsequently acts in the action reader and the return exponent. The central selection of the electron belongs to the involution of the APP chart, whose unique fixed point is $(5, 5)$. The centrality of that cell and its scalar evaluation are two results linked by the route and its record.

29.1 Reversible coordinates of the material event record

The material record assigns to a route γ twelve oriented events $\mathbf{e}(\gamma) = (e_0, \dots, e_{11})$, one for each nonadic window of the one-hundred-and-eight-position calendar. The same temporal order allows these events to be realized in the twelve-coordinate carrier used by K . This identification requires preserving the origin and orientation of the calendar.

Let $\mathbf{1}$ be the uniform vector, $P_{11} = I - \mathbf{1}\mathbf{1}^*/12$, and let P_3 be the three-dimensional projector of the declared incidence chart. The geometric construction of the record determines

$$u_K = P_3 P_{11} K, \quad \|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0, \quad u_K \perp \mathbf{1},$$

and the transverse projector

$$P_{10}(K) = P_{11} - \frac{u_K u_K^*}{\|u_K\|^2}.$$

Proposition 29.1 (Material decomposition relative to the direction of K). *For every event vector $\mathbf{e}$, the map*

$$\mathbf{e} \mapsto \left(s_C = \mathbf{1}^* \mathbf{e}, \quad a_K = \frac{u_K^* \mathbf{e}}{\|u_K\|^2}, \quad v_K = P_{10}(K) \mathbf{e} \right) \quad (29.4)$$

is invertible onto its image and satisfies

$$\mathbf{e} = \frac{s_C}{12} \mathbf{1} + a_K u_K + v_K, \quad \|\mathbf{e}\|^2 = \frac{|s_C|^2}{12} + |a_K|^2 \|u_K\|^2 + \|v_K\|^2. \quad (29.5)$$

Proof. The projectors $\mathbf{1}\mathbf{1}^*/12$, $u_K u_K^*/\|u_K\|^2$ and $P_{10}(K)$ are pairwise orthogonal and sum to the identity. Applying this equality to $\mathbf{e}$ proves the inversion. Orthogonality of the three summands gives the norm identity. $\square$

This chart preserves the complete event vector. It therefore also preserves the functionals that depend on its signs, sums and correlations, once reconstruction (29.5) has been performed. The record of predecessors of the nines, the action count and the tower coordinates remain in the enriched trajectory: the preceding chart modifies only the coordinates of the twelve events. The character is evaluated on the same recovered signature. The construction thus provides a geometric representation of the information used by mass, with a direction distinguished by K and ten transverse components.

The amount of information retained by each readout can be established exactly. In the algebraic domain of records, $K' = K + \mathbf{1}$ satisfies

$$u_{K'} = u_K, \quad P_{10}(K') = P_{10}(K), \quad \kappa_{\text{ag}}(K') - \kappa_{\text{ag}}(K) = \frac{1}{999}.$$

The last identity follows by summing the geometric progression of the twelve positional weights. If the regional readouts are held fixed, the reading of α changes by $-1/999$. This algebraic check determines the precise scope of the direction: its projector preserves a polarization, whereas scalar closure also uses the uniform and positional information of the record. The admissibility of K' as another terminal history requires its own APP-TRIT-TPK construction.

29.2 Positive congruence and spectrum of the material operator

For a multisection M , let $\mathcal{H}_M$ be the finite-dimensional space with an orthonormal basis labelled by its routes. The return reader $r \in \{D, \Omega\}$ produces a self-adjoint operator $B_M^r e_\gamma = \kappa_\gamma^r e_\gamma$. The refined character, evaluated on differences relative to the electron, gives the basal operator

$$M_M^{(0)} e_\gamma = m_e^{(0)} \chi_{\text{ref}}(\sigma_\gamma - \sigma_e, \eta_\gamma - \eta_e) e_\gamma.$$

The tower coefficients and their convergence belong to the reader of each route. Its beta readout is the congruence

$$D_M^r = R_{\text{act}}^{B_M^r/2}, \quad M_M^{\beta, r} = D_M^r M_M^{(0)} D_M^r. \quad (29.6)$$

Proposition 29.2 (Exact invariants of material congruence). *The operator D_M^r is positive and invertible. The congruence (29.6) preserves rank and inertia; in particular, it preserves positivity. If $M_M^{(0)}$ and B_M^r are diagonal in the route basis, then*

$$m_\gamma^\beta = m_\gamma^{(0)} R_{\text{act}}^{\kappa_\gamma^r}.$$

For two routes in the same dimensional chart, one obtains

$$\frac{m_Y^\beta}{m_X^\beta} = \chi_{\text{ref}}(\sigma_Y - \sigma_X, \eta_Y - \eta_X) R_{\text{act}}^{\kappa_Y^r - \kappa_X^r}.$$

Proof. Functional calculus gives $D_M^r = \exp((\log R_{\text{act}}) B_M^r / 2)$, with strictly positive spectrum. Its inverse is the exponential of the negative exponent. For every v , the transformed quadratic form equals $(D_M^r v)^* M_M^{(0)} (D_M^r v)$. The invertible change of variable preserves the maximal dimensions of positive, negative and null subspaces; these determine inertia and rank. In a common diagonal basis, the two factors jointly contribute $R_{\text{act}}^{\kappa_\gamma^r}$. Dividing two values cancels the common dimensional section and uses the multiplicative law of the character. $\square$

The distinction between transporting a form and modifying an operator with respect to a fixed metric can be checked without approximations. For

$$M_0 = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \quad M' = D^* M_0 D = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix},$$

the determinant of D is one and the values associated with the two routes are interchanged. Even the unordered spectrum changes if $M_0 = I$: it changes from $\{1, 1\}$ to $\{4, 1/4\}$. Both checks preserve positivity, rank and inertia.

Proposition 29.3 (Spectral invariance with a transported metric). *Let $G_0 > 0$ be a metric and let D be invertible. If*

$$M' = D^* M_0 D, \quad G' = D^* G_0 D,$$

the pairs (M', G') and (M_0, G_0) have the same generalized spectrum, with the same multiplicities.

Proof. One has

$$\det(M' - \lambda G') = |\det D|^2 \det(M_0 - \lambda G_0), \quad (G')^{-1} M' = D^{-1} G_0^{-1} M_0 D.$$

The first equality preserves the roots and their multiplicities; the second exhibits a similarity of the associated operators. $\square$

The positive diagonal flow induced by K therefore has two precise mathematical realizations on a material form. When it transports the form and its metric simultaneously, it describes the same object in another chart. When it acts on the form while preserving the route metric, it produces an operator collection that may change the relative masses. Its physical identification is determined by the route reader and the declared action.

29.3 Compatibility of the dodecaphasic flow with material return

Let

$$A_K = (\log \rho_A) \operatorname{diag}(\hat{u}_{K,1}, \dots, \hat{u}_{K,12}), \quad \hat{u}_K = u_K / \|u_K\|, \quad \rho_A = \pi / \varphi^2,$$

so that $g_K(t) = e^{tA_K}$. The material return collection on the multisection M has generator

$$C_M^r = \frac{1}{2} (\log R_{\text{act}}) B_M^r, \quad S_M^r(t) = e^{tC_M^r}.$$

Proposition 29.4 (Transport of congruence by an intertwining reader). *A declared linear map $J_M : \mathbb{C}^{12} \rightarrow \mathcal{H}_M$ satisfying*

$$C_M^r J_M = J_M A_K \tag{29.7}$$

satisfies

$$S_M^r(t) J_M = J_M g_K(t),$$

and, for $M_M(t) = S_M^r(t) M_M^{(0)} S_M^r(t)$,

$$J_M^* M_M(t) J_M = g_K(t)^* (J_M^* M_M^{(0)} J_M) g_K(t). \tag{29.8}$$

Proof. Identity (29.7) is iterated to each power and summed in the convergent exponential series. Taking adjoints and substituting in the material form proves (29.8). $\square$

The criterion is finite and admits a direct operator-level stress test. In the declared diagonal bases, if a_j and c_γ are the eigenvalues of A_K and C_M^r , respectively, one must have

$$(c_\gamma - a_j)(J_M)_{\gamma j} = 0 \quad \text{for each } (\gamma, j).$$

Each nonzero coefficient of the reader requires the corresponding equality of eigenvalues. The recovered composition in (29.1)–(29.3) and the reversible chart (29.4) are effective results. The additional identification of their two flows requires a reader J_M with property (29.7); this proposition determines exactly what a realization supporting that identification must verify.

29.4 Elliptic transport and regular Legendre transformation

The variational realization of transport has an explicit antecedent in the action ellipse. The angular pair already obtained determines $\eta = \operatorname{artanh}(C^*/A)$, in the chamber $A \neq 0$, $|C^*/A| < 1$. The coordinates (Q, P) , both with the dimension of the square root of action, realize the form

$$J_\eta(Q, P) = \frac{1}{2}(e^{-\eta}Q^2 + e^\eta P^2), \quad \lambda = P dQ, \quad \Omega = dQ \wedge dP.$$

The transport between forms, with $M_\eta = \operatorname{diag}(e^{\eta/2}, e^{-\eta/2})$ and $J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, is $M_\eta e^{-\vartheta J_2} M_\eta^{-1}$. An interpolation $\eta(t)$, $\vartheta(t)$, with $\dot{\vartheta} = \omega(t) > 0$, has the Hamiltonian recovered from Article X:

$$H_{\text{tr}}(Q, P, t) = \omega(t)J_{\eta(t)}(Q, P) + \frac{\dot{\eta}(t)}{2}QP. \quad (29.9)$$

The second term comes from $\dot{M}_\eta M_\eta^{-1}$. The equations

$$\dot{Q} = \omega e^\eta P + \frac{\dot{\eta}}{2}Q, \quad \dot{P} = -\omega e^{-\eta}Q - \frac{\dot{\eta}}{2}P$$

preserve $J_{\eta(t)}(Q(t), P(t))$: the explicit variation of η and the contribution of the connection term cancel. This result and the identity $P\dot{Q} - H_{\text{tr}} = p\dot{q} - \omega(q^2 + p^2)/2$, for $Q = e^{\eta/2}q$, $P = e^{-\eta/2}p$, are proved in the section on variational conservation in the work on the holographic extreme and mean ratio.

Proposition 29.5 (Regular Lagrangian of elliptic transport). *Suppose $\eta \in C^2$, $\omega \in C^1$ and $\omega > 0$. The Legendre transformation with respect to P is invertible and yields*

$$L_{\text{tr}}(Q, \dot{Q}, t) = \frac{(\dot{Q} - \dot{\eta}Q/2)^2}{2\omega e^\eta} - \frac{\omega e^{-\eta}}{2}Q^2. \quad (29.10)$$

Proof. Write $a = \dot{\eta}/2$, $b = \omega e^\eta > 0$ and $c = \omega e^{-\eta}$. Then $H_{\text{tr}} = bP^2/2 + aQP + cQ^2/2$, and

$$\dot{Q} = \partial_P H_{\text{tr}} = bP + aQ, \quad P = \frac{\dot{Q} - aQ}{b}.$$

Substituting this expression in $P\dot{Q} - H_{\text{tr}}$, the terms containing P combine into $P(\dot{Q} - aQ) - bP^2/2 = (\dot{Q} - aQ)^2/(2b)$, which proves (29.10). Moreover,

$$\partial_{\dot{Q}} L_{\text{tr}} = P, \quad \partial_P^2 H_{\text{tr}} = \omega e^\eta > 0, \quad \partial_Q^2 L_{\text{tr}} = \frac{1}{\omega e^\eta} > 0.$$

The transformation is regular and its Euler–Lagrange equations are equivalent to the preceding Hamiltonian equations. $\square$

This algebraic elimination joins the Hamiltonian already constructed to its second-order Lagrangian form. Regularity follows from the quadratic term of the elliptic realization. For a purely linear cotangent lift $H_K(q, p) = p^\top A_K q$, the equation $\dot{q} = A_K q$ is independent of p , and the Hessian $\partial_p^2 H_K$ vanishes. Its variational formulation uses the first-order action $\int p^\top (\dot{q} - A_K q) dt$. The corpus also contains its discrete antecedent $L_n = p_{n+1}^\top (q_{n+1} - U_n q_n)$, whose equations yield the transport $q_{n+1} = U_n q_n$ and the dual transport $p_{n+1} = U_n^{-\top} p_n$. Both formulations have specified domains and operators. The temporal interpolation and the action section are retained as part of the declared physical realization.

30 Grassmannian representation of readers and forms valued in the action line

Positive realizations preserve information that can subsequently be used by physical readers. This preservation has a precise formulation: a marked representation of the record allows its coordinates to be reconstructed and every function using them to be transported by composition. The geometry represents an outcome of the construction; its choice of coordinates preserves the provenance of that outcome.

The starting domain consists of the states produced by APP, TRIT, and TPK in the preceding sections. APP preserves its additive and multiplicative sheets through remainder and quotient; TRIT selects the local regime with its orientation; the composition $U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t$ updates position, record, and memory. The prolongation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

preserves the prefix and distinguishes phase return from state return. The spectral, solenoidal, gauge, cohomological, and determinantal realizations remain correlative operations of the joint discrete structure of the continuum. Record extraction and regional representations take place in that same domain. In what follows, all of them are previously constructed HMT outputs.

30.1 The marked collection and its inverse

Let $L \geq 2$, and let $K = (K_1, \dots, K_L)$ be a positive record with charge $Q = \sum_j K_j$. In the dodecaphasic application, $L = 12$ and $Q = 6263$. Define

$$d_j = K_j/Q, \quad t_0 = 0, \quad t_j = \sum_{a=1}^j d_a, \quad B(K) = \begin{pmatrix} 1 & \cdots & 1 \\ t_0 & \cdots & t_L \end{pmatrix}.$$

The collection $\mathcal{G}_L^{\text{aff}}$ is the image of these row planes in $\text{Gr}_{>0}(2, L+1)$, retaining the ordered columns $0, \dots, L$, the endpoint markings, and the charge Q . Affine normalization fixes $t_0 = 0, t_L = 1$. We write Δ_{ij} for the minor of columns i, j of any representative of the plane.

Proposition 30.1 (Inverse of the marked realization). *The map $K \mapsto ([B(K)], Q)$ is injective. Its inverse on the image is given by*

$$K_j = Q \frac{\Delta_{j-1,j}}{\Delta_{0L}}, \quad t_j = \frac{\Delta_{0j}}{\Delta_{0L}}, \quad 1 \leq j \leq L. \quad (30.1)$$

The ratios are invariant under any invertible change of basis of the two rows.

Proof. In the declared representative, $\Delta_{ij} = t_j - t_i$, and $\Delta_{0L} = 1$. The formulas recover the consecutive differences and their sums. If B is replaced by AB , with A invertible, all minors are multiplied by $\det A$; their ratios remain unchanged. The charge recovers the scale preceding normalization. Therefore two records with equal marked data have the same coordinates. $\square$

The condition of membership in $\mathcal{G}_L^{\text{aff}}$ matters. The consecutive ratios of an arbitrary positive plane may fail the identity $\sum_j \Delta_{j-1,j} = \Delta_{0L}$. In the constructed collection this identity is telescoping. The inverse acts on the marked image just defined.

30.2 Reader algebra and changes of chart

Let $\mathcal{D}$ be the image of an enriched-state representation of the form $x \mapsto (K(x), \xi(x))$. The symbol ξ collects exactly the remaining coordinates used by the reader: regional representations, origin and

orientation, route, signature, boundary conditions, and dimensional sections, according to the object considered. These coordinates are not assumed to be independent. Define

$$\mathcal{F} : \mathcal{D} \longrightarrow \tilde{\mathcal{G}}, \quad (K, \xi) \longmapsto ([B(K)], Q, \xi), \quad \tilde{\mathcal{G}} := \mathcal{F}(\mathcal{D}).$$

The preceding proposition provides the inverse $\mathcal{I}$ of $\mathcal{F}$ on its image.

Theorem 30.2 (Faithful representation of the algebra of retained readers). *For each reader $\mathcal{R} : \mathcal{D} \rightarrow V$, with declared codomain V , there exists a unique geometric reader $\mathcal{R}^{\text{geom}} : \tilde{\mathcal{G}} \rightarrow V$ such that*

$$\mathcal{R}^{\text{geom}} = \mathcal{R} \circ \mathcal{I}, \quad \mathcal{R}^{\text{geom}} \circ \mathcal{F} = \mathcal{R}.$$

For scalar readers, this correspondence preserves sums, products, the unit, and conjugation. It is an isomorphism from the algebra of functions on $\mathcal{D}$ to the corresponding algebra on $\tilde{\mathcal{G}}$.

Proof. The composition is defined because $\mathcal{I}\mathcal{F}$ is the identity. If two geometric readers have the same composition with $\mathcal{F}$, they agree on its image, which is all of $\tilde{\mathcal{G}}$. The algebraic operations hold pointwise; for example, $(\mathcal{R}_1\mathcal{R}_2) \circ \mathcal{I} = (\mathcal{R}_1 \circ \mathcal{I})(\mathcal{R}_2 \circ \mathcal{I})$. The inverse composition is restriction along $\mathcal{F}$. $\square$

Faithfulness refers to the representation (K, ξ) . For an observable $\mathcal{O}$ of the full enriched state, the descent criterion is that $\mathcal{O}(x) = \mathcal{O}(x')$ whenever $(K(x), \xi(x)) = (K(x'), \xi(x'))$. Necessity follows by composition; sufficiency allows the value on a fibre to be defined using any of its representatives. This criterion explicitly preserves which information must accompany the record.

Now let $\mathcal{T}$ be a triangulation of the labelled polygon determining a positive Plücker chart. Its boundary coordinates and a homogeneous normalization are included, for example $\Delta_{0L} = 1$. If the diagonal ac is exchanged for bd , with $a < b < c < d$, the change of chart is

$$\Delta_{bd} = \frac{\Delta_{ab}\Delta_{cd} + \Delta_{ad}\Delta_{bc}}{\Delta_{ac}}. \quad (30.2)$$

The denominator is positive, and the Plücker relation proves that both charts evaluate the same point.

Corollary 30.3 (Compatibility of readers with mutations). *Let $r_{\mathcal{T}}$ and $r_{\mathcal{T}'}$ be the expressions of $\mathcal{R}^{\text{geom}}$ in two charts preserving the markings, and let $\mu_{\mathcal{T}\mathcal{T}'}$ be their positive transition. Then*

$$r_{\mathcal{T}'} \circ \mu_{\mathcal{T}\mathcal{T}'} = r_{\mathcal{T}}$$

on the common domain. A closed sequence of chart changes that preserves the marked point preserves all these readers.

Proof. The two reconstructions through (30.1) recover the same K . The coordinates ξ are transported with their markings. Applying $\mathcal{R}$ proves the equality; iterating it proves the claim about a closed sequence. $\square$

Readers can be expressed in different charts while remaining functions of the same object. A permutation of labels acts additionally on the positional marking; a transformation changing the point acts on the reader's argument. Each operation has its own transport law.

The diagonal action on columns provides an explicit check. For $\lambda_i > 0$, $\Delta_{ij} \mapsto \lambda_i \lambda_j \Delta_{ij}$. The cross-ratio $\Delta_{ab}\Delta_{cd}/(\Delta_{ad}\Delta_{bc})$ is preserved, while $\Delta_{j-1,j}/\Delta_{0L}$ acquires the factor $\lambda_{j-1}\lambda_j/(\lambda_0\lambda_L)$. With $t = (0, 1/4, 1/2, 1)$ and $\lambda = (1, 2, 1, 1)$, the reconstructed gaps would be $(1/2, 1/2, 1/2)$, whose sum is $3/2$. Cross-ratios alone preserve a quotient different from the one required by the marked inverse.

30.3 Explicit composition with action and constitutive response

In the dodecaphasic chart, we write $b = 1000$ and

$$\kappa^{\text{geom}} = \frac{Q}{b^{12} - 1} \sum_{j=1}^{12} b^{12-j} \frac{\Delta_{j-1,j}}{\Delta_{0,12}}, \quad \alpha^{\text{geom}} = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa^{\text{geom}}. \quad (30.3)$$

These are the geometric expressions of the closure reader (29.1). The three regional representations and K share a coinductive generation; the expression uses their internal constructive order. The action reader (29.2), evaluated at α^{geom} , determines $\eta_{\text{ret}}^{\text{geom}}$, and therefore

$$\hbar_{\text{ret}}^{\text{geom}} = \eta_{\text{ret}}^{\text{geom}} 10^{-34} \mathcal{U}_S = \hbar_{\text{ret}}.$$

The equality holds in the action line $\mathcal{L}_S$ with its section $\mathcal{U}_S$; it preserves the decade arising from the determinantal reader $\varphi_{\text{HMT}} \alpha^{16}$. The composition neither changes the section nor applies it a second time.

In the lifted angular chart preserving the origin and real branch of the representation, the subsequent angular pair determines

$$x = \frac{\pi_{\text{HMT}}}{180} (1000\alpha), \quad y = \frac{\pi_{\text{HMT}}}{90} (\eta_{\text{ret}} + \alpha), \quad q_{\pm} = e^{-x \mp y}.$$

The branch is part of the retained markings: reduction of an angle modulo 360 alone preserves different information. The equalities below use the declared lift, in which the first coordinate is 1000α . In the chamber $x > |y|$, with fixed sheet projectors $P_{\pm}$, let $T = q_+ P_+ + q_- P_-$. The response of the corpus compares incidences 90 and 120 through

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1} = r_+ P_+ + r_- P_-, \quad r_{\pm} = f(q_{\pm}^{30}), \quad f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}.$$

The rule satisfies $\mathcal{V}_{\Sigma_4}(T^{30}) = \Sigma_3(T^{30})$, where $\Sigma_j(S) = \sum_{a=0}^{j-1} S^a$. Its uniqueness follows from invertibility of $\Sigma_4(T^{30})$. The four normalized readings are

$$\hat{\varepsilon} = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{Z} = r_-/r_+, \quad \hat{c} = (r_+ r_-)^{-1}.$$

In the dimensional lines of action, charge, and speed they are realized as

$$Z = \hat{Z} u_S u_Q^{-2}, \quad c = \hat{c} u_C, \quad \mu = \hat{\mu} u_S u_C^{-1} u_Q^{-2}, \quad \varepsilon = \hat{\varepsilon} u_S^{-1} u_C^{-1} u_Q^2.$$

The identities $\mu\varepsilon = c^{-2}$ and $\mu/\varepsilon = Z^2$ follow by multiplication and division. The chart-change corollary preserves this complete response, with the sheet markings and dimensional sections included in ξ . The same argument applies to a mass reader with its route, signature, and return operator retained, on the domain of Section 29.

Theorem 30.4 (Recovery of the record from the labelled response). *With the regional representations, sheet orientation, base $b = 1000$, origin, length twelve, and declared angular lift fixed, the map $K \mapsto (\hat{\varepsilon}, \hat{\mu})$ is injective on the integer records of its physical image satisfying $0 \leq K_j < b$ and the stated contractive chamber. Its exact inverse is the composition of the following operations:*

$$\begin{aligned} r_+ &= \sqrt{\hat{\varepsilon}}, & r_- &= \sqrt{\hat{\mu}}, \\ r_{\pm} s_{\pm}^3 + (r_{\pm} - 1)(1 + s_{\pm} + s_{\pm}^2) &= 0, & 0 < s_{\pm} < 1, \\ x &= -\frac{1}{60} \log(s_+ s_-), & \alpha &= \frac{180x}{1000\pi_{\text{HMT}}}, \\ \kappa &= \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \alpha, & N &= (b^{12} - 1)\kappa, \\ K_j &= \lfloor N/b^{12-j} \rfloor \bmod b. \end{aligned}$$

Proof. The positive roots recover the labelled coefficients. The derivative

$$f'(s) = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0 \quad (0 < s < 1)$$

and its limits 1 and $3/4$ prove that the cubic equation has exactly one root in that interval. Since $s_{\pm} = e^{-30x \mp 30y}$, their product is e^{-60x} . This recovers x , α , and κ . On the image considered, $N = \sum_j K_j b^{12-j}$ is an integer between zero and $b^{12} - 1$. Successive Euclidean division uniquely recovers its twelve digits, including leading zeros. $\square$

The precision needed for numerical recovery can also be expressed. With exact regional representations, an error $|\tilde{\alpha} - \alpha| < 1/[2(b^{12} - 1)]$ implies $|\tilde{N} - N| < 1/2$; rounding to the nearest integer recovers N . If the regional representations have errors, the same condition applies to the sum of errors in $\pi + e - \varphi - \alpha$. Exact injectivity and metrological resolution are thus distinct properties.

The integer domain is essential. In an analytic prolongation of positive records, perturbing three consecutive positions proportionally to $(1, -(b+1), b)$ preserves both the sum and the positional reading, since $1 - (b+1) + b = 0$ and $b^2 - b(b+1) + b = 0$. For sufficiently small perturbations it preserves positivity and changes the Grassmannian point. The preceding injectivity uses the condition of twelve integer digits; the continuous analytic collection has different fibres.

30.4 Refinement and inherited readers

If each gap d_j is subdivided into positive gaps $d_{j,a}$ with $\sum_a d_{j,a} = d_j$, the original endpoints retain their positions. Let ρ be the operation of summing the successors of each interval and H the inclusion of its endpoints in the new ordered set. Then

$$J_{\text{heredada}}(B(\vec{d}), Q) = (Q \sum_a d_{j,a})_j = K, \quad \mathcal{R}_{\text{heredada}} = \mathcal{R}(K, \xi).$$

The equality proves commutativity between inherited reconstruction and subdivision. In a sequence of subdivisions, sums associate; the result persists in every finite composition. This preserves the length-twelve reader and its markings. A reader defined on the new blocks will have its own domain and comparison law.

30.5 Canonical forms valued in a dimensional line

Let P be one of the rank-one polytopes constructed in the manuscript and $P = \bigcup_{\sigma} P_{\sigma}$ an oriented subdivision of the same support, with multiplicity one and compatible faces. We write Ω_{σ} and Ω_P for their canonical forms. Their identities are $\Omega_P = \sum_{\sigma} \Omega_{\sigma}$ and $\text{Res}_F \Omega_P = \Omega_F$, with the orientation convention fixed in the amplituhedral development. The action line $\mathcal{L}_S$ is taken to be constant over this polytope; its section comes from the fixed HMT state.

Theorem 30.5 (Transport of the form by an action section). *For $a \in \mathcal{L}_S$, constant with respect to the coordinates of the polytope,*

$$\Omega_P := a \otimes \Omega_P = \sum_{\sigma} a \otimes \Omega_{\sigma}, \quad \text{Res}_F \Omega_P = a \otimes \Omega_F.$$

The identities are preserved under successive subdivisions and iterated residues on flags of faces. They apply, in particular, to $a = \hbar_{\text{ret}}$.

Proof. The tensor product with a line is linear. The residue acts on the differential factor and commutes with the constant coefficient. Compatibility of subdivisions and iterated residues is transported by the same linear operations. Under a dimensional change of basis $u'_S = \lambda u_S$, with $\lambda > 0$, the numerical coefficient changes by λ^{-1} ; the tensor and its residues remain unchanged. $\square$

Theorem 30.6 (Gluing condition for boundary coefficients). *Let a_σ be sections of the same line, regular in a neighbourhood of the faces and meromorphic in a common ambient realization. Suppose their products with the canonical forms have poles of at most first order along the faces considered. In the relative interior of a common codimension-one face F of two cells σ, τ , with opposite orientations,*

$$\text{Res}_F(a_\sigma \Omega_\sigma + a_\tau \Omega_\tau) = (a_\sigma|_F - a_\tau|_F) \Omega_F. \quad (30.4)$$

Thus the pole of this pair on the divisor of F disappears exactly when the two traces agree. If the coefficients are constant and the cell adjacency graph is connected, cancellation for each pair of cells across their common face is equivalent to the existence of a single common coefficient a .

Proof. With a transverse local coordinate z , a logarithmic form is written $dz/z \wedge \omega + \eta$, where η is regular with respect to that divisor. Regularity of a_σ gives $\text{Res}_F(a_\sigma \Omega_\sigma) = a_\sigma|_F \text{Res}_F \Omega_\sigma$. The two orientations produce Ω_F and $-\Omega_F$. The form Ω_F is nonzero in the relative interior of the face; equality of residues is equivalent to equality of traces. For a simple pole, vanishing of the residue removes the pole. In the constant case, equality propagates along every edge of the connected adjacency graph. $\square$

For the global sum, all terms with a pole on the same ambient divisor must be counted. If H is the hyperplane containing F , and I_H collects those terms, the general identity is

$$\text{Res}_H \left(\sum_\nu a_\nu \Omega_\nu \right) = \sum_{\nu \in I_H} a_\nu|_H \text{Res}_H \Omega_\nu,$$

provided the coefficients are regular on H . A coplanar facet, even if it is not adjacent to the real face F , contributes to this sum. The two-term formula represents the global residue when the remaining terms are regular on that divisor. Pairwise cancellation of all interior facets removes their contributions; the exterior facets preserve the corresponding boundary residues.

On a face of higher codimension, this formula is applied successively with the induced orientations. The signs are those of cellular incidence and satisfy cancellation of the boundary of a boundary. The collection of local readers defines pairwise compatible gluing when their traces agree after transport to the same coefficient line. The global residue is obtained by summing the contributions of each divisor. This statement gives an effective criterion for distributing a physical reading over charts or cells without introducing spurious discontinuities.

30.6 Pole cancellation and global normalization

Interior cancellation determines a local condition. Identification with a global canonical form also requires the pole divisor and the normalized exterior residues. In projective space, two rational top-degree forms with the same logarithmic poles and the same residues differ by a holomorphic top-degree form; $H^0(\mathbb{P}^m, \Omega^m) = 0$ proves their equality. This is the global condition used in the polytope proof [13].

An interval example shows why variable coefficients require this condition. For $0 < c < 1$, let

$$\Omega_{0c} = \frac{c dx}{x(c-x)}, \quad \Omega_{c1} = \frac{(1-c) dx}{(x-c)(1-x)}, \quad \Omega_{01} = \frac{dx}{x(1-x)}.$$

With $a_1(x) = 1 + x(c-x)$ and $a_2(x) = 1$, their traces agree at all relevant endpoints, and yet

$$a_1 \Omega_{0c} + a_2 \Omega_{c1} = \Omega_{01} + c dx.$$

The additional term is regular in the affine chart and has a pole at projective infinity. The residues at the finite endpoints and cancellation at c are preserved, while the resulting form has different global behaviour. Multiplying the example by a section of $\mathcal{L}_S$ gives the same dimensional check.

The overall consequence is precise: the marked Grassmannian representation transports the retained readers; mutations preserve their values when they change only the chart; refinement preserves their inherited evaluations; and forms valued in the action line are assembled through equality of traces and control of poles. Each equality expresses the operation preserving the corresponding object.

31 Hexadic–octadic incidence, area operator and sectoral information

The geometric recovery of the record acquires physical content when composed with observables whose construction preserves the incidences and state conditions used. The passage from a hexad to its marked octad provides a precise case: enlarging the support transforms two incidence spaces, determines their multiplicities and allows that distinction to be preserved in an area operator and in the logarithm of a reduced state. The following exposition brings together that construction from the second article in the series [3] and the marked Grassmannian representation of the preceding section.

The starting point remains the enriched state produced by APP–TRIT–TPK. The two APP operations preserve their complete divisions, TRIT determines regime and orientation, and the composition $U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t$ transports and updates the information on position, carry and memory. Compatible prolongation and nonadic holonomy jointly preserve the arithmetic and incidence realizations of the discrete structure of the continuum. The record K , the regional coordinates $\pi_{\text{HMT}}, \varphi_{\text{HMT}}, e_{\text{HMT}}$, the coordinate α_{HMT} and the exceptional configuration arise from that common construction. Here we use its results and the markings that allow them to be prolonged towards boundary observables.

31.1 Incidence lifting and difference of multiplicities

Let $\mathcal{H}$ be the Witt hexad system on the twelve positions of the record. Fix the incidence chart $\ell : [12] \rightarrow \mathcal{D}$, where $\mathcal{D}$ is a dodecad of the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. The chart transports the hexads of $\mathcal{H}$ to the residual design on $\mathcal{D}$. This realization datum preserves the marked configuration determined in the exceptional development. The weight-eight supports of G_{24} constitute the octads of the system $S(5, 8, 24)$.

Proposition 31.1 (Unique lift of a residual hexad). *The sets $O \cap \mathcal{D}$, with O an octad and $|O \cap \mathcal{D}| = 6$, form a system $S(5, 6, 12)$. Each hexad of that system is the intersection of $\mathcal{D}$ with a unique octad. In particular, the marked hexad $H \subset \mathcal{D}$ has a determined lift O_H and an inclusion $H \subset O_H$.*

Proof. The code weights are 0, 8, 12, 16, 24. If $j = |O \cap \mathcal{D}|$, the binary sum of the vectors with supports O and $\mathcal{D}$ has weight $20 - 2j$; hence, $j \in \{2, 4, 6\}$. Every five-element subset $T \subset \mathcal{D}$ belongs to a unique octad. Since its intersection with $\mathcal{D}$ contains T , that intersection has six elements. This yields a unique residual hexad on each five-element subset. If two octads lifted the same hexad, both would contain any of its five-element subsets and would coincide by the Steiner property. $\square$

In what follows, $\binom{H}{2}$ denotes the set of two-element subsets of H . We preserve that second factor when enlarging the first:

$$\mathcal{F}_6 = H \times \binom{H}{2}, \quad \mathcal{F}_8 = O_H \times \binom{H}{2}, \quad j(h, P) = (h, P). \quad (31.1)$$

An element of these spaces is a point–pair incidence. The distinguished point ranges first over the hexad and then over its octad, while the pair continues to belong to the original hexad. This specification fixes the operation responsible for the increase.

Proposition 31.2 (Exact enlargement of the incidence space). *The map j is injective and its complement is $(O_H \setminus H) \times \binom{H}{2}$. Consequently,*

$$|\mathcal{F}_6| = 6 \binom{6}{2} = 90, \quad |\mathcal{F}_8| = 8 \binom{6}{2} = 120, \quad |\mathcal{F}_8 \setminus j(\mathcal{F}_6)| = 30. \quad (31.2)$$

Proof. The inclusion $H \subset O_H$ preserves both coordinates and proves injectivity. The disjoint partition $O_H = H \sqcup (O_H \setminus H)$ induces the partition of the Cartesian product. Since $|O_H \setminus H| = 2$ and $\binom{6}{2} = 15$, the complement has $2 \cdot 15 = 30$ elements. $\square$

The two multiplicities are obtained before any probability distribution on the boundary. The normalized difference $(90 - 120)/(24\binom{6}{2}) = -1/12$ preserves the sign of the oriented comparison. Its interpretation depends on the incidence space in which the operation has been performed.

31.2 Collective transport and the Barbero–Immirzi parameter

The boundary representation retains the labels of both sectors. To make its support explicit, write $x_j = r_{2j-1} + 3r_{2j}$, with $r_k \in \{0, 1, 2\}$, in three coordinates of $\mathbb{Z}/9\mathbb{Z}$. The positional regrouping

$$\beta_{729}(x_1, x_2, x_3) = (r_1 + 3r_2 + 9r_3, r_4 + 3r_5 + 9r_6) \quad (31.3)$$

is a bijection onto $(\mathbb{Z}/27\mathbb{Z})^2$. Uniqueness of the ternary expansion proves that its inverse extracts the six digits and restores the three initial pairs. Positional order is preserved with its carry information; the respective additions remain defined in their own groups.

In the block $B_\partial = \{0, \dots, 8\}^2$ of the torus of order twenty-seven, we distinguish the oriented links

$$E_{90} = \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \leq j \leq 8\}, \quad (31.4)$$

$$E_{120} = \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \leq i \leq 8\}. \quad (31.5)$$

The coordinates are interpreted modulo twenty-seven. The two sets are disjoint and contain eighteen links each: two sides of nine links per direction. The labels 90 and 120 arise from the incidence spaces; the cardinalities of the link sets are 18 and 18.

Let u_m be the normalized uniform vectors of $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$, with $m = 90, 120$, and let v_m be the corresponding normalized uniform vectors of $\ell^2(E_{90}) \oplus \ell^2(E_{120})$. For example, $u_{90} = 90^{-1/2} \sum_{f \in \mathcal{F}_6} (\delta_f, 0)$ and $v_{90} = 18^{-1/2} \sum_{e \in E_{90}} (\delta_e, 0)$. Each pair is orthonormal.

Proposition 31.3 (Preservation of the collective spectral structure). *The linear map $\mathcal{J}u_m = v_m$ is a surjective isometry between the two collective planes. If*

$$D_{\text{inc}} = 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \quad D_\partial = 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|,$$

then $\mathcal{J}D_{\text{inc}} = D_\partial \mathcal{J}$. The sectoral projectors and their oriented combinations are transported by the same map.

Proof. The map takes one orthonormal basis to another and is determined by those images. The operator equality holds on u_m , since both sides have value mv_m . The projectors are $(120I - D)/30$ and $(D - 90I)/30$ in each plane, so that their combinations are preserved by linear composition. $\square$

The preceding transport acts on the uniform modes. The intrinsic fluctuations of each incidence space remain in their orthogonal complements. Area weighting uses precisely the two labels preserved by this isometry.

The Barbero–Immirzi parameter is determined from those data and the previously constructed angular orientation. Let A and C^* be the angular coordinates in degrees, in the lift fixed in the preceding section, and set

$$x = \frac{\pi_{\text{HMT}} A}{180}, \quad y = \frac{\pi_{\text{HMT}} C^*}{180}, \quad q_\pm = e^{-x \mp y}, \quad S_m(q) = \frac{q^m}{1 - q^{3m}}.$$

The chamber $x > |y|$ is preserved, so that $0 < q_\pm < 1$. The oriented chain of one turn assigns twelve sectors to $\mathcal{F}_6$ and an oppositely oriented return seam to $\mathcal{F}_8$. Its evaluation is $12S_{90} - S_{120}$. The coefficient of $(1 - q)^{-1}$ is determined by

$$\lim_{q \uparrow 1} (1 - q)(12S_{90}(q) - S_{120}(q)) = \frac{12}{270} - \frac{1}{360} = \frac{1}{24},$$

because $1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$. The conjugate evaluation of the chain, together with the angular component, defines the structural parameter

$$\gamma_{\text{HMT}} = \frac{\pi_{\text{HMT}} A C^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)]. \quad (31.6)$$

The origin of the coefficients $12 : -1$ lies in the oriented chain. The preceding factorization determines its pole coefficient. The inversion $C^* \mapsto -C^*$ interchanges the channels and changes the sign of γ_{HMT} . In what follows, we retain the branch $C^* > 0$, $\gamma_{\text{HMT}} > 0$, with the area normalization fixed by that same orientation.

31.3 Longitudinal return and determination of the area scale

The dimensional unit of area admits an explicit construction from the same enriched state. We write L_α for the resulting physical length; the integer L denoting the length of a record in Section 30 retains its combinatorial meaning. This distinction separates the cardinality of a representation, the extent of an edge, and the radius of its realisation.

The composition uses the regional coordinates and the record already generated by APP-TRIT-TPK. APP retains both sheets with their remainders and quotients; TRIT orients the local regime; TPK prolongs the state, retaining carry, boundary, route, and memory. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ and the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ retain phase return with memory advancement. The joint construction of the continuum retains its spectral, solenoidal, gauge, cohomological, and determinantal aspects. The length constructed below is a subsequent dimensional reading of that structure and its correlated outputs $\pi_{\text{HMT}}, \varphi_{\text{HMT}}, e_{\text{HMT}}, \alpha_{\text{HMT}}$.

Fix the marked record

$$\Omega = \{(j, k, q, u, v) : j, k \in \mathbb{Z}/12\mathbb{Z}, \quad k - j \in \{0, 3, 6, 9\}, \quad 1 \leq q \leq 8, \quad u < v, \quad u, v \in \{1, 2, 4, 5, 7, 8\}\}.$$

Subtraction is taken in that cyclic group. Its cardinality is $N_\Omega = 12 \cdot 4 \cdot 8 \binom{6}{2} = 5760$. The factor 120 comes from the previously constructed octadic point-pair incidence space; the twelve positions and the four positions in each orbit retain their respective markings. This resolution extends the carrier of the Hadamard return. Between the initial and final Hermitian spaces, each of dimension N_Ω , let B be the transported inverse, with $B^*B = BB^* = I/4$. The local identity $H_4^*H_4 = 4I$ produces this normalisation for the inverse; its orthogonal extension and the transport of the markings preserve it.

In the final space, let $u_\Omega = N_\Omega^{-1/2} \mathbf{1}$, $P = u_\Omega u_\Omega^*$, $C = (I - P)B$, and $M = PB$. The complete transport $C + M = B$ is retained. The atomic projectors $P_i = e_i e_i^*$, determined by the inscription, define the diagonal expectation $\Delta_\Omega(T) = \sum_i P_i T P_i$.

Proposition 31.4 (Recorded response and retained complement). *Among the linear maps into the diagonal algebra that fix the P_i and are bimodular over that algebra, Δ_Ω is unique. Moreover,*

$$\Delta_\Omega(CC^*) = r_\Omega I, \quad r_\Omega = \frac{N_\Omega - 1}{4N_\Omega} = \frac{5759}{23040}, \quad \Delta_\Omega(MM^*) = \frac{I}{4N_\Omega}. \quad (31.7)$$

The two responses sum to $I/4$.

Proof. For the matrix unit E_{ij} , bimodularity imposes $E(E_{ij}) = P_i E(E_{ij}) P_j$. Since the image is diagonal, this term is zero when $i \neq j$; when $i = j$, fixing P_i determines its image. The matrix units form a basis and determine the map. Furthermore, $CC^* = (I - P)/4$, $MM^* = P/4$, and every diagonal entry of P is $1/N_\Omega$. Applying the expectation proves the three identities. $\square$

The isometric inscription $W e_i = e_i \otimes e_i$ realises $W^*(T \otimes I)W = \Delta_\Omega(T)$. The complete inscription retains the information in the record; the expectation extracts its diagonal response. The complement M remains in the transport and has the specific response given in (31.7). This separation identifies the observable entering the length and the complementary component that is retained.

The Smith normal form of H_4 is $\text{diag}(1, 2, 2, 4)$; its cokernel has order sixteen. The product of the generated scalar $\alpha = \alpha_{\text{HMT}}$, indexed by the classes of that cokernel, defines the finite multiplicative reader

$$n_H := \prod_{\bar{v} \in \text{coker}_{\mathbb{Z}}(H_4)} \alpha = \alpha^{16}.$$

Let $L_* > 0$ be a radial reference section of the dimensional line of length. The longitudinal composition acts linearly on an oriented cochain β_* :

$$\mathcal{T}_L \beta_* = \alpha^{16} (\Delta_{\Omega}(CC^*) \otimes I) \beta_* = q_{\alpha} \beta_*, \quad q_{\alpha} = \alpha^{16} r_{\Omega}, \quad L_{\alpha} = q_{\alpha} L_*. \quad (31.8)$$

The formula specifies a concrete composition of readers. Linearity preserves the length type and its orientation; the coefficient comes from the recorded return and the local norm. For example, the diagonal response to a source ae_i is ar_{Ω} , whereas the norm $\|C^* ae_i\| = |a| \sqrt{r_{\Omega}}$ constitutes another reading of the same transport.

If $\beta_*(\bar{a}) = -U(a)\beta_*(a)$, multiplication by q_{α} preserves oriented inversion. If an edge is subdivided into nine compatible segments, $\beta_*(a) = \sum_j U_j^{-1} \beta_*(a_j)$, the same multiplication preserves that identity and memory concatenation. In an auxiliary extension of the record, P , CC^* , and Δ_{Ω} are transported by tensoring with the identity. Thus r_{Ω} is preserved; the cardinality of a refined mesh and the cardinality of the marked record have distinct functions.

31.4 Radial reciprocity and recognition of the Planck length

The circular reading of the one-hundred-and-eight-step calendar retains $108\ell_0 = 2\pi_{\text{HMT}} L_{\alpha}$, with $\ell_0 = ct_0$. Consequently,

$$t_0 = \frac{\pi_{\text{HMT}} L_{\alpha}}{54c}, \quad \omega = \frac{c}{L_{\alpha}}, \quad Q_{\text{clk}} = \hbar_{\text{ret}} \omega = \frac{\hbar_{\text{ret}} c}{L_{\alpha}}. \quad (31.9)$$

The returned action is obtained from its full HMT equation in (29.2); the speed is obtained from the two constitutive channels of Section 30. The circular relation converts the lifted advance into a radius. If $c = \hat{c} u_C$, $t_0 = u_T$, and $u_L = u_C u_T$, then $L_*/u_L = 54\hat{c}/(\pi_{\text{HMT}} q_{\alpha})$. The radial reference L_* and the edge unit u_L are related with their dimensional types made explicit.

In a finite fibre, let $X = X^* > 0$ be an invertible positive character, $U = 2B$, and $Y = UXU^*$. The longitudinal transport is $D_L = L_{\alpha} U$. Its positive radial response is defined by the metric in the final space, $\|R_X v\|^2 = \|XD_L^* v\|^2$ for every v . The energy and conjugate length retain the same character:

$$H_X = Q_{\text{clk}} Y, \quad M_X = H_X/c^2, \quad \Lambda_X = L_{\alpha} Y^{-1}.$$

Theorem 31.5 (Common radial coefficient and area scale). *The preceding conditions determine a unique positive response $R_X = L_{\alpha} Y$. The following identities hold:*

$$R_X \Lambda_X = L_{\alpha}^2 I, \quad H_X \Lambda_X = \hbar_{\text{ret}} c I, \quad R_X = \frac{L_{\alpha}^2}{\hbar_{\text{ret}} c} H_X. \quad (31.10)$$

In the physical realisation identifying R_X with the reduced gravitational radius $R_X = (G/c^2)M_X$, the coefficient is unique and common to all admissible positive characters:

$$G = \frac{c^3 L_{\alpha}^2}{\hbar_{\text{ret}}}, \quad \frac{\hbar_{\text{ret}} G}{c^3} = L_{\alpha}^2. \quad (31.11)$$

Proof. Polarisation of the equality of norms determines $R_X^2 = D_L X^2 D_L^* = L_{\alpha}^2 U X^2 U^*$. The operator $L_{\alpha} U X U^*$ is positive and its square is the specified operator. Uniqueness of the positive square root proves the first assertion. The two product identities follow by substitution. The physical convention gives $L_{\alpha} Y = (G Q_{\text{clk}}/c^4) Y$. Invertibility of Y allows its cancellation, and the expression for Q_{clk} yields (31.11). If another coefficient preserves the same data, its difference multiplied by M_X is zero and is therefore itself zero. $\square$

The proof applies to the finite positive fibre; it also extends to bounded uniformly positive operators satisfying the same identities. The reduced-gravitational-radius reading is the declared physical convention of this realisation. The subsequent conventional recognition $\ell_P^2 = \hbar_{\text{ret}} G/c^3$ identifies $\ell_P = L_\alpha$. The length is first obtained through (31.8); this identification preserves its construction. The equivalent circular formula is

$$G = \left(\frac{54}{\pi_{\text{HMT}}} \right)^2 \frac{c^5 t_0^2}{\hbar_{\text{ret}}}.$$

The circular factor is already included when L_α is used. The length and action section also determine

$$t_P = \frac{L_\alpha}{c}, \quad m_P = \frac{\hbar_{\text{ret}}}{cL_\alpha}, \quad E_P = \frac{\hbar_{\text{ret}} c}{L_\alpha}.$$

These identities follow by substituting (31.11) into the conventional expressions for the three scales; positivity determines their roots. For a mode of character $y > 0$, the readings are $m(y) = m_P y$, $r_g(y) = L_\alpha y$, and $\bar{\lambda}(y) = L_\alpha/y$. Their longitudinal product is $r_g(y)\bar{\lambda}(y) = L_\alpha^2$. Equality of the two lengths is equivalent to $y = y^{-1}$ and, by positivity, characterises exactly $y = 1$. The calendar retains $t_0 = (\pi_{\text{HMT}}/54)t_P$ and $108t_0 = 2\pi_{\text{HMT}}t_P$.

A horizon radius $R_h = L_\alpha \zeta_h$ retains its own factor $\zeta_h = R_h/L_\alpha$. Its spatial area belongs to the specified horizon; the elementary scale of the following sectorial operator is L_α^2 . In the Schwarzschild specialisation for the same mass, the radius is $2r_g(y)$ and thus $\zeta_h = 2y$; a horizon in another realisation retains its own geometric formation condition.

31.5 Boundary occupations and sectorial reconstruction

Each link of $E_{90} \sqcup E_{120}$ determines a factor $\mathbb{C}^3$, with basis $|0\rangle, |1\rangle, |2\rangle$ and number operator $N_e = \text{diag}(0, 1, 2)$. The boundary state space is $\mathcal{H}_A = (\mathbb{C}^3)^{\otimes 18} \otimes (\mathbb{C}^3)^{\otimes 18}$. We define N_m as the sum of the number operators of the eighteen links in sector m . The two operators commute and each has integer spectrum between zero and thirty-six.

The angular scale arises from the clock and the normalized character of the conjugate triplet:

$$\theta_{\text{clk}} = \frac{C^*}{6} \frac{\pi_{\text{HMT}}}{180}, \quad \chi_{10} = \frac{1}{3} \text{Tr} \text{diag}(1, e^{i\pi_{\text{HMT}}/18}, e^{-i\pi_{\text{HMT}}/18}) = \frac{1 + 2 \cos(\pi_{\text{HMT}}/18)}{3}, \quad a_0 = \chi_{10} \theta_{\text{clk}}. \quad (31.12)$$

The character identity is obtained by summing the conjugate phases. On the branch considered, $a_0 > 0$. The area unit is $L_\alpha^2 = \ell_P^2$, constructed and recognized in (31.8)–(31.11). Write $a_\partial = \gamma_{\text{HMT}} a_0 > 0$. The area operator is

$$\hat{\mathcal{A}} = L_\alpha^2 a_\partial N, \quad N = N_{90} + N_{120}, \quad D = 90N_{90} + 120N_{120}. \quad (31.13)$$

Its spectrum is $\{nL_\alpha^2 a_\partial : 0 \leq n \leq 72\}$. The multiplicity of n is the coefficient of z^n in $(1 + z + z^2)^{36}$, since each tensor factor contributes one of the three occupations.

Theorem 31.6 (Reconstruction of the two sectorial occupations). *The pair of observables (N, D) determines*

$$N_{90} = 4N - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3N. \quad (31.14)$$

In the joint spectrum, their values (n, d) satisfy $n \in \mathbb{Z}$, $d \in 30\mathbb{Z}$, $90n \leq d \leq 120n$ and $0 \leq 4n - d/30 \leq 36$, $0 \leq d/30 - 3n \leq 36$. These conditions characterize the image of the integer occupations.

Proof. The transformation from (N_{90}, N_{120}) to (N, D) has matrix $\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}$, whose determinant is 30. Its inverse gives the operator identities. Commutativity allows them to be applied in each joint eigenspace. The inequalities express nonnegativity and the maximum capacities of the sectors; divisibility makes both occupations integers. Conversely, each integer between zero and thirty-six can be expressed as a sum of eighteen terms from $\{0, 1, 2\}$, and therefore every pair satisfying these conditions belongs to the joint spectrum. $\square$

Area preserves the sum of occupations; the weighted observable preserves their sectoral composition. Thus, $(N_{90}, N_{120}) = (1, 0)$ and $(0, 1)$ have the same area $L_\alpha^2 a_\partial$ and weights 90 and 120. Complementarily, $(4, 0)$ and $(0, 3)$ have the same weight $D = 360$, but different areas. The pair distinguishes both situations. Within each joint eigenspace, the microscopic degeneracies corresponding to the distributions among links are preserved.

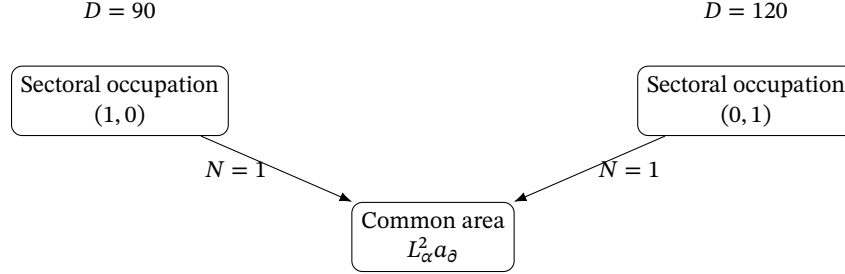


Figure 14: Equality of area preserves total occupation. Incidence weighting distinguishes the two compositions; their joint reading allows the occupations of both sectors to be reconstructed.

31.6 Reduced state, purification and entanglement entropy

The informational interpretation requires the state to be fixed. For finite $\Delta \in \mathbb{R}$, with the normalization of the modular parameter of the sectoral construction, we define

$$x_m = e^{-m\Delta}, \quad z_m = 1 + x_m + x_m^2, \quad Z = z_{90}^{18} z_{120}^{18}, \quad \rho_A(\Delta) = Z^{-1} e^{-\Delta D}. \quad (31.15)$$

The sum of commuting occupations and the tensor decomposition prove the expression for Z . All eigenvalues of ρ_A are strictly positive for finite real Δ . Its modular Hamiltonian, denoted by K_A to distinguish it from the arithmetic record K , is

$$K_A = -\log \rho_A = (\log Z)I + \Delta D. \quad (31.16)$$

This operator characterizes the spectral distribution of the reduced state. The time-evolution generator is defined in the dynamical development with its own parameters and domain. For $\Delta \neq 0$, with Δ and the normalization known, the pair $(\hat{\mathcal{A}}, K_A)$ recovers (N, D) and, through (31.14), both occupations. For $\Delta = 0$, the state is uniform and $K_A = 36 \log 3 I$.

We construct the bipartition explicitly. Let $\mathcal{H}_{\bar{A}}$ be a copy of $\mathcal{H}_A$, with the conjugate bases fixed. For a link of weight w , the vector

$$|\text{TFD}(w)\rangle = \frac{1}{\sqrt{1 + e^{-w} + e^{-2w}}} \sum_{n=0}^2 e^{-wn/2} |n\rangle_A \otimes |n\rangle_{\bar{A}}$$

has norm one. The pure state

$$|\Psi_\Delta\rangle = |\text{TFD}(90\Delta)\rangle^{\otimes 18} \otimes |\text{TFD}(120\Delta)\rangle^{\otimes 18} \quad (31.17)$$

determines the bipartition $A | \bar{A}$.

Proposition 31.7 (Sectoral reduction and entropy of the purification). *The partial trace of $|\Psi_\Delta\rangle\langle\Psi_\Delta|$ over $\mathcal{H}_{\bar{A}}$ is $\rho_A(\Delta)$. Defining*

$$f_m = \frac{x_m + 2x_m^2}{z_m}, \quad s(\Delta) = -\text{Tr}(\rho_A(\Delta) \log \rho_A(\Delta)),$$

we have

$$\langle N \rangle_\Delta = 18(f_{90} + f_{120}), \quad (31.18)$$

$$\langle D \rangle_\Delta = 18(90f_{90} + 120f_{120}), \quad (31.19)$$

$$\langle \hat{A} \rangle_\Delta = 18L_\alpha^2 a_\partial (f_{90} + f_{120}), \quad (31.20)$$

$$s(\Delta) = 18 \log z_{90} + 18 \log z_{120} + \Delta \langle D \rangle_\Delta. \quad (31.21)$$

The last expression is the entanglement entropy of the purification (31.17), in dimensionless units.

Proof. In the partial trace of a link, the orthogonality of $|n\rangle_{\bar{A}}$ eliminates terms with distinct indices and leaves the weights $(1, x_m, x_m^2)/z_m$. The partial trace of a tensor product is the product of the partial traces; this yields (31.15). The first moment of each link is f_m . Additivity of the number operators gives the first three identities. The last follows by substituting (31.16) into $s = \text{Tr}(\rho_A K_A)$. The coefficients of the preceding construction are Schmidt coefficients of the stated bipartition, so this entropy coincides with its entanglement entropy. $\square$

The maximum entropy of this space is $36 \log 3$, attained at $\Delta = 0$. The quantity $36 \log 3 - s(\Delta)$ is the relative entropy of $\rho_A(\Delta)$ with respect to $I/3^{36}$. Incidence fixes the sectors and their multiplicities; the state parameter fixes the probabilities; the dimensional chart converts occupation into area. The physical identification of $\mathcal{H}_A$ preserves the bipartition and the algebra of observables just specified.

31.7 Finite modular identity and transport of geometric information

Fix the reference state $\rho_0 = \rho_A(\Delta_0)$, with finite real Δ_0 , and hold $\Delta_0, a_\partial, L_\alpha$ constant. The normalized sectoral areas $\mathcal{A}_m^n = a_\partial N_m^n$ satisfy

$$K_0 = -\log \rho_0 = c_0 I + \beta_{90} \mathcal{A}_{90}^n + \beta_{120} \mathcal{A}_{120}^n, \quad c_0 = \log Z(\Delta_0), \quad \beta_m = \frac{m \Delta_0}{a_\partial}. \quad (31.22)$$

The relation $\beta_{120} = \frac{4}{3} \beta_{90}$ preserves the incidence ratio. For $\Delta_0 \neq 0$, the ratio of the two coefficients is defined and equals $4/3$.

Theorem 31.8 (Finite identity of area and relative information). *For every normalized state σ of $\mathcal{H}_A$, let $\delta_\sigma \langle B \rangle = \text{Tr}[(\sigma - \rho_0)B]$. Then*

$$s(\sigma) - s(\rho_0) = \sum_{m \in \{90, 120\}} \beta_m \delta_\sigma \langle \mathcal{A}_m^n \rangle - D(\sigma \| \rho_0), \quad (31.23)$$

where $D(\sigma \| \rho_0) = \text{Tr}[\sigma(\log \sigma - \log \rho_0)]$ is finite and nonnegative, with the convention $0 \log 0 = 0$.

Proof. The definition yields $D(\sigma \| \rho_0) = -s(\sigma) + \text{Tr}(\sigma K_0)$, while $s(\rho_0) = \text{Tr}(\rho_0 K_0)$. Subtracting the identities and substituting (31.22) proves (31.23); the scalar term vanishes by normalization of the traces. For positivity, let p_i and u_i be the eigenvalues and eigenvectors of σ , and $q_j > 0, v_j$ those of ρ_0 . With $r_i = \langle u_i, \rho_0 u_i \rangle$, concavity of $\log$ gives $\langle u_i, \log \rho_0 u_i \rangle \leq \log r_i$. Thus, $D(\sigma \| \rho_0) \geq \sum_i p_i \log(p_i/r_i) \geq 0$, by convexity of $t \log t$, since $\sum_i r_i = 1$. Full rank of ρ_0 guarantees finiteness. $\square$

Its infinitesimal version keeps the reference state fixed. If $\sigma(\varepsilon)$ is differentiable, normalized and $\sigma(0) = \rho_0$, then

$$\left. \frac{ds(\sigma(\varepsilon))}{d\varepsilon} \right|_0 = \sum_m \beta_m \left. \frac{d\langle \mathcal{A}_m^n \rangle_{\sigma(\varepsilon)}}{d\varepsilon} \right|_0.$$

Indeed, the derivative of the trace of $-\sigma \log \sigma$ is $-\text{Tr}[\dot{\sigma}(\log \rho_0 + I)]$; normalization eliminates $\text{Tr} \dot{\sigma}$ and substitution of K_0 gives the formula. The finite identity adds the exact relative-entropy correction to that tangent response. The coefficients of the physical areas are β_m/L_α^2 , preserving the same dimensional chart.

Composition with the Grassmannian representation finally specifies what information is preserved. In $\mathcal{F}(K, \xi) = ([B(K)], Q, \xi)$, the additional coordinates ξ retain the regional magnitudes, angular orientation, marked inclusion, sector identification, occupation basis, parameter Δ and relevant area unit. The inverse of Proposition 30.1 recovers K ; the other conditions are preserved through their markings. By Theorem 30.2, composition with that inverse transports $\gamma_{\text{HMT}}, \hat{\mathcal{A}}, \rho_A, K_A$ and their evaluations on the stated domain. Changes of chart of the same marked point preserve these objects, and refinements preserve the inherited evaluations when they maintain the same sectoral data.

This articulation preserves three distinguishable levels of information. The marked positive representation recovers the record with the retained data; the area-modular-Hamiltonian pair recovers the two occupations under the conditions of (31.16); the purification determines the entanglement of a specific bipartition. Information loss in a projection is computed through its fibers: area identifies occupations with equal sums, while the joint reading distinguishes their composition and preserves the remaining microscopic degeneracy. The link between geometry and information is thus expressed by verifiable maps and identities on the same construction.

31.8 Naturality of the scale and operator-valued canonical forms

The generated length makes the compatibility between the positive realisation, the area operator, and sectorial information explicit. The marked representation $\mathcal{F}(K, \xi)$ retains, among the relevant data in ξ , the resolution Ω , its projectors, the transported return, and the section L_* . The generated regional coordinates, the action, and the constitutive channels retain their preceding genealogy. The marked inverse therefore recovers every argument of (31.8) and (31.11). In particular, the previously constructed geometric reader of α^{geom} determines the expressions

$$L_{\alpha}^{\text{geom}} = L_*(\alpha^{\text{geom}})^{16} \frac{5759}{23040}, \quad G^{\text{geom}} = \frac{(c^{\text{geom}})^3 (L_{\alpha}^{\text{geom}})^2}{h_{\text{ret}}^{\text{geom}}}.$$

The speed and action readers are obtained by the same composition with the marked inverse. These expressions retain the return data and the radial reference section.

Corollary 31.9 (Preservation under changes of chart and inherited refinement). *Positive changes of chart preserving the marked point, and refinements recovering the initial record, preserve L_{α} , G , γ_{HMT} , and $\hat{\mathcal{A}} = L_{\alpha}^2 a_{\partial} N$, provided they also transport the declared sectorial and dimensional data. The normalised spectrum $\hat{\mathcal{A}}/L_{\alpha}^2 = a_{\partial} N$, its multiplicities, and the state $\rho_A(\Delta)$ remain unchanged.*

Proof. Theorem 30.2 transports each reader by composition with the inverse. Mutations of the same point reconstruct the same record. Under inherited refinement, summing the subdivided separations reconstructs each initial coordinate. Tensor extension of the inscription preserves r_{Ω} , while its remaining arguments are fixed by the markings. Substitution therefore preserves the length, the gravitational coefficient, and the area operator. The occupation projectors preserve the multiplicity polynomial $(1 + z + z^2)^{36}$ and the normalised expression of the state. $\square$

The inherited refinement considered here retains the same space of thirty-six boundary links. An extension introducing a different physical boundary space requires specified intertwining maps for N_{90} and N_{120} , together with the transport of the declared sectorial data.

Let P now be a rank-one polytope constructed in this manuscript, with an oriented subdivision $P = \bigcup_{\sigma} P_{\sigma}$ having the same support, multiplicity one, and compatible faces. The same sectorial fibre $\mathcal{H}_A$ is fixed over all its cells. The operator-valued top-degree form with the dimension of area is defined by

$$\Omega_P^{\mathcal{A}} = \hat{\mathcal{A}} \otimes \Omega_P.$$

Here $\hat{\mathcal{A}}$ is constant with respect to the differential coordinates of the polytope: it is determined by the fixed HMT state, whereas Ω_P describes its geometric support. The coefficient $\hat{\mathcal{A}}$ is positive semidefinite. Its component in the vacuum sector $N = 0$ vanishes. On an eigenspace with occupation $n > 0$, division by $nL_{\alpha}^2 a_{\partial}$ recovers the canonical form Ω_P multiplied by the identity on that eigenspace.

Proposition 31.10 (Areal additivity and boundary residues). *In the common sectorial trivialisation,*

$$\Omega_P^A = \sum_{\sigma} \hat{\mathcal{A}} \otimes \Omega_{P_{\sigma}}, \quad \text{Res}_F \Omega_P^A = \hat{\mathcal{A}} \otimes \Omega_F. \quad (31.24)$$

These identities persist under successive subdivisions and iterated residues. For a fixed state ρ , taking the trace gives $\text{Tr}(\rho \hat{\mathcal{A}}) \Omega_P$ and preserves the same scalar identities.

Proof. The operator space of $\mathcal{H}_A$ is finite-dimensional. In any basis, the assertion is the additivity identity for canonical forms multiplied by each constant matrix entry of $\hat{\mathcal{A}}$. Both residue and trace are linear. Cancellation at interior facets retains their orientations and the same operator entry on both sides. Iteration applies these operations to each subdivision and each flag of faces. $\square$

If distinct cells carry regular operator-valued coefficients A_{σ} , the defect of a pair along its common face is $(A_{\sigma}|_F - A_{\tau}|_F) \otimes \Omega_F$. Equality of the operator-valued restrictions to the boundary provides the gluing condition, under the simple-pole hypotheses of Theorem 30.6. The global sum must also retain every contribution along each ambient divisor and the control of exterior poles. Dimensional normalisation, residue cancellation, and coverage of the support therefore have distinct and compatible criteria.

The finite modular identity likewise admits a direct expression in physical areas $\mathcal{A}_m = L_{\alpha}^2 a_{\partial} N_m$:

$$s(\sigma) - s(\rho_0) = \sum_{m=90,120} \frac{m \Delta_0}{L_{\alpha}^2 a_{\partial}} \text{Tr}[(\sigma - \rho_0) \mathcal{A}_m] - D(\sigma \| \rho_0). \quad (31.25)$$

This is (31.23) after substitution of the constructed length. The angular factor a_0 , contained in $a_{\partial} = \gamma_{\text{HMT}} a_0$, is retained in full. When dimensional sections satisfying $L'_{\alpha} = s L_{\alpha}$ are compared while preserving the same state and the same dimensionless data, areas are multiplied by s^2 and their modular coefficients by s^{-2} . Entropy and relative entropy remain unchanged. This comparison of sections does not constitute a second solution for the same fixed longitudinal data.

For a spherical horizon of radius $R_h = L_{\alpha} \zeta_h$, the geometric area is $4\pi_{\text{HMT}} L_{\alpha}^2 \zeta_h^2$. Its identification with the expectation of the sectorial operator, when required for the same cross-section, imposes the explicit condition

$$a_{\partial} \text{Tr}(\rho N) = 4\pi_{\text{HMT}} \zeta_h^2.$$

The horizon geometry and the boundary state must satisfy this relation through their respective constructions. The factor ζ_h characterises that horizon; the longitudinal scale of the coupling remains L_{α} .

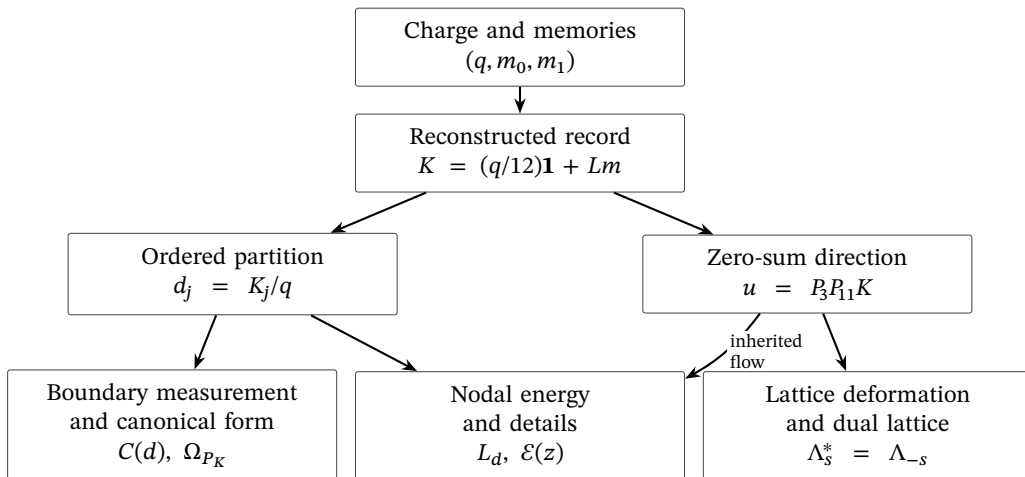


Figure 15: Subsequent realizations of the recoverable record. The interval branch determines path minors, the polygon's canonical form and the nodal Laplacian. The algebraic direction determines the interval flow and, together with the marked lattice realization, the twenty-four-dimensional dual collection. Each arrow preserves the domain and hypotheses of its construction; the lower codomains are distinct.

Part III

Gauge realization, refinement and spectral gap

32 Positivities, physical spaces, and spectral transport

Positive geometry and gauge theory receive distinct structures from the same genealogical process. The former realizes the weights and their incidences through ordered minors, cells, and canonical forms. The latter preserves gauge incidence, holonomy, representations, routes, and memory within a compatible measure. Its form of positivity relevant to physical reconstruction is reflection positivity. The conceptual connection acquires mathematical content when the maps producing each form are preserved.

In the geometric part, it has been proved that subdividing a cell preserves its canonical form through cancellation of interior poles. For nodal energies, Schur reduction preserves an effective form and records the detail. In the gauge part, refinement compatibility must also preserve the vacuum, the invariant subspace, the generator, and the spectral witness. This is the chain developed below: arithmetic–geometric scale, common measure, semigroup, physical projection, refinement, and continuous reconstruction.

The norm of a vector, the positivity of a moment matrix, and the existence of a Hamiltonian gap have different domains. A congruence with an invertible factor preserves the positivity and inertia of a form; an isometric intertwining of the generator, together with its domain, transports its realization onto the reduced image of the inclusion. The uniform bound and the persistent witness then determine the limiting gap. The development therefore keeps explicit the inclusions and semigroup identities that justify passage to the limit. The gap scale originates in areal–volumetric incidence and its declared dimensional unit.

The following sections bring together the projective construction and its subsequent development depending on the coupling g . The initial comparisons with a Gibbs parametrization are subsequently completed through the explicit action, measure, and transport. The expository order preserves the proofs of the process and places the complete formulation after its dependencies. The four reflections correspond to the four Euclidean directions of the reconstruction. The development assembled here concerns this gauge theory and its gap, with the compact, connected, simple group declared in the theorems.

The planar supersymmetric sector in which amplitudes are studied through amplituhedra retains a specific physical question. Connecting it with the present realization requires the supercharge operators, their domains, and the corresponding representation. The dimension four of the Euclidean chart and the difference four of the modular block remain previously defined results of their respective operators.

33 Areal–volumetric transition and spectral scale

The constant determining the gauge semigroup is obtained before the Hamiltonian is introduced. Its origin combines two independent readers of the same genealogical state: the return frequency yielding π_{HMT} and the stationary distribution of the two areal–volumetric sheets.

33.1 Dynamics of the two sheets

Let

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad e_{\text{ar}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_{\text{vol}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (33.1)$$

The Fibonacci recurrence gives, by induction,

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}. \quad (33.2)$$

The Perron vector is proportional to $(1, \varphi)$. Since F_{av} is symmetric, the Parry measure assigns the two sheets weights proportional to $(1, \varphi^2)$, and hence

$$\frac{\mu_{ar}}{\mu_{vol}} = \varphi^{-2}, \quad \mu_{vol} = \frac{\varphi^2}{1 + \varphi^2}. \quad (33.3)$$

Here μ_{vol} is a dimensionless frequency character; it does not denote a volume of dimension L^4 .

The regional reader of π is introduced exactly once through

$$D_\pi = \pi_{HMT} |e_{ar}\rangle \langle e_{ar}| + |e_{vol}\rangle \langle e_{vol}|. \quad (33.4)$$

The ratio obtained at depth n is

$$\Theta_n := \frac{\langle e_{ar}, D_\pi F_{av}^n e_{ar} \rangle}{\langle e_{vol}, F_{av}^n e_{vol} \rangle} = \pi_{HMT} \frac{F_{n-1}}{F_{n+1}} \longrightarrow r_{av} := \frac{\pi_{HMT}}{\varphi_{HMT}^2}. \quad (33.5)$$

The matrix, route, and quotient are fixed before consulting a mass gap or a gauge-theory parameter. At the native depth $n = 108$,

$$F_{107} = 10284720757613717413913, \quad F_{109} = 26925748508234281076009, \quad (33.6)$$

and

$$\left| \Theta_{108} - \frac{\pi_{HMT}}{\varphi_{HMT}^2} \right| < 2 \cdot 10^{-45}. \quad (33.7)$$

33.2 Dimensional transduction

Let ℓ_0 be the length unit of the dimensional reader. The inverse length scale and the parameter of a slab of thickness a_m are

$$\kappa_{av} := \frac{\mu_{vol}}{\ell_0} \left(\frac{\pi_{HMT}}{\varphi_{HMT}^2} - 1 \right) > 0, \quad q_m := e^{-3a_m \kappa_{av}}. \quad (33.8)$$

Thus $[\kappa_{av}] = L^{-1}$, and the energy gap is

$$\Delta_E = 3\hbar c \kappa_{av}; \quad (33.9)$$

in units $\hbar = c = 1$, this is written as $\Delta_E = 3\kappa_{av}$. The electronic memory Δ_4 belongs to a different domain and a different reader; it does not enter (33.8).

In the semigroups of the following sections, t, s, a_m are Euclidean lengths and the generators D and H^{ov} have dimension L^{-1} . The energy Hamiltonian is $H_E = \hbar_{int} c_{int} D$. If $\tau = t/c_{int}$, then

$$e^{-tD} = e^{-\tau H_E / \hbar_{int}}, \quad \Delta = 3\kappa_{av}, \quad \Delta_E = \hbar_{int} c_{int} \Delta, \quad m_{gap} = \frac{\hbar_{int} \Delta}{c_{int}}.$$

The same gap is thus expressed, respectively, as inverse length, energy, and mass in the declared dimensional chart.

Since $a_m = 9a_{m+1}$, the definition immediately gives

$$q_{m+1}^9 = q_m. \quad (33.10)$$

This identity is the nonadic root that will transport the same semigroup and the same spectral scale through every refinement.

34 Projective construction of a Euclidean Yang–Mills gauge process, Osterwalder–Schrader reconstruction, and collective gap in dimension 4

The specialization of the enriched nonadic dynamics, constructed in the preceding sections, is

$$(\hat{X}_\infty, \Gamma_9) \xrightarrow{\mathcal{R}_{\text{YM}}} (\mu_m, T_m, H_m, \Omega_m) \longrightarrow (\mu_\infty, \text{OS}, E(4)) \longrightarrow \text{gap}(H_{\text{phys}}) > 0.$$

The clover reader yielding F^2 recognizes curvature on this same projective process; it does not construct μ_m retrospectively. The gauge theory, OS reconstruction, simple vacuum, and gap are already determined by this chain. The identity (34.44) subsequently compares an optional parametrization by g_R ; it is a non-governing check, not a condition for identifying the constructed Yang–Mills theory.

Gauge object and refinement domain

Let G be a compact, connected, simple Lie group. For each $m \geq 0$, consider the Euclidean lattice

$$\Lambda_m = a_m \mathbb{Z}^4, \quad a_{m+1} = \frac{a_m}{9}. \quad (34.1)$$

The cylindrical gauge algebra is generated by link variables in G , representation intertwining operators, and the enriched coordinates of color, orientation, holonomy, fusion carry, route, and memory. Let μ_m be the common invariant measure and let

$$\mathcal{H}_m = L^2(\mathcal{A}_m, \mu_m)^{\text{gauge}}.$$

The constant vector Ω_m defines

$$P_{\Omega_m} = |\Omega_m\rangle\langle\Omega_m|, \quad Q_m^{\text{phys}} = I - P_{\Omega_m}. \quad (34.2)$$

The second projector is the physical complement of the vacuum; it is not identified with a residual used solely to organize the renormalization scale.

The update circuit is local:

$$K_m^{\text{loc}} = \prod_{r=1}^{9 \cdot 81} \prod_{B \in \mathcal{B}_{m,r}} K_{m,B}. \quad (34.3)$$

The partition $\{\mathcal{B}_{m,r}\}_r$ enumerates each coordinate exactly once; the circuit factors will be identified below with the semigroup of their conditional expectations, not with nine independent dynamics. Blocks of the same color have disjoint supports, and the physical diameter of each support is uniformly bounded as m varies. This locality is part of the object, so that refinement does not turn an elementary update into an instantaneously global operation.

Let T_m be the transfer induced by (34.3). The isometric inclusions $J_m : \mathcal{H}_m \rightarrow \mathcal{H}_{m+1}$ preserve ancestry, representation, orientation, carry, and memory. For the four Euclidean reflections $\Theta_{\nu,m}$, $\nu = 1, \dots, 4$, one has

$$(T_{m+1})^9 J_m = J_m T_m, \quad \Theta_{\nu,m+1} J_m = J_m \Theta_{\nu,m} \quad (\nu = 1, \dots, 4). \quad (34.4)$$

The four directions therefore belong to the same refinement, the same measure, and the same dynamics.

Common semigroup, relative corona, and four reflection forms

In the triangular chart of the complete state, let $E_{m,c}$ be the commuting orthogonal conditional expectations associated with independent coordinates. The positive overlap generator is first constructed at a seed level m_0 :

$$\hat{H}_{m_0}^{\text{ov}} = 3\kappa_{\text{av}} \sum_{c \in \mathcal{I}_{m_0}^{\text{seed}}} (I - E_{m_0,c}), \quad (34.5)$$

and is extended by

$$\begin{aligned}\hat{H}_{m+1}^{\text{ov}} &= \hat{\Gamma}_m^* (\hat{H}_m^{\text{ov}} \otimes I + I \otimes \hat{H}_{m+1}^{\text{det}}) \hat{\Gamma}_m, \\ \hat{H}_{m+1}^{\text{det}} &= 3\kappa_{\text{av}} \sum_{t \in \mathcal{J}_{m+1/m}^{\text{det}}} (I - E_{m+1,t}^{\text{det}}),\end{aligned}\quad (34.6)$$

where

$$\kappa_{\text{av}} = \frac{\mu_{\text{vol}}}{\ell_0} \left(\frac{\pi_{\text{HMT}}}{\varphi_{\text{HMT}}^2} - 1 \right) > 0. \quad (34.7)$$

The volumetric frequency μ_{vol} is dimensionless; the dimension comes from ℓ_0 .

The semigroup, the transfer over one physical step, and their scale compatibility are defined with explicit types:

$$\hat{K}_{m,t}^{\text{ov}} = e^{-t\hat{H}_m^{\text{ov}}}, \quad \hat{T}_m = \hat{K}_{m,a_m}^{\text{ov}}, \quad a_{m+1} = a_m/9, \quad (34.8)$$

Since the partition enumerates each coordinate exactly once and the conditional expectations of its blocks commute, for each block and for the complete circuit one obtains

$$K_{m,B} = e^{-3\kappa_{\text{av}} a_m (I - E_{m,B})}, \quad K_m^{\text{loc}} = e^{-a_m \hat{H}_m^{\text{ov}}} = \hat{T}_m. \quad (34.9)$$

$$\hat{H}_{m+1}^{\text{ov}} \hat{J}_m = \hat{J}_m \hat{H}_m^{\text{ov}}, \quad (\hat{T}_{m+1})^9 \hat{J}_m = \hat{J}_m \hat{T}_m. \quad (34.10)$$

Thus, the ninth power is compared through $\hat{J}_m$; operators acting on different Hilbert spaces are not equated.

The common eight-face measure originates in this same reversible kernel:

$$\hat{\Omega}_m^{(8)}(dy_1 \cdots dy_8) = \int \hat{\mu}_m^{\text{ov}}(dz) \prod_{r=1}^8 \hat{K}_{m,a_m/2}^{\text{ov}}(z, dy_r). \quad (34.11)$$

For each reflection $\nu = 1, \dots, 4$ and every cylindrical observable F in the positive half-space,

$$\int \overline{F(\Theta_{\nu,m} y)} F(y) d\hat{\Omega}_m^{(8)} = \|\hat{\mathcal{B}}_{m,\nu} F\|^2 \geq 0, \quad (34.12)$$

and the marginal on opposite faces gives the connecting identity

$$\hat{T}_{m,\nu}^{\text{OS}} = \hat{K}_{m,a_m}^{\text{ov}} = e^{-a_m \hat{H}_m^{\text{ov}}} = \hat{T}_m, \quad \nu = 1, \dots, 4. \quad (34.13)$$

The four reflections therefore yield the same transfer-measure pair.

Cellular persistence of the terminal and exact gap

Let

$$K_9 = C_*(Q_{9,3}; \mathbb{Z}), \quad K_{10} = C_*(Q_{10,3}; \mathbb{Z}).$$

The relative corona has dimensions

$$C_*(Q_{10,3}, Q_{9,3}) = (271, 756, 702, 217), \quad 271 + 702 = 756 + 217 = 973.$$

The inclusion $j : K_9 \rightarrow K_{10}$, cellular collapse $r : K_{10} \rightarrow K_9$, and integral homotopy H_{cell} satisfy

$$rj = I, \quad \partial H_{\text{cell}} + H_{\text{cell}} \partial = I - jr, \quad rH_{\text{cell}} = 0, \quad H_{\text{cell}} j = 0, \quad H_{\text{cell}}^2 = 0. \quad (34.14)$$

For every chain map $F : K_{10} \otimes V_{q=6} \rightarrow \mathcal{T}$, with $P = (jr) \otimes I_{q=6}$,

$$F - FP = d_{\mathcal{T}}(F(H_{\text{cell}} \otimes I)) + F(H_{\text{cell}} \otimes I)(\partial \otimes I). \quad (34.15)$$

The complete corona is thus null-homotopic, and the terminal FP remains literally unchanged. The 271 vertices do not replace the edges, faces, and cubes of the corona; nor is the three-dimensional cellular factor identified with the physical gauge factor $q = 6$.

For each persistent coordinate c , in the local chart $q = (q_1, \dots, q_6) \in \mathbb{F}_3^6$, define the centered witness

$$\bigcap_c \text{Ran } E_{m,c} = \mathbb{C}\Omega_m, \quad W_c(q) = \mathbf{1}_{\{q_1+\dots+q_6=0\}} - \frac{1}{3}, \quad (34.16)$$

where the sum in the indicator is taken in $\mathbb{F}_3$. The Hoeffding decomposition of the commuting expectations proves

$$\ker \hat{H}_m^{\text{ov}} = \mathbb{C}\Omega_m, \quad \hat{H}_m^{\text{ov}} \geq 3\kappa_{\text{av}}(I - P_{\Omega_m}). \quad (34.17)$$

Equality does not follow merely from the triviality of the kernel: each nonconstant sector of the decomposition receives at least one projector $I - E_{m,c}$, and by (34.5) its minimum cost is $3\kappa_{\text{av}}$. The explicit witness W_c satisfies

$$\hat{H}_m^{\text{ov}} W_c = 3\kappa_{\text{av}} W_c, \quad \|W_c\|^2 = \frac{2}{9}, \quad (34.18)$$

so that the complement is nonempty and the gap is exactly $3\kappa_{\text{av}}$.

The scale is expressed by

$$q_m = e^{-3\kappa_{\text{av}} a_m}, \quad q_{m+1}^9 = q_m, \quad -a_m^{-1} \log q_m = 3\kappa_{\text{av}}. \quad (34.19)$$

Although $q_m \rightarrow 1$ under refinement, the spectral quotient per physical unit remains constant. The cellular contraction (34.15) preserves the terminal; the gap originates in the positive overlap generator and its Hoeffding decomposition, not in acyclicity alone.

The inclusions in (34.10) form the inductive limit

$$\mathcal{H}_\infty = \varinjlim (\mathcal{H}_m, \hat{J}_m). \quad (34.20)$$

On the algebraic inductive union, define the form

$$\mathfrak{h}_\infty[\hat{J}_{m,\infty}\psi] := \langle \psi, \hat{H}_m^{\text{ov}} \psi \rangle_{\mathcal{H}_m}. \quad (34.21)$$

Intertwining of the generators makes this definition independent of the representative. The form is positive and closable; its closure determines the self-adjoint generator $\hat{H}_\infty^{\text{ov}}$. Compatibility of the forms transports

$$\bar{\mathfrak{h}}_\infty[\psi] \geq 3\kappa_{\text{av}} \|(I - P_{\Omega_\infty})\psi\|^2, \quad (34.22)$$

and the persistent vector $\hat{J}_{m,\infty} W_c$ attains equality. Consequently, the exact gap of the limiting generator is $3\kappa_{\text{av}}$.

Exact hypotheses and logical chain of the theorem

The gauge realization uses:

1. the compact, connected, simple group and the four-dimensional lattice (34.1);
2. the local circuit (34.3), with uniform physical diameter, and a common invariant measure;
3. the isometric inclusions and the four reflections intertwined by (34.4) and (34.10);
4. the four factorizations (34.12) and the exact connection (34.13) on the same transfer–measure pair;
5. the commuting-expectation generator (34.5)–(34.6) and its Hoeffding decomposition;
6. the integral SDR of the corona and preservation of the terminal (34.14)–(34.15);
7. the terminal P_{Ω_m} , the physical complement Q_m^{phys} , and the scaling law (34.19);
8. the inductive limit and the compatible closure of forms (34.20)–(34.22);
9. the internal action and velocity charts $\hbar_{\text{int}}, c_{\text{int}}$, used solely to express the mass dimension.

Theorem 34.1 (Local gauge theory with a simple vacuum and collective gap). *Under the preceding premises, the limit of the gauge collection is local and reflection-positive in all four Euclidean directions. Its generator has a simple vacuum and satisfies, in the gauge-invariant collective sector,*

$$\Delta_{\text{YM}}^{\text{HMT}} = 3\kappa_{\text{av}} > 0, \quad m_{\text{gap}}^{\text{HMT}} = \frac{\hbar_{\text{int}}}{c_{\text{int}}} \Delta_{\text{YM}}^{\text{HMT}} > 0. \quad (34.23)$$

The gap belongs to the collective sector; it does not assign an elementary mass to the gluon.

Proof. The uniform locality of the circuit preserves the net of local algebras. The equalities (34.10) identify the ninth power after applying the correct inclusion. The construction (34.11) uses this same semigroup on the eight faces, and (34.13) identifies its four OS marginals with $\hat{T}_m$. Thus, the four positive forms (34.12) are compatible in the limit and preserve a single measure and a single dynamics.

The expectations in (34.5) are orthogonal and commuting. Their Hoeffding decomposition separates the constant mode from every nonconstant sector: the former is $\mathbb{C}\Omega_m$, whereas every sector of the complement receives at least one factor $I - E_{m,c}$ with weight $3\kappa_{\text{av}}$. This proves (34.17). The witness (34.18) attains the bound, so it is not merely a lower bound: it is the first nonzero energy.

The homotopy (34.15) contracts the cellular complement and leaves FP intact; the corona therefore introduces no second vacuum and erases no carry. Isometric compatibility transports the vacuum and its complement to the next level without mixing them with the scale residual. The law (34.19) shows that the spectral quotient per physical unit is exactly $3\kappa_{\text{av}}$ at every level. The forms (34.21)–(34.22) transport the bound and the witness to the self-adjoint limiting generator, so the gap remains exact. Reconstruction by Osterwalder–Schrader positivity identifies the Hamiltonian with the generator of this same transfer, preserves the simple vacuum, and produces (34.23). The final dimensional chart transforms the gap into the stated collective mass. $\square$

Gauge compression and physical gap witness

Let

$$\hat{\mathcal{H}}_m := L^2(\hat{X}_m, \hat{\mu}_m^{\text{ov}})$$

be the space of the complete material realization. In addition to the usual gauge action on links and frames, each collar c carries the finite incidence action

$$q_c \mapsto q_c + Bx_c, \quad x_c \in \mathbb{F}_3^4, \quad (34.24)$$

where the six rows of B are indexed by (01, 02, 03, 12, 13, 23). The product of both actions preserves $\hat{\mu}_m^{\text{ov}}$. With $\mathcal{C}_m$ denoting the finite set of collars present at this level, write

$$\mathcal{G}_m^{\text{full}} := G^{V(\Lambda_m)} \times \prod_{c \in \mathcal{C}_m} \mathbb{F}_3^4.$$

Its Haar average defines the orthogonal projection

$$\mathbb{P}_m^{\text{phys}} F := \int_{\mathcal{G}_m^{\text{full}}} F(g^{-1} \cdot) dg, \quad \hat{\mathcal{H}}_m^{\text{phys}} := \text{Ran } \mathbb{P}_m^{\text{phys}}. \quad (34.25)$$

This is a compression onto the gauge subalgebra of the complete state; the marginal forgetting the incidence coordinates is a different map and is not confused with $\mathbb{P}_m^{\text{phys}}$.

Lemma 34.2 (Reduction of the Hamiltonian to the physical subalgebra). *The projection $\mathbb{P}_m^{\text{phys}}$ reduces the generator and its semigroup. The refinement inclusions simultaneously reduce the physical subalgebras:*

$$\begin{aligned} [\mathbb{P}_m^{\text{phys}}, \hat{H}_m^{\text{ov}}] &= 0, & [\mathbb{P}_m^{\text{phys}}, e^{-t\hat{H}_m^{\text{ov}}}] &= 0, \\ \mathbb{P}_{m+1}^{\text{phys}} \hat{J}_m &= \hat{J}_m \mathbb{P}_m^{\text{phys}}, & H_m^{\text{phys}} &:= \hat{H}_m^{\text{ov}}|_{\hat{\mathcal{H}}_m^{\text{phys}}}, \\ H_{m+1}^{\text{phys}} \hat{J}_m &= \hat{J}_m H_m^{\text{phys}}. \end{aligned} \quad (34.26)$$

Proof. The physical generator on links is gauge-equivariant by the bi-invariance of Haar measure. In the product chart, E_{q_c} is uniform expectation on $\mathbb{F}_3^6$; invariance of the uniform measure under the translations in (34.24) implies that it commutes with this action. Expectations on the marking subtails act on disjoint factors. Thus every summand of

$$\hat{H}_m^{\text{ov}} = H_m^{\text{ov}} \otimes I + 3\kappa_{\text{av}} \sum_c [(I - E_{q_c}) + (I - E_{u_c^p})]$$

commutes with the average (34.25). The functional calculus gives commutation with the semigroup. Finally, $\hat{J}_m$ preserves the ancestral factors and inserts the constant in the new factors; averaging before or after this inclusion produces the same function. This proves the four identities in (34.26). $\square$

Lemma 34.3 (Nonzero intertwined gauge witness). *For every collar c , the function*

$$W_c(q) := \mathbf{1}_{\{q_1 + \dots + q_6 = 0\}} - \frac{1}{3} \quad (34.27)$$

belongs to $\hat{\mathcal{H}}_m^{\text{phys}}$, is nonzero, and satisfies

$$\mathbb{E} W_c = 0, \quad \|W_c\|^2 = \frac{2}{9}, \quad \mathbb{E}(W_c^3) = \frac{2}{27}. \quad (34.28)$$

Moreover,

$$H_m^{\text{phys}} W_c = 3\kappa_{\text{av}} W_c, \quad H_{m+1}^{\text{phys}} \hat{J}_m W_c = 3\kappa_{\text{av}} \hat{J}_m W_c. \quad (34.29)$$

Consequently, the value $3\kappa_{\text{av}}$ belongs to the spectrum of the physical restriction, not merely to that of a material dilation.

Proof. The gauge action on links acts on the first factor and fixes W_c . For the incidence action, each x_i appears in three of the six coordinates, hence

$$\sum_{i < j} (Bx)_{ij} = 3 \sum_i x_i = 0 \quad \text{in } \mathbb{F}_3.$$

The condition in (34.27) is therefore invariant; it is also invariant under permutations of the six incidences. Thus $\mathbb{P}_m^{\text{phys}} W_c = W_c$.

Under the uniform measure, the sum of the six trits is uniform on $\mathbb{F}_3$. The function takes the value $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$, giving (34.28). The first summand of $\hat{H}_m^{\text{ov}}$ annihilates W_c ; for $c' \neq c$, $(I - E_{q_{c'}})W_c = 0$, whereas $E_{q_c} W_c = 0$. All marking expectations annihilate the corresponding difference because W_c is constant on these subtails. What remains is exactly $\hat{H}_m^{\text{ov}} W_c = 3\kappa_{\text{av}} W_c$. The first identity in (34.29) follows from reduction; the second, from the intertwining in (34.26). $\square$

Gauge semigroup, vacuum, and exact spectral bound

On the physical subalgebra, write

$$D_m^{\text{YM}} := H_m^{\text{phys}}, \quad K_{m,t} := \exp(-tD_m^{\text{YM}}), \quad P_{\Omega_m} := |\Omega_m\rangle\langle\Omega_m|.$$

The Hoeffding decomposition of the same generator and the functional calculus give, without changing the measure or process,

$$K_{m,t} P_{\Omega_m} = P_{\Omega_m}, \quad \|K_{m,t}(I - P_{\Omega_m})\| \leq e^{-3\kappa_{\text{av}} t}. \quad (34.30)$$

The witness in (34.27) is nonzero and attains the bound:

$$D_m^{\text{YM}} W_c = 3\kappa_{\text{av}} W_c, \quad \|W_c\|^2 = \frac{2}{9}. \quad (34.31)$$

Thus the vacuum is simple, and the gap is not merely bounded below but is exactly

$$\Delta_{\text{YM}}^{\text{HMT}} = 3\kappa_{\text{av}} > 0. \quad (34.32)$$

The semigroup, vacuum projector, contraction of the complement, and W_c belong to the same projective collection and constitute the underlying chain establishing the result.

Euclidean reconstruction, Schwinger fields, and curvature reader

The common projective measure in the theorem determines isometric inclusions $\hat{J}_m$ between the level spaces. For each of the four coordinate reflections θ_μ , $\mu = 1, \dots, 4$, there is a factorization

$$\int \overline{F(\theta_\mu y)} F(y) d\hat{\Omega}_m(y) = \|\mathcal{B}_{m,\mu} F\|_2^2 \geq 0. \quad (34.33)$$

The four forms originate in the same measure and semigroup. The OS connecting maps

$$\mathcal{J}_{m,\mu}^{\text{OS}} = \mathcal{V}_{m+1,\mu}^{-1} \hat{J}_m \mathcal{V}_{m,\mu}$$

intertwine their Hamiltonians. In the colimit,

$$\mathcal{V}_{\infty,\mu} H_{\text{OS},\infty}^{(\mu)} \mathcal{V}_{\infty,\mu}^{-1} = \hat{H}_\infty, \quad \mu = 1, \dots, 4. \quad (34.34)$$

The limiting form is diagonalized by the Hoeffding decomposition:

$$\mathcal{E}_\infty(u) = 3\kappa_{\text{av}} \sum_{S \in \mathcal{J}_\infty} |S| \|u_S\|^2. \quad (34.35)$$

The partial sums provide a recovery sequence, and semicontinuity yields the lower limit. Mosco convergence of the closed forms preserves the one-dimensional kernel and spectral threshold. The explicit vector W_c satisfies

$$\hat{H}_m W_c = 3\kappa_{\text{av}} W_c, \quad \|W_c\|^2 = \frac{2}{9}, \quad \mathbb{E}(W_c^3) = \frac{2}{27}, \quad (34.36)$$

so the lower bound is attained and the gap is exactly $3\kappa_{\text{av}}$.

Theorem 34.4 (Euclidean completion and curvature output). *The projective collection in Theorem 34.1 determines a strongly continuous action of $E(4)$, a nonscalar local gauge net, compatible Schwinger functionals, and a field output in $H_{\text{loc}}^{-s}(\mathbb{R}^4)$ for $s > 2$. The surface-ordered product of the links defines clover and F^2 readers; on the smooth domain, the action reader converges to*

$$\mathcal{S}_{\text{YM}}(A) = \frac{1}{4g_R^2} \int_{\mathbb{R}^4} \|F_A(x)\|^2 d^4x. \quad (34.37)$$

These outputs preserve the simple vacuum and exact gap of the same common Hamiltonian.

Proof. The cofinal Gram matrices of the smoothed readers and commutativity of the rotation and refinement squares preserve the null ideals; their colimit constructs the strong representation of $E(4)$ and the local net. Martingale estimates for the link increments give tightness in H_{loc}^{-s} , $s > 2$, and the mixed characteristic functionals identify the Schwinger functionals with the same cylindrical process. The oriented product around each plaquette constructs the clover. Its expansion in the smooth regime and strong convergence of the marginal reader give (34.37). None of these operations changes the measure, semigroup, or limiting form used to obtain (34.36); hence the vacuum and gap are preserved. $\square$

Exact identification of the gauge limit

Identification of the theory does not rely on four reflections generating the Euclidean group by themselves. The reflections provide OS positivity; the continuous action originates, separately, in the screen groupoid and refinement compatibility. For $g = (g_v)_{v \in \Lambda_m} \in G^{\Lambda_m}$, the action on a link variable is

$$(g \cdot U)_e = g_{s(e)} U_e g_{t(e)}^{-1}. \quad (34.38)$$

Haar invariance, triangular disintegration of the kernel, and cancellation of interior edges prove

$$(g \cdot)_* \hat{\mu}_m^{\text{ov}} = \hat{\mu}_m^{\text{ov}}, \quad K_{m,t}^{\text{ov}}(g \cdot U, g \cdot dV) = K_{m,t}^{\text{ov}}(U, dV). \quad (34.39)$$

By differentiation on the cylindrical core, every vertical generator X satisfies the Ward identity

$$\int \nabla_X F d\hat{\mu}_m^{\text{ov}} = 0. \quad (34.40)$$

Curvature is obtained from the same link system. If a coarse plaquette p is subdivided into its 81 subplaquettes p' , then

$$\text{Hol}_p^{(m)} = \prod_{p' \subset p}^{\rightarrow} T_{p'} \text{Hol}_{p'}^{(m+1)} T_{p'}^{-1}. \quad (34.41)$$

The contributions of every interior edge occur with opposite orientations and cancel exactly. The expression is gauge-covariant and commutes with the inclusions $\hat{J}_m$. In a smooth chart, define

$$F_{\text{cl},m}(x) = \frac{1}{4a_m^2} \sum_{p \ni x} \text{Ad}_{T_p} \log \text{Hol}_p, \quad F_{\text{cl},m} = F_A + O(a_m^2). \quad (34.42)$$

Hence, for $A \in C_{\text{loc}}^4$ and $a_m^2/g_m^2 \rightarrow 0$, Riemann summation gives

$$\frac{a_m^4}{4g_m^2} \sum_{x \in \Lambda_m} \|F_{\text{cl},m}(x)\|^2 \longrightarrow \frac{1}{4g_R^2} \int_{\mathbb{R}^4} \|F_A(x)\|^2 d^4x. \quad (34.43)$$

Theorem 34.5 (Euclidean gauge process and curvature reader for Yang–Mills). *The limit constructed in Theorem 34.1 is a Euclidean gauge process in dimension 4 for the fixed compact, connected, simple group. It has local gauge covariance, a nonscalar local net, reflection positivity, a strongly continuous representation of $E(4)$, compatible Schwinger functionals, gauge-covariant curvature with classical limit (34.43), a simple vacuum, and spectral gap $3\kappa_{\text{av}} > 0$.*

Proof. The identities (34.39)–(34.40) construct gauge symmetry and its Ward identities at every level; their naturality transports them to the limit. The uniform locality of the kernels produces the isotonic local net. The four forms (34.33), together with the screen groupoid action on the Weyl core, produce OS positivity and the strong action of $E(4)$, respectively. Tightness in H_{loc}^{-s} , $s > 2$, identifies a unique limiting law and its Schwinger functionals. Finally, (34.41)–(34.43) identify the field reader and its classical limit, whereas (34.35)–(34.36) prove a simple vacuum and exact gap for the same law.

The measure is constructively determined by its projective marginals, reversible kernel, and cylindrical functionals. The clover reader does not reweight this measure: it yields a function of the same process and its classical limit. $\square$

Proposition 34.6 (Non-governing Gibbs–action comparison check). *The projective law, its simple vacuum, and its exact gap have already been constructed by (34.30)–(34.32). As an optional external Gibbs–action comparison, a parametrization by g_R may study a collection*

$$g_R \longmapsto (\hat{\mu}_{m,g_R}^{\text{ov}}, H_{m,g_R}^{\text{phys}})$$

and a relation between the measure or its Dirichlet form and the action functional, for example

$$\frac{d\hat{\mu}_{m,g_R}^{\text{ov}}}{d\nu_m}(U) = Z_{m,g_R}^{-1} e^{-S_{m,g_R}^{F^2}(U)}, \quad \mathfrak{h}_{m,g_R}[F] = \langle F, H_{m,g_R}^{\text{phys}} F \rangle, \quad (34.44)$$

with refinement compatibility and passage to the limit. Here ν_m may be the product Haar measure of the finite lattice. This check is designated `NON_GOVERNING_CHECK`: it is not a premise of HMT gauge theory or its gap.

Proof. In (34.43), varying g_R rescales the curvature reader, whereas the formulas constructing $\hat{\mu}_m^{\text{ov}}$, $\hat{H}_m^{\text{ov}}$, and the gap $3\kappa_{\text{av}}$ do not contain g_R . They are therefore independent constructions. The compression (34.26) and witness (34.29) prove the physical persistence of the gap in the already constructed process; the check only subsequently compares an additional dynamical dependence on g_R . $\square$

The sequence proved in this section is

common measure $\rightarrow$ four OS $\rightarrow$ common Hamiltonian $\rightarrow E(4)$, Schwinger, $H_{\text{loc}}^{-s} \rightarrow$ clover and F^2 .

This sequence proves a four-dimensional Yang–Mills theory with a simple vacuum and exact positive gap $3\kappa_{\text{av}}$ in the recorded HMT domain. The identity (34.44) remains outside the governing chain as an independent subsequent comparison.

The g -dependent realization is developed in (35.46) and (35.53): the measure, conditional expectations, and generator are transported jointly, and (35.58) preserves the bound and its witness.

Wilson reading

The Wilson observable is obtained after the generator. For an Archimedean link chart,

$$A_e = \sum_a \text{ev}_{\mathbb{R}}(x_{e,a})T_a, \quad U_e = e^{a_m A_e}, \quad W_\gamma = \text{Tr} \prod_{e \in \gamma} U_e. \quad (34.45)$$

The reading preserves the representation, intertwining operator, and holonomy order. It serves as a terminal observable of the resulting theory; it does not define the measure, semigroup, or gap retrospectively.

Necessity of the hypotheses and exact obstructions

Structural necessity test 34.7 (Necessity of the common dynamics and measure). *Theorem 34.1 does not hold if the nine subphases are replaced by independent updates, different measures or kernels are used for the four reflections, uniform physical locality is lost, $\hat{J}_m$ ceases to intertwine transfer and reflection, carry or memory is erased, P_{Ω_m} is removed, Q_m^{phys} is identified with a scale residual, $q_m \leq q_* < 1$ is required in lattice units, the complete corona is replaced by its 271 vertices, FP is erased, or (34.45) is used to select the generator.*

Proof. Different updates, kernels, or measures destroy the simultaneous factorization (34.12). Without locality or intertwining, positivity at one stage does not define a compatible local net. Erasing carry, memory, or FP removes information preserved by (34.15). Using only the vertices changes the relative complex and can create zero modes. Confusing the two complements allows the physical sector to disappear through a purely scalar operation. A fixed bound in lattice units contradicts (34.19). Finally, Wilson is only a reading of the constructed state and does not by itself prove either (34.12) or (34.17). $\square$

The independent residuals are: failure of $\hat{T}_{m,\nu}^{\text{OS}} = e^{-a_m \hat{H}_m^{\text{ov}}}$, failure of the intertwining (34.10), a negative OS form, a kernel vector orthogonal to Ω_m , or a nonconstant Hoeffding sector with energy below $3\kappa_{\text{av}}$.

Scope of the gauge construction

The proof combines the joint connection, preservation of the terminal, the double reading, the local circuit, the common measure, the four reflection factorizations, intertwined refinement, and the positive generator. In the declared regular gauge domain, locality, four-dimensional Yang–Mills theory, the simple vacuum, and the collective gap are exact results. Osterwalder–Schrader reconstruction and the Wilson observable are subsequent outputs of the same state; neither acts as a retrospective selector of the measure, semigroup, vacuum, or gap. The optional comparison with a collection parametrized by g_R is studied through (34.44); this check neither governs nor reopens the already established measure, semigroup, vacuum, or gap.

34.1 Projective gauge incidence of the common constructor and descent to the quotient

The gauge image is not the link marginal. Its complete object at each level is

$$\begin{aligned} \mathfrak{G}_{\text{gau}}^{\text{int}} : \\ \Lambda_m, \hat{X}_m, \hat{\mu}_m^{\text{ov}}, \{\hat{\mu}_{m,g}^{\text{YM}}, \nu_{m,g}, K_{m,g}^{\text{phys}}\}_{g>0}, \\ \mathfrak{G}_{\text{gau},m}(d_{\text{gau}}) = (\mathcal{G}_m^{\text{full}}, \mathbb{P}_m^{\text{phys}}, \hat{f}_m, \hat{\Gamma}_m, \{E_{m,c}\}_c, \hat{H}_m^{\text{ov}}, D_m^{\text{YM}}, \\ \{\Theta_{\nu,m}\}_{\nu=1}^4, \{\mathfrak{A}_{+, \nu, m}, \text{ev}_t^{(\nu)}, \mathcal{V}_{m,\nu}^{\text{OS}}\}_{\nu=1}^4, \\ \hat{\Omega}_m^{(8)}, P_{\Omega_m}, W_c, \text{Carr}_m, \text{Mem}_m, \text{Surv}_m, \tau_m, \mathcal{P}_m^{F^2}). \end{aligned} \quad (34.46)$$

The group $\mathcal{G}_m^{\text{full}}$ acts simultaneously on links, frames, and incidence coordinates $q_c \in \mathbb{F}_3^6$. The physical projection is its Haar average. The reduced generator is

$$D_m^{\text{YM}} = 3\kappa_{\text{av}} \sum_c (I - E_{m,c}) \Big|_{\text{Ran } \mathbb{P}_m^{\text{phys}}}, \quad (34.47)$$

and W_c belongs to this subalgebra. The four reflections act on the same measure, the same kernel, and the same terminal. The preceding exact source construction establishes the incidence gauge action (34.24), the physical Haar projection (34.25), and the intertwined compression (34.26). This construction yields the complete equivariant kernel $\hat{K}_{m,t}^{\text{ov}}$, its descent to the physical quotient $K_{m,t}^{\text{phys}}$, the four reflection operators, and the action $U_m(g)$ of $E(4)$ on the smooth cylindrical core. This block assembles, without introducing them as data, the generator, vacuum, curvature witness, and OS forms:

$$D_{m+1}^{\text{YM}} \hat{f}_m = \hat{f}_m D_m^{\text{YM}}, \quad \mathbb{P}_{m+1}^{\text{phys}} \hat{f}_m = \hat{f}_m \mathbb{P}_m^{\text{phys}}, \quad \hat{f}_m W_{m,c} = W_{m+1,c}. \quad (34.48)$$

The first and third identities express, in the common notation, (34.26) and (34.29); the semigroup, simple vacuum, and exact gap have already been obtained in (34.30)–(34.32). Here $\nu_{m,g} := (\pi_m^{\text{phys}})_* \hat{\mu}_{m,g}^{\text{YM}}$ and $K_{m,g}^{\text{phys}}(t) = e^{-tD_{m,g}^{\text{YM}}}$ on the physical quotient. Gauge completeness also includes disintegration with respect to each collection of Euclidean hyperplanes. If $A_j \in \mathfrak{A}_{+, \nu, m}$ and $t_1 \leq \dots \leq t_n$, the marginal of the same physical measure satisfies

$$\begin{aligned} & \int \prod_{j=1}^n (\tau_{t_j e_\nu} A_j)(x) d\hat{\mu}_{m,g}^{\text{YM}}(x) \\ &= \int \prod_{j=1}^n A_j(y_j) \nu_{m,g}(dy_1) \prod_{j=1}^{n-1} K_{m,g}^{\text{phys}}(t_{j+1} - t_j; y_j, dy_{j+1}) \\ &= \langle \mathbf{1}, M_{A_1} e^{-(t_2-t_1)D_{m,g}^{\text{YM}}} M_{A_2} \dots e^{-(t_n-t_{n-1})D_{m,g}^{\text{YM}}} M_{A_n} \mathbf{1} \rangle. \end{aligned} \quad (34.49)$$

The preceding reconstruction proves the identity in all four directions through (34.33) and constructs a unique intertwined Hamiltonian in (34.34). Consequently, OS translation is expressed here as

$$\mathcal{V}_{m,\nu}^{\text{OS}} T_{m,\nu}^{\text{OS}}(s) (\mathcal{V}_{m,\nu}^{\text{OS}})^{-1} = e^{-sD_{m,g}^{\text{YM}}}. \quad (34.50)$$

Thus, the expectation generator and the translation Hamiltonian are not identified merely by name: they are two already constructed realizations of the same disintegration, whose limiting form and curvature output are given in (34.35) and (34.37).

35 Gauge image and composition towards Yang–Mills

35.1 Enriched gauge state and transfer generators

The already generated object received by this chapter is the complete gauge structure $\mathcal{G}_{\text{gau}}$; it is neither an external datum nor a terminal hypothesis. At every level m , it preserves the lattice $\Lambda_m = a_m \mathbb{Z}^4$, $a_{m+1} = a_m/9$, the measure μ_m , the intertwiners J_m , four reflections $\Theta_{v,m}$, holonomy, color, representation, fusion carry, route, memory, terminal, and curvature reader. Its two dynamics are typed with distinct symbols before any limit.

The semigroup of a single memory fiber is

$$K_{m,t}^{\text{fib}} = P_{\Omega_m} + e^{-3\kappa_{\text{av}} t} (I - P_{\Omega_m}) = e^{-t D_m^{\text{fib}}}, \quad D_m^{\text{fib}} := 3\kappa_{\text{av}} (I - P_{\Omega_m}). \quad (35.1)$$

Its spectrum is contained in $\{0, 3\kappa_{\text{av}}\}$. The generator of the complete product chart is, by contrast,

$$D_m^{\text{prod}} := 3\kappa_{\text{av}} \sum_{c \in \mathcal{I}_m} (I - E_{m,c}), \quad K_{m,t}^{\text{prod}} := e^{-t D_m^{\text{prod}}}, \quad (35.2)$$

where the orthogonal expectations $E_{m,c}$ commute. With two or more nonconstant coordinates, D_m^{prod} has levels $6\kappa_{\text{av}}, 9\kappa_{\text{av}}, \dots$; thus (35.1) cannot be declared equal to (35.2). The former is the simple fiber; the latter, the overlap product.

35.2 Finite Hoeffding theorem, vacuum, and attained gap

In the product chart constructed in (35.21), the $E_{m,c}$ are commuting orthogonal conditional expectations, and the proof of (35.59) establishes

$$\bigcap_{c \in \mathcal{I}_m} \text{Ran } E_{m,c} = \mathbb{C} \Omega_m. \quad (35.3)$$

The Hoeffding decomposition is the orthogonal sum

$$\mathcal{H}_m = \bigoplus_{S \subseteq \mathcal{I}_m} \mathcal{H}_{m,S}, \quad u = \sum_S u_S,$$

where $E_{m,c} u_S = 0$ if $c \in S$ and $E_{m,c} u_S = u_S$ if $c \notin S$. Then

$$D_m^{\text{prod}} u_S = 3\kappa_{\text{av}} |S| u_S, \quad \langle u, D_m^{\text{prod}} u \rangle = 3\kappa_{\text{av}} \sum_S |S| \|u_S\|^2. \quad (35.4)$$

The component $S = \emptyset$ is exactly the vacuum in (35.3). Therefore

$$D_m^{\text{prod}} \geq 3\kappa_{\text{av}} (I - P_{\Omega_m}), \quad \|K_{m,t}^{\text{prod}} (I - P_{\Omega_m})\| \leq e^{-3\kappa_{\text{av}} t}. \quad (35.5)$$

On an incidence coordinate $q = (q_1, \dots, q_6) \in \mathbb{F}_3^6$, the witness

$$W_c(q) := \mathbf{1}_{\{q_1 + \dots + q_6 = 0\}} - \frac{1}{3} \quad (35.6)$$

satisfies

$$\mathbb{E} W_c = 0, \quad \|W_c\|^2 = \frac{2}{9}, \quad D_m^{\text{prod}} W_c = 3\kappa_{\text{av}} W_c. \quad (35.7)$$

Thus, at every finite level, the threshold in (35.5) is attained. This calculation determines the spectrum of the finite product generator; the continuous four-dimensional Yang–Mills measure is explicitly constructed in (35.46)–(35.53).

35.3 Physical gauge compression

Let G be compact, connected, and simple. The group acting at level m is

$$\mathcal{G}_m^{\text{full}} = G^{V(\Lambda_m)} \times \prod_{c \in \mathcal{C}_m} \mathbb{F}_3^4,$$

where the second factor acts on the six incidences through a matrix that must remain explicit. If the rows are indexed by $(01, 02, 03, 12, 13, 23)$ and the columns by $0, 1, 2, 3$, fix

$$B_{(ij),k}^+ := \delta_{ik} + \delta_{jk} \in \mathbb{F}_3, \quad (B^+ x)_{ij} = x_i + x_j. \quad (35.8)$$

This is the unoriented incidence of the collar; it is not confused with the signed coboundary of an edge. The finite action is

$$q_c \mapsto q_c + B^+ x_c, \quad x_c \in \mathbb{F}_3^4. \quad (35.9)$$

The Haar projection

$$\mathbb{P}_m^{\text{phys}} F := \int_{\mathcal{G}_m^{\text{full}}} F(g^{-1} \cdot) dg, \quad \mathcal{H}_m^{\text{phys}} := \text{Ran } \mathbb{P}_m^{\text{phys}} \quad (35.10)$$

differs from the marginal that forgets incidence. Invariance of the expectations $E_{m,c}$ gives

$$[\mathbb{P}_m^{\text{phys}}, D_m^{\text{prod}}] = 0, \quad [\mathbb{P}_m^{\text{phys}}, K_{m,t}^{\text{prod}}] = 0. \quad (35.11)$$

Moreover, each variable of $x_c \in \mathbb{F}_3^4$ occurs three times in the sum of the six incidences, so that

$$\sum_{i < j} (B^+ x_c)_{ij} = 3 \sum_{i=0}^3 x_{c,i} = 0 \quad \text{in } \mathbb{F}_3. \quad (35.12)$$

Hence W_c is gauge-invariant and belongs to $\mathcal{H}_m^{\text{phys}}$. The finite gap is not an artifact of a nonphysical dilation.

35.4 Reflection positivity at every level

For a common reversible kernel, define the eight-face measure

$$\Omega_m^{(8)}(dy_1 \cdots dy_8) := \int \mu_m(dz) \prod_{r=1}^8 K_{m,a_m/2}^{\text{prod}}(z, dy_r). \quad (35.13)$$

If $\Theta_{\nu,m}$ interchanges the two halves with respect to direction ν , the Markov property and reversibility factorize, for every cylindrical observable F supported in the positive half-space,

$$\int \overline{F(\Theta_{\nu,m} y)} F(y) d\Omega_m^{(8)} = \|\mathcal{B}_{m,\nu} F\|_2^2 \geq 0, \quad \nu = 1, \dots, 4. \quad (35.14)$$

The opposite-face marginal recovers the same K_{m,a_m}^{prod} . Thus the four finite OS forms follow from a single measure and a single semigroup, not from four independent processes.

35.5 Complete gauge state and physical incidence coordinates

The current source construction does not take the marginal retaining only the links as its physical space. Its domain is $\mathcal{G}_{\text{gau}}$, with link, frame, representation, incidence, color, route, orientation, carry, memory, survival, and terminal. Projection onto links is a subsequent forgetful map. Assessing the physical witness after applying this forgetful map replaces the object and does not transfer the conclusion.

35.5.1 Projective measure and link output

For a finite screen $\Gamma_m = (V_m, E_m)$, start from the lifted space

$$\tilde{X}_m := G^{V_m} \times G^{E_m}, \quad d\tilde{\mu}_m(h, z) = \prod_{v \in V_m} dh_v \prod_{e \in E_m} k_{\tau_{m,e}}(z_e) dz_e, \quad (35.15)$$

and obtain each link through

$$U_e = h_{s(e)} z_e h_{t(e)}^{-1}. \quad (35.16)$$

A coarse edge e is refined into the ordered chain $e_1 \cdots e_9$. Set $\tau = (\tau_1, \dots, \tau_9)$, with $\tau_j = \tau_{m+1,e_j} > 0$ and $\tau = \tau_{m,e} = \sum_{j=1}^9 \tau_j$. Let $d\lambda_z^{(9)}$ be the Haar disintegration over the product fiber, characterized by

$$\int_{G^9} F(z_1, \dots, z_9) dz_1 \cdots dz_9 = \int_G \int_{z_1 \cdots z_9 = z} F d\lambda_z^{(9)} dz.$$

The correct bridge for times that need not be equal is

$$dB_{z,\tau}^{(9)}(z_1, \dots, z_9) = \frac{\prod_{j=1}^9 k_{\tau_j}(z_j)}{k_\tau(z)} d\lambda_z^{(9)}, \quad z_1 \cdots z_9 = z, \quad (35.17)$$

which is a probability measure because $k_{\tau_1} * \cdots * k_{\tau_9} = k_\tau$. Uniform nonadic refinement recovers $\tau_j = \tau/9$, but the formula does not presuppose this equality.

To make the innovation independent without erasing the coarse product, fix, for each (e, z) , a Borel transport

$$\chi_{m,e,z} : ((0, 1)^\mathbb{N}, du^{\otimes \mathbb{N}}) \longrightarrow (\{(z_j) : z_1 \cdots z_9 = z\}, B_{z,\tau}^{(9)}) \quad (35.18)$$

that is an isomorphism modulo null sets. It exists because both spaces are standard nonatomic realizations; one choice is fixed once and transported under edge relabelings. The coarse variable z and the innovation $u_{m+1,e}^{\det}$ are then independent factors in the triangular chart. The new frames are integrated with Haar measure. Route, carry, orientation, and memory are transported by the morphisms Ext_{gau} of the gauge datum itself.

The coarse projection multiplies the nine increments and forgets only the new innovations. This definition yields

$$(\hat{\pi}_{m+1,m})_* \hat{\mu}_{m+1}^{\text{ov}} = \hat{\mu}_m^{\text{ov}}, \quad \hat{J}_m F = F \circ \hat{\pi}_{m+1,m}. \quad (35.19)$$

The same edge occurs exactly once in the measure; the six incident faces are six oriented readers of that edge. Overlapping routes preserve the shared increment and are not replaced by independent copies.

35.5.2 The incidence fiber is a physical coordinate of the complete state

The incidence tail included in $\mathfrak{G}_{\text{gau}}$ decomposes without loss of information:

$$\Sigma : (\mathbb{F}_3^2)^\mathbb{N} \longrightarrow (\mathbb{F}_3^2)^\mathbb{N} \times \mathbb{F}_3^6, \quad \Sigma_* \sigma = \sigma \otimes u_6. \quad (35.20)$$

The first factor continues to supply the innovation charts, and the second preserves the six incidences $q_c = (q_{01}, q_{02}, q_{03}, q_{12}, q_{13}, q_{23})$ of collar c . Rather than leaving implicit which coordinates the generator governs, fix a recursive triangular chart

$$\hat{X}_m = \hat{X}_{m_0}^{\text{seed}} \times \prod_{\ell=m_0+1}^m u_{\ell/\ell-1}^{\det}, \quad \hat{\mu}_m^{\text{ov}} = \hat{\mu}_{m_0}^{\text{seed}} \otimes \bigotimes_{\ell=m_0+1}^m v_{\ell/\ell-1}^{\det}. \quad (35.21)$$

The seed factor enumerates once the coarse frames and links, the incidences q_c , and their marking charts. Each $u_{\ell/\ell-1}^{\det}$ enumerates once the new frames, the bridges (35.18), the new incidences, and the marking subtails of that refinement. The factorization is internal to $\mathfrak{G}_{\text{gau}}$ and adds no further initial datum.

Record the exhaustive, disjoint, countable index

$$\mathcal{J}_m := \mathcal{J}_{m_0}^{\text{frame}} \sqcup \mathcal{J}_{m_0}^{\text{link}} \sqcup \mathcal{J}_{m_0}^q \sqcup \mathcal{J}_{m_0}^{\text{mark}} \sqcup \bigsqcup_{\ell=m_0+1}^m (\mathcal{J}_{\ell/\ell-1}^{\text{frame}} \sqcup \mathcal{J}_{\ell/\ell-1}^{\text{bridge}} \sqcup \mathcal{J}_{\ell/\ell-1}^q \sqcup \mathcal{J}_{\ell/\ell-1}^{\text{mark}}). \quad (35.22)$$

Each independent factor of (35.21) occurs exactly once; an infinite subtail is enumerated through its scalar cylinders. No genealogical factor remains outside $\mathcal{J}_m$. In particular, the abbreviations in the recurrence literally mean

$$\begin{aligned} \mathcal{J}_{m_0}^{\text{seed}} &:= \mathcal{J}_{m_0}^{\text{frame}} \sqcup \mathcal{J}_{m_0}^{\text{link}} \sqcup \mathcal{J}_{m_0}^q \sqcup \mathcal{J}_{m_0}^{\text{mark}}, \\ \mathcal{J}_{\ell/\ell-1}^{\text{det}} &:= \mathcal{J}_{\ell/\ell-1}^{\text{frame}} \sqcup \mathcal{J}_{\ell/\ell-1}^{\text{bridge}} \sqcup \mathcal{J}_{\ell/\ell-1}^q \sqcup \mathcal{J}_{\ell/\ell-1}^{\text{mark}}. \end{aligned} \quad (35.23)$$

The link marginal forgets q_c ; the physical subalgebra, by contrast, is obtained by averaging the complete gauge action

$$\mathcal{G}_m^{\text{full}} := G^{V(\Lambda_m)} \times \prod_{c \in \mathcal{C}_m} \mathbb{F}_3^4, \quad q_c \mapsto q_c + B^+ x_c. \quad (35.24)$$

Thus the physical projection is

$$\mathbb{P}_m^{\text{phys}} F := \int_{\mathcal{G}_m^{\text{full}}} F(g^{-1} \cdot x) dg, \quad \hat{\mathcal{H}}_m^{\text{phys}} := \text{Ran } \mathbb{P}_m^{\text{phys}}. \quad (35.25)$$

It does not coincide with the forgetful map $\hat{X}_m \rightarrow X_m^{\text{link}}$. This distinction is the type safeguard that prevents exclusion of W_c from the physical sector.

35.6 Gauge generator and local transport circuit

The complete triangular chart provides a unitary map

$$\hat{\Gamma}_m : L^2(\hat{X}_{m+1}, \hat{\mu}_{m+1}^{\text{ov}}) \longrightarrow L^2(\hat{X}_m, \hat{\mu}_m^{\text{ov}}) \hat{\otimes} \mathcal{K}_{m+1}^{\text{det}}, \quad \hat{\Gamma}_m \hat{J}_m F = F \otimes \mathbf{1}_{\text{det}}. \quad (35.26)$$

For $\iota \in \mathcal{J}_m$, let $E_{m,\iota}$ be the expectation integrating exactly factor ι of (35.21) and fixing all the others. These are orthogonal Markov projections, preserve the unit, and commute. On the dense core $\mathcal{C}_m$ of cylinders depending on finitely many coordinates, define the form

$$\hat{\mathcal{E}}_m[F] := 3\kappa_{\text{av}} \sum_{\iota \in \mathcal{J}_m} \|(I - E_{m,\iota})F\|_2^2. \quad (35.27)$$

The sum is finite for each cylinder. Its closure is a Dirichlet form and determines the self-adjoint Markov generator $\hat{H}_m^{\text{ov}}$. At a seed level this is written, in the sense of forms, as

$$\hat{H}_{m_0}^{\text{ov}} = 3\kappa_{\text{av}} \sum_{\iota \in \mathcal{J}_{m_0}^{\text{seed}}} (I - E_{m_0,\iota}), \quad (35.28)$$

and each refinement adds exactly the expectations of its new coordinates:

$$\begin{aligned} \hat{H}_{m+1}^{\text{ov}} &= \hat{\Gamma}_m^* (\hat{H}_m^{\text{ov}} \otimes I + I \otimes \hat{H}_{m+1}^{\text{det}}) \hat{\Gamma}_m, \\ \hat{H}_{m+1}^{\text{det}} &= 3\kappa_{\text{av}} \sum_{\iota \in \mathcal{J}_{m+1/m}^{\text{det}}} (I - E_{m+1,\iota}^{\text{det}}). \end{aligned} \quad (35.29)$$

The color partition is made explicit. If a cell is indexed by $n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$, fix

$$\chi_m(n) := (n_1 \bmod 9, n_2 \bmod 9, n_3 + 3n_4 \bmod 9) \in (\mathbb{Z}/9\mathbb{Z})^3, \quad |(\mathbb{Z}/9\mathbb{Z})^3| = 9 \cdot 81. \quad (35.30)$$

If two closed collars of cellular radius one intersect, their difference δ satisfies $|\delta_j| \leq 2$. Equality of colors forces $\delta_1 = \delta_2 = 0$ and $\delta_3 + 3\delta_4 \equiv 0 \pmod{9}$. Since the latter integer belongs to $[-8, 8]$, it is zero; the bounds $|\delta_3|, |\delta_4| \leq 2$ then give $\delta_3 = \delta_4 = 0$. Thus distinct cells of the same color have disjoint collars. Define explicitly

$$\mathcal{B}_{m,r} := \{\iota \in \mathcal{J}_m : \text{cell}(\iota) = n, \chi_m(n) = r\}, \quad r \in (\mathbb{Z}/9\mathbb{Z})^3. \quad (35.31)$$

Transport of a screen by $g \in E(4)$ also transports this partition; it does not require a rotation to preserve the label coordinates. Thus $\{\mathcal{B}_{m,r}\}_{r=1}^{9 \cdot 81}$ groups disjoint spatial supports; the internal index of each block enumerates its factors in (35.22). For a cylinder F , the product contains only the blocks on which F depends; closure defines the strong product for the entire form. A block generator preserves the Hoeffding multiplicities; it is not replaced by a single projector that would erase them. By commutativity in each chart,

$$\begin{aligned} H_{m,B} &:= 3\kappa_{av} \sum_{t \in B} (I - E_{m,t}), \\ K_{m,B} &:= e^{-a_m H_{m,B}} = \prod_{t \in B} e^{-3\kappa_{av} a_m (I - E_{m,t})}, \\ K_m^{\text{loc}} &:= \prod_{r,B}^{\text{strong}} K_{m,B} = e^{-a_m \hat{H}_m^{\text{ov}}}. \end{aligned} \quad (35.32)$$

The physical diameter of every block remains uniformly bounded. Locality follows: the generator introduces no instantaneous update over the entire lattice. The partition is compatible with Bianchi, not merely with geometric disjointness. For a cubic cell c , fix the oriented product, with all factors transported to the same vertex,

$$\mathcal{B}_{m,c}(x) := \prod_{p \in \partial c}^{\rightarrow} T_p \text{Hol}_p(x) T_p^{-1}. \quad (35.33)$$

Every interior edge of the boundary occurs twice with opposite orientations; hence $\mathcal{B}_{m,c} = e$ pointwise. A block $\mathcal{B}_{m,r}$ contains the complete collar of every cell it updates: if it meets a face of c , it simultaneously updates the inverse pair sharing that edge. The preceding cancellation remains literal after every circuit factor:

$$K_{m,B} \mathbf{1}_{\{\mathcal{B}_{m,c}=e\}} = \mathbf{1}_{\{\mathcal{B}_{m,c}=e\}}, \quad K_m^{\text{loc}} \mathbf{1}_{\{\mathcal{B}_{m,c}=e\}} = \mathbf{1}_{\{\mathcal{B}_{m,c}=e\}}. \quad (35.34)$$

Thus the calendar of $9 \cdot 81$ layers preserves Bianchi at every step and in the strong product, not merely at the end of a sweep.

The gauge action permutes or translates the coordinates integrated by each $E_{m,t}$ and preserves their measures. Hence

$$U_m(g) \hat{H}_m^{\text{ov}} = \hat{H}_m^{\text{ov}} U_m(g) \quad (g \in \mathcal{G}_m^{\text{full}}), \quad [\mathbb{P}_m^{\text{phys}}, \hat{H}_m^{\text{ov}}] = 0, \quad \mathbb{P}_{m+1}^{\text{phys}} \hat{J}_m = \hat{J}_m \mathbb{P}_m^{\text{phys}}. \quad (35.35)$$

Denote the complete semigroup and its Markov kernel by

$$\hat{K}_{m,t}^{\text{ov}} := e^{-t \hat{H}_m^{\text{ov}}}, \quad \hat{K}_{m,t}^{\text{ov}}(x, dy) \quad (x, y \in \hat{X}_m). \quad (35.36)$$

The first identity in (35.35) implies

$$\hat{K}_{m,t}^{\text{ov}}(gx, gA) = \hat{K}_{m,t}^{\text{ov}}(x, A) \quad (g \in \mathcal{G}_m^{\text{full}}). \quad (35.37)$$

To type the reduction without using a restricted kernel on the wrong space, let $q_m : \hat{X}_m \rightarrow Y_m := \hat{X}_m / \mathcal{G}_m^{\text{full}}$ be the orbit map and $\nu_m = (q_m)_* \hat{\mu}_m^{\text{ov}}$. The pullback

$$\mathcal{W}_m : L^2(Y_m, \nu_m) \longrightarrow \hat{\mathcal{H}}_m^{\text{phys}}, \quad \mathcal{W}_m \phi = \phi \circ q_m \quad (35.38)$$

is unitary. Equivariance of the complete kernel allows the definition

$$K_{m,t}^{\text{phys}}(q_m x, A) := \hat{K}_{m,t}^{\text{ov}}(x, q_m^{-1} A), \quad A \subseteq Y_m, \quad (35.39)$$

independently of the representative x . The physical object is then typed by

$$D_m^{\text{YM}} := \hat{H}_m^{\text{ov}}|_{\hat{\mathcal{H}}_m^{\text{phys}}}, \quad e^{-t D_m^{\text{YM}}} = \mathcal{W}_m K_{m,t}^{\text{phys}} \mathcal{W}_m^{-1}. \quad (35.40)$$

This is the complete product generator, not the single-fiber generator in (35.1).

35.7 Yang–Mills action, g -dependent measure, and exact transport

So far, the reference kinematic chart has been written. Denote it by $\hat{\mu}_m^0 := \hat{\mu}_m^{\text{ov}}$ and $\hat{H}_m^0 := \hat{H}_m^{\text{ov}}$, with $\hat{K}_{m,t}^0 := e^{-t\hat{H}_m^0} = \hat{K}_{m,t}^{\text{ov}}$. To turn the structural reader $\mathcal{P}_m^{F^2}$ into a law, rather than a subsequent comparison, first realize its positive square root on the triangular factorization. Polarization of $\mathcal{P}_m^{F^2}$ on a collar b is the positive Gram form

$$Q_{m,b}(u, v) := \frac{1}{4}(\mathcal{P}_{m,b}^{F^2}(u + v) - \mathcal{P}_{m,b}^{F^2}(u - v)).$$

Quotient the finite reader space of the collar by $\ker Q_{m,b}$, complete it with respect to $Q_{m,b}$, and denote the resulting Hilbert space by $\mathcal{E}_{m,b}^F$. If $\zeta_{m,b}(x_b)$ is the vector of centered readers of the triangular coordinate x_b , define explicitly

$$\xi_{m,b}(x_b) := Q_{m,b}^{1/2} \zeta_{m,b}(x_b) \in \mathcal{E}_{m,b}^F, \quad p_{m,b}(x_b) := \|\xi_{m,b}(x_b)\|^2. \quad (35.41)$$

The index b ranges over the Gram collars of minimal support. A collar is assigned to the first color in (35.30) containing its base cell; each term thus occurs once, and collars of the same color are disjoint. More precisely, positive diagonalization groups the triangular factors into a disjoint partition

$$\mathcal{J}_m^F = \bigsqcup_{r \in (\mathbb{Z}/9\mathbb{Z})^3} \bigsqcup_{b \in \mathcal{B}_{m,r}^F} \mathcal{J}_{m,b}, \quad x_b := (x_t)_{t \in \mathcal{J}_{m,b}}, \quad \mathcal{B}_{m,r}^F := \{b : Q_{m,b} \neq 0, \chi_m(b) = r\}. \quad (35.42)$$

No triangular coordinate belongs to two $\mathcal{J}_{m,b}$; if a geometric reader meets two collars, its Gram mode is assigned to the first and its orthogonal complement to the second. This is the finite spectral decomposition of the positive form, not a duplication of the physical variable. The square root of the orthogonal sum, computed color by color, gives the identity, not an approximation,

$$\mathcal{P}_m^{F^2}(x) = \sum_{r \in (\mathbb{Z}/9\mathbb{Z})^3} \sum_{b \in \mathcal{B}_{m,r}^F} p_{m,b}(x_b), \quad S_{m,g}(x) := \frac{1}{4g^2} \mathcal{P}_m^{F^2}(x), \quad g \in (0, \infty). \quad (35.43)$$

Here $x_b : \hat{X}_m \rightarrow X_{m,b}$ is the coordinate projection of the collar, so the domain, action, and codomain of each summand are fixed. The reader is gauge-invariant because $Q_{m,b}$ uses the invariant inner product on $\mathfrak{g}$; it is reflection-invariant because a reflection merely permutes the six oriented pairs; and it is natural under Euclidean relabeling. The partition (35.42) and the product chart give explicitly

$$\hat{\mu}_m^0 = \lambda_m^{\text{inert}} \otimes \bigotimes_{r,b} \lambda_{m,b}^0, \quad d\lambda_{m,b}^g := (Z_{m,b}^g)^{-1} e^{-p_{m,b}/(4g^2)} d\lambda_{m,b}^0, \quad \hat{\mu}_{m,g}^{\text{YM}} = \lambda_m^{\text{inert}} \otimes \bigotimes_{r,b} \lambda_{m,b}^g. \quad (35.44)$$

The inert factor contains exactly those coordinates on which the reader is zero; it is not removed from the theory.

The coarse–detail separation in (35.21) is orthogonal for the same square root: direct polarization gives $Q_{m+1} = Q_m \oplus Q_{m+1/m}^{\text{det}}$ and $\zeta_{m+1} = \zeta_m \oplus \zeta_{m+1/m}^{\text{det}}$. If $x_{m+1} = \hat{\Gamma}_m^{-1}(x_m, u)$, then

$$\xi_{m+1}(x_{m+1}) = \xi_m(x_m) \oplus \xi_{m+1/m}^{\text{det}}(u), \quad S_{m+1,g} \circ \hat{\Gamma}_m^{-1} = S_{m,g} + S_{m+1/m,g}^{\text{det}}. \quad (35.45)$$

This is precisely how carry and memory prevent an ancestral face from being counted twice: the coarse part is not added again in the detail. In particular, $Z_{m+1,g} = Z_{m,g} Z_{m+1/m,g}^{\text{det}}$ for the normalizers, and one obtains the collection of probability measures genuinely depending on the coupling

$$d\hat{\mu}_{m,g}^{\text{YM}}(x) := Z_{m,g}^{-1} e^{-S_{m,g}(x)} d\hat{\mu}_m^0(x), \quad (\hat{\pi}_{m+1,m})_* \hat{\mu}_{m+1,g}^{\text{YM}} = \hat{\mu}_{m,g}^{\text{YM}}. \quad (35.46)$$

To write Schwinger–Dyson correctly with respect to Haar, rather than pretending that the reference measure is flat, let λ_m be the product of Haar, Lebesgue, and uniform counting measures in the lifted chart, and let $R_m := d\hat{\mu}_m^0/d\lambda_m > 0$. The total regulated action is

$$S_{m,g} := S_{m,g} - \log R_m, \quad d\hat{\mu}_{m,g}^{\text{YM}} = Z_{m,g}^{-1} e^{-S_{m,g}} d\lambda_m. \quad (35.47)$$

The term $-\log R_m$ contains the heat kernels and bridge charts of the initial gauge structure; it is independent of g . The Yang–Mills term that changes the coupling and whose classical limit is calculated below is exactly $S_{m,g} = \mathcal{P}_m^{F^2}/(4g^2)$. Consistency of the second equality and standardness of the finite spaces allow application of Kolmogorov’s theorem and define the unique cylindrical measure $\hat{\mu}_{\infty,g}^{\text{YM}}$ whose marginals are $\hat{\mu}_{m,g}^{\text{YM}}$. No eigenvalue has been used here: g , the action, and the measure have been fixed first.

To define dynamics on this measure without losing the product, gauge, or surface identities, the transport is constructed, not postulated. On each nonatomic factor $X_{m,b}$, fix a Borel chart $\rho_{m,b} : X_{m,b}/\mathcal{G}_{m,b} \rightarrow (0, 1)$ and first disintegrate over the gauge quotient, then over the Haar orbit. Let $F_{m,b}^0$ and $F_{m,b}^g$ be the continuous distribution functions of $\rho_{m,b}$ for the reference and reweighted measures. The triangular transport is

$$t_{m,b,g} := (F_{m,b}^g)^{-1} \circ F_{m,b}^0, \quad T_{m,b,g}(y, h) := (t_{m,b,g}(y), h), \quad (T_{m,b,g})_* \lambda_{m,b}^0 = \lambda_{m,b}^g. \quad (35.48)$$

On a purely atomic factor where $p_{m,b} = 0$, take $T_{m,b,g} = I$. In the presence of atoms and a continuous coordinate, the chart is the lexicographic atom–interval order, and the same formula applies to the full standard nonatomic space. The transports of the $9 \cdot 81$ colors are composed in the order fixed by (35.30); within one color, they commute because the collars are disjoint. One obtains

$$T_{m,g} := \prod_{r \in (\mathbb{Z}/9\mathbb{Z})^3} \prod_{b \in \mathcal{B}_{m,r}^F} T_{m,b,g}, \quad (T_{m,g})_* \hat{\mu}_m^0 = \hat{\mu}_{m,g}^{\text{YM}}, \quad \hat{\pi}_{m+1,m} T_{m+1,g} = T_{m,g} \hat{\pi}_{m+1,m}. \quad (35.49)$$

The last equality is checked factor by factor using (35.45): the transport of the coarse factor is the one already fixed, and the new transports act only on the innovation. The quotient charts are chosen on a representative screen and transported; therefore $T_{m,g}$ commutes with gauge and reflection and, for $a \in E(4)$, satisfies $T_{a\alpha,g} = U_\alpha(a) T_{\alpha,g} U_\alpha(a)^{-1}$.

The composition

$$\mathcal{T}_{m,g} : L^\infty(\hat{X}_m, \hat{\mu}_{m,g}^{\text{YM}}) \longrightarrow L^\infty(\hat{X}_m, \hat{\mu}_m^0), \quad \mathcal{T}_{m,g} F := F \circ T_{m,g} \quad (35.50)$$

is, by construction, an isomorphism of probability spaces and $*$ -algebras, and extends unitarily to L^2 . In particular,

$$\mathcal{T}_{m,g}(FG) = \mathcal{T}_{m,g} F \mathcal{T}_{m,g} G, \quad \mathcal{T}_{m,g}(F^*) = (\mathcal{T}_{m,g} F)^*, \quad \int F d\hat{\mu}_{m,g}^{\text{YM}} = \int \mathcal{T}_{m,g} F d\hat{\mu}_m^0. \quad (35.51)$$

The domain of each $T_{m,b,g}$ is the holonomy space itself, satisfying the subdivision law and Bianchi constraint. Its image therefore remains in that space. Applying the preceding multiplicativity generator by generator gives the pointwise identities

$$\begin{aligned} \mathcal{T}_{m,g}(\text{Hol}_p) &= \overrightarrow{\prod_{p' \subset p}} \mathcal{T}_{m,g}(T_{p'} \text{Hol}_{p'} T_{p'}^{-1}), \\ \mathcal{T}_{m,g}(F_{\text{cl},m}) &= F_{\text{cl},m} \circ T_{m,g}, \\ \overrightarrow{\prod_{p \subset \partial c}} \mathcal{T}_{m,g}(T_p \text{Hol}_p T_p^{-1}) &= e, \quad \text{Alt}_{\lambda\mu\nu} D_\lambda(F_{\text{cl},m} \circ T_{m,g})_{\mu\nu} = 0. \end{aligned} \quad (35.52)$$

The third line is the edge-by-edge cancellation of the oriented boundary of the boundary of c ; the fourth is its infinitesimal polarization. Thus transport preserves products, state, clover, and Bianchi, not merely the first chaos.

Finally, transport the expectations, generator, kernel, projection, and inclusion:

$$\begin{aligned} E_{m,t}^g &:= \mathcal{T}_{m,g}^{-1} E_{m,t} \mathcal{T}_{m,g}, & \hat{H}_{m,g}^{\text{YM}} &:= \mathcal{T}_{m,g}^{-1} \hat{H}_m^0 \mathcal{T}_{m,g}, \\ \hat{K}_{m,t}^g &:= \mathcal{T}_{m,g}^{-1} \hat{K}_{m,t}^0 \mathcal{T}_{m,g}, & \mathbb{P}_{m,g}^{\text{phys}} &:= \mathcal{T}_{m,g}^{-1} \mathbb{P}_m^{\text{phys}} \mathcal{T}_{m,g}, \\ \hat{J}_{m,g} &:= \mathcal{T}_{m+1,g}^{-1} \hat{J}_m \mathcal{T}_{m,g}, & D_{m,g}^{\text{YM}} &:= \hat{H}_{m,g}^{\text{YM}}|_{\text{Ran } \mathbb{P}_{m,g}^{\text{phys}}}. \end{aligned} \quad (35.53)$$

The first line is not merely an abstract conjugation. Using (35.44), its action on a cylinder is the heat-bath update of the action:

$$(E_{m,b}^g F)(x_{\neq b}) = \frac{\int_{X_{m,b}} F(x_{\neq b}, u) e^{-p_{m,b}(u)/(4g^2)} d\lambda_{m,b}^0(u)}{\int_{X_{m,b}} e^{-p_{m,b}(u)/(4g^2)} d\lambda_{m,b}^0(u)}, \quad \hat{H}_{m,g}^{\text{YM}} = 3\kappa_{\text{av}} \sum_b (I - E_{m,b}^g). \quad (35.54)$$

Thus both the measure and every local transition contain g and the action $p_{m,b}/(4g^2)$ explicitly. They are Markov because $\mathcal{T}_{m,g}\mathbf{1} = \mathbf{1}$, local because each $T_{m,b,g}$ has the same collar as $p_{m,b}$, and depend on g through (35.48). The equivalence is spectral and does not select the measure from the spectrum: To establish the dynamical equations as well, let $X_{m,b}^a$ be the left-invariant field on link factor b associated with $e_a \in \mathfrak{g}$; on a continuous innovation chart, use the ordinary directional derivative, and treat finite coordinates by exact differences. For every smooth cylinder F , the common domain is $\mathcal{C}_m^1 := \bigcap_{b,a} \text{Dom } X_{m,b}^a$, with compact support in the open innovation charts; group factors have no boundary. Haar invariance and the product rule give

$$\int X_{m,b}^a F d\hat{\mu}_{m,g}^{\text{YM}} = \int F X_{m,b}^a S_{m,g} d\hat{\mu}_{m,g}^{\text{YM}}, \quad F \in \mathcal{C}_m^1. \quad (35.55)$$

On a finite coordinate, for $\Delta_\varepsilon F(q) := F(q + \varepsilon) - F(q)$, the change of variable $q \mapsto q + \varepsilon$ gives the exact version, without a formal derivative,

$$\int \Delta_\varepsilon F(q) d\hat{\mu}_{m,g}^{\text{YM}} = \int F(q) (e^{S_{m,g}(q) - S_{m,g}(q-\varepsilon)} - 1) d\hat{\mu}_{m,g}^{\text{YM}}(q). \quad (35.56)$$

If δ_ε is the oriented sum of these fields on the edges incident to a vertex, gauge invariance of the reference measure and of $S_{m,g}$ gives $\delta_\varepsilon S_{m,g} = 0$ and hence the Ward identity

$$\int \delta_\varepsilon F d\hat{\mu}_{m,g}^{\text{YM}} = 0. \quad (35.57)$$

For the finite gauge factor $\mathbb{F}_3^4$, invariance $S_{m,g}(q + \varepsilon) = S_{m,g}(q)$ reduces (35.56) to the same Ward identity with δ_ε replaced by Δ_ε . Both expressions are compatible with $\hat{J}_{m,g}$ because (35.45) separates coarse derivatives from detail derivatives. Every cylinder of the limit resides at some level; thus (35.55) and (35.57) pass literally to the cofinal state. Neither the Schwinger–Dyson nor the Ward equation contains the gap as a premise.

$$D_{m,g}^{\text{YM}} \geq 3\kappa_{\text{av}}(I - P_{\Omega_{m,g}}), \quad W_{c,g} := \mathcal{T}_{m,g}^{-1} W_c, \quad D_{m,g}^{\text{YM}} W_{c,g} = 3\kappa_{\text{av}} W_{c,g}. \quad (35.58)$$

Hereafter fix $g \in (0, \infty)$ and suppress that subscript; the measures, expectations, inclusions, and generators without superscripts are the objects in (35.53). The geometric holonomies remain canonical coordinate functions on configuration space, and their law therefore does change with g ; $\mathcal{T}_{m,g}$ takes them to composite functions of the reference chart. Only the spectral witness is transported explicitly in (35.58). This convention does not erase g : its explicit occurrence is (35.46)–(35.53).

35.8 Simple vacuum and a witness attaining the gap

The intersection of all expectations of the complete state consists of constants:

$$\bigcap_{t \in \mathcal{I}_m} \text{Ran } E_{m,t} = \mathbb{C} \Omega_m. \quad (35.59)$$

This identity is not introduced as a hypothesis. In the product chart (35.21), each $E_{m,t}$ integrates one coordinate of the exhaustive index (35.22) and fixes the others. If u belongs to the intersection, its conditional expectations with respect to every finite cylinder are constant. Martingale convergence of

these cylinders to the complete σ -algebra implies that u is constant almost everywhere. The reverse inclusion is immediate because every expectation preserves the unit. This proves (35.59) and, after gauge projection, leaves a one-dimensional physical vacuum. The Hoeffding decomposition of (35.28)–(35.29) gives

$$\hat{\mathcal{E}}_m(u) = 3\kappa_{\text{av}} \sum_{S \in J_m} |S| \|u_S\|^2, \quad D_m^{\text{YM}} \geq 3\kappa_{\text{av}}(I - P_{\Omega_m}). \quad (35.60)$$

The function

$$W_c(q) := \mathbf{1}_{\{q_1 + \dots + q_6 = 0\}} - \frac{1}{3} \quad (35.61)$$

is invariant under $G^{V(\Lambda_m)}$ because it depends on neither links nor frames. For the incidence action, each x_i occurs three times, hence

$$\sum_{i < j} (B^+ x)_{ij} = 3 \sum_i x_i = 0 \quad \text{in } \mathbb{F}_3. \quad (35.62)$$

Thus $\mathbb{P}_m^{\text{phys}} W_c = W_c$. Under u_6 , the sum of the six trits is uniform; by direct calculation,

$$\mathbb{E} W_c = 0, \quad \|W_c\|^2 = \frac{2}{9}, \quad \mathbb{E}(W_c^3) = \frac{2}{27}. \quad (35.63)$$

Only the projector $I - E_{m,q_c}$ acts nontrivially on W_c . Therefore

$$D_m^{\text{YM}} W_c = 3\kappa_{\text{av}} W_c. \quad (35.64)$$

The lower bound in (35.60) is not merely a bound: the physical complement contains a nonzero vector attaining the threshold.

35.9 Nonadic refinement and persistence in the limit

The physical inclusions satisfy

$$D_{m+1}^{\text{YM}} \hat{J}_m = \hat{J}_m D_m^{\text{YM}}, \quad (e^{-a_{m+1} D_{m+1}^{\text{YM}}})^9 \hat{J}_m = \hat{J}_m e^{-a_m D_m^{\text{YM}}}, \quad a_{m+1} = a_m/9. \quad (35.65)$$

In the inductive limit

$$\hat{\mathcal{H}}_\infty^{\text{phys}} = \varinjlim (\hat{\mathcal{H}}_m^{\text{phys}}, \hat{J}_m) \quad (35.66)$$

define on the algebraic union

$$\mathcal{E}_\infty[\hat{J}_{m,\infty} u] := \langle u, D_m^{\text{YM}} u \rangle. \quad (35.67)$$

Denote its generator by $D_{\infty,g}^{\text{YM}}$; following the convention after (35.58), also write $D_\infty^{\text{YM}} := D_{\infty,g}^{\text{YM}}$. The first intertwining in (35.65) makes the definition independent of the representative. Hoeffding truncations form a form core and give a recovery sequence; Fatou's lemma applied to the nonnegative sum gives lower semicontinuity. The closure therefore satisfies

$$\bar{\mathcal{E}}_\infty[u] \geq 3\kappa_{\text{av}} \|(I - P_{\Omega_\infty})u\|^2. \quad (35.68)$$

The inclusion preserves the ancestral coordinates, so $\hat{J}_{m,\infty} W_c$ remains nonzero and satisfies

$$D_\infty^{\text{YM}} \hat{J}_{m,\infty} W_c = 3\kappa_{\text{av}} \hat{J}_{m,\infty} W_c. \quad (35.69)$$

Consequently,

$$\inf(\text{Spec } D_\infty^{\text{YM}} \setminus \{0\}) = 3\kappa_{\text{av}} > 0, \quad \ker D_\infty^{\text{YM}} = \mathbb{C}\Omega_\infty. \quad (35.70)$$

35.10 Four OS reconstructions of the same dynamics

The complete kernel in (35.36) defines a reversible law on $\hat{X}_m$. Physical reconstruction uses its correctly typed descent (35.39): on the quotient Y_m , define

$$\Omega_{m,\text{phys}}^{(8)}(d\bar{y}_1 \cdots d\bar{y}_8) = \int \nu_m(d\bar{z}) \prod_{r=1}^8 K_{m,a_m/2}^{\text{phys}}(\bar{z}, d\bar{y}_r). \quad (35.71)$$

For each direction $\nu = 1, \dots, 4$, reflection interchanges the four positive faces with the negative ones. Conditioning on the central state z and using reversibility,

$$\int \overline{F(\Theta_{\nu,m,\nu})} F(y) d\Omega_{m,\text{phys}}^{(8)}(y) = \|\mathcal{B}_{m,\nu} F\|_2^2 \geq 0. \quad (35.72)$$

The marginal of the opposite pair normal to ν is

$$\nu_m(d\bar{x}) K_{m,a_m}^{\text{phys}}(\bar{x}, d\bar{y}). \quad (35.73)$$

To identify the Hamiltonian of one direction, take the OS subspace of observables on a single positive face, $F(y) = F_\nu(\bar{y}_{\nu,+})$; multiple-face observables retain the preceding positivity but are not confused with this transfer space. On the single-face quotient, the map

$$\mathcal{V}_{m,\nu}[F] := e^{-a_m D_m^{\text{YM}}/2} \mathcal{W}_m F \quad (35.74)$$

is isometric for the OS norm and has dense range. Its extension identifies the OS quotient with $\hat{\mathcal{H}}_m^{\text{phys}}$. Set $\bar{D}_m := \mathcal{W}_m^{-1} D_m^{\text{YM}} \mathcal{W}_m$ and $\bar{J}_m := \mathcal{W}_{m+1}^{-1} \hat{J}_m \mathcal{W}_m$. The intertwining and $a_m = 9a_{m+1}$ give

$$\bar{J}_m e^{-a_m \bar{D}_m/2} = e^{-a_m \bar{D}_{m+1}/2} \bar{J}_m = (e^{-a_{m+1} \bar{D}_{m+1}/2})^9 \bar{J}_m. \quad (35.75)$$

The last member belongs to the range of $e^{-a_{m+1} \bar{D}_{m+1}/2}$; hence the inverse of $\mathcal{V}_{m+1,\nu}$ appearing below is applied only on its range and is well defined. The connecting maps

$$\mathcal{J}_{m,\nu}^{\text{OS}} := \mathcal{V}_{m+1,\nu}^{-1} \hat{J}_m \mathcal{V}_{m,\nu} \quad (35.76)$$

are isometries and intertwine the four reconstructed Hamiltonians:

$$H_{\text{OS},m+1}^{(\nu)} \mathcal{J}_{m,\nu}^{\text{OS}} = \mathcal{J}_{m,\nu}^{\text{OS}} H_{\text{OS},m}^{(\nu)}, \quad \mathcal{V}_{\infty,\nu} H_{\text{OS},\infty}^{(\nu)} \mathcal{V}_{\infty,\nu}^{-1} = D_{\infty}^{\text{YM}}. \quad (35.77)$$

The four reflections do not produce four measures: they are four outputs of the same state, the same kernel, and the same generator.

35.11 Locality, Euclidean action, and curvature output

The directed category of screens includes refinements, overlays, and Euclidean transports. If $a \in E(4)$ and α is a screen, complete relabeling defines

$$U_\alpha(a) : \hat{\mathcal{H}}_\alpha^{\text{phys}} \longrightarrow \hat{\mathcal{H}}_{a\alpha}^{\text{phys}}, \quad U_\beta(a) \hat{J}_{\alpha\beta} = \hat{J}_{a\alpha,a\beta} U_\alpha(a). \quad (35.78)$$

Transport preserves measure, generator, null ideals, and supports. The naturality identity makes it possible to define, without choosing a representative, a unitary map on the algebraic colimit:

$$U(a) \hat{J}_{\alpha,\infty} F := \hat{J}_{a\alpha,\infty} U_\alpha(a) F. \quad (35.79)$$

Relabelings satisfy $U_{a\alpha}(b) U_\alpha(a) = U_\alpha(ba)$ and preserve the norm; hence (35.79) extends to a unitary representation of $E(4)$.

35.11.1 Local curvature of the incidence channel

Every compact, connected, simple group has a nontrivial maximal torus. Fix $t_3 \in G$ of order exactly three. For a collar c , let

$$\omega_c(q) := \sum_{i < j} q_{ij} \in \mathbb{F}_3. \quad (35.80)$$

Identity (35.12) proves that ω_c is invariant under the finite action. If h_c is the frame at the base vertex of the collar, define the geometric holonomy and, for the spectral witness, its transported image under the law at coupling g :

$$\text{Hol}_{m,c}^{\text{inc,geom}} := h_c t_3^{\omega_c(q)} h_c^{-1} \in G, \quad \text{Hol}_{m,c,g}^{\text{inc,spec}} := \mathcal{T}_{m,g}^{-1} \text{Hol}_{m,c}^{\text{inc,geom}}, \quad \text{Hol}_{m,c}^{\text{inc}} := \text{Hol}_{m,c,g}^{\text{inc,spec}}. \quad (35.81)$$

Under G^{V_m} it transforms by conjugation, and under $\prod_c \mathbb{F}_3^4$ it remains fixed. Since t_3 has order three, the functional calculus of this three-valued variable gives the literal identity

$$W_{c,g} := \mathcal{T}_{m,g}^{-1} W_c^0 = \mathbf{1}_{\{e\}}(\text{Hol}_{m,c}^{\text{inc}}) - \frac{1}{3}. \quad (35.82)$$

Write $W_c := W_{c,g}$ throughout the rest of the chapter. Since $T_{m,b,g}$ acts within the quotient of the same collar, the spectral holonomy is a function in its local algebra of geometric holonomies; it adds no variable and does not change the support. The geometric holonomies, used in $S_{m,g}$ and in the clover, retain their g -dependent distribution. The projector on the right-hand side is a local class function; hence W_c is not a decoupled auxiliary coordinate but an observable of the incidence holonomy of the same gauge restriction.

For a bounded open set $O \subset \mathbb{R}^4$, let $\mathfrak{A}_{\text{YM}}(O)$ be the algebra generated by bounded class functions of (35.81), plaquette holonomies, and clover readers supported in O . Define the physical curvature sector by

$$\mathcal{H}_{\text{YM}}^{\text{curv}} := \overline{\bigcup_{O \in \mathbb{R}^4} \mathfrak{A}_{\text{YM}}(O) \Omega_\infty} \subseteq \hat{\mathcal{H}}_\infty^{\text{phys}}. \quad (35.83)$$

By (35.82), each $\hat{J}_{m,\infty} W_c$ belongs explicitly to this sector. Coordinate expectations map curvature cylinders to curvature cylinders; the semigroup therefore leaves $\mathcal{H}_{\text{YM}}^{\text{curv}}$ invariant. Since it is self-adjoint, the sector is reducing. The bound in (35.68) restricts to it, and the preceding vector attains equality:

$$D_{\text{curv},g}^{\text{YM}} := D_{\infty,g}^{\text{YM}}|_{\mathcal{H}_{\text{YM}}^{\text{curv}}}, \quad D_{\text{curv}}^{\text{YM}} := D_{\text{curv},g}^{\text{YM}}, \quad \inf(\text{Spec } D_{\text{curv},g}^{\text{YM}} \setminus \{0\}) = 3\kappa_{\text{av}}. \quad (35.84)$$

35.11.2 Cofinal output system and cross products

The inclusion $\hat{J}_{\alpha\beta}$ preserves an ancestral coordinate; it must not be confused with the block average yielding a continuous field. To write this second map without altering the genealogy, let $\mathfrak{P}$ be the directed set of bounded cellular screens, closed under overlay, nonadic refinement, and Euclidean transport. For $\alpha \in \mathfrak{P}$, set

$$V_\alpha := \text{span}\{e_{\alpha,c} := |c|^{-1/2} \mathbf{1}_c : c \in \mathcal{C}_\alpha^{\text{cell}}\} \subset L^2(\mathbb{R}^4), \quad P_\alpha : L^2(\mathbb{R}^4) \rightarrow V_\alpha. \quad (35.85)$$

Overlays ensure that $P_\alpha h \rightarrow h$ for every $h \in L^2$ along $\mathfrak{P}$. On the gauge side, normalize the witness by

$$Z_{\alpha,c} := \sqrt{\frac{9}{2}} W_{\alpha,c}, \quad \langle Z_{\alpha,c}, Z_{\alpha,d} \rangle = \delta_{cd}, \quad D_\alpha^{\text{YM}} Z_{\alpha,c} = 3\kappa_{\text{av}} Z_{\alpha,c}. \quad (35.86)$$

The second equality uses exactly the moments in (35.63) and the factorization of distinct coordinates. Let $\mathcal{K}_\alpha^{\text{cell}}$ be the space generated by these vectors. If $\beta \geq \alpha$, define the output intertwiner, distinct from $\hat{J}_{\alpha\beta}$, generator by generator through

$$I_{\alpha\beta}^{\text{curv}} Z_{\alpha,c} := \sum_{\substack{d \in \mathcal{C}_\beta^{\text{cell}} \\ d \subset c}} \sqrt{\frac{|d|}{|c|}} Z_{\beta,d}. \quad (35.87)$$

Since the subcells are disjoint and their volumes sum to $|c|$, one obtains, without taking limits,

$$(I_{\alpha\beta}^{\text{curv}})^* I_{\alpha\beta}^{\text{curv}} = I, \quad I_{\beta\gamma}^{\text{curv}} I_{\alpha\beta}^{\text{curv}} = I_{\alpha\gamma}^{\text{curv}}, \quad D_{\beta}^{\text{YM}} I_{\alpha\beta}^{\text{curv}} = I_{\alpha\beta}^{\text{curv}} D_{\alpha}^{\text{YM}}. \quad (35.88)$$

Cellular relabeling preserves volumes and descendants; for $a \in E(4)$, one also has

$$U_{\beta}(a) I_{\alpha\beta}^{\text{curv}} = I_{a\alpha, a\beta}^{\text{curv}} U_{\alpha}(a). \quad (35.89)$$

For $h \in C_c^{\infty}(\mathbb{R}^4)$, now define the block reader

$$\Phi_{\alpha}(h) := \sum_{c \in \mathcal{C}_{\alpha}^{\text{cell}}} \langle h, e_{\alpha, c} \rangle_{L^2} Z_{\alpha, c}. \quad (35.90)$$

On a uniform mesh, this formula is $\sqrt{9/2} a_{\alpha}^2 \sum_c (P_{\alpha} h)(x_c) W_{\alpha, c}$; the cellular average, not a point evaluation, fixes the cross product. Indeed, if γ refines two screens α and β , the preceding definitions give the computed identity

$$\langle I_{\alpha\gamma}^{\text{curv}} \Phi_{\alpha}(h), I_{\beta\gamma}^{\text{curv}} \Phi_{\beta}(k) \rangle = \langle P_{\alpha} h, P_{\beta} k \rangle_{L^2}. \quad (35.91)$$

By polarization, with $k = h$,

$$\|I_{\alpha\gamma}^{\text{curv}} \Phi_{\alpha}(h) - I_{\beta\gamma}^{\text{curv}} \Phi_{\beta}(h)\|_2^2 = \|P_{\alpha} h - P_{\beta} h\|_{L^2}^2 \longrightarrow 0. \quad (35.92)$$

Thus the Cauchy property follows from strong convergence of projections, rather than being an additional postulated identity. The colimit of (35.87) therefore contains

$$\Phi(h) := \lim_{\alpha} I_{\alpha, \infty}^{\text{curv}} \Phi_{\alpha}(h), \quad \langle \Phi(h), \Phi(k) \rangle = \langle h, k \rangle_{L^2}, \quad D_{\text{pub}}^{\text{YM}} \Phi(h) = 3\kappa_{\text{av}} \Phi(h). \quad (35.93)$$

This output adds no degrees of freedom. At each step, the decomposition $V_{\beta} = V_{\alpha} \oplus (V_{\beta} \ominus V_{\alpha})$ has a nonadic Haar basis. The exhaustive order of new incidences in (35.22) pairs this basis, label by label, with coordinates $Z_{\beta, d}$ of the innovation factor, preserving the support of the Haar function. The triangular unitaries $\hat{\Gamma}_{\alpha}$ glue these pairings and produce a based isometry

$$\mathcal{U}_{\text{curv}} : \varinjlim (\mathcal{K}_{\alpha}^{\text{cell}}, I_{\alpha\beta}^{\text{curv}}) \longrightarrow \mathcal{H}_{\text{YM}}^{\text{curv}}, \quad D_{\text{curv}}^{\text{YM}} \mathcal{U}_{\text{curv}} = \mathcal{U}_{\text{curv}} D_{\text{pub}}^{\text{YM}}, \quad U(a) \mathcal{U}_{\text{curv}} = \mathcal{U}_{\text{curv}} U_{L^2}(a). \quad (35.94)$$

Hereafter $\Phi(h)$ also denotes its image under $\mathcal{U}_{\text{curv}}$. Compatibility in (35.49) defines $\mathcal{T}_{\infty, g}$ in the limit. The physical output at the fixed coupling is not a second isometry chosen only on the first chaos, but the restriction

$$\mathcal{U}_{\text{curv}, g} := \mathcal{T}_{\infty, g}^{-1} \mathcal{U}_{\text{curv}, 0}, \quad \tilde{\mathcal{U}}_g := \mathcal{T}_{\infty, g}^{-1} : L^{\infty}(\hat{X}_{\infty}, \hat{\mu}_{\infty}^0) \longrightarrow L^{\infty}(\hat{X}_{\infty}, \hat{\mu}_{\infty, g}^{\text{YM}}). \quad (35.95)$$

On the polynomial $*$ -algebra generated by holonomies, plaquettes, clover, and $\Phi(h)$, one verifies, generator by generator and then by linearity,

$$\tilde{\mathcal{U}}_g(AB) = \tilde{\mathcal{U}}_g(A) \tilde{\mathcal{U}}_g(B), \quad \tilde{\mathcal{U}}_g(A^*) = \tilde{\mathcal{U}}_g(A)^*, \quad \omega_g(\tilde{\mathcal{U}}_g A) = \omega_0(A). \quad (35.96)$$

The surface, clover, and Bianchi identities are preserved by (35.52); by monotone density, the extension holds on the entire bounded local algebra. Thus (35.91) is the restriction of a complete probabilistic and algebraic isomorphism, not a substitute for it.

35.11.3 Dense core, strong continuity, and local net

Let $M_{\Phi(h)}$ be the multiplication operator affiliated with the local algebra, and

$$W(h, t) := \exp(itM_{\Phi(h)}), \quad h \in C_c^{\infty}(\mathbb{R}^4), \quad t \in \mathbb{R}. \quad (35.97)$$

Complete $\mathfrak{A}_{\text{YM}}(O)$ with these unitaries for $\text{supp } h \subset O$ and with the bounded class functions of plaquette holonomies. Denote by $\mathfrak{A}_{\text{YM}}(O)^{\text{sm}}$ the $*$ -algebra thus generated with smooth test functions. The core

$$\mathcal{D}_{\text{sm}} := \text{span} \{A_1 \cdots A_n \Omega_{\infty} : A_j \in \mathfrak{A}_{\text{YM}}(O_j)^{\text{sm}}, \quad O_j \in \mathbb{R}^4\} \quad (35.98)$$

is dense by the cyclic definition in (35.83). Equations (35.89) and (35.94) give, for $a \in E(4)$,

$$U(a)\Phi(h) = \Phi(a \cdot h), \quad \|(U(a) - I)\Phi(h)\| = \|a \cdot h - h\|_{L^2}. \quad (35.99)$$

Moreover, $|e^{ix} - e^{iy}| \leq |x - y|$ and (35.93) imply

$$\|(W(a \cdot h, t) - W(h, t))\Omega_\infty\| \leq |t| \|a \cdot h - h\|_{L^2}. \quad (35.100)$$

Continuity is also checked for the holonomy generators of the core; (35.100) alone does not include them. For an oriented polygonal route γ , a plaquette p , and $f \in C^1(G)^{\text{class}}$, set

$$A_{\gamma,f} := f(\text{Hol}_\gamma), \quad A_{p,f} := f(\text{Hol}_p). \quad (35.101)$$

In an overlay of γ and $a\gamma$, with $a \in E(4)$, the bridge law (35.17) cancels the common increments. For each remaining increment z , use $|f(xz) - f(x)| \leq \|\nabla f\|_\infty d_G(z, e)$ and the second-moment bound for the heat kernel $\int_G d_G(z, e)^2 k_\tau(z) dz \leq C_G \tau$. For the reweighted law, $0 < e^{-p/(4g^2)} \leq 1$ and continuity of p on the compact factor give

$$\frac{\int_G d_G(z, e)^2 e^{-p(z)/(4g^2)} k_\tau(z) dz}{\int_G e^{-p(z)/(4g^2)} k_\tau(z) dz} \leq C_{G,g} \tau, \quad (35.102)$$

uniformly over screens in a compact set and for $0 < \tau \leq \tau_0$. Summing the times of the increments in the symmetric difference gives the computed estimate

$$\begin{aligned} \|U(a)A_{\gamma,f}\Omega_\infty - A_{\gamma,f}\Omega_\infty\|_2^2 &\leq C_{G,g} \|\nabla f\|_\infty^2 \tau (\gamma \triangle a\gamma), \\ \|U(a)A_{p,f}\Omega_\infty - A_{p,f}\Omega_\infty\|_2^2 &\leq C_{G,g} \|\nabla f\|_\infty^2 \tau (\partial p \triangle \partial(a p)). \end{aligned} \quad (35.103)$$

Here the symmetric difference is not measured as two disjoint curves: the surface connectors fill it by the minimal oriented ribbon $\Sigma(\gamma, a\gamma)$, and

$$\tau(\gamma \triangle a\gamma) := \sum_{p \subset \Sigma(\gamma, a\gamma)} \tau_p, \quad \tau(\gamma \triangle a\gamma) \leq C_\gamma d_{E(4)}(a, e), \quad (35.104)$$

with the same definition for plaquette boundaries. The inequality follows from summing the additive times in (35.17) over a ribbon of width $d_{E(4)}(a, e)$ and length bounded by that of γ . The bridge-time measure on the right-hand side thus tends to zero as $a \rightarrow e$; this is exactly the continuity of the route connectors in the gauge image. Every bounded clover reader $\chi(F_{\text{cl}})$, $\chi \in C_b^1$, is a function of a finite sum of transported plaquettes; the same bound, multiplied by $\|\chi'\|_\infty$ and the number of terms, also proves its continuity.

For merely bounded class functions, the regularization entering the core is made explicit. If db is left Haar measure on $E(4)$ and $\eta \in C_c^\infty(E(4))$, define the Bochner integral

$$A[\eta] := \int_{E(4)} \eta(b) U(b) A U(b)^* db, \quad A \in \{A_{\gamma,f}, A_{p,f}, \chi(F_{\text{cl}})\}, \quad \chi \in C_b^1. \quad (35.105)$$

With $(L_a \eta)(b) := \eta(a^{-1}b)$, the change of variable $b \mapsto ab$ gives

$$U(a)A[\eta]U(a)^* = A[L_a \eta], \quad \|(A[L_a \eta] - A[\eta])\Omega_\infty\| \leq \|A\| \|L_a \eta - \eta\|_{L^1(E(4))} \longrightarrow 0. \quad (35.106)$$

From now on, the holonomy, plaquette, and clover generators of $\mathfrak{A}_{\text{YM}}(O)^{\text{sm}}$ are understood to be those in (35.101) with $f \in C^1$, or their regularizations (35.105), with the swept support remaining compact within O . Approximate identities on $E(4)$, together with (35.103), recover the unregularized generators in L^2 ; hence the cyclic sector is not reduced.

A telescoping product argument using (35.100) and (35.106) proves continuity of all generators of $\mathcal{D}_{\text{sm}}$; by density and unitarity,

$$\lim_{a \rightarrow e} \|(U(a) - I)\Psi\| = 0 \quad (\Psi \in \mathcal{H}_{\text{YM}}^{\text{curv}}). \quad (35.107)$$

Disjoint supports use disjoint factors and commute. This gives the net

$$\begin{aligned} O_1 \subseteq O_2 &\Rightarrow \mathfrak{A}_{\text{YM}}(O_1) \subseteq \mathfrak{A}_{\text{YM}}(O_2), \\ [\mathfrak{A}_{\text{YM}}(O_1), \mathfrak{A}_{\text{YM}}(O_2)] &= 0 \quad (O_1 \cap O_2 = \emptyset), \\ U(a)\mathfrak{A}_{\text{YM}}(O)U(a)^* &= \mathfrak{A}_{\text{YM}}(aO). \end{aligned} \quad (35.108)$$

The Gram identity for $h \neq 0$ and (35.82) prove that the net is nonscalar.

35.11.4 Osterwalder–Schrader axioms and continuous reconstruction

For a direction ν , let $\mathfrak{A}_{+, \nu}$ be the closure of the union of the preceding algebras compactly supported in $\{x_\nu > 0\}$. The static state, stack law, and transfer operator are not defined separately. On a screen α , let $\hat{\mu}_{\alpha, g}^{\text{YM}}$ be the marginal of (35.46), let $\nu_{\alpha, g} := (\pi_\alpha^{\text{phys}})_* \hat{\mu}_{\alpha, g}^{\text{YM}}$ be its law on the quotient, and let $K_{\alpha, g}(t) = e^{-tD_{\alpha, g}^{\text{YM}}}$ be its kernel. For ordered times $t_1 \leq \dots \leq t_n$, the Euclidean marginal of the same measure is

$$\mathbb{P}_{\alpha, g}^{E, (\nu; t_1, \dots, t_n)} := (\text{ev}_{t_1}^{(\nu)}, \dots, \text{ev}_{t_n}^{(\nu)})_* \hat{\mu}_{\alpha, g}^{\text{YM}}. \quad (35.109)$$

The disintegration identity forming part of (34.49) calculates it, rather than replacing it with an auxiliary chain:

$$d\mathbb{P}_{\alpha, g}^{E, (\nu; t_1, \dots, t_n)} = \nu_{\alpha, g}(dy_1) \prod_{j=1}^{n-1} K_{\alpha, g}(t_{j+1} - t_j; y_j, dy_{j+1}). \quad (35.110)$$

If M_A denotes multiplication by a bounded reader A , its correlations are therefore

$$\begin{aligned} S_{\alpha, n}^{(\nu, g)}(A_1, t_1; \dots; A_n, t_n) &:= \int \prod_{j=1}^n (\tau_{t_j e_\nu} A_j)(x) d\hat{\mu}_{\alpha, g}^{\text{YM}}(x) \\ &= \langle \mathbf{1}, M_{A_1} K_{\alpha, g}(t_2 - t_1) M_{A_2} \cdots K_{\alpha, g}(t_n - t_{n-1}) M_{A_n} \mathbf{1} \rangle_{L^2(\nu_{\alpha, g})}. \end{aligned} \quad (35.111)$$

The law of the stationary stack at those same times is this Euclidean marginal, written through its disintegration:

$$d\mathbb{P}_{\alpha, g}^{(\nu; t_1, \dots, t_n)} := d\mathbb{P}_{\alpha, g}^{E, (\nu; t_1, \dots, t_n)} = \nu_{\alpha, g}(dy_1) \prod_{j=1}^{n-1} K_{\alpha, g}(t_{j+1} - t_j; y_j, dy_{j+1}). \quad (35.112)$$

Integrating successively over $y_n, y_{n-1}, \dots, y_1$ gives the exact identity

$$S_{\alpha, n}^{(\nu, g)}(A_1, t_1; \dots; A_n, t_n) = \int \prod_{j=1}^n A_j(y_j) d\mathbb{P}_{\alpha, g}^{(\nu; t_1, \dots, t_n)}(y). \quad (35.113)$$

If all times coincide, the kernels are the identity, and

$$S_{\alpha, n}^{(\nu, g)}(A_1, t; \dots; A_n, t) = \int A_1 \cdots A_n d\nu_{\alpha, g}. \quad (35.114)$$

Thus the static Schwinger state is the equal-time marginal of the same Euclidean measure, and its disintegration is the Markov law; they are not two functionals subsequently identified by name. Compatibility of $\hat{J}_{\alpha, g}$ with the measure and kernel makes (35.113) compatible with every overlay. Its limit defines

$$S_n^{\text{YM}}(A_1, t_1; \dots; A_n, t_n) := \lim_{\alpha} S_{\alpha, n}^{(\nu, g)}(A_{1, \alpha}, t_1; \dots; A_{n, \alpha}, t_n). \quad (35.115)$$

Insertions of Φ are obtained by differentiating the unitaries (35.97) at $t = 0$. Since the transported $Z_{\alpha, c}$ are centered, independent, and uniformly bounded, Hoeffding's lemma gives, level by level,

$$\langle \Omega_\alpha, e^{t\Phi_\alpha(h)} \Omega_\alpha \rangle \leq \exp\left(\frac{9}{16} t^2 \|P_\alpha h\|_{L^2}^2\right). \quad (35.116)$$

The bound passes to the limit by (35.92); all derivatives are continuous in the Schwartz topology and define tempered distributions.

The eight-face form first extends to every finite stack. If F depends on the N positive layers of a screen α , the reflection in (35.112) and conditioning on the central layer give

$$\begin{aligned} & \int \overline{F(\Theta_{\nu} y_{-N}, \dots, \Theta_{\nu} y_{-1})} F(y_1, \dots, y_N) d\Omega_{\alpha, N}^{(\nu, g)}(y) \\ &= \int \nu_{\alpha, g}(dz) \left| \int K_{\alpha, g}(a_{\alpha}/2; z, dy_1) \prod_{j=1}^{N-1} K_{\alpha, g}(a_{\alpha}; y_j, dy_{j+1}) F(y_1, \dots, y_N) \right|^2 \geq 0. \end{aligned} \quad (35.117)$$

Every half-space cylinder belongs to a finite stack; hence (35.117) realizes the OS form before closure. In the OS quotient, if F has insertions at positive times, let $(\tau_s F)(A_1, t_1; \dots; A_n, t_n) := F(A_1, t_1 + s; \dots; A_n, t_n + s)$. Define the translation operator by

$$T_{\text{OS}}^{(\nu)}(s)[F] := [\tau_s F], \quad s \geq 0. \quad (35.118)$$

For half-space cylinders F, G , the Euclidean–Markov equality (35.110) and the semigroup property give, generator by generator,

$$\begin{aligned} \langle [F], T_{\text{OS}}^{(\nu)}(s)[G] \rangle_{\text{OS}} &= S_g^{\text{YM}}((\Theta_{\nu} F)^* \tau_s G) \\ &= \langle \mathcal{V}_{\infty, \nu}[F], e^{-s D_{\text{curv}, g}^{\text{YM}}} \mathcal{V}_{\infty, \nu}[G] \rangle. \end{aligned} \quad (35.119)$$

The identity shows both that null classes remain null and that, by density of the cylinders,

$$\mathcal{V}_{\infty, \nu} T_{\text{OS}}^{(\nu)}(s) \mathcal{V}_{\infty, \nu}^{-1} = e^{-s D_{\text{curv}, g}^{\text{YM}}}. \quad (35.120)$$

This is the explicit arrow Euclidean measure $\rightarrow$ disintegration $\rightarrow$ OS transfer $\rightarrow$ Hamiltonian.

Proposition 35.1 (Continuous Osterwalder–Schrader package). *The functionals (35.115) satisfy:*

1. *normalization, Hermiticity, and bosonic symmetry;*
2. *Euclidean covariance and tempered regularity;*
3. *reflection positivity in each of the four directions,*

$$\sum_{i,j} \bar{a}_i a_j S^{\text{YM}}((\Theta_{\nu} F_i)^* F_j) \geq 0, \quad F_i \in \mathfrak{A}_{+, \nu}; \quad (35.121)$$

4. *locality and exponential clustering: if A, B are centered local observables and a translation of length r separates their supports, then*

$$|S^{\text{YM}}(A \tau_r B)| \leq e^{-3\kappa_{\text{av}} r} \|A \Omega_{\infty}\| \|B \Omega_{\infty}\|. \quad (35.122)$$

OS reconstruction from any of the four half-nets produces the same cyclic space, the same vacuum, and the Hamiltonian $D_{\text{curv}, g}^{\text{YM}}$. OS continuation produces local Wightman fields and a Poincaré representation whose spectrum satisfies

$$\text{Spec}(P^0, \mathbf{P}) \subseteq \bar{V}_+, \quad P^0 = D_{\text{curv}, g}^{\text{YM}} \geq 0. \quad (35.123)$$

Proof. Normalization and Hermiticity originate in the state; symmetry and locality, in Euclidean commutativity for disjoint supports. Covariance and continuity are (35.99)–(35.107); regularity is (35.116). For cylindrical F_i , all lie in a finite overlay and stack; (35.117) and polarization prove (35.121). Approximation in the OS norm and Cauchy–Schwarz extend it to $\mathfrak{A}_{+, \nu}$.

After a Euclidean rotation, spatial separation becomes positive time. The stack–transfer identity (35.113) and the operator (35.118) express the centered correlation as

$$\langle A^* \Omega_{\infty}, e^{-r D_{\text{curv}, g}^{\text{YM}}} B \Omega_{\infty} \rangle.$$

The spectral bound (35.84) gives (35.122). Finally, the connecting maps (35.76) form a directed system; their union is dense by (35.98). The identity (35.77) identifies the four closures and their generators with $D_{\text{curv}, g}^{\text{YM}}$. The already verified hypotheses of regularity, covariance, reflection positivity, and clustering are the inputs to OS continuation; its reconstruction theorem gives Wightman locality and (35.123). $\square$

35.11.5 Four-dimensional reconstruction of Yang–Mills theory

Identification does not rest solely on evaluating a clover on a given connection. Let $\mathcal{R}_G^{\text{loc}}$ be the $*$ -algebra of readers formed by all class functions of incidence and plaquette holonomies, their transports to a base vertex, and polarizations of the reader $\mathcal{P}^{F^2}$. Let

$$\mathcal{E}_G := \Lambda^2(\mathbb{R}^4)^* \otimes \mathfrak{g} \quad (35.124)$$

with the invariant inner product. In each cell, polarization of $\mathcal{P}^{F^2}$ gives a positive Gram form $G_{\alpha,c}^F$ on $\mathcal{E}_G$. Quotient by its kernel and apply its positive square root; denote the resulting normalized reader by $Y_{\alpha,c}(v)$, which satisfies

$$\langle Y_{\alpha,c}(v), Y_{\alpha,d}(w) \rangle = \delta_{cd} \langle v, w \rangle_{\mathcal{E}_G}. \quad (35.125)$$

If $d \subset c$, let $T_{d \leftarrow c}$ be the frame transport already present in the surface holonomy. The complete-curvature intertwiner is

$$I_{\alpha\beta}^F Y_{\alpha,c}(v) := \sum_{d \subset c} \sqrt{\frac{|d|}{|c|}} Y_{\beta,d}(\text{Ad}_{T_{d \leftarrow c}} v). \quad (35.126)$$

Invariance of the inner product on $\mathfrak{g}$, disjointness of subcells, and composition of transports prove

$$(I_{\alpha\beta}^F)^* I_{\alpha\beta}^F = I, \quad I_{\beta\gamma}^F I_{\alpha\beta}^F = I_{\alpha\gamma}^F. \quad (35.127)$$

For $h \in C_c^\infty(\mathbb{R}^4; \mathcal{E}_G)$, set

$$\mathbf{F}_\alpha(h) := \sum_{c \in \mathcal{C}_\alpha^{\text{cell}}} Y_{\alpha,c} \left(|c|^{-1/2} \int_c h(x) d^4x \right). \quad (35.128)$$

In a common overlay, the same calculation as in (35.91) now gives

$$\langle I_{\alpha\gamma}^F \mathbf{F}_\alpha(h), I_{\beta\gamma}^F \mathbf{F}_\beta(k) \rangle = \langle P_\alpha h, P_\beta k \rangle_{L^2(\mathcal{E}_G)}. \quad (35.129)$$

Thus $\mathbf{F}_\alpha(h)$ is Cauchy by strong convergence of the projections, not by a new hypothesis. The same pairing of Haar functions and innovations in (35.94), now applied to the orthonormalized basis of $G_{\alpha,c}^F$, realizes the colimit through a local, gauge-covariant isometry into $L^2(\hat{X}_\infty, \hat{\mu}_{\infty,g}^{\text{YM}})$. Its limit defines the gauge-covariant random distribution

$$\begin{aligned} \mathbf{F}_{\mu\nu} &\in \mathcal{S}'(\mathbb{R}^4; \mathfrak{g}) \hat{\otimes} L^2(\hat{X}_\infty, \hat{\mu}_{\infty,g}^{\text{YM}}), \\ U(a) \mathbf{F}(h) U(a)^* &= \mathbf{F}(a \cdot h), \\ u \mathbf{F}(h) u^{-1} &= \mathbf{F}(\text{Ad}_u h), \quad a \in E(4), u \in G. \end{aligned} \quad (35.130)$$

Its order-three spectral class channel is precisely Φ . The finite surface relation

$$\text{Hol}_p^{(\alpha)} = \overrightarrow{\prod_{p' \subset p}} T_{p'} \text{Hol}_{p'}^{(\beta)} T_{p'}^{-1} \quad (35.131)$$

passes to the cofinal closure because all its cross products are calculated in an overlay. In particular, the random clover reader, not merely its evaluation on a smooth connection, satisfies the small-loop identity

$$\mathbf{F}(h) = L^2\text{-}\lim_{\alpha} a_\alpha^4 \sum_x \langle (P_\alpha h)(x), F_{\text{cl},\alpha}(x) \rangle, \quad \delta_{\text{gauge}} S_n^{\text{YM}} = 0, \quad \text{Alt}_{\lambda\mu\nu} D_\lambda \mathbf{F}_{\mu\nu} = 0. \quad (35.132)$$

The first equality is (35.129) written in the clover representative; the second is the Ward identity (35.57) passed to the limit; and the third is cancellation of the boundary of a boundary in the surface product. None uses the mass term.

To avoid defining the target by the conclusion sought, let $\mathbf{YM}_4(G, g)$ be the class of local quintuples $(\mathfrak{A}, \mu_g, K_g, \text{Hol}, \mathbf{F})$ satisfying the following checkable equations: (i) the Gibbs density is (35.46); (ii) K_g is Markov, reversible, and satisfies (35.55)–(35.57); (iii) holonomy, clover, and curvature satisfy (35.52)

and (35.132); and (iv) their functionals are the same stack law as in (35.113) and satisfy the OS package. The constructed object is, component by component,

$$\begin{aligned} \mathfrak{Q}_{\text{YM}}^{(4)}(G, g) &:= (\{\mathfrak{A}_{\text{YM}}(O)\}_O, S_g^{\text{YM}}, U, \Theta, \Omega_{\infty, g}, \\ &\quad \hat{\mu}_{\infty, g}^{\text{YM}}, \{S_{m, g}, \mathcal{S}_{m, g}\}_m, K_g, \text{Hol}, \mathbf{F}, \\ &\quad \mathcal{H}_{\text{OS}}, D_{\text{curv}, g}^{\text{YM}}) \in \mathbf{YM}_4(G, g). \end{aligned} \quad (35.133)$$

The output map omits none of these components:

$$\begin{aligned} \mathcal{P}_{\text{YM}, g} : \mathfrak{G}_{\text{gau}}(G) &\longrightarrow \mathbf{YM}_4(G, g), \\ (\hat{\mu}^0, \mathbb{P}^{\text{phys}}, D^0, \Theta, W, \text{Hol}, \mathcal{P}^{F^2}) &\longmapsto (\hat{\mu}_g^{\text{YM}}, S_g, \mathcal{S}_g, K_g, \mathbb{P}_g^{\text{phys}}, S_g^{\text{YM}}, \mathfrak{A}_{\text{YM}}, D_{\text{curv}, g}^{\text{YM}}, \mathbf{F}, \text{Hol}). \end{aligned} \quad (35.134)$$

The S_g^{YM} are thus exactly the Schwinger functions of the g -dependent measure, Markov law, and transfer operator already constructed.

It remains to check that $S_{m, g}$ is the Yang–Mills action, not merely a function bearing that name. Let $A \in C_c^3(\mathbb{R}^4; T^*\mathbb{R}^4 \otimes \mathfrak{g})$ be a smooth connection in a chart where the logarithm of small loops is unique. For the plaquette $p_{\mu\nu}(x)$, the ordered Magnus expansion gives

$$\log \text{Hol}_{p_{\mu\nu}(x)}(A) = a_m^2 F_{A, \mu\nu}(x) + \frac{a_m^3}{2} (D_\mu + D_\nu) F_{A, \mu\nu}(x) + R_{m, \mu\nu}(x), \quad (35.135)$$

with $\|R_{m, \mu\nu}(x)\| \leq C_G a_m^4 (\|D^2 F_A\|_{\infty, p} + \|F_A\|_{\infty, p}^2)$. Averaging over the four orientations cancels the odd term and defines

$$F_{\text{cl}, m}(x) := \frac{1}{4a_m^2} \sum_{p \ni x} \text{Ad}_{T_p} \log \text{Hol}_p, \quad \|F_{\text{cl}, m}(x) - F_A(x)\| \leq C_G a_m^2 (\|D^2 F_A\|_{\infty} + \|F_A\|_{\infty}^2). \quad (35.136)$$

The definition of the square root (35.41) is precisely the polarization of these six components. Substituting (35.135) into each generator of this polarization gives

$$\xi_{m, b}(A) = a_m^2 \text{vec}_{\mu < \nu} F_{A, \mu\nu}(x_b) + r_{m, b}(A), \quad \|r_{m, b}(A)\| \leq C_{G, A} a_m^4. \quad (35.137)$$

The odd cross terms cancel through the four clover orientations; the even terms are included exactly once by (35.42). By Riemann summation and Cauchy–Schwarz,

$$\left| S_{m, g}(A) - \frac{1}{4g^2} a_m^4 \sum_x \|F_{\text{cl}, m}(x)\|^2 \right| \leq C_{G, A, g} a_m^2, \quad (35.138)$$

and, using the bound in (35.136),

$$S_{m, g}(A) \longrightarrow S_{\text{YM}, g}(A) := \frac{1}{4g^2} \int_{\mathbb{R}^4} \|F_A(x)\|^2 d^4 x. \quad (35.139)$$

The Schwinger–Dyson and Ward equations have already been calculated on the measure using this action. The gap, by contrast, follows subsequently from the unitary conjugation in (35.53) and the witness (35.58); it was not used to choose g , $S_{m, g}$, or $\hat{\mu}_{m, g}^{\text{YM}}$. This is the noncircular connection between action, law, and gap.

Theorem 35.2 (Yang–Mills closure for every compact, connected, simple Lie group). *The complete structure $\mathfrak{G}_{\text{gau}}$ constructs a local projective collection for every compact, connected, simple Lie group. The map (35.134) yields a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$: its functionals satisfy the continuous OS package, and their continuation satisfies the local Wightman axioms by Proposition 35.1; its holonomies and distributional curvature satisfy (35.132), and the four reconstructions coinduce the same Hamiltonian $D_{\infty, g}^{\text{YM}}$, with a simple vacuum. Its law is $\hat{\mu}_{\infty, g}^{\text{YM}}$, constructed by the compatible collection of local actions $\{S_{m, g}, \mathcal{S}_{m, g}\}_m$, and satisfies the Schwinger–Dyson and Ward equations (35.55)–(35.57). The cyclic curvature sector (35.83) is reducing, contains the witness (35.82), and has*

$$\boxed{\Delta_{\text{YM}}^{\text{HMT}} = 3\kappa_{\text{av}} > 0}, \quad m_{\text{gap}}^{\text{HMT}} = \frac{\hbar_{\text{int}}}{c_{\text{int}}} \Delta_{\text{YM}}^{\text{HMT}} > 0. \quad (35.140)$$

The local net, strong action of $E(4)$, Schwinger distributions, holonomy, and $\mathbf{F}$ originate in the same complete state. The exact reader $\mathcal{P}^{F^2}$ defines the density of the law; the clover is its smooth chart, and (35.139) identifies its classical action. The gap belongs to the collective gauge-invariant curvature sector and does not assign an elementary mass to the gluon.

Proof. Kinematic projectivity is (35.19); the square root (35.41), cocycle (35.45), and density (35.46) first construct the action and measure at coupling g . The factor-by-factor transport (35.48)–(35.51) proves projectivity and preservation of the state, products, and $*$; (35.52) establishes clover and Bianchi. The exhaustive index (35.22), the form (35.27), and the circuit (35.32) construct the common generator. Physical reduction and its quotient kernel are (35.35)–(35.40). The conjugation (35.53) transports them to the g -dependent measure without altering their spectrum. The Hoeffding decomposition proves the vacuum and bound, and (35.62)–(35.64) exhibit a physical vector attaining the threshold. Nonadic intertwining transports form, vacuum, and witness to the limit, where (35.68)–(35.70) determine the exact gap. The four factorizations (35.72), the range inclusion (35.75), and the connecting maps (35.76) identify their reconstructions with that same generator. The holonomy (35.81) places W_c explicitly in the curvature sector, where (35.84) proves the exact gap. The system (35.87) controls cross products through (35.91); (35.95)–(35.96) lift that first chaos to the isomorphism of the entire algebra. Strong convergence of P_α produces Φ ; the bounds (35.103) and (35.106) also prove $E(4)$ -continuity of holonomies, plaquettes, and clover. Euclidean disintegration (35.110), the exact identity (35.113), and the intertwining (35.119) connect the same measure, Schwinger state, Markov stack, OS transfer, and gap generator. Proposition 35.1 extends the four finite forms to the complete half-nets, proves regularity and clustering, and reconstructs the Hamiltonian. Finally, (35.130)–(35.134) identify those same functionals as the Schwinger functionals of the local gauge theory; the integrations by parts (35.55)–(35.57) prove Schwinger–Dyson and Ward. Lastly, (35.135)–(35.139) identify the classical action before spectral equivalence is applied. The chain does not use the gap to choose the measure. $\square$

Falsification criterion 35.3 (Collapse of the product dynamics). *If D_m^{prod} is replaced by D_m^{fib} , all nonconstant sectors receive the same eigenvalue. A Hoeffding vector with $|S| = 2$ should have energy $6\kappa_{\text{av}}$ by (35.4), but the simple semigroup would assign it $3\kappa_{\text{av}}$. Identifying these two generators is false.*

Falsification criterion 35.4 (Confusion between genealogical inclusion and block averaging). *If $I_{\alpha\beta}^{\text{curv}}$ is replaced by $\hat{J}_{\alpha\beta}$ in (35.91), a coarse coordinate persists with coefficient one while the fine coefficient changes by the scale ratio. For $a_{m+1} = a_m/9$, the ancestral cross product carries a factor $1/81$ and does not converge to the diagonal norm. The identity (35.92) fails; this is why the two maps are kept separate.*

Falsification criterion 35.5 (Ablation of the incidence fiber). *If $\hat{X}_m$ is replaced by its link marginal, q_c is erased, along with the physical observable W_c . The forgetful map does not reflect the complete gauge subalgebra; it therefore cannot be used to deny the eigenvalue of the source object. The object has been replaced; the conclusion is not transferable.*

Falsification criterion 35.6 (Replacement of the exact reader by its smooth chart). *If the asymptotic term in (35.139) is used directly as a density on a finite screen, the controlled remainder in (35.138) is erased and the exact cocycle (35.45) is lost. The governing finite density is $Z_{m,g}^{-1} e^{-\mathcal{P}_m^{F^2}/(4g^2)}$; the clover and smooth limit identify it with the classical action but do not replace it.*

Articulation of the proofs. The distinction between simple fiber and product generator, the common measure, gauge compression, the four Osterwalder–Schrader conditions, the limiting form, and the physical witness are exact results previously established in the regular HMT gauge domain. Disintegration with nonuniform times, the exhaustive index of expectations, the quotient kernel, the matrix B^+ , the explicit action on the colimit, the cofinal block system, the continuous OS package, distributional curvature, and incidence holonomy are explicit formalizations: they determine the domains and prove the connections previously stated, retaining the gap as a result. Their complete composition establishes the preceding theorem. The positive square root of the F^2 reader, its Gibbs density, compatible local transport, the Schwinger–Dyson/Ward equations, the complete algebraic isomorphism, the holonomy

continuity bounds, and the stack–transfer identity are explicit formalizations of the four correspondences. Identification of the clover observable with the classical action follows as a consequence of the Magnus expansion.

36 Tritic generators, quadratic actions, and energy transport

Dynamic geometry preserves the distinction between the state generating a realization and the coordinates in which that realization expresses its evolution. APP constructs and preserves the additive and multiplicative sheets; the TRIT orients the operating mode; the TPK composes selection, transport, and update on the enriched state through

$$U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Nonadic holonomy extends this state through $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$. Phase return preserves the advance of memory. Its outputs $\pi_{\text{HMT}}, \varphi_{\text{HMT}}, e_{\text{HMT}}, \alpha_{\text{HMT}}$ belong to the same correlated generation, with the readers and internal constructive order already established. The joint discrete structure of the continuum subsequently supports the geometric and physical representations. At this stage of the construction, a Hamiltonian is the functional realizing an already typed transport on a symplectic fiber; its domain, orientation, and time parameter are part of the statement.

The article on the holographic extreme and mean ratio recovers three concrete operators. Multiplication by 4 modulo 9 acts on the ternary orbits of units; on its augmentation module, its matrix is

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

Parabolic transport and areal–volumetric incidence are represented, on their respective planes, by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Incidence originates in the sheet calendar; its powers count compositions of these modes. The resulting real generators are

$$J_+ = \frac{2C + I}{\sqrt{3}}, \quad J_0 = N, \quad J_- = \frac{2F_{\text{av}}^2 - 3I}{\sqrt{5}}, \quad J_\tau^2 = -\tau I. \quad (36.1)$$

Here $\tau = +1, 0, -1$ distinguishes the elliptic, parabolic, and hyperbolic regimes. The fibers have their own origins and bases. The formulas $e^{\tau J_+} = -I$, $e^{J_0} = I + J_0$, and $e^{2 \log \varphi J_-} = F_{\text{av}}^2$ evaluate previously generated HMT coordinates on these operators [2].

36.1 Alternating form and Hamiltonian of the generator

On an oriented two-dimensional real fiber, fix

$$\omega = dQ \wedge dP, \quad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad z = (Q, P)^T.$$

We adopt $\iota_{X_H} \omega = dH$; therefore $\dot{Q} = \partial_P H$, $\dot{P} = -\partial_Q H$. This convention also fixes the sign of the oriented action.

Proposition 36.1 (Hamiltonian realization of a traceless generator). *For a real two-by-two traceless matrix J , the matrix $G_J = -\Omega J$ is symmetric, and the Hamiltonian*

$$H_J(z) = \frac{1}{2} z^T G_J z \quad (36.2)$$

produces exactly $\dot{z} = Jz$. If $J^2 = -\tau I$, then $\det G_J = \tau$. For a symplectic isomorphism $B : V \rightarrow V'$, the transport

$$J' = BJB^{-1}, \quad H'(z') = H_J(B^{-1}z')$$

satisfies $G' = B^{-T}G_J B^{-1}$ and intertwines both flows.

Proof. Writing $J = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ gives $G_J = \begin{pmatrix} -c & a \\ a & b \end{pmatrix}$. Thus $\Omega \nabla H_J = \Omega(-\Omega J)z = Jz$. Cayley–Hamilton gives $J^2 = -\det(J)I$, and $\det(-\Omega) = 1$. Finally, $B^T \Omega B = \Omega$ allows the substitution $z = B^{-1}z'$ in both the form and the equation. $\square$

Applied to (36.1), the proposition gives

$$H_{J_+} = -\frac{Q^2 - QP + P^2}{\sqrt{3}}, \quad H_{J_0} = \frac{P^2}{2}, \quad H_{J_-} = \frac{-Q^2 - QP + P^2}{\sqrt{5}}. \quad (36.3)$$

The elliptic sign records the orientation of C in the augmentation basis. The reverse flow $-J_+$ has the opposite positive form. The action ellipse of the corpus uses precisely a decreasing phase orientation for its positive Hamiltonian. Preserving this sign makes the cyclic operator compatible with the action PdQ .

Conjugation also provides an exact comparison criterion. If $BJ_\tau = J_{\tau'}B$ and B is invertible, squaring gives $(\tau - \tau')B = 0$, hence $\tau = \tau'$. A connection may preserve ω between fibers in different regimes and preserve their respective memories; conjugation of the generators additionally requires equality of the regime. This distinction makes it possible to study a tritic transition without replacing it by a renaming of the same flow.

36.2 Deformation of the ellipse and Legendre transformation

The coordinate $\eta = \operatorname{artanh}(C^*/A)$, on the domain $|C^*/A| < 1$, defines the symplectic frame $M_\eta = \operatorname{diag}(e^{\eta/2}, e^{-\eta/2})$. Deformation in the two directions is reciprocal and preserves their product. In the normalized chart $(q, p) = M_\eta^{-1}(Q, P)$, consider

$$L_\tau = \begin{pmatrix} 0 & 1 \\ -\tau & 0 \end{pmatrix}, \quad I_{\tau, \eta} = \frac{1}{2}(\tau e^{-\eta} Q^2 + e^\eta P^2).$$

These representatives classify oriented quadratic flows. Their use on a native fiber proceeds through a declared symplectic frame, in accordance with Proposition 36.1. The positive elliptic representative is the one used by the action ellipse; for this representative, I has the dimension of action.

The frame changes can be written explicitly. If $b_+ = \sqrt{\sqrt{3}/2}$ and $b_- = \sqrt{\sqrt{5}/2}$, the matrices

$$B_+ = \begin{pmatrix} b_+ & b_+/\sqrt{3} \\ 0 & 2b_+/\sqrt{3} \end{pmatrix}, \quad B_- = \begin{pmatrix} b_- & -b_-/\sqrt{5} \\ 0 & 2b_-/\sqrt{5} \end{pmatrix}$$

have determinant one and satisfy $(-J_+)B_+ = B_+L_{+1}$ and $J_-B_- = B_-L_{-1}$. For the parabolic regime, take $B_0 = I$. Multiplication of the two columns verifies the equalities, and determinant one proves preservation of the alternating form.

Theorem 36.2 (Variable transport and regular Lagrangian). *Let $\eta(t)$ be of class C^1 and $\omega(t) > 0$. The Hamiltonian*

$$H_{\tau, \text{tr}} = \omega I_{\tau, \eta} + \frac{\dot{\eta}}{2} QP \quad (36.4)$$

preserves $I_{\tau, \eta(t)}$ along its solutions. Its Legendre transformation with respect to P is invertible and gives

$$\mathcal{L}_{\tau, \text{tr}}(Q, \dot{Q}, t) = \frac{(\dot{Q} - \dot{\eta}Q/2)^2}{2\omega e^{\eta}} - \frac{\omega \tau e^{-\eta}}{2} Q^2. \quad (36.5)$$

Proof. The equations are

$$\dot{Q} = \omega e^{\eta} P + \frac{\dot{\eta}}{2} Q, \quad \dot{P} = -\omega \tau e^{-\eta} Q - \frac{\dot{\eta}}{2} P.$$

Differentiating $I_{\tau, \eta(t)}$ and substituting these equations cancels separately the ωQP terms and the terms involving variation of η . The second derivative $\partial_P^2 H = \omega e^{\eta} > 0$ allows one to solve for $P = (\dot{Q} - \dot{\eta}Q/2)/(\omega e^{\eta})$. Substitution into $P\dot{Q} - H$ proves (36.5). Moreover,

$$P\dot{Q} - H_{\tau, \text{tr}} = p\dot{q} - \frac{\omega}{2}(\tau q^2 + p^2),$$

which proves that the mixed term is the variational connection of the moving frame. Conservation follows from the complete transport, not from keeping the semiaxes constant. $\square$

The elliptic result and its action are developed in the variational corpus [2]; the same proof exhibits its two conjugate quadratic realizations. The vanishing determinant of the parabolic form coexists with a regular Legendre transformation: the latter depends on the second derivative with respect to momentum, not on the determinant of the complete form. By contrast, for the linear cotangent lift $H_A(q, p) = p^T A q$, one has $\partial_p^2 H_A = 0$. Its first-order action preserves $\dot{q} = Aq$ through a multiplier; it is a variational object distinct from (36.5).

36.3 Memory return and conserved variational charge

In the calendar memory record, the counters (g, s) receive the increment $(4, 72)$ upon a complete return. The relation $72 = 18 \cdot 4$ fixes

$$v = (1, 18)^T, \quad I_{\text{mem}} = s - 18g, \quad B = \begin{pmatrix} 1 & 0 \\ 18 & 1 \end{pmatrix}.$$

The native stabilizer is

$$N_v = BNB^{-1} = \begin{pmatrix} -18 & 1 \\ -324 & 18 \end{pmatrix}, \quad N_v m = v I_{\text{mem}}(m), \quad N_v^2 = 0. \quad (36.6)$$

On the realified plane $m = (g, s)^T$, equipped with $dg \wedge ds$, the Hamiltonian $H_{\text{mem}} = I_{\text{mem}}^2/2$ produces $dm/du = N_v m$. The parameter u belongs to this dimensionless realization of the record. The chart $q = g$, $p = s - 18g$ is symplectic and transforms the action one-form according to

$$s dg = p dq + d(9q^2). \quad (36.7)$$

Thus, eliminating s from the first-order action gives

$$\mathcal{L}_{\text{mem}} = \frac{1}{2} \left(\frac{dg}{du} \right)^2 + 18g \frac{dg}{du} = \frac{1}{2} \left(\frac{dg}{du} \right)^2 + \frac{d}{du} (9g^2). \quad (36.8)$$

The variational equation is $d^2g/du^2 = 0$, and the momentum conserved in the symplectic chart is $p = I_{\text{mem}}$. The complete affine return originates in the linear Hamiltonian $H_{\text{ret}} = 4I_{\text{mem}}$: its time- $u = 1$ map is $(g, s) \mapsto (g + 4, s + 72)$. Both Hamiltonians commute because they are functions of the same charge, but they realize distinct operations: parabolic stabilization and translation of the record. These formulas recover the variational mechanism of memory, including its boundary term [2].

36.4 Metric curvature and geodesic energy

The metric realization of the regimes uses a radial chart and an inverse length scale $\rho > 0$:

$$g_{\tau,\rho} = dr^2 + S_{\tau,\rho}(r)^2 d\theta^2, \quad S_{\tau,\rho}(r) = \begin{cases} \sin(\rho r)/\rho, & \tau = +1, \\ r, & \tau = 0, \\ \sinh(\rho r)/\rho, & \tau = -1. \end{cases} \quad (36.9)$$

The Gaussian curvature is $K_G = -S''/S = \tau\rho^2$. The chart is considered in its regular interior, $S > 0$; for the spherical realization, $0 < r < \pi/\rho$. The poles are treated using different regular charts. The parameter ρ and the dimensional realization of the metric retain their own specification.

For a realized mass $m > 0$, the geodesic Lagrangian and its Hamiltonian are

$$\mathcal{L}_{\text{geo}} = \frac{m}{2}(\dot{r}^2 + S^2\dot{\theta}^2), \quad H_{\text{geo}} = \frac{1}{2m}\left(p_r^2 + \frac{p_\theta^2}{S^2}\right). \quad (36.10)$$

Indeed, $p_r = m\dot{r}$, $p_\theta = mS^2\dot{\theta}$; the velocity Hessian is $m \text{diag}(1, S^2)$ and is positive and invertible. The radial equation is $\ddot{r} - SS'\dot{\theta}^2 = 0$, whereas $d(mS^2\dot{\theta})/dt = 0$ conserves angular momentum. Positivity of this energy holds for all three curvatures. The indefinite form of the hyperbolic generator in (36.3) and the positive geodesic energy of a negatively curved metric are functions on different spaces. This comparison determines precisely which meaning of curvature enters each statement.

36.5 Constitutive response and modal realization of Maxwell theory

The action sections and angular coordinates precede the constitutive response [15]. In the fixed real orientation $x > y > 0$, with the projectors of the two sheets,

$$T = q_+P_+ + q_-P_-, \quad q_\pm = e^{-x \mp y}, \quad \mathcal{V} = (I - T^{90})(I - T^{120})^{-1}, \quad 0 < q_+ < q_- < 1,$$

the functional calculus produces

$$r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}}, \quad \varepsilon = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{c} = (r_+r_-)^{-1}, \quad \hat{Z} = r_-/r_+.$$

The corresponding dimensional charts preserve exactly $\varepsilon\mu = c^{-2}$ and $\mu/\varepsilon = Z^2$. Permittivity, permeability, and velocity are thus outputs of the same operator, with the incidence sectors fixed by the TPK; the following dynamics uses them as already constructed results.

To exhibit transport to a field Hamiltonian, fix an oriented spatial realization, boundary conditions allowing integration by parts, and a real transverse mode f normalized by

$$\nabla \cdot f = 0, \quad \int |f|^2 = 1, \quad \nabla \times \nabla \times f = k^2 f, \quad k > 0.$$

In transverse gauge and without sources, $A = af$ and the momentum $\Pi = pf$ define an immersion of the modal plane into field space. For homogeneous constitutive constants, the Maxwell Hamiltonian and symplectic form restrict to

$$H_{\text{EM}}(a, p) = \frac{p^2}{2\varepsilon} + \frac{k^2 a^2}{2\mu}, \quad \omega_{\text{EM}} = da \wedge dp. \quad (36.11)$$

Integration by parts gives $\int |\nabla \times f|^2 = k^2$, proving this restriction. With $\omega_k = ck$, the change

$$Q = \sqrt{\varepsilon\omega_k} a, \quad P = \frac{p}{\sqrt{\varepsilon\omega_k}} \quad (36.12)$$

preserves $da \wedge dp = dQ \wedge dP$ and gives $H_{\text{EM}} = \omega_k(Q^2 + P^2)/2$. Its inverse and its equations $\dot{Q} = \omega_k P$, $\dot{P} = -\omega_k Q$ constitute the modal intertwining with the positive elliptic regime. The restriction is invariant

because the Maxwell operator preserves the transverse eigenmode. The Legendre transformation produces $\mathcal{L}_{\text{EM}} = \varepsilon \dot{a}^2/2 - k^2 a^2/(2\mu)$.

Two transverse modes of equal frequency, combined in quadrature, realize the two circular orientations of the corpus. The field satisfies $\mathbf{B} = c^{-1} \hat{\mathbf{k}} \times \mathbf{E}$ and $\omega = ck$. The relation to the oscillator is proved here through the field immersion, modal normalization, and transport of the symplectic form, with the boundary conditions kept explicit. A change in k , cavity geometry, or preparation modifies the realized mode; it leaves the genealogy of ε, μ, c, Z intact.

In the Lorentzian chart of the same sector, coupling to a material section is written using $F = dA$ and an already typed charge q_e . For this variational realization, ∇ is taken to be the Levi-Civita connection of g . On $T^*\mathcal{M}$, the covariant Hamiltonian

$$H_A = \frac{1}{2m} g^{\mu\nu} (p_\mu - q_e A_\mu)(p_\nu - q_e A_\nu)$$

is the Legendre transform of $\mathcal{L}_A = \frac{m}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu + q_e A_\mu \dot{x}^\mu$. The Hessian mg is invertible. The Euler-Lagrange equation gives $m \nabla_u u^\mu = q_e \tilde{F}^\mu{}_\nu u^\nu$. Antisymmetry of F preserves $g(u, u)$. This is the covariant realization with an affine parameter, whereas (36.11) uses the time of a transverse spatial decomposition: both parameters and both domains are defined.

36.6 Graded channels, Clifford representation, and mass operator

The material atlas first produces complete channels of route, sheet, flavor, color, and representation. At finite depth, its Hermitian realization is $\mathcal{H}_K = \bigoplus_{s \in S_K} V_s$, with an orthonormal channel basis and transported parity grading. The exterior algebra of the odd sector generates the CAR: insertion of a mode a_j^* and its adjoint removal a_j satisfy $\{a_i, a_j^*\} = \delta_{ij} I$ and $\{a_i, a_j\} = 0$. The proof counts the exchange signs of complete modes; the masses are evaluated subsequently [4].

In the single-mode exterior sector, the basis $(|0\rangle, |1\rangle)$ identifies the removal operator with $a = N$. The map $e_1 \mapsto |0\rangle, e_2 \mapsto |1\rangle$ is an isometry of the normalized complex plane and intertwines both nilpotent operators. This identification uses the declared inner product and grading; it preserves the distinct meanings of their bases. With the internal i already realized, one obtains

$$\sigma_1 = N + N^*, \quad \sigma_2 = -i(N - N^*), \quad \sigma_3 = NN^* - N^*N. \quad (36.13)$$

Direct multiplication gives $\sigma_j^* = \sigma_j$ and $\sigma_i \sigma_j = \delta_{ij} I + i \varepsilon_{ijk} \sigma_k$. In particular, the nilpotent pair and its adjoint produce a Clifford representation; the adjoint is that of the Hermitian fiber and is transported together with it.

The mass operator on a finite collection of channels $\mathcal{F}$ is constructed by the record, character, and their refinements:

$$M_\beta = S M_0 S, \quad S = R_{\text{act}}^{B_M/2}, \quad R_{\text{act}} = \hbar_{\text{pre}}/\hbar_{\text{ret}} > 0, \quad M_0 > 0. \quad (36.14)$$

Here B_M is the diagonal route-exponent operator, and M_0 contains the refined character relative to the electronic origin. Positivity follows from $\langle v, M_\beta v \rangle = \langle S v, M_0 S v \rangle > 0$ for $v \neq 0$. The sheet product preserves the null sector through its own realization; the exponential of the material character remains positive.

In this realization, write $\hbar := \hbar_{\text{ret}}$, the already constructed action section, and $c := c_{\text{vac}}$ in the declared dimensional chart. On $\mathbb{C}^4 \otimes \mathcal{F}$, define

$$\alpha_j^D = \sigma_1 \otimes \sigma_j, \quad \beta^D = \sigma_3 \otimes I_2.$$

The superscript distinguishes these matrices from the fine-structure constant. The spatial chart 1 + 3 assigns three momentum components $\mathbf{p}$. The multiplication map

$$H_D(\mathbf{p}) = c \sum_{j=1}^3 \alpha_j^D p_j \otimes I_{\mathcal{F}} + c^2 \beta^D \otimes M_\beta \quad (36.15)$$

is the free Dirac realization of this material operator.

Theorem 36.3 (Spinorial factorization of material dispersion). *The matrix (36.15) is self-adjoint and satisfies*

$$H_D(\mathbf{p})^2 = I_4 \otimes (c^2 |\mathbf{p}|^2 I_{\mathcal{F}} + c^4 M_{\beta}^2). \quad (36.16)$$

On an eigenspace $M_{\beta} v = mv$, its energies are $\pm E_m(\mathbf{p})$, where $E_m = \sqrt{c^2 |\mathbf{p}|^2 + m^2 c^4}$.

Proof. The relations in (36.13) give $\{\alpha_i^D, \alpha_j^D\} = 2\delta_{ij}I_4$, $\{\alpha_i^D, \beta^D\} = 0$, and $(\beta^D)^2 = I_4$. The flavor factor commutes with the spinorial matrices. The cross terms vanish, and the squares give (36.16). The vanishing trace and Clifford relations produce both branches, with spinorial multiplicity two when $E_m > 0$. $\square$

In momentum representation, the natural domain is $\{\psi \in L^2(\mathbb{R}^3; \mathbb{C}^4 \otimes \mathcal{F}) : H_D \psi \in L^2\}$. As a measurable Hermitian matrix acting by multiplication, the operator is self-adjoint on this domain. The spatial representation originates in the unitary Fourier transform subsequent to the construction of the channels. The theorem determines the free realization and its exact dependence on HMT mass; the interaction retains its connection operators and its own domain.

On the positive branch $m > 0$, the ordinary Legendre transformation of the symbol $E_m(\mathbf{p})$ is

$$\mathbf{v} = \frac{c^2 \mathbf{p}}{E_m}, \quad \mathcal{L}_m = -mc^2 \sqrt{1 - |\mathbf{v}|^2/c^2}, \quad |\mathbf{v}| < c.$$

Solving $\mathbf{p} = m\mathbf{v}/\sqrt{1 - |\mathbf{v}|^2/c^2}$ and substituting into $\mathbf{p} \cdot \mathbf{v} - E_m$ proves the equality. The spinorial action, by contrast, is first order in the field: $\mathcal{L}_D = \frac{i\hbar}{2}(\psi^* \dot{\psi} - \dot{\psi}^* \psi) - \psi^* H_D \psi$. Its variation gives $i\hbar \dot{\psi} = H_D \psi$; its Hessian in the field velocities is singular. Both statements are compatible because they act on different variational spaces.

For a static Abelian connection in a flat spatial chart, with $\Pi_j = -i\hbar \partial_j - q_e A_j$, constant mass, and vanishing scalar potential, the following holds on smooth compactly supported functions:

$$[\Pi_i, \Pi_j] = iq_e \hbar \epsilon_{ijk} B_k,$$

and the same factorization produces

$$H_D(A)^2 = c^2 \Pi^2 + c^4 M_{\beta}^2 - q_e \hbar c^2 \Sigma \cdot \mathbf{B}, \quad \Sigma_j = I_2 \otimes \sigma_j. \quad (36.17)$$

The flavor identity factors are implicit. The curvature term is the product of two antisymmetric objects: the commutator of covariant derivatives and Pauli multiplication. Its coefficient is fixed when the charge representation and action chart are chosen; it is not selected through a target spectral value.

If V is the already constructed unitary flavor transformation, then

$$(I_4 \otimes V) H_D(M_{\beta}) (I_4 \otimes V^*) = H_D(V M_{\beta} V^*).$$

Unitary mixing preserves the complete spectrum. The positive congruence in (36.14) transports a different structure and may modify the eigenvalues in a fixed metric. For example, $M_0 = \text{diag}(1, 2)$, $S = \text{diag}(2, 1)$ gives $S M_0 S = \text{diag}(4, 2)$. If the inner product is also transported to $G' = S^* S$, the generalized problem $S M_0 S v = \lambda G' v$ recovers the eigenvalues of M_0 through $w = S v$. In this way, a change of flavor basis, material deformation, and metric transport have precise spectral consequences.

36.7 Curvature of the scalar potential in the electroweak realization

The electroweak sector uses the determinant of the same angular operator, $D_A = \det T = e^{-2A_{\text{rad}}}$, together with the already constructed material signatures. In the angular realization with common refinements of W and Z , their signatures $(94, -1)$ and $(95, -1)$ give $m_W^{\leq}/m_Z^{\leq} = e^{-A_{\text{rad}}}$. In the rationalized tree-level convention of the corpus,

$$g^2 = \frac{4\pi_{\text{HMT}} \alpha_{\text{HMT}}}{1 - D_A}, \quad \lambda_H = \frac{\pi_{\text{HMT}} \alpha_{\text{HMT}}}{2(1 - D_A)} \exp\left(6A_{\text{rad}} + \frac{10}{3} C_{\text{rad}}^*\right) > 0.$$

The second formula follows from the scalar signature (97, 9) and $m_H^2/m_W^2 = 8\lambda_H/g^2$. The potential normalization is $V(H) = -\mu_H^2 H^*H + \lambda_H(H^*H)^2$. These compositions preserve their scale and tree-level convention; the refinements that cancel in a ratio are declared in the realization itself.

In a real radial chart $H^*H = h^2/2$, the potential is $V(h) = -\mu_H^2 h^2/2 + \lambda_H h^4/4$. For $\mu_H^2 > 0$, its critical points are 0 and $\pm v$, with $v^2 = \mu_H^2/\lambda_H$, and

$$V''(0) = -\mu_H^2, \quad V''(\pm v) = 2\mu_H^2.$$

In the natural units of this chart, the normalized homogeneous mode has $H_h = p_h^2/2 + V(h)$. Its linearization at a critical point h_0 has generator

$$A_{h_0} = \begin{pmatrix} 0 & 1 \\ -V''(h_0) & 0 \end{pmatrix}, \quad A_{h_0}^2 = -V''(h_0)I.$$

Thus the origin is hyperbolic and the minima are elliptic; the boundary $V'' = 0$ is of parabolic type. This statement follows from the second derivative of the potential and an explicit modal normalization. The positive material energy of the minimum and the unstable direction of another background thus belong to the same function with different expansion points. The linearized mode is a particular transport of the three quadratic types; its frequencies and units originate in the potential, not in identifying its Hessian with the Gaussian curvature in (36.9).

36.8 Yang–Mills curvature mode and elliptic oscillation

The Yang–Mills realization constructed in the preceding sections has a positive generator D^{YM} of inverse-length dimension. The scale

$$\kappa_{\text{av}} = \frac{\mu_{\text{vol}}}{\ell_0} \left(\frac{\pi_{\text{HMT}}}{\varphi_{\text{HMT}}^2} - 1 \right) > 0, \quad \delta = 3\kappa_{\text{av}}$$

descends from sheet incidence and the regional reader, with the declared length unit ℓ_0 . The vacuum is simple, and the curvature witness in the product chart is

$$W(q) = \mathbf{1}_{\{q_1 + \dots + q_6 = 0\}} - \frac{1}{3}, \quad \mathbb{E}W = 0, \quad \|W\|^2 = \frac{2}{9}, \quad D^{\text{YM}}W = \delta W.$$

Unitary transport to the interaction measure takes W to W_g and preserves these identities. The refinement inclusions J_m intertwine the generators and preserve the witness, so it persists in the physical limit. Here the complete result of the vacuum, uniform-gap, and reconstruction proofs is used, rather than an isolated eigenvalue of a finite matrix [14].

Let $w = W_g/\sqrt{2/9}$ be the unit vector in the physical fiber and $\hbar = \hbar_{\text{ret}} > 0$. The energy Hamiltonian is $H_E = \hbar c D^{\text{YM}}$. On the realified Hilbert space, adopt the form $\omega_{\mathcal{H}}(u, v) = 2\hbar \text{Im}\langle u, v \rangle$, with an inner product antilinear in the first argument. Define

$$\mathfrak{F}(Q, P) = \frac{Q + iP}{\sqrt{2\hbar}} w. \quad (36.18)$$

Theorem 36.4 (Symplectic immersion of the physical curvature mode). *The map $\mathfrak{F} : \mathbb{R}^2 \rightarrow \mathcal{H}^{\text{phys}}$ is a real immersion whose image is invariant under physical evolution and satisfies*

$$\mathfrak{F}^* \omega_{\mathcal{H}} = dQ \wedge dP, \quad \langle \mathfrak{F}, H_E \mathfrak{F} \rangle = \frac{c\delta}{2} (Q^2 + P^2). \quad (36.19)$$

The restricted Schrödinger equation exactly intertwines the elliptic flow $\dot{Q} = c\delta P$, $\dot{P} = -c\delta Q$. The map is compatible with the refinement inclusions.

Proof. For two variations $(a, b), (a', b')$, the inner product of their images has imaginary part $(ab' - ba')/(2\hbar)$. This proves the first identity. The eigenvalue equation and $\|w\| = 1$ give the second. Since $\dot{\psi} = -icD^{\text{YM}}\psi$, the coefficient $Q + iP$ satisfies $\dot{Q} + i\dot{P} = -ic\delta(Q + iP)$. Finally, $J_m w_m = w_{m+1}$ implies $J_m \mathfrak{F}_m = \mathfrak{F}_{m+1}$. These identities pass to the limit through the already constructed isometries. $\square$

The complete plane describes amplitudes of one mode; its circle $Q^2 + P^2 = 2\hbar$ represents the normalized vectors of this complex eigenspace. On this circle, the global phase evolves and the quantum ray remains fixed. Superpositions of distinct eigenspaces also retain their relative phases. The immersion preserves this distinction between amplitude space and projective space.

The energy associated with the spectral gap and the mass gap are $\Delta_E = \hbar c\delta$ and $m_{\text{gap}} = \hbar\delta/c$. The modal frequency $c\delta$ has inverse-time dimension. The Euclidean lengths in $e^{-sD^{\text{YM}}}$ and the physical time in $e^{-itH_E/\hbar}$ are related through the chart and reconstruction, preserving their distinct groups and semigroups. The composition $\mathfrak{F} \circ M_\eta^{-1}$ transports this same energy to the action ellipse; if the frame varies, the connection term in (36.4) completes its evolution.

The resulting connection is an exact realization on the invariant curvature mode and its extensions. Positivity of the quantum Hamiltonian generates a rotation in its symplectic realification; the designation of a spatial metric as hyperbolic pertains to the metric tensor, and the unipotent transport of the record pertains to its memory. In finite dimension, a group $I + tN$ unitary for every $t \in \mathbb{R}$, with $N^2 = 0$, forces $N = 0$: differentiating unitarity gives $N^* = -N$, and then $N^*N = -N^2 = 0$. The classical parabolic flow preserves its symplectic form and has a quantization on suitable spaces; this identity only delimits its identification with a finite unitary matrix.

A collection of concrete transports is thereby articulated. Memory produces a charge and a parabolic action; the ellipse frame produces the variational connection and its Lagrangian; the constitutive responses determine the Maxwell coefficients; the channel grading constructs Clifford and allows factorization of the material operator; Yang–Mills reconstruction supplies a symplectic modal immersion preserving exactly its gap and refinement. Each identification has a map, a preserved form, and an intertwined equation.

37 Discussion: reconstruction, invariants and duality

The principal structural result is recoverability of form. The uniform charge and the two nonadic memories reconstruct the record through an explicit inverse; the positive cone describes exactly the domain in which that reconstruction induces strictly positive minors. The stability bound quantifies the persistence of this domain. Thus, the geometric weights admit an operator-theoretic origin and recovery, in addition to their numerical evaluation.

The composition developed here has an explicit sequence. The terminal record generates twelve positive weights; their partial sums give thirteen ordered points; a planar network realizes their minors; the moment curve constructs polygons and polytopes; their triangulations provide coverage and forms with oriented residues. Refinement of a fixed cell preserves the sum of its forms. Introducing new vertices on the moment curve, by contrast, produces another geometry and requires comparing domains.

The algebraic direction of the record distinguishes two operations. Positive column rescaling preserves the relevant projective ratios. Its normalized action on the intervals instead generates an effective deformation, certified by an exact nonzero derivative. This deformation admits a convex potential whose Legendre transform is preserved by subdivision with inherited direction. Invariance thus has a verifiable mechanism: each subcell preserves its predecessor cell's generator and the sum of its weights recovers the preceding weight.

Energy refinement provides a second exact conservation law. Schur reduction recovers the effective Laplacian and the minimizing prolongation separates each configuration into its observed part and a reconstructible detail. The detail energies add orthogonally. With decreasing meshes and uniformly bounded energy, this sequence converges in the declared Sobolev realization. Invariance of the canonical

form, identity of the convex potential and conservation of energy are three distinct compatibility results, proved for the specified refinements.

Prolongation to the twelve lattice planes preserves ternary symmetry and volume. The decisive result is the reciprocity $\Lambda_s^* = \Lambda_{-s}$, which induces the inversion $\Theta_s(t) = t^{-12}\Theta_{-s}(t^{-1})$. The split realization of signature $(24, 24)$ preserves integrality, evenness and self-duality throughout the flow. In the Euclidean projection, the explicit variation of a norm distinguishes deformation from an automorphism of the initial lattice. This distinction determines the precise content of the connection with momentum–winding duality.

The material realization preserves its own readers. The action readout and the mass character receive the record through already constructed compositions; congruence preserves positivity and inertia, while the metric determines the interpretation of their eigenvalues. These differences specify the meaning of multimorphology: several mathematical forms arise from a common genealogy and are related by verifiable operations. Full identification of higher-rank amplituhedra will require coverage and residue proofs for those specific maps. The rank-one result proved here establishes a complete case, with rational coordinates, triangulation and reproducible checks.

The temporal orientation of TRIT integrates return with memory, threshold update and conjugate prolongations. Its elliptic, nilpotent and hyperbolic generators provide distinct evolution laws; TPK composition preserves the genealogy that concatenates them. In particular, the activity of the zero regime is expressed by unipotent transport, and the double projection simultaneously preserves coordinate convergence and exceptional incidence. The article’s geometric thesis lies in this articulation: the initial structure generates recoverable records, and these support positive realizations whose transformations, boundaries and memories can be calculated and compared precisely.

The intertwining relation $B_F M = M B_V$ adds exact transport from the modular block to the incidence quotient. In it, multiplicities one and three follow from the Gram matrix, and the declared signed form is expressed as $J_L = (7I_Q - \bar{B}_F)/2$. Vandermonde and Hankel matrices, in turn, provide explicit realizations $Y^{(k)}$ for $2 \leq k \leq 9$, with positivity and rank proved on the initial partition. Nonadic prolongation subsequently extends this construction to every admissible finite pair of rank and external dimension, through the node and moment systems developed here. These results extend the system of realizations without identifying the dimension of a quotient, the rank of a matrix and the number of generators of a superalgebra.

In the gauge realization, the areal–volumetric scale precedes the Hamiltonian. The product generator preserves all sectors of the orthogonal decomposition, while its physical restriction contains a witness attaining the first positive level. The refinement intertwiners preserve coercivity and this witness. The continuum development also brings together the common measure, curvature, Euclidean continuity, reflection forms and the reconstructed Hamiltonian. Thus, the gap argument is situated alongside the construction that gives it a domain and meaning, while Plücker positivity retains its own geometric realization. Their articulation requires these explicit transports; the term positivity denotes typed properties that the common state allows one to compare without substituting one for another.

Part IV

Exact verification and enumeration of emissions

38 Exact verification of the finite constructions

Reproduction distinguishes the general proof from the evaluation of a specialization. The preceding propositions use algebraic identities, positivity and explicit convergence hypotheses. Their computational checks use integers and rational fractions; the lattice development additionally preserves its exact representation in the declared algebraic extension. The record data are outputs of the preceding construction. Executions in this section check their geometric and operator-theoretic transport, while keeping the provenance of their generation separate.

The first check recovers separations from minors, verifies the Plücker relation in the triangulations and compares chart changes with effective deformations. An invertible change of rows multiplies every minor by the same determinant; the marked ratios remain equal. Independent column rescaling preserves certain cross-ratios and, in general, changes the reconstructed separations. Evaluating both operations on the same data provides a negative control for the equivalence asserted.

The second check subdivides lengths by nine, preserves the inherited markings and verifies the averaging and Gram identities. Determinants are computed by exact elimination and compared with their Vandermonde products. Positivity in any finite dimension follows from the proved formula; the finite cases executed check the implementation of that formula, its indices and its normalizations. Counting measures and chronological measures are kept separate so that equality of moments does not depend on silently changing normalization.

The third check reproduces the first eight moments through the order-four Jacobi operator. The symmetric matrix is represented rationally in the monic-polynomial basis; its metric is $\text{diag}(h_0, \dots, h_3)$. Self-adjointness in that metric, the three-term recurrence and agreement between the matrix resolvent and its continued fraction at $z = 2$ are verified. The result keeps explicit the observed constant vector and the charge Q , which enter the marked spectral fidelity theorem.

The fourth group checks symplectic changes and their Hamiltonians. Differentiation with respect to momentum recovers velocity; the Legendre transform is checked as a polynomial or Laurent identity. The inertia of a quadratic form and positivity of a quantum Hamiltonian belong to their respective spaces. The Clifford representation is checked through its anticommutators, and composition with the mass operator is proved by squaring the Dirac operator. These equalities identify each term and its cross-products.

Finally, the preceding checks of triangulation, canonical form, Schur reduction, mass congruence and lattice construction are preserved. In Yang–Mills, the propositions on the common measure, refinement and reconstruction are set out in the text with their premises and falsifiers. A finite matrix check does not replace passage to the limit or field reconstruction: their domains are recorded separately in the reproducible delivery.

The source package links each program to the statement it evaluates and preserves its complete output. Conditions are checked through explicit branches, so that they remain active when the interpreter is run in optimized mode. They are also reproduced in isolated mode, without loading user packages. The correspondence between construction, proof and execution can be inspected step by step without turning a program's output into an assertion of broader scope than what it checks.

39 Census of residual-emitter signatures

This census specifies the finite fibers used in Section 4. Each initial condition is identified by

$$\nu = 4(9i + j) + d, \quad 0 \leq i, j \leq 8, \quad d = 0, 1, 2, 3 \longleftrightarrow N, E, S, O.$$

Euclidean division recovers $d = [\nu]_4$, $j = [\lfloor \nu/4 \rfloor]_9$ and $i = \lfloor \nu/36 \rfloor$. Thus $0 \leq \nu < 324$ enumerates each condition exactly once, and pairs of these identifiers enumerate the 104 976 conditions in Ω_{seed} .

Let $f_+(\nu)$ and $f_\times(\nu)$ be the six-component signatures obtained from the emission recurrence. At every active step the cell is read before the cursor advances; the direction vectors are

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

Coordinates are reduced modulo nine and the vector is reversed after steps 27 and 54. Zero-phase steps do not move either cursor. Together with (4.2), these rules determine f_+ and $f_\times$ without a target-value catalog.

The table gives their complete images. The multiplicity m_σ counts identifiers in the fiber $f_\sigma^{-1}(a)$; $\nu_{\min}$ is its least element. Each word retains its initial zeros. The complete fiber is determined by

$$f_\sigma^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_\sigma(\nu) = a\}.$$

f_+	m_+	$\nu_{\min}$	$f_\times$	$m_\times$	$\nu_{\min}$
122564	18	17	000000	108	8
122898	18	24	177393	8	1
212645	18	5	177555	8	54
212988	18	0	177771	8	11
221456	18	29	339555	8	6
221889	18	12	339771	8	15
456221	18	28	339933	8	59
465898	18	21	393177	8	0
546988	18	9	393393	8	45
564122	18	16	393555	8	40
645212	18	4	555177	8	52
654889	18	33	555339	8	4
889221	18	13	555393	8	41
889654	18	32	555555	24	84
898122	18	25	555717	8	14
898465	18	20	555771	8	18
988212	18	1	555933	8	24
988546	18	8	717555	8	12
			717717	8	21
			717933	8	19
			771177	8	9
			771339	8	13
			771555	8	16
			933339	8	57
			933555	8	26
			933717	8	17

Evaluating the recurrence for the 324 identifiers gives the two partitions in the table: $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. Their fiber product yields 432 fibers of size 144, 18 of size 432, and 18 of size 1944, by injectivity of (4.3). Finally, applying (4.4) to that product of images gives 243 distinct words. This finite census belongs to the residual emitter; subsequent coinductive refinement retains its own law.

40 Conclusions

The geometric thesis is established through reconstruction and transport: a positive readout of the discrete structure of the continuum preserves, with its markings, the record from which it arises and the readers that record determines. Total charge and ratios of minors recover each coordinate; the two nonadic memories provide another operator-theoretic reconstruction. Value and form are thus articulated by explicit inverses, after APP-TRIT-TPK generation and its compatible prolongations. The double projection keeps the arithmetic realization and exceptional incidence together within that genealogy.

Refinement overcomes the limitation of a matrix with thirteen columns. For any finite rank and external dimension, a sufficiently high nonadic level produces the positive minors and the corresponding amplituhedral image. Inherited nodes preserve their flags; cell averages preserve the measure and record the orthogonal detail. In rank one, simplicial coverage and residue recursion extend to every finite dimension. In higher ranks, the result determines compatible positive collections with proved rank and limit; their dimensional scope is distinguished from the global classification of all amplituhedral cells.

Conservation admits three complementary expressions. The canonical form adds under subdivision of a fixed support; effective energy is preserved by Schur reduction; and the chronological measure determines moments whose increments decompose into recoverable details. These moments construct Jacobi operators whose finite compressions have simple spectrum and positive residues. This yields an effective passage from arithmetic separations to a spectral representation, with a continued fraction determined by the same measure. Mutations changing only the chart preserve the readers of the marked point.

The physical realization gains precision through its operators. The Hamiltonians of the three tritic regimes have explicit symplectic forms and Legendre transforms. The variable elliptic chart incorporates its connection term; parabolic memory preserves its invariant; hyperbolic opening has its own evolution. The readers of action and constitutive response are transported through the Grassmannian inverse. With the declared regional outputs and integer markings, the labelled constitutive response in turn recovers the record. The equality-of-traces condition governs the gluing of forms valued in the action line.

The exceptional deformation preserves volume and ternary symmetry, and converts parameter inversion into lattice duality. Its theta reciprocity and split-signature realization link geometric variation with lattice duality. In Yang-Mills, the areal-volumetric scale, common measure, uniform coercivity and persistent witness place the gap alongside the Hamiltonian realizing it. The development preserves the physical space and the intertwiners required to transport that assertion through refinement.

The joint result is an architecture of forms with memory: dimension can increase, representation can change and the relevant information remains recoverable through specified maps. This is the article's distinctive content. Positive geometry, lattice duality and the spectral realization are linked by constructions specifying what they generate, what they preserve and how they are reconstructed from the same discrete origin. The incidence representation also preserves the uniform component of the record: its inverse reconstructs all twelve coordinates and allows the geometric direction to be recovered after memory transport. In the sectorial realization, total area and weighted occupation jointly determine the hexad-octad distribution. The specified purification converts that distribution into an entanglement structure, with a finite identity relating entropy, sector areas and relative entropy. Information conservation is thus specified for each application: the full record, direction, area and composition have explicitly distinct inverses or fibres. The area scale is determined by a length generated through the linear composition of recorded return and the local Hadamard norm. Radial reciprocity determines the gravitational coefficient in the declared reduced realization and subsequently identifies that length as the Planck length. Mutations of the same marked point and inherited refinements preserve the area operator, its multiplicities and the modular identity. Canonical forms valued in that operator extend conservation to residues, while each horizon radius retains its own geometric factor. This composition makes the dependence between length, incidence and area explicit without altering the sectorial states.

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